

Project Data Page:





Table of Contents

Cover Page	1
Vessel Design Summary:	2
Nozzle Summary:	5
Nozzle Schedule:	6
Bill of Material:	7
MDMT Summary:	8
Warnings and Errors:	9
Input Echo:	10
XY Coordinate Calculations:	18
Internal Pressure Calculations:	19
Element: Channel head	20
Element: Channel shell	22
Element: Shell	24
Element: Shell head	27
Shop Pressure Test Results:	30
Field Pressure Test Results:	32
ASME TS Calc: Pressure Test	34
External Pressure Calculations:	52
Element and Detail Weights:	53
Nozzle Flange MAWP:	56
Wind Load Calculation:	57
Earthquake Load Calculation:	59
Center of Gravity Calculation:	61
Saddle Calcs: Operating Case:	62
Saddle Calcs: Test Case:	79
Saddle Calcs: Ext. Pressure:	94
Nozzle Calcs.: Manhole	107
Nozzle Calcs.: N1- Tube inlet	113
Nozzle Calcs.: N2- Tube outlet	124
Nozzle Calcs.: Shell outlet-1	135
Nozzle Calcs.: Shell inlet-1	145
ASME TS Calc: DESIGN CASE	155



DESIGN CALCULATION

In Accordance with ASME Section VIII Division 1

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\103-C2.pvdb

Date of Analysis : Apr 10,2026 3:13pm

PV Elite 28, January 2026



PV Elite 28 Licensee: SL User

FileName : 103-C2

Page 2 of 192

Vessel Design Summary: Step: 28 3:14pm Apr 10,2026

Vessel Design Summary:

ASME Code, Section VIII Division 1, 2025

Diameter Spec : 2666.000 x 2660.000 mm ID
Vessel Design Length, Tangent to Tangent 8.80 m
Specified Datum Line Distance 0.00 m
Shell Side Design Temperature 343 °C
Channel Side Design Temperature 343 °C
Shell Side Design Pressure 14.281 N/mm²
Channel Side Design Pressure 4.950 N/mm²
Shop Shell Side Test Pressure 19.279 N/mm²
Field Shell Side Test Pressure 18.565 N/mm²
Shop Channel Side Test Pressure 7.032 N/mm²
Field Channel Side Test Pressure 6.435 N/mm²
Wind Design Code ASCE/SEI 7-16
Earthquake Design Code ASCE/SEI 7-16

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-387 11	2	...	K11789	+IT
Head	SA-387 11	2	...	K11789	
Nozzle	SA-336 F11	3	...	K11572	+IT
Nozzle	SA-105	...	...	K03504	
Nozzle Flg	SA-105	...	...	K03504	
Tubes	SA-213 TP316L	...	...	S31603	
Tubesheet	SA-336 F11	3	...	K11572	
Hrz Bolting	SA-193 B7	...	<= 64	G41400	
Saddles	SA-516 70	...	...	K02700	

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized
+Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (N/mm² & mm):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
Channel head	4.951	0.00	5.350	2.0000	N/A	No
Channel shell	4.951	0.00	5.210	2.0000	N/A	No
Shell	14.307	0.10	14.500	3.0000	N/A	No
Shell head	14.307	0.10	14.700	3.0000	N/A	No

Liquid Level: 2.67 m Dens.: 999.542 kg/m³ Sp. Gr.: 1.000

Element Types and Properties:

Element Type	"To" Elev m	Element Length m	Nominal Thickness mm	Finished Thickness mm	Reqd Thk Internal mm	Reqd Thk External mm	Long Eff	Circ Eff
Sphere	0.0	0.0	34.0	28.0	26.0	...	1.00	1.00
Cylinder	2.5	2.5	50.0	50.0	47.6	...	1.00	1.00



PV Elite 28 Licensee: SL User

FileName : 103-C2

Page 3 of 192

Vessel Design Summary: Step: 28 3:14pm Apr 10,2026

Cylinder	8.8	5.8	142.0	142.0	139.8	...	1.00	1.00
Sphere	8.8	0.0	82.0	70.0	68.1	...	1.00	1.00

External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	External Design Pressure N/mm ²	External M.A.W.P. N/mm ²
10	20	28	No Calc	...	3.77
20	Tsht	50	No Calc	...	5.01
Tsht	40	50	No Calc	...	No Calc
40	50	142	15.1577	0.10131	14.8
50	60	70	6.70047	0.10131	12.1

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners m	Allowable Length Bet. Stiffeners m	Ring Inertia Required mm ⁴	Ring Inertia Available mm ⁴
10	20	No Calc	No Calc	No Calc	No Calc
20	Tsht	2.9443	No Calc	No Calc	No Calc
Tsht	40	No Calc	No Calc	No Calc	No Calc
40	50	6.24336	25400	No Calc	No Calc
50	60	No Calc	No Calc	No Calc	No Calc

Local Stress Analysis Results:

Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
N1- Tube inlet	WRC-297	0.947	Shell	Passed
N2- Tube outle	WRC-297	0.947	Shell	Passed
Shell outlet-1	WRC-297	0.976	Shell	Passed
Shell inlet-1	WRC-297	0.978	Shell	Passed

Foundation/Support Summary

Saddle Parameters:

Saddle Width	254.000 mm
Saddle Bearing Angle	180.000 deg.
Centerline Dimension	2300.000 mm
Wear Pad Width	304.000 mm
Wear Pad Thickness	30.000 mm
Wear Pad Bearing Angle	180.000 deg.
Distance from Saddle to Tangent	1150.000 mm

Baseplate Length	2500.000 mm
Baseplate Thickness	100.000 mm
Baseplate Width	552.000 mm
Number of Ribs (including outside ribs)	5
Rib Thickness	35.000 mm
Web Thickness	35.000 mm
Height of Center Web	306.000 mm
Number of Bolts in Baseplate	8

Baseplate Sketch

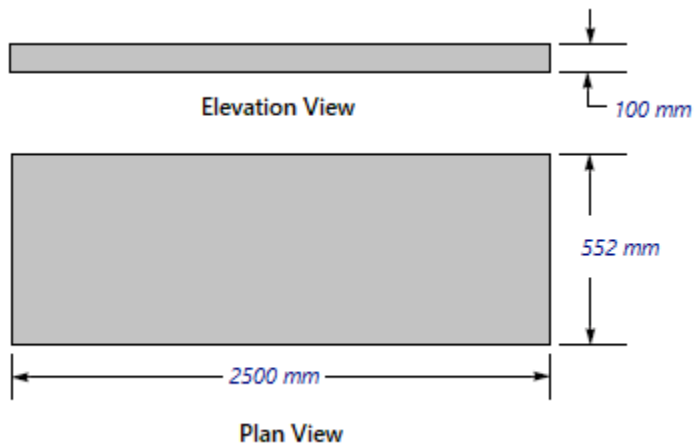
PV Elite 28 Licensee: SL User

FileName : 103-C2

Page 4 of 192

Vessel Design Summary:

Step: 28 3:14pm Apr 10, 2026



Maximum Tensile Bolt Load	239. N
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Summary of Maximum Saddle Loads, Operating Case, Un-Factored:

Maximum Vertical Saddle Load	3205543.75	N
Maximum Transverse Saddle Shear Load	285195.38	N
Maximum Longitudinal Saddle Shear Load	352826.41	N

Summary of Maximum Saddle Loads, Pressure Test Case:

Maximum Vertical Saddle Load	3704416.50	N
Maximum Transverse Saddle Shear Load	7223.34	N
Maximum Longitudinal Saddle Shear Load	4129.96	N

Weights:

Fabricated	- Bare w/o removable internals	193815.6	kgm
Shop Test	- Fabricated + water (full)	443948.5	kgm
Shipping	- Fab. + rem. intls.+ shipping app.	193815.6	kgm
Erected	- Fab. + rem. intls.+ insul. (etc)	344615.6	kgm
Empty	- Fab.+ intls. + details + wghts	344615.6	kgm
Operating	- Empty + operating liquid (no ca)	383289.0	kgm
Field Test	- Empty weight + test medium (full)	443948.5	kgm

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FileName : 103-C2

Page 5 of 192

Nozzle Summary:

Step: 26 3:14pm Apr 10,2026

Nozzle Calculation Summary (N/mm² & mm):

Description	MAWP	Ext	MAPNC	UG-45	[tr]	Weld Path	Areas or Stresses
Manhole	4.95	...	...	OK	13.51	OK	Passed
N1- Tube inlet	4.95	...	...	OK	18.71	OK	Passed
N2- Tube outlet	4.95	...	...	OK	18.71	OK	Passed
Shell outlet-1	14.3	OK	...	OK	17.11	OK	Passed
Shell inlet-1	14.3	OK	...	OK	17.13	OK	Passed

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 14.3 Nozzle : Shell outlet-1 [Shellside]
 Minimum MAWP Nozzles : 4.95 Nozzle : N1- Tube inlet [Tubeside]

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	X Coordinate mm	Layout Angle deg	Dia. Limit mm
10	Manhole	0.000	0.000	1231.200
20	N1- Tube inlet	1100.000	0.000	1840.800
20	N2- Tube outlet	1100.000	180.000	1840.800
40	Shell outlet-1	4200.000	0.000	532.000
40	Shell inlet-1	5500.000	180.000	532.000

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.

lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

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FileName : 103-C2

Page 6 of 192

Nozzle Schedule:

Step: 25 3:14pm Apr 10,2026

Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia in	Wall Thk mm	Reinforcing Diameter	Pad Thk mm	Cut Length mm	Flg Class
Shell outlet-1	10.000 in	Actual	None	10.000	28.575	...	...	548.08	...
Shell inlet-1	10.000 in	Actual	None	10.000	28.575	...	...	548.08	...
Manhole	24.000 in	Actual	WNF	25.938	24.610	...	...	503.99	900
N1- Tube inlet	36.000 in	Actual	None	38.750	34.924	...	...	666.83	...
N2- Tube outlet	36.000 in	Actual	None	38.750	34.925	...	...	666.83	...

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
Shell outle	SA-105	145.000	10.000	...	...	...
Shell inlet	SA-105	145.000	10.000	...	...	...
Manhole	SA-336 F11	50.000	10.000	...	...	...
N1- Tube in	SA-336 F11	50.000	15.000	...	...	...
N2- Tube ou	SA-336 F11	50.000	15.000	...	...	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum m	Layout Angle deg	Proj Outside mm	Proj Inside mm	Installed in Component
Shell outlet-1	4.200	0.0	400.00	0.00	Shell
Shell inlet-1	5.500	180.0	400.00	0.00	Shell
Manhole	...	0.0	396.00	0.00	Channel head
N1- Tube inlet	1.100	0.0	432.00	0.00	Channel shell
N2- Tube outlet	1.100	180.0	432.00	0.00	Channel shell

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FileName : 103-C2

Page 7 of 192

Bill of Material:

Step: 24 3:14pm Apr 10,2026

Bill of Materials:

QTY	DESCRIPTION	MATERIAL
1	SPHERICAL HEAD: 34.0mm THK X 2666.0mm ID X 0.0m	SA-387 11
1	SPHERICAL HEAD: 82.0mm THK X 2660.0mm ID X 0.0m	SA-387 11
1	CLADDING: 1333.000mm X 3.000mm THK	...
1	CLADDING: 2500.000mm X 3.000mm THK	...
7	BAFFLES: 2648.0mm DIA X 19.0mm THK	SA-516 70
20	TIE RODS: 19mm DIA X 5500mm	SA-193 B7M
1	TUBESHEET: 2980.0mm X 500.0mm THK	SA-336 F11
5546	TUBES: 5500.0mm X 25.4mm DIA X 3.0mm THK	SA-213 TP316L
2	SADDLE: 254.0mm X 180 DEG	SA-516 70
1	CLASS 900 GR 1.1, 24.0" BLIND FLANGE(S)	SA-105
2	CYLINDRICAL SEGMENT 50.0mm THK X 3326.9mm X 2.5m	SA-387 11
1	CYLINDRICAL SEGMENT 50.0mm THK X 1721.6mm X 2.5m	SA-387 11
2	CYLINDRICAL SEGMENT 142.0mm THK X 3217.3mm X 5.8m	SA-387 11
1	CYLINDRICAL SEGMENT 142.0mm THK X 1921.9mm X 5.8m	SA-387 11
1	NAMEPLATE	...

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FileName : 103-C2

Page 8 of 192

MDMT Summary:

Step: 27 3:14pm Apr 10,2026

Minimum Design Metal Temperature Results Summary (°C):

Description	Curve	Basic MDMT	Reduced MDMT	UG-20(f) MDMT	Thickness ratio	Gov Thk mm	E*	PWHT reqd
Notes								
Shell head [9]	A	43	41		0.969	70.000	1.00	Yes
Left Saddle [24]	A	7						No
Right Saddle [24]	A	7						No
Channel head [9]	A	22	18		0.924	28.000	1.00	Yes
Channel shell [8]	!	-12	-15		0.949	50.000	1.00	Yes
Manhole [1]	A	22	18		0.924	28.000	1.00	Yes
Nozzle Flg [5]	A	-18	-104					
N1- Tube inlet[1]	!	-12	-15		0.949	50.000	1.00	No
N2- Tube outle[1]	!	-12	-15		0.949	50.000	1.00	No
Warmest MDMT:		22	18					

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-c).
- [5] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-b).
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per UCS-66.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [18] - Impact Testing not required, see UCS-66(b)(3)
- [20] - Cylinder/Cone Junction MDMT based on Longitudinal Stress considerations
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld

UG-84(b)(2) was not considered.

UCS-66(g) was not considered.

UCS-66(i) was not considered.

Notes:

- Impact test temps were not entered in and not considered in the analysis.
- UCS-66(i) applies to impact tested materials not by specification and
- UCS-66(g) applies to materials impact tested per UG-84.1 General Note (c).
- The Basic MDMT includes the (30F) PWHT credit if applicable.

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FileName : 103-C2

Page 9 of 192

Warnings and Errors:

Step: 0 3:13pm Apr 10,2026

Class From To : Basic Element Checks.

Class From To: Check of Additional Element Data

There were no geometry errors or warnings.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



One or more nozzles in this model was specified using the actual thickness basis and a standard flange is also specified. In many cases a matching nominal diameter cannot be determined and the weight of the flange and optional blind flange cannot be determined. When actual thickness basis is specified, you must calculate the weight of the assembly (by hand) and enter it into the 'Overriding Weight' field in the nozzle dialog when there is a standard flange specified. However, for FVC nozzles that have their nominal diameter specified in the FVC dialog, entering the overriding weight for that nozzle is not required.



The ASME Code Section VIII Division 1 does not explicitly require stresses on elements to be computed under test conditions. Please see interpretation VIII-1-10-12 for more information. PV Elite also computes weight due to test fluid and uses the weight in various calculations such as saddle and leg loads. PV Elite also computes static pressure due to operating and test liquids. The static pressure is used in conjunction with the design and test pressures to determine a total pressure in the determination of required thickness, MAWP etc. This ensures compliance with UG-22 requirements.

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FileName : 103-C2

Page 10 of 192

Input Echo:

Step: 1 3:13pm Apr 10,2026

Units used in this Analysis (SI):

Name	System Unit	Constant	User Unit
Length	Feet	0.3048	m
Force	Pounds	4.4480	N
Mass	Pounds	0.4536	kgm
Area	sq. inches	0.0006	m ²
Moment	ft. lbs.	1.3563	N-m
Stress	lbs./sq.in.	6.8946	kPa
Temperature	Degrees F	0.5556	°C
Pressure	psig	0.0069	N/mm ²
Elast. Modulus	lbs./sq.in.	0.0069	N/mm ²
Pipe Density	lbs./cu.in.	27680.0000	kg/m ³
Ins. Density	lbs./cu.ft.	16.0183	kg/m ³
Fluid Density	lbs./cu.ft.	16.0183	kg/m ³
Wind Speed	miles/hr	0.4470	m/sec
Tray Weight	lbs./sq.ft.	47.8790	N/m ²
Inertia	in.4	416231.0000	mm ⁴
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	0.0479	kPa
Elevation	Feet	0.3048	m
Volume	in.3	0.1600E-04	m ³
Diameter	inches	25.4000	mm
Thickness	inches	25.4000	mm

PV Elite Vessel Analysis Program: Input Data

Exchanger Design Pressures and Temperatures

Shell Side Design Pressure	14.281	N/mm ²
Channel Side Design Pressure	4.9503	N/mm ²
Shell Side Design Temperature	343.0	°C
Channel Side Design Temperature	343.0	°C
Radiography, Shell Side	RT-1	
Radiography, Channel Side	RT-1	
Service Type, Shell Side	Air/Water/Steam	
Service Type, Channel Side	Air/Water/Steam	
MDMT (CET), Shell Side	-12.0	°C
MDMT (CET), Tube Side	-12.0	°C
User defined MAWP, Shell Side	14.28	N/mm ²
User defined MAWP, Channel Side	4.95	N/mm ²
User defined MAPnc, Shell Side	0	N/mm ²
User defined MAPnc, Channel Side	0	N/mm ²
User defined Test Pres., Shell Side	0	N/mm ²
User defined Test Pres., Channel Side	0	N/mm ²

Projection of Nozzle from Vessel Top	0	mm
Projection of Nozzle from Vessel Bottom	0	mm
Type of Construction	Welded	
Use Higher Longitudinal Stresses (Flag)	Y	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	
Consider Vortex Shedding	N	

Shop Pressure Test:

Type of Pressure Test	UG-99(c)
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	No



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FileName : 103-C2

Page 11 of 192

Input Echo:

Step: 1 3:13pm Apr 10,2026

Field Pressure Test:

Type of Pressure Test UG-99(b) Note [35]
 Pressure Test Position Horizontal
 Test Performed in Corroded Condition Yes

Load Case 1	NP+EW+WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS
Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE
Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW
Load Case 16	IP+VE+EW
Load Case 17	NP+VO+OW
Load Case 18	FS+BS+IP+OW
Load Case 19	FS+BS+EP+OW
Load Case 20	BL+IP+OW

Wind Design Code	ASCE/SEI 7-16
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed [V]	27 m/sec
Surface Roughness Category	C: Open Terrain
Type of Surface	Moderately Smooth
Base Elevation	0.3 m
Percent Wind for Hydrotest	50.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 m
Distance Upwind of Crest Lh	0 m
Distance from Crest to the Vessel x	0 m
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 m

Seismic Design Code	ASCE/SEI 7-16
Seismic Load Reduction Scale Factor	1.000
Importance Factor	1.250
Table Value Fa	1.600
Table Value Fv	2.400
Max. Mapped Res. Acceleration [Ss]	0.147
Max. Eff. Ground Acceleration [S]	0.065
Force Modification Factor R	3.000
Site Class	D
Component Elevation Ratio z/h	0.000
Amplification Factor Ap	1.000
Force Factor	0.000
Consider Vertical Acceleration	Yes
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Moment Reduction Factor Tau	1.000

M.A.W.P + Static Head to Nozzle	Y
Consider MAP New and Cold in Noz. Design	N
Consider External Loads for Nozzle Des.	Y



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FileName : 103-C2

Page 12 of 192

Input Echo:

Step: 1 3:13pm Apr 10,2026

Use ASME VIII-1 Appendix 1-9	N
Perform Blast Load Analysis	No
Material Database Year	Current w/Addenda or Code Year

Configuration Directives:

g0 is taken as min(g0, attached shell thk)	Yes
Do not use Nozzle MDMT Interpretation VIII-1 01-37	No
Use Table G instead of exact equation for "A"	Yes
Shell Head Joints are Tapered	Yes
Compute "K" in corroded condition	Yes
Use Code Case 2286	Yes
Use the MAWP to compute the MDMT	Yes
For thickness ratios <= 0.35, MDMT will be -155F (-104C)	Yes
For PWHT & P1 Materials the MDMT can be < -55F (-48C)	No
Using Metric Material Databases, ASME II D	Yes
Calculate B31.3 type stress for Nozzles with Loads	Yes
Reduce the MDMT due to lower membrane stress	Yes
Consider Longitudinal Stress in MDMT Calculations	Yes

Complete Listing of Vessel Elements and Details:

Element#	1/4
Element From Node	10
Element To Node	20
Element Type	Spherical
Description	Channel head
Distance "FROM" to "TO"	0 m
Inside Diameter	2666 mm
Element Thickness	28 mm
Internal Corrosion Allowance	2 mm
Nominal Thickness	34 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	4.9503 N/mm ²
Design Temperature Internal Pressure	343 °C
Design External Pressure	0 N/mm ²
Design Temperature External Pressure	24 °C
Effective Diameter Multiplier	1.2
Material Name	SA-387 11
Allowable Stress, Ambient	137997 kPa
Allowable Stress, Operating	137997 kPa
Allowable Stress, Hydrotest	278979 kPa
Material Density	7750.4 kg/m ³
P Number Thickness	16.002 mm
Yield Stress, Operating	239106 kPa
UCS-66 Chart Curve Designation	A
External Pressure Chart Name	CS-3
UNS Number	K11789
Class / Thickness / Grade	2 ::
Product Form	Plate
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No
Element From Node	10
Detail Type	Liquid
Detail ID	Liquid: 10
Dist. from "FROM" Node / Offset dist	0 m
Height/Length of Liquid	2.666 m
Liquid Density	18.492 kg/m ³
Element From Node	10
Detail Type	Lining



PV Elite 28 Licensee: SL User

FileName : 103-C2

Page 13 of 192

Input Echo:

Step: 1 3:13pm Apr 10,2026

Detail ID	316SS weld over
Dist. from "FROM" Node / Offset dist	-1.333 m
Height/Length of Lining	1.333 m
Thickness of Lining	3 mm
Density	8009.1 kg/m ³

Element From Node	10
Detail Type	Nozzle
Detail ID	Manhole
Dist. from "FROM" Node / Offset dist	0 mm
Nozzle Diameter	24 in.
Nozzle Schedule	None
Nozzle Class	900
Layout Angle	0.0
Blind Flange (Y/N)	Y
Weight of Nozzle (Used if > 0)	21429 N
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-336 F11 +IT

Element From Node	10
Detail Type	Weight
Detail ID	Davit
Dist. from "FROM" Node / Offset dist	-1.33 m
Miscellaneous Weight	4903 N
Offset from Element Centerline	0 mm

Element#	2/4
Element From Node	20
Element To Node	40
Element Type	Cylinder
Description	Channel shell
Distance "FROM" to "TO"	2.5 m
Inside Diameter	2666 mm
Element Thickness	50 mm
Internal Corrosion Allowance	2 mm
Nominal Thickness	50 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	4.9503 N/mm ²
Design Temperature Internal Pressure	343 °C
Design External Pressure	0 N/mm ²
Design Temperature External Pressure	24 °C
Effective Diameter Multiplier	1.2
Material Name	SA-387 11 +IT
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	20
Detail Type	Liquid
Detail ID	Liquid: 20
Dist. from "FROM" Node / Offset dist	0 m
Height/Length of Liquid	2.666 m
Liquid Density	18.492 kg/m ³

Element From Node	20
Detail Type	Lining
Detail ID	316SS Weld over
Dist. from "FROM" Node / Offset dist	0 m
Height/Length of Lining	2.5 m
Thickness of Lining	3 mm
Density	8009.1 kg/m ³

Element From Node	20
-------------------	----



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FileName : 103-C2

Page 14 of 192

Input Echo:

Step: 1 3:13pm Apr 10,2026

Detail Type	Nozzle
Detail ID	N1- Tube inlet
Dist. from "FROM" Node / Offset dist	1.1 m
Nozzle Diameter	36 in.
Nozzle Schedule	None
Nozzle Class	None
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	15221 N
Grade of Attached Flange	None
Nozzle Matl	SA-336 F11 +IT

Element From Node	20
Detail Type	Nozzle
Detail ID	N2- Tube outlet
Dist. from "FROM" Node / Offset dist	1.1 m
Nozzle Diameter	36 in.
Nozzle Schedule	None
Nozzle Class	None
Layout Angle	180.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	15221 N
Grade of Attached Flange	None
Nozzle Matl	SA-336 F11 +IT

Element From Node	20
Detail Type	Weight
Detail ID	PP gura
Dist. from "FROM" Node / Offset dist	1.25 m
Miscellaneous Weight	21573 N
Offset from Element Centerline	0 mm

Element#	3/4
Element From Node	40
Element To Node	50
Element Type	Cylinder
Description	Shell
Distance "FROM" to "TO"	5.8 m
Inside Diameter	2660 mm
Element Thickness	142 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	142 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	14.281 N/mm ²
Design Temperature Internal Pressure	343 °C
Design External Pressure	0.1013 N/mm ²
Design Temperature External Pressure	200 °C
Effective Diameter Multiplier	1.2
Material Name	SA-387 11
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	40
Detail Type	Saddle
Detail ID	Left Saddle
Dist. from "FROM" Node / Offset dist	1.15 m
Width of Saddle	254 mm
Height of Saddle at Bottom	2300 mm
Saddle Contact Angle	180.0
Height of Composite Ring Stiffener	0 mm
Width of Wear Plate	304 mm
Thickness of Wear Plate	30 mm



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FileName : 103-C2

Page 15 of 192

Input Echo:

Step: 1 3:13pm Apr 10,2026

Contact Angle, Wear Plate (degrees)	180.0	
Friction coefficient	0.0	
Moment Factor	3.0	
Dimension E at base (optional)	0	mm
Circumferential Eff. over Saddle	1.0	
Circumferential Eff. at Midspan	1.0	
Tangent to Tangent dist. (optional)	0	m
Element From Node	40	
Detail Type	Saddle	
Detail ID	Right Saddle	
Dist. from "FROM" Node / Offset dist	4.55	m
Width of Saddle	254	mm
Height of Saddle at Bottom	2300	mm
Saddle Contact Angle	180.0	
Height of Composite Ring Stiffener	0	mm
Width of Wear Plate	304	mm
Thickness of Wear Plate	30	mm
Contact Angle, Wear Plate (degrees)	180.0	
Friction coefficient	0.40000001	
Moment Factor	3.0	
Dimension E at base (optional)	0	mm
Circumferential Eff. over Saddle	1.0	
Circumferential Eff. at Midspan	1.0	
Tangent to Tangent dist. (optional)	0	m
Element From Node	40	
Detail Type	Liquid	
Detail ID	Liquid: 40	
Dist. from "FROM" Node / Offset dist	0	m
Height/Length of Liquid	2.66	m
Liquid Density	999.54	kg/m³
Element From Node	40	
Detail Type	Nozzle	
Detail ID	Shell outlet-1	
Dist. from "FROM" Node / Offset dist	1.2	m
Nozzle Diameter	10	in.
Nozzle Schedule	None	
Nozzle Class	None	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	4137.8	N
Grade of Attached Flange	None	
Nozzle Matl	SA-105	
Element From Node	40	
Detail Type	Nozzle	
Detail ID	Shell inlet-1	
Dist. from "FROM" Node / Offset dist	2.5	m
Nozzle Diameter	10	in.
Nozzle Schedule	None	
Nozzle Class	None	
Layout Angle	180.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	4417.1	N
Grade of Attached Flange	None	
Nozzle Matl	SA-105	
Element From Node	40	
Detail Type	Weight	
Detail ID	103-C1 WEI (Ope	
Dist. from "FROM" Node / Offset dist	1.15	m
Miscellaneous Weight	837609	N
Offset from Element Centerline	4990	mm



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FileName : 103-C2

Page 16 of 192

Input Echo:

Step: 1 3:13pm Apr 10,2026

Element From Node	40
Detail Type	Weight
Detail ID	103-C1 WEI (Ope
Dist. from "FROM" Node / Offset dist	4.55 m
Miscellaneous Weight	837609 N
Offset from Element Centerline	4990 mm

Element From Node	40
Detail Type	Weight
Detail ID	103-C1 WEI (Hyd
Dist. from "FROM" Node / Offset dist	1.15 m
Miscellaneous Weight	985336 N
Offset from Element Centerline	4990 mm

Element From Node	40
Detail Type	Weight
Detail ID	103-C1 WEI (Hyd
Dist. from "FROM" Node / Offset dist	4.55 m
Miscellaneous Weight	985336 N
Offset from Element Centerline	4990 mm

Element From Node	40
Detail Type	Weight
Detail ID	103-C1 WEI (Emp
Dist. from "FROM" Node / Offset dist	1.15 m
Miscellaneous Weight	735450 N
Offset from Element Centerline	4990 mm

Element From Node	40
Detail Type	Weight
Detail ID	103-C1 WEI (EMp
Dist. from "FROM" Node / Offset dist	4.55 m
Miscellaneous Weight	735450 N
Offset from Element Centerline	4990 mm

Element From Node	40
Detail Type	Weight
Detail ID	Fireproofing 1
Dist. from "FROM" Node / Offset dist	1.15 m
Miscellaneous Weight	3922.4 N
Offset from Element Centerline	2300 mm

Element From Node	40
Detail Type	Weight
Detail ID	Fireproofing 2
Dist. from "FROM" Node / Offset dist	4.55 m
Miscellaneous Weight	3922.4 N
Offset from Element Centerline	-2300 mm

Element From Node	40
Detail Type	For./Mom.
Detail ID	Left saddle- wi
Dist. from "FROM" Node / Offset dist	1.15 m
Force in X Direction	21827 N
Force in Y Direction	0 N
Force in Z Direction	21827 N
Moment about X Axis	0 N-m
Moment about Y Axis	0 N-m
Moment about Z Axis	0 N-m
Force/Moment Combination Method	SRSS

Element From Node	40
Detail Type	For./Mom.
Detail ID	Right saddle-wi



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FileName : 103-C2

Page 17 of 192

Input Echo:

Step: 1 3:13pm Apr 10,2026

Dist. from "FROM" Node / Offset dist	4.55	m
Force in X Direction	21827	N
Force in Y Direction	0	N
Force in Z Direction	21827	N
Moment about X Axis	0	N-m
Moment about Y Axis	0	N-m
Moment about Z Axis	0	N-m
Force/Moment Combination Method	SRSS	

Element From Node	40	
Detail Type	For./Mom.	
Detail ID	Left sadd.-seis	
Dist. from "FROM" Node / Offset dist	1.15	m
Force in X Direction	154586	N
Force in Y Direction	0	N
Force in Z Direction	154586	N
Moment about X Axis	0	N-m
Moment about Y Axis	0	N-m
Moment about Z Axis	0	N-m
Force/Moment Combination Method	SRSS	

Element From Node	40	
Detail Type	For./Mom.	
Detail ID	Right sadd. sei	
Dist. from "FROM" Node / Offset dist	4.55	m
Force in X Direction	154586	N
Force in Y Direction	0	N
Force in Z Direction	154586	N
Moment about X Axis	0	N-m
Moment about Y Axis	0	N-m
Moment about Z Axis	0	N-m
Force/Moment Combination Method	SRSS	

Element#	4/4
Element From Node	50
Element To Node	60
Element Type	Spherical
Description	Shell head
Distance "FROM" to "TO"	0 m
Inside Diameter	2660 mm
Element Thickness	70 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	82 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	14.281 N/mm ²
Design Temperature Internal Pressure	343 °C
Design External Pressure	0.1013 N/mm ²
Design Temperature External Pressure	200 °C
Effective Diameter Multiplier	1.2
Material Name	SA-387 11
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No
Element From Node	50
Detail Type	Liquid
Detail ID	Liquid: 50
Dist. from "FROM" Node / Offset dist	0 m
Height/Length of Liquid	2.66 m
Liquid Density	999.54 kg/m ³



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FileName : 103-C2

Page 18 of 192

XY Coordinate Calculations: Step: 2 3:13pm Apr 10,2026

XY Coordinate Calculations:

From	To	X (Horiz.) m	Y (Vert.) m	DX (Horiz.) m	DY (Vert.) m
Channel head		...	...	...	...
Channel shell		2.5	...	2.5	...
Shell		8.8	...	5.8	...
Shell head		8.8	...	...	...

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

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FileName : 103-C2

Page 19 of 192

Internal Pressure Calculations: Step: 3 3:13pm Apr 10,2026

Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd N/mm ²	Nominal Thickness mm	Total Corr Allowance mm	Element Diameter mm	Allowable Stress(SE) kPa	Element#
Channel head		4.9508	34	2	2666	137997	1
Channel shell		4.9508	50	2	2666	147988	2
Shell		14.307	142	3	2660	147989	3
Shell head		14.307	82	3	2660	147989	4

Element Required Thickness and MAWP :

From	To	Design Pressure N/mm ²	M.A.W.P. Corroded N/mm ²	M.A.P. New & Cold N/mm ²	Minimum Thickness mm	Required Thickness mm	Element#
Channel head		4.95029	5.35	5.77	28	26.032	1
Channel shell		4.95029	5.21	5.43	50	47.5729	2
Shell		14.2808	14.5	14.8	142	139.796	3
Shell head		14.2808	14.7	15.4	70	68.0596	4

Summary of Heat Exchanger Maximum Allowable Working Pressures :

Note:

For Exchanger designs, the following values include MAWPs that consider the tubesheet, tubes, tube/tubesheet joint etc. These values were determined by iteration. Review the tubesheet analysis report for more information.

Shell Side MAWP = 14.3 N/mm²
 Shell Side MAPnc = 14.8 N/mm²
 Channel Side MAWP = 4.95 N/mm²
 Channel Side MAPnc = 5.43 N/mm²

Elements Suitable for Design Internal Pressure.

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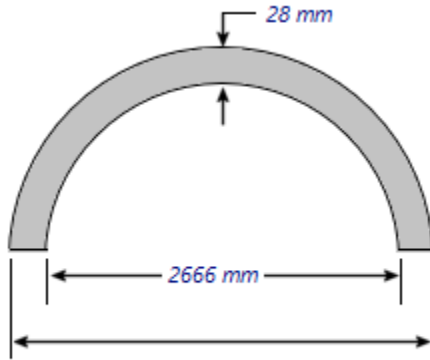
Page 20 of 192

Element: Channel head

Step: 4 3:13pm Apr 10,2026

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Channel head

Spherical Head Design Information:			
Design Pressure P	4.951 N/mm ²	Int. Design Temperature	343 °C
Material	SA-387 11	Ext. Design Temperature	24 °C
Length	0.000 m	External Pressure Chart	CS-3
		UNS Number	K11789
Int. Corr. All. c	2.0000 mm	Allowable Stress (ope) S	137996.8 kPa
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	137996.8 kPa
Inside Diameter Di	2666.000 mm	Actual Stress	127589.2 kPa
Spec. Min. Thk t _e , or s _a	28.0000 mm	Spec. Nominal Thk.	34.000 mm
Surface Area	11.7 m ²	SG of Contents	0.019
Element Volume	4.8 m ³	Weight of Contents	899.5 N
Empty Weight	29591.7 N	Operating Weight	30491.3 N
		Comp. All. Ope. Stress	84499.6 kPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff. E or v	100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 33.325 mm

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 16.663 mm

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ per UG-27 (d)}$$

$$= (4.95 \cdot 1335) / (2 \cdot 137997 \cdot 1 - 0.2 \cdot 4.95)$$

$$= 24.0320 + 2.0000 = 26.0320 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.000 N/mm²

$$= (2 \cdot S \cdot E \cdot t) / (R + 0.2 \cdot t) \text{ per UG-27 (d)}$$

$$= (2 \cdot 137997 \cdot 1 \cdot 26) / (1335 + 0.2 \cdot 26)$$

$$= 5.35 - 0.00048 = 5.35 \text{ N/mm}^2$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (R + 0.2 \cdot t) \text{ per UG-27 (d)}$$



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FileName : 103-C2

Page 21 of 192

Element: Channel head

Step: 4 3:13pm Apr 10,2026

$$= (2 \cdot 137997 \cdot 1 \cdot 28) / (1333 + 0.2 \cdot 28)$$

$$= 5.77 \text{ N/mm}^2$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (4.95 \cdot (1335 + 0.2 \cdot 26)) / (2 \cdot 1 \cdot 26)$$

$$= 127589.188 \text{ kPa}$$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 1.889 %

MDMT Calculations in the Spherical Portion:

Govrn. thk, $t_g = 28$, $t_r = 24$, $c = 2$ mm, $E^* = 1$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.92$, Temp. Reduction = 4 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 22 °C

Min Metal Temp. at Required thickness (UCS 66.1) 18 °C

Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	A	Press. at MDMT	4.950 N/mm ²
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	28.000 mm	Basic MDMT	22 °C
Stress/Thk. Ratio	0.924	Temperature Difference	4 °C
UCS-66(c) red.	No	Computed Min. Temp.	18 °C
		Joint Efficiency E*	100 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 33.325 mm

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 16.663 mm

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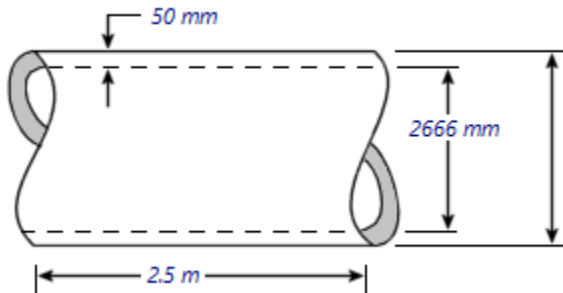
Page 22 of 192

Element: Channel shell

Step: 5 3:13pm Apr 10,2026

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Channel shell

Cylindrical Shell Design Information:			
Design Pressure P	4.951 N/mm ²	Int. Design Temperature	343 °C
Material	SA-387 11	Ext. Design Temperature	24 °C
Length	2.500 m	External Pressure Chart	CS-3
		UNS Number	K11789
Int. Corr. All. c	2.0000 mm	Allowable Stress (ope) S	147988.4 kPa
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	147988.4 kPa
Inside Diameter Di	2666.000 mm	Actual Stress	140655.6 kPa
Spec. Min. Thk t _e , or sa	50.0000 mm	Spec. Nominal Thk.	50.000 mm
Surface Area	21.7 m ²	SG of Contents	0.019
Element Volume	13.6 m ³	Weight of Contents	2530.6 N
Empty Weight	81060.9 N	Operating Weight	83591.5 N
		Comp. All. Ope. Stress	93946.5 kPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff. E or v	100 %

Material treatment options: +IT

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (4.95 \cdot 1335) / (147988 \cdot 1 - 0.6 \cdot 4.95) \\
 &= 45.5729 + 2.0000 = 47.5729 \text{ mm}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$\begin{aligned}
 &\text{Less Operating Hydrostatic Head Pressure of } 0.000 \text{ N/mm}^2 \\
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (147988 \cdot 1 \cdot 48) / (1335 + 0.6 \cdot 48) \\
 &= 5.21 - 0.00048 = 5.21 \text{ N/mm}^2
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (147988 \cdot 1 \cdot 50) / (1333 + 0.6 \cdot 50) \\
 &= 5.43 \text{ N/mm}^2
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (4.95 \cdot (1335 + 0.6 \cdot 48)) / (1 \cdot 48)
 \end{aligned}$$



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FileName : 103-C2

Page 23 of 192

Element: Channel shell

Step: 5 3:13pm Apr 10,2026

= 140655.625 kPa

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f (1 - R_f / R_o)$) 1.841 %

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Material treatment options: +IT

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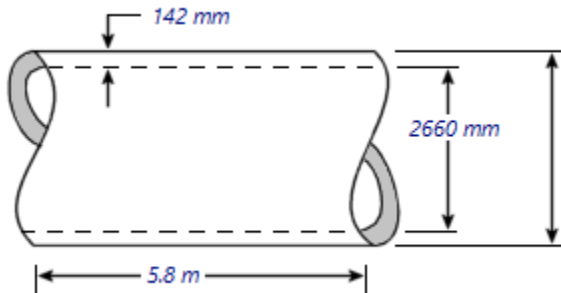
Page 24 of 192

Element: Shell

Step: 6 3:13pm Apr 10,2026

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Shell

Cylindrical Shell Design Information:			
Design Pressure P	14.307 N/mm ²	Int. Design Temperature	343 °C
Material	SA-387 11	Ext. Design Temperature	200 °C
Length	5.800 m	External Pressure Chart	CS-3
		UNS Number	K11789
Int. Corr. All. c	3.0000 mm	Allowable Stress (ope) S	147988.7 kPa
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	147988.7 kPa
Inside Diameter Di	2660.000 mm	Actual Stress	145778.2 kPa
Spec. Min. Thk t,e, or sa	142.0000 mm	Spec. Nominal Thk.	142.000 mm
Surface Area	53.6 m ²	SG of Contents	1.000
Element Volume	12.0 m ³	Weight of Contents	120934.5 N
Empty Weight	551005.9 N	Operating Weight	671940.3 N
		Comp. All. Ope. Stress	109389.2 kPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (14.3 \cdot 1333) / (147989 \cdot 1 - 0.6 \cdot 14.3) \\
 &= 136.7959 + 3.0000 = 139.7959 \text{ mm}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.026 N/mm²

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (147989 \cdot 1 \cdot 139) / (1333 + 0.6 \cdot 139) \\
 &= 14.5 - 0.026 = 14.5 \text{ N/mm}^2
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (147989 \cdot 1 \cdot 142) / (1330 + 0.6 \cdot 142) \\
 &= 14.8 \text{ N/mm}^2
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (14.3 \cdot (1333 + 0.6 \cdot 139)) / (1 \cdot 139) \\
 &= 145778.156 \text{ kPa}
 \end{aligned}$$



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FileName : 103-C2

Page 25 of 192

Element: Shell

Step: 6 3:13pm Apr 10,2026

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 5.068 %

Note: Please Check Requirements of UCS-79 as Elongation is > 5%.

Minimum Design Metal Temperature Results:

Warning:

The minimum metal temperature could not be computed because the governing thickness for this component was greater than 100.000 mm. This component must be impact tested.

Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	A	Press. at MDMT	14.280 N/mm ²
UCS-66(b) red.	No	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	142.000 mm	Basic MDMT	-17 °C
Stress/Thk. Ratio	-1.000	Temperature Difference	0 °C
UCS-66(c) red.	No	Computed Min. Temp.	-17 °C
		Joint Efficiency E*	100 %

EXTERNAL PRESSURE RESULTS for: Shell

Cylindrical Shell From 40 to 50 Ext. Chart: CS-2 at 200 °C

Maximum External Pressure per Code Case 2286:

Determine Parameter [Mx]:

$$= L / \sqrt{(R_o \cdot t)}$$

$$= 6243 / \sqrt{(1472 \cdot 139)}$$

$$= 13.802$$

Compute the Factor [Ch]:

$$\text{Since } M_x > 13 \text{ and } M_x < 2 \cdot (D_o/t)^{0.94}, Ch = 1.12 \cdot M_x - 1.058$$

$$= 1.12 \cdot 13.8 - 1.058$$

$$= 0.0697$$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$$= 1.6 \cdot Ch \cdot E_y (t / D_o)$$

$$= 1.6 \cdot 0.07 \cdot 193337 (139/2944)$$

$$= 1017723.438 \text{ kPa}$$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$$= Fhe / F_y$$

$$= 1017723.4/261995.9$$

$$= 3.885$$

Predicted Buckling Stress [Fic]:

$$\text{Since } Fr \geq 2.439, Fic = F_y$$

$$= 261995.875 \text{ kPa}$$

Factor of Safety [FS]:

$$\text{Since } Fic [261996] \geq F_y [261996], FS = 1.667$$

Allowable Circumferential Compressive Stress [Fha]:

$$\text{Since } Fr > 2.439, Fha = F_y / FS$$

$$= 261995.875 / 1.667$$

$$= 157166.078 \text{ kPa}$$

Allowable External Pressure at the given Thickness [Pa]:



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FileName : 103-C2

Page 26 of 192

Element: Shell

Step: 6 3:13pm Apr 10,2026

= 2 • Fha(t / Do)
= 2 • 157 (139/2944)
= 14.842 N/mm²

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FileName : 103-C2

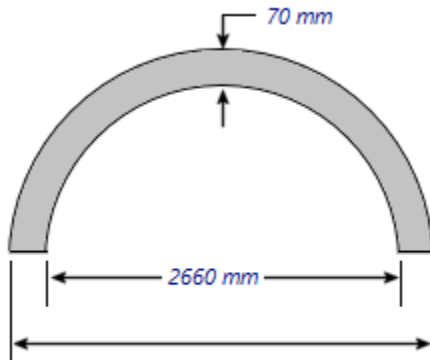
Page 27 of 192

Element: Shell head

Step: 7 3:14pm Apr 10,2026

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Shell head

Spherical Head Design Information:			
Design Pressure P	14.307 N/mm ²	Int. Design Temperature	343 °C
Material	SA-387 11	Ext. Design Temperature	200 °C
Length	0.000 m	External Pressure Chart	CS-3
		UNS Number	K11789
Int. Corr. All. c	3.0000 mm	Allowable Stress (ope) S	147988.7 kPa
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	147988.7 kPa
Inside Diameter Di	2660.000 mm	Actual Stress	143744.3 kPa
Spec. Min. Thk t _e , or s _a	70.0000 mm	Spec. Nominal Thk.	82.000 mm
Surface Area	12.5 m ²	SG of Contents	1.000
Element Volume	4.8 m ³	Weight of Contents	48296.8 N
Empty Weight	73624.1 N	Operating Weight	121920.9 N
		Comp. All. Ope. Stress	109389.2 kPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff. E or v	100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 33.250 mm

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 16.625 mm

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ per UG-27 (d)}$$

$$= (14.3 \cdot 1333) / (2 \cdot 147989 \cdot 1 - 0.2 \cdot 14.3)$$

$$= 65.0596 + 3.0000 = 68.0596 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.026 N/mm²

$$= (2 \cdot S \cdot E \cdot t) / (R + 0.2 \cdot t) \text{ per UG-27 (d)}$$

$$= (2 \cdot 147989 \cdot 1 \cdot 67) / (1333 + 0.2 \cdot 67)$$

$$= 14.7 - 0.026 = 14.7 \text{ N/mm}^2$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (R + 0.2 \cdot t) \text{ per UG-27 (d)}$$



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FileName : 103-C2

Page 28 of 192

Element: Shell head

Step: 7 3:14pm Apr 10,2026

$$= (2 \cdot 147989 \cdot 1 \cdot 70) / (1330 + 0.2 \cdot 70)$$

$$= 15.4 \text{ N/mm}^2$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (14.3 \cdot (1333 + 0.2 \cdot 67)) / (2 \cdot 1 \cdot 67)$$

$$= 143744.281 \text{ kPa}$$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 4.486 %

MDMT Calculations in the Spherical Portion:

Govrn. thk, $t_g = 70$, $t_r = 64.9$, $c = 3 \text{ mm}$, $E^* = 1$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.97$, Temp. Reduction = 2 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 43 °C

Min Metal Temp. at Required thickness (UCS 66.1) 41 °C

Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	A	Press. at MDMT	14.280 N/mm ²
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	70.000 mm	Basic MDMT	43 °C
Stress/Thk. Ratio	0.969	Temperature Difference	2 °C
UCS-66(c) red.	No	Computed Min. Temp.	41 °C
		Joint Efficiency E*	100 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 33.250 mm

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 16.625 mm

EXTERNAL PRESSURE RESULTS for: Shell head

Spherical Head From 50 to 60 Ext. Chart: CS-2 at 200 °C

Maximum External Pressure per Code Case 2286:

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$$= 0.075 \cdot E_y (t / R_o)$$

$$= 0.075 \cdot 193337 (67 / 1400)$$

$$= 693901.125 \text{ kPa}$$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$$= F_{he} / S_y$$

$$= 693901.1 / 261995.9$$

$$= 2.649$$

Predicted Buckling Stress [Fic]:

$$\text{Since } F_r > 1.6 \text{ and } F_r < 6.25, F_{ic} = 1.31 \cdot F_y / (1.15 + S_y / F_{he})$$

$$= 1.31 \cdot 261996 / (1.15 + 261996 / 693901)$$

$$= 224680.188 \text{ kPa}$$

Factor of Safety [FS]:

$$\text{Since } F_{ic} > 0.55 F_y \text{ and } F_{ic} < F_y$$

$$= 2.407 - 0.741 \cdot F_{ic} / F_y$$

$$= 2.407 - 0.741 \cdot 224680 / 261996$$



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FileName : 103-C2

Page 29 of 192

Element: Shell head

Step: 7 3:14pm Apr 10,2026

= 1.772

Allowable Circumferential Compressive Stress [Fha]:

Since $Fr > 1.6$ and $Fr < 6.25$, $Fha = 1.31 \cdot Fy / (FS \cdot (1.15 + Sy/Fhe))$
= $1.31 \cdot 261996 / (1.77 \cdot (1.15 + 261996/693901))$
= 126827.633 kPa

Allowable External Pressure at the given Thickness [Pa]:

= $2 \cdot Fha[t/Ro]$
= $2 \cdot 126828 [67/1400]$
= 12.140 N/mm²

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FileName : 103-C2

Page 30 of 192

Shop Pressure Test Results: Step: 8 3:14pm Apr 10,2026

Hydrostatic Test Pressure Results:

Exchanger Shell Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	18.564 N/mm ²
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	18.565 N/mm ²
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	19.279 N/mm ²
UG100	= 1.10 * M.A.W.P. * Sa/S	15.708 N/mm ²
PED	= max(1.43*DP, 1.25*DP*ratio)	20.422 N/mm ²
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	20.422 N/mm ²
App 27-4	= M.A.W.P.	14.280 N/mm ²

Exchanger Channel Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	6.435 N/mm ²
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	6.435 N/mm ²
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	7.032 N/mm ²
UG100	= 1.10 * M.A.W.P. * Sa/S	5.445 N/mm ²
PED	= max(1.43*DP, 1.25*DP*ratio)	7.079 N/mm ²
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	7.079 N/mm ²
App 27-4	= M.A.W.P.	4.950 N/mm ²

Horizontal Test performed per: UG-99c

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (kPa & N/mm²):

From To	Stress	Allowable	Ratio	Pressure
Channel head	168697.2	278978.8	0.605	7.06
Channel shell	192384.9	278978.8	0.690	7.06
Shell	192384.8	278978.8	0.690	19.30
Shell head	185315.8	278978.8	0.664	19.30

Stress ratios for Nozzle and Pad Materials (kPa):

Description	Pad/Nozzle	Ambient	Operating	Ratio
Manhole	Nozzle	147991.42	147991.42	1.000
N1- Tube inlet	Nozzle	147988.41	147988.41	1.000
N2- Tube outlet	Nozzle	147988.41	147988.41	1.000
Shell outlet-1	Nozzle	137989.50	122834.11	1.123
Shell inlet-1	Nozzle	137989.50	122834.11	1.123

Minimum

1.000

Stress ratios for Pressurized Vessel Elements (kPa):

Description	Ambient	Operating	Ratio
Channel head	137996.80	137996.80	1.000
Channel shell	147988.41	147988.41	1.000
Shell	147988.73	147988.73	1.000
Shell head	147988.73	147988.73	1.000

Minimum

1.000

Stress ratios for Exchanger Materials (kPa):

Description	Ambient	Operating	Ratio
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PV Elite 28 Licensee: SL User

FileName : 103-C2

Page 31 of 192

Shop Pressure Test Results: Step: 8 3:14pm Apr 10,2026

Tube Material	114991.25	94542.65	1.216
Tubesheet Material	147988.73	147988.73	1.000

Minimum 1.000

Hoop Stress in Nozzle Wall during Pressure Test (kPa):

Description	Ambient	Operating	Ratio
Manhole	91641.91	278983.81	0.328
N1- Tube inlet	96624.97	278978.81	0.346
N2- Tube outlet	96622.33	278978.81	0.346
Shell outlet-1	78072.87	179386.34	0.435
Shell inlet-1	78072.87	179386.34	0.435

Elements Suitable for Test Pressure.

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FileName : 103-C2

Page 32 of 192

Field Pressure Test Results: Step: 9 3:14pm Apr 10, 2026

Hydrostatic Test Pressure Results:

Exchanger Shell Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	18.564 N/mm ²
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	18.565 N/mm ²
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	19.279 N/mm ²
UG100	= 1.10 * M.A.W.P. * Sa/S	15.708 N/mm ²
PED	= max(1.43*DP, 1.25*DP*ratio)	20.422 N/mm ²
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	20.422 N/mm ²
App 27-4	= M.A.W.P.	14.280 N/mm ²

Exchanger Channel Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	6.435 N/mm ²
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	6.435 N/mm ²
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	7.032 N/mm ²
UG100	= 1.10 * M.A.W.P. * Sa/S	5.445 N/mm ²
PED	= max(1.43*DP, 1.25*DP*ratio)	7.079 N/mm ²
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	7.079 N/mm ²
App 27-4	= M.A.W.P.	4.950 N/mm ²

UG-99(b)(35), Test Pressure Calculation [Shell Side]:

= Test Factor • Design Pressure • Stress Ratio

= 1.3 • 14.3 • 1

= 18.565 N/mm²

UG-99(b) Note 35, Test Pressure Calculation [Channel Side]:

= Test Factor • Design Pressure • Stress Ratio

= 1.3 • 4.95 • 1

= 6.435 N/mm²

Horizontal Test performed per: UG-99b (Note 35)

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (kPa & N/mm²):

From To	Stress	Allowable	Ratio	Pressure
Channel head	166523.2	278978.8	0.597	6.46
Channel shell	183576.9	278978.8	0.658	6.46
Shell	189431.5	278978.8	0.679	18.59
Shell head	186788.6	278978.8	0.670	18.59

Stress ratios for Nozzle and Pad Materials (kPa):

Description	Pad/Nozzle	Ambient	Operating	Ratio
Manhole	Nozzle	147991.42	147991.42	1.000
N1- Tube inlet	Nozzle	147988.41	147988.41	1.000
N2- Tube outlet	Nozzle	147988.41	147988.41	1.000
Shell outlet-1	Nozzle	137989.50	122834.11	1.123
Shell inlet-1	Nozzle	137989.50	122834.11	1.123

Minimum

1.000

Stress ratios for Pressurized Vessel Elements (kPa):

Description	Ambient	Operating	Ratio
-------------	---------	-----------	-------



PV Elite 28 Licensee: SL User

FileName : 103-C2

Page 33 of 192

Field Pressure Test Results: Step: 9 3:14pm Apr 10, 2026

Channel head	137996.80	137996.80	1.000
Channel shell	147988.41	147988.41	1.000
Shell	147988.73	147988.73	1.000
Shell head	147988.73	147988.73	1.000

Minimum 1.000

Stress ratios for Exchanger Materials (kPa):

Description	Ambient	Operating	Ratio
Tube Material	114991.25	94542.65	1.216
Tubesheet Material	147988.73	147988.73	1.000

Minimum 1.000

Hoop Stress in Nozzle Wall during Pressure Test (kPa):

Description	Ambient	Operating	Ratio
Manhole	83899.16	278983.81	0.301
N1- Tube inlet	88461.19	278978.81	0.317
N2- Tube outlet	88458.77	278978.81	0.317
Shell outlet-1	75186.52	179386.34	0.419
Shell inlet-1	75186.52	179386.34	0.419

Elements Suitable for Test Pressure.

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FileName : 103-C2

Page 34 of 192

ASME TS Calc: Pressure Test

Case: 1 3:14p Apr 10,2026

Input Echo, Tubesheet Number 1, Description: Tubesheet

Shell Data:

Main Shell Description: Shell

Shell Test Pressure		18.57 N/mm ²
Shell Thickness	ts	142.0000 mm
Shell Internal Corrosion Allowance	cas	0.0000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	2660.000 mm
Shell Temperature for Internal Pressure	Ts	21.11 °C
Shell Material		SA-387 11

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K11789
Shell Allowable Stress at Temperature	Ss	147988.73 kPa
Shell Allowable Stress at Ambient		147988.73 kPa

Channel Data:

Channel Description: Channel shell

Channel Type:		Cylinder
Channel Test Pressure		6.44 N/mm ²
Channel Thickness	tc	50.0000 mm
Channel Internal Corrosion Allowance	cac	0.0000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	2666.000 mm
Channel Design Temperature	TEMPC	21.11 °C
Channel Flange Matl		SA-387 11

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K11789
Channel Allowable Stress at Temperature	Sc	147988.73 kPa
Channel Allowable Stress at Ambient		147988.73 kPa

Tube Data:

Number of Tube Holes	Nt	5546
Tube Wall Thickness	et	3.0480 mm
Tube Outside Diameter	D	25.4000 mm
Total Straight Tube Length	Lt	5500.00 mm
Straight Tube Length (bet. inner tubsht faces) L		5000.00 mm
Design Temperature of the Tubes		21.11 °C
Tube Material		SA-213 TP316L
Tube Material UNS Number		S31603
Is this a Welded Tube		No
Tube Material Specification used	Smls. tube	
Tube Allowable Stress at Temperature		114991.25 kPa
Tube Allowable Stress At Ambient		114991.25 kPa
Tube Yield Stress At design Temperature	Syt	169987.06 kPa
Tube Pitch (Center to Center Spacing)	P	31.7500 mm
Tube Layout Pattern		Triangular

Fillet Weld Leg	af	3.0000 mm
Groove Weld Leg	ag	3.0000 mm
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3
Is the Tube Inset		No

Radius to Outermost Tube Hole Center	ro	1272.760 mm
Largest Center-to-Center Tube Distance	U1	101.6000 mm
Length of Expanded Portion of Tube	ltx	50.0000 mm
Tube-side pass partition groove depth	hg	5.0000 mm
Area of Tube Layout	Ap	5.0 m ²
Perimeter of Tube Layout	Cp	8219.2 mm



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FileName : 103-C2

Page 35 of 192

ASME TS Calc: Pressure Test

Case: 1 3:14p Apr 10,2026

Tubesheet Data:

Tubesheet TYPE: U-tube, Integral both sides, Conf. a

Tubesheet Design Metal Temperature T 21.11 °C
Tubesheet Material SA-336 F11

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Tubesheet Material UNS Number K11572
Tubesheet Allowable Stress at Temperature S 147988.73 kPa
Tubesheet Allowable Stress at Ambient Tt 147988.73 kPa
Thickness of Tubesheet h 500.0000 mm
Tubesheet Corr. Allowance (Shell side) Cats 0.0000 mm
Tubesheet Corr. Allowance (Channel side) Catc 0.0000 mm
Tubesheet Outside Diameter A 2980.000 mm

Area of the Untubed Lanes AL 0.203 m²

Junction Stress Reduction option Perform Plastic Calculation

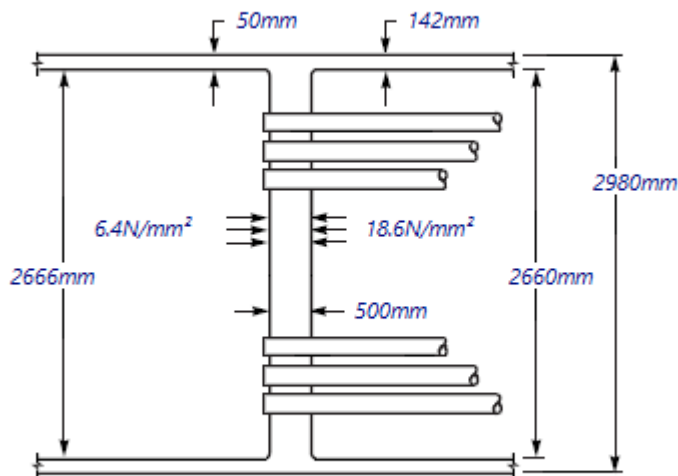
Perform Differential Pressure Design No

Run Multiple Load Cases Yes

Tubesheet Integral with Both

Tubesheet Extended as Flange No

Tubesheet Integral w/Shell and Channel



Configuration a:

ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties :

Shell - TM-1 Material Group C

Elastic Mod. at Design Temperature 21.1 °C 0.20409E+06 N/mm²
Elastic Mod. at Ambient Temperature 21.1 °C 0.20409E+06 N/mm²

Channel - TM-1 Material Group C



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FileName : 103-C2

Page 36 of 192

ASME TS Calc: Pressure Test

Case: 1 3:14p Apr 10,2026

Elastic Mod. at Design Temperature	21.1 °C	0.20409E+06 N/mm ²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20409E+06 N/mm ²

Tubes - TM-1 Material Group G

Elastic Mod. at Tubsht. Design Temp.	21.1 °C	0.19513E+06 N/mm ²
Elastic Mod. at Ambient Temperature	21.1 °C	0.19513E+06 N/mm ²

TubeSheet - TM-1 Material Group C

Elastic Mod. at Design Temperature	21.1 °C	0.20409E+06 N/mm ²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20409E+06 N/mm ²

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:

$$= (P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P) \text{ per Appendix 1-1 (a)(1)}$$

$$= (6.44 \cdot (25.4/2 - 0)) / (114991 \cdot 1 + 0.4 \cdot 6.44)$$

$$= 0.6951 + 0.0000 = 0.6951 \text{ mm}$$

Tube Required Thickness under External Pressure (Shellside pressure) :

Determine Parameter [Mx]:

$$= L / \sqrt{(Ro \cdot t)}$$

$$= 6273 / \sqrt{(12.7 \cdot 3.05)}$$

$$= 1008.206$$

Compute the Factor [Ch]:

$$\text{Since } Mx > 2 \cdot (Do/t)0.94, Ch = 0.55 / (Do/t)$$

$$= 0.55 / (25.4/3.05)$$

$$= 0.0660$$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$$= 1.6 \cdot Ch \cdot Ey (t / Do)$$

$$= 1.6 \cdot 0.066 \cdot 195128 (3.05/25.4)$$

$$= 2472524.500 \text{ kPa}$$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$$= Fhe / Fy$$

$$= 2472524.5 / 169987.1$$

$$= 14.545$$

Predicted Buckling Stress [Fic]:

$$\text{Since } Fr \geq 2.439, Fic = Fy$$

$$= 169987.062 \text{ kPa}$$

For non-Carbon and Low Alloy Steel Materials re-compute [Fic]:

Factor A [A]:

$$= F / Emod, \text{ see 2286-6 Step 1}$$

$$= 2472524 / 195128$$

$$= 0.01267$$

From the External Pressure Chart, HA-4, determine the Stress [B]:

$$= 98376.195 \text{ kPa}$$

The Tangent Modulus from 3-D.5.2, [Et]:

$$= 2B/A$$

$$= 2 \cdot 98376 / 0.013$$

$$= 15527.4 \text{ N/mm}^2$$

Predicted Buckling Stress [Fic]:

$$= (F \cdot Et) / (FS \cdot Emod)$$

$$= (2472524 \cdot 15527) / (1 \cdot 195128)$$



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FileName : 103-C2

Page 37 of 192

ASME TS Calc: Pressure Test

Case: 1 3:14p Apr 10, 2026

= 196752.391 kPa

Factor of Safety [FS]:

Since $Fic [196752] \geq Fy[169987]$, FS = 1.667

Allowable Circumferential Compressive Stress [Fha]:

Since $Fr > 2.439$, $Fha = Fy/FS$

= 169987.062/1.667

= 101971.836 kPa

Allowable External Pressure at the given Thickness [Pa]:

= $2 \cdot Fha(t / Do)$

= $2 \cdot 102 (3.05/25.4)$

= 24.475 N/mm²

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	2.3120 mm
Allowable Internal Pressure at Corroded thickness	30.53 N/mm ²
Required Internal Design Pressure	6.44 N/mm ²
Allowable External Pressure at Corroded thickness	24.47 N/mm ²
Required External Design Pressure	18.57 N/mm ²
Required Thickness due to Shell Side pressure	2.3120 mm

Detailed Results for load Case D1 un-corr. (Psd,min + Ptd,max)

Results for ASME U-tube Tubesheet Calculations for Configuration a:

Per Edition 2025, Original Thickness :

ASME Section VIII-2 Part 4.18.7.4 Step 1:

Equivalent outer tube limit circle diameter [Do]:

= $2 \cdot ro + dt$

= $2 \cdot 1273 + 25.4$

= 2570.920 mm

Basic ligament efficiency for shear [μ]:

= $(p - dt) / p$

= $(31.8 - 25.4) / 31.8$

= 0.200

ASME Section VIII-2 Part 4.18.7.4 Step 2:

Ratio [ps]:

= Ds / Do (Configurations a, b, c)

= 2660/2571

= 1.0346

Ratio [pc]:

= Dc / Do (Configurations a, e, f)

= 2666/2571

= 1.0370

ASME Section VIII-2 Part 4.18.7.4 Step 3:

Ratio [p]:

= ltx / h

= 50/500

= 0.100 (must be $0 \leq p \leq 1$)

Effective tube hole Diameter [d*]:

= $\max(dt - 2tt(EtT/E) \cdot (StT/S) \cdot (p); dt - 2tt)$



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FileName : 103-C2

Page 38 of 192

ASME TS Calc: Pressure Test

Case: 1 3:14p Apr 10, 2026

$$= \max(25.4 - 2 \cdot 3.05 (195128/204092) \cdot (114991/147989) \cdot (0.1); 25.4 - 2 \cdot 3.05)$$

$$= 24.9471 \text{ mm}$$

Effective tube pitch [p*]:

$$= p / \sqrt{(1 - 4 \cdot \min(A; 4 \cdot Do \cdot p) / (\pi \cdot Do^2))}$$

$$= 31.8 / \sqrt{(1 - 4 \cdot \min(0.2; 4 \cdot 2571 \cdot 31.8) / (\pi \cdot 2571^2))}$$

$$= 32.3886 \text{ mm}$$

Effective ligament efficiency for bending [μ*]:

$$= (p^* - d^*) / p^*$$

$$= (32.4 - 24.9) / 32.4$$

$$= 0.2298$$

$$h/p = 15.748032; \mu^* = 0.229756; E^* = 39041 \text{ N/mm}^2$$

$$E^*/E = 0.191293; \nu^* = 0.419013$$

Note: As h/p (15.748) is > 2, data values for h/p = 2 were used.

ASME Section VIII-2 Part 4.18.7.4 Step 4:

Shell coefficient [βs]:

$$= ((12(1 - \nu_s^2))^{1/4}) / \sqrt{(Ds + ts)ts}$$

$$= ((12(1 - 0.3^2))^{1/4}) / \sqrt{(2660 + 142)142}$$

$$= 0.0029 \text{ 1/mm}$$

Shell coefficient [ks]:

$$= \beta_s \cdot E_s \cdot ts^3 / (6(1 - \nu_s^2))$$

$$= 0.0029 \cdot 204092 \cdot 142^3 / (6(1 - 0.3^2))$$

$$= 308443584.0000 \text{ N/mm}^2 \cdot \text{mm}^2$$

Shell coefficient [λs]:

$$= 6(Ds + ts) \cdot ks / h^3 \cdot (1 + h \cdot \beta_s + (h \cdot \beta_s)^2/2)$$

$$= 6(2660 + 142) \cdot 308443584 / 500^3 \cdot (1 + 500 \cdot 0.0029 + (500 \cdot 0.0029)^2/2)$$

$$= 144328.7 \text{ N/mm}^2$$

Shell coefficient [δs]:

$$= Ds^2 / (4 \cdot E_s \cdot ts) \cdot (1 - Ds \cdot \nu_s / (2(Ds + ts)))$$

$$= 2660^2 / (4 \cdot 204092 \cdot 142) \cdot (1 - 2660 \cdot 0.3 / (2(2660 + 142)))$$

$$= 0.52348 \text{E-04 mm/kPa}$$

Parameter [ws]:

$$= \rho_s \cdot ks \cdot \beta_s \cdot \delta_s (1 + h \cdot \beta_s)$$

$$= 1.03 \cdot 308443584 \cdot 0.0029 \cdot 0.52348 \text{E-04} (1 + 500 \cdot 0.0029)$$

$$= 117511.0781 \text{ mm}^2$$

Channel coefficient [βc]:

$$= ((12(1 - \nu_c^2))^{1/4}) / \sqrt{(Dc + tc)tc}$$

$$= ((12(1 - 0.3^2))^{1/4}) / \sqrt{(2666 + 50)50}$$

$$= 0.0049 \text{ 1/mm}$$

Channel coefficient [kc]:

$$= \beta_c \cdot E_c \cdot tc^3 / (6(1 - \nu_s^2))$$

$$= 0.0049 \cdot 204092 \cdot 50^3 / (6(1 - 0.3^2))$$

$$= 23048842.0000 \text{ N/mm}^2 \cdot \text{mm}^2$$

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:

$$CqDelta = CqTheta = CmTheta = CmDelta = 1$$

Channel coefficient [λc]:

$$= 6(Dc + tc / \cos(\theta)) \cdot kc / h^3 \cdot (CmTheta + h \cdot \beta_c \cdot (Cm\Delta + CqTheta) / 2 + (h \cdot \beta_c)^2/2 \cdot Cq\Delta)$$

$$= 6(2666 + 50 / \cos(0)) \cdot 23048842 / 500^3 \cdot (1 + 500 \cdot 0.0049 \cdot (1 + 1) / 2 + (500 \cdot 0.0049)^2/2 \cdot 1)$$

$$= 19556.1 \text{ N/mm}^2$$



PV Elite 28 Licensee: SL User

FileName : 103-C2

Page 39 of 192

ASME TS Calc: Pressure Test

Case: 1 3:14p Apr 10, 2026

Channel coefficient [δc]:

$$= Dc^2 / (4 \cdot Ec \cdot tc) \cdot (1 - Dc \cdot vc / (2(Dc + tc)))$$

$$= 2666^2 / (4 \cdot 204092 \cdot 50) \cdot (1 - 2666 \cdot 0.3 / (2(2666 + 50)))$$

$$= 0.14850E-03 \text{ mm/kPa}$$

Channel coefficient [zetac]:

$$\text{zetac} = 0$$

Parameter [ωc]:

$$= \rho c \cdot kc [\delta C (Cm\Theta + \beta C + Cq\Delta \cdot h \cdot \beta C^2) - \zeta c (Cm\Theta + h \cdot Cq\Theta + h \cdot \beta C / 2)]$$

$$= 1.04 \cdot 23048842 [0.000148497 (1 + 0.0049 \cdot 1 \cdot 500) - 0.000 (1/500 + 1 \cdot 0.0049/2)]$$

$$= 60688.4023 \text{ mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 5:

Diameter ratio [K]:

$$= A / Do$$

$$= 2980/2571$$

$$= 1.1591$$

Coefficient [F]:

$$= (1 - v^*) / E^* \cdot (\lambda s + \lambda c + E \cdot \ln(K))$$

$$= (1 - 0.42) / 39041 \cdot (144329 + 19556 + 204092 \cdot \ln(1.16))$$

$$= 2.8873$$

ASME Section VIII-2 Part 4.18.7.4 Step 6:

Moment on tubesheet due to pressures (Ps, Pt) [Mts]:

$$= Do^2/16 \cdot [(ps-1) \cdot (ps^2+1) \cdot Ps - (pc-1) \cdot (pc^2+1) \cdot Pt]$$

$$= 2571^2/16 \cdot [(1.03-1) \cdot (1.03^2+1) \cdot 0 - (1.04-1) \cdot (1.04^2+1) \cdot 6.44]$$

$$= -204042.0469 \text{ N/mm}^2 \cdot \text{mm}^2$$

Moment acting on unperforated tubesheet rim [M*]:

$$= Mts + \omega c \cdot Pt - \omega s \cdot Ps$$

$$= -204042 + 60688 \cdot 6.44 - 117511 \cdot 0$$

$$= 186510.5000 \text{ N/mm}^2 \cdot \text{mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 7:

Maximum bending moment acting on periphery of tubesheet [Mp]:

$$= (M^* - Do^2/32 \cdot F(Ps - Pt)) / (1 + F)$$

$$= (186510 - 2571^2/32 \cdot 2.89 (0 - 6.44)) / (1 + 2.89)$$

$$= 1035268.3750 \text{ N/mm}^2 \cdot \text{mm}^2$$

Maximum bending moment acting on center of tubesheet [Mo]:

$$= Mp + Do^2/64 \cdot (3 + v^*) \cdot (Ps - Pt)$$

$$= 1035268 + 2571^2/64 \cdot (3 + 0.42) \cdot (0 - 6.44)$$

$$= -1237063.0000 \text{ N/mm}^2 \cdot \text{mm}^2$$

Maximum bending moment acting on tubesheet [M]:

$$= \max(|Mp|; |Mo|)$$

$$= \max(|1035268|; |-1237063|)$$

$$= 1237063.0000 \text{ N/mm}^2 \cdot \text{mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 8:

Tubesheet Bending Stress at original thickness:

$$= 6 \cdot M / (\mu^* (h - hg')^2)$$

$$= 6 \cdot 1237063 / (0.23 (500 - 5)^2)$$

$$= 131837.9375 \text{ kPa}$$



PV Elite 28 Licensee: SL User

FileName : 103-C2

Page 40 of 192

ASME TS Calc: Pressure Test

Case: 1 3:14p Apr 10, 2026

Allowable Tubesheet Bending Stress [σ_{All}]:

$$= 2 \cdot \beta T \cdot S_y$$

$$= 2 \cdot 0.95 \cdot 309976$$

$$= 588955.12 \text{ kPa}$$

Tubesheet bending stress at final thickness [σ]:

$$= 6 \cdot M / (\mu \cdot (h - h_g')^2)$$

$$= 6 \cdot 1234275 / (0.25 \cdot (231 - 5)^2)$$

$$= 588919.6250 \text{ kPa}$$

Required tubesheet thickness for Bending Stress [HreqB]:

$$= h + C_{ats} + C_{atc}$$

$$= 231 + 0 + 0$$

$$= 231.0725 \text{ mm}$$

Required tubesheet thickness for given loadings (includes CA) [Hreq]:

$$= \max(H_{reqB}; H_{reqS})$$

$$= \max(231; 82.3)$$

$$= 231.0725 \text{ mm}$$

ASME Section VIII-2 Part 4.18.7.4 Step 9:

Absolute pressure differential [$|P_s - P_t|$]:

$$= |P_s - P_t|$$

$$= |0 - 6.44|$$

$$= 6.435 \text{ N/mm}^2$$

Shear Stress check [τ_{limit}]:

$$= 4 \cdot \mu \cdot h / D_o \cdot 0.8 \cdot \beta T \cdot S_y$$

$$= 4 \cdot 0.2 \cdot 500 / 2571 \cdot 0.8 \cdot 0.95 \cdot 309976$$

$$= 36653.348 \text{ kPa}$$

$|P_s - P_t|$ is $\leq$ the Shear Stress limit (τ_{limit}). Therefore, the required thickness for shear is for information only.

Minimum required thickness for Shear [HreqS]:

$$= 1 / (4 \cdot \mu \cdot (0.8 \cdot \beta T \cdot S_y)) \cdot (4 \cdot A_p / C_p) \cdot |P_s - P_t| + C_{ats} + C_{atc}$$

$$= 1 / (4 \cdot 0.2 \cdot 235582) \cdot (4 \cdot 4.95 / 8219) \cdot |0 - 6.44| + 0$$

$$= 82.3264 \text{ mm}$$

ASME Section VIII-2 Part 4.18.7.4 Step 10:

Note: For a given Shell thickness of: 142.000mm

Min. Shell len. adjacent to the tubesheet is: 1106.261 mm

Note: For a given Channel thickness of: 50.000 mm

Min. Channel len. adjacent to the tubesheet is: 657.185mm

Tube Metal Area [At]:

$$= \pi (d_o - t) t$$

$$= \pi (25.4 - 3.05) 3.05$$

$$= 0.000 \text{ m}^2$$

Tube Weld Size Results per UW-20.3:

Tube Strength [Ft]:

$$= \pi \cdot t (d_o - t) S$$

$$= \pi \cdot 3.05 (25.4 - 3.05) 161488$$

$$= 34562.801 \text{ N}$$

Groove Weld Strength [Fg]:

$$= 0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w \text{ (but not } > F_t)$$

$$= 0.85 \cdot \pi \cdot 3 (25.4 + 0.67 \cdot 3) 161488$$

$$= 34562.8008 \text{ N}$$



PV Elite 28 Licensee: SL User

FileName : 103-C2

Page 41 of 192

ASME TS Calc: Pressure Test

Case: 1 3:14p Apr 10, 2026

Weld Strength Factor [fw]:

= S / S_w where $S_w = \min(S \text{ and } S_t)$

= 161488/161488

= 1.000

Design Strength Ratio [fd]:

= 1 for full strength welds

Fillet Weld Strength [Ff]:

= $0.55 \cdot \pi \cdot a_f (d_o + 0.67 \cdot a_f) S_w$ (but not > F_t)

= $0.55 \cdot \pi \cdot 3 (25.4 + 0.67 \cdot 3) 161488$

= 22944.0781 N

Minimum Length of Weld [ar]:

= $2 \left(\sqrt{((0.75 \cdot d_o)^2 + 1.07 \cdot t \cdot (d_o - t) f_w \cdot f_d)} - 0.75 \cdot d_o \right)$

= $2 \left(\sqrt{((0.75 \cdot 25.4)^2 + 1.07 \cdot 3.05 \cdot (25.4 - 3.05) \cdot 1 \cdot 1)} - 0.75 \cdot 25.4 \right)$

= 3.652 mm

Minimum Required Fillet Weld Leg [afr]: 1.8288 mm

Minimum Required Groove Weld Leg [agr]: 1.8288 mm

Tube-Tubesheet Jt allowable, 34563 is >= tube strength 34563 N

Note: This tube-tubesheet joint is a Full Strength joint

Stress/Force summary for loadcase D1 un-corr. (Psd,min + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet bend. stress	131837.9 <=	588955.1 kPa	Ok
Tubesheet shear stress	38789.2 <=	235582.0 kPa	Ok

Thickness results for loadcase D1 un-corr. (Psd,min + Ptd,max):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	231.072	500.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.829	3.000	Ok
Tube-Tubesheet Groove Weld Leg :	1.829	3.000	Ok

Detailed Results for load Case D2 un-corr. (Psd,max + Ptd,min)

Results for ASME U-tube Tubesheet Calculations for Configuration a:

Per Edition 2025, Original Thickness :

ASME Section VIII-2 Part 4.18.7.4 Step 1:

Equivalent outer tube limit circle diameter [Do]:

= $2 \cdot r_o + d_t$

= $2 \cdot 1273 + 25.4$

= 2570.920 mm

Basic ligament efficiency for shear [μ]:

= $(p - d_t) / p$

= $(31.8 - 25.4) / 31.8$

= 0.200

ASME Section VIII-2 Part 4.18.7.4 Step 2:



PV Elite 28 Licensee: SL User

FileName : 103-C2

Page 42 of 192

ASME TS Calc: Pressure Test

Case: 1 3:14p Apr 10, 2026

Ratio [ps]:

$$= D_s / D_o \text{ (Configurations a, b, c)}$$

$$= 2660/2571$$

$$= 1.0346$$

Ratio [pc]:

$$= D_c / D_o \text{ (Configurations a, e, f)}$$

$$= 2666/2571$$

$$= 1.0370$$

ASME Section VIII-2 Part 4.18.7.4 Step 3:

Ratio [p]:

$$= l_{tx} / h$$

$$= 50/500$$

$$= 0.100 \text{ (must be } 0 \leq p \leq 1 \text{)}$$

Effective tube hole Diameter [d*]:

$$= \max(dt - 2t_t (E t_T / E) \cdot (S t_T / S) \cdot (p); dt - 2t_t)$$

$$= \max(25.4 - 2 \cdot 3.05 (195128/204092) \cdot (114991/147989) \cdot (0.1); 25.4 - 2 \cdot 3.05)$$

$$= 24.9471 \text{ mm}$$

Effective tube pitch [p*]:

$$= p / \sqrt{ (1 - 4 \cdot \min(A_l; 4 \cdot D_o \cdot p) / (\pi \cdot D_o^2)) }$$

$$= 31.8 / \sqrt{ (1 - 4 \cdot \min(0.2; 4 \cdot 2571 \cdot 31.8) / (\pi \cdot 2571^2)) }$$

$$= 32.3886 \text{ mm}$$

Effective ligament efficiency for bending [μ*]:

$$= (p^* - d^*) / p^*$$

$$= (32.4 - 24.9) / 32.4$$

$$= 0.2298$$

$h/p = 15.748032$; $\mu^* = 0.229756$; $E^* = 39041 \text{ N/mm}^2$

$E^*/E = 0.191293$; $\nu^* = 0.419013$

Note: As h/p (15.748) is > 2 , data values for $h/p = 2$ were used.

ASME Section VIII-2 Part 4.18.7.4 Step 4:

Shell coefficient [βs]:

$$= ((12 (1 - \nu_s^2))^{1/4}) / \sqrt{ (D_s + t_s) t_s }$$

$$= ((12 (1 - 0.3^2))^{1/4}) / \sqrt{ (2660 + 142) 142 }$$

$$= 0.0029 \text{ 1/mm}$$

Shell coefficient [ks]:

$$= \beta_s \cdot E_s \cdot t_s^3 / (6 (1 - \nu_s^2))$$

$$= 0.0029 \cdot 204092 \cdot 142^3 / (6 (1 - 0.3^2))$$

$$= 308443584.0000 \text{ N/mm}^2 \cdot \text{mm}^2$$

Shell coefficient [λs]:

$$= 6 (D_s + t_s) \cdot k_s / h^3 \cdot (1 + h \cdot \beta_s + (h \cdot \beta_s)^2 / 2)$$

$$= 6 (2660 + 142) \cdot 308443584 / 500^3 \cdot (1 + 500 \cdot 0.0029 + (500 \cdot 0.0029)^2 / 2)$$

$$= 144328.7 \text{ N/mm}^2$$

Shell coefficient [δs]:

$$= D_s^2 / (4 \cdot E_s \cdot t_s) \cdot (1 - D_s \cdot \nu_s / (2 (D_s + t_s)))$$

$$= 2660^2 / (4 \cdot 204092 \cdot 142) \cdot (1 - 2660 \cdot 0.3 / (2 (2660 + 142)))$$

$$= 0.52348 \text{E-04 mm/kPa}$$

Parameter [ws]:

$$= p_s \cdot k_s \cdot \beta_s \cdot \delta_s (1 + h \cdot \beta_s)$$

$$= 1.03 \cdot 308443584 \cdot 0.0029 \cdot 0.52348 \text{E-04} (1 + 500 \cdot 0.0029)$$

$$= 117511.0781 \text{ mm}^2$$



PV Elite 28 Licensee: SL User

FileName : 103-C2

Page 43 of 192

ASME TS Calc: Pressure Test

Case: 1 3:14p Apr 10, 2026

Channel coefficient [β_c]:

$$= \left(\left(12(1 - v_c^2) \right)^{1/4} \right) / \sqrt{\left((D_c + t_c) t_c \right)}$$

$$= \left(\left(12(1 - 0.3^2) \right)^{1/4} \right) / \left((2666 + 50) 50 \right)$$

$$= 0.0049 \text{ 1/mm}$$

Channel coefficient [k_c]:

$$= \beta_c \cdot E_c \cdot t_c^3 / \left(6(1 - v_s^2) \right)$$

$$= 0.0049 \cdot 204092 \cdot 50^3 / \left(6(1 - 0.3^2) \right)$$

$$= 23048842.0000 \text{ N/mm}^2 \cdot \text{mm}^2$$

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:

$$CqDelta = CqTheta = CmTheta = CmDelta = 1$$

Channel coefficient [λ_c]:

$$= 6(D_c + t_c / \cos(\theta)) \cdot k_c / h^3 \cdot (CmTheta + h \cdot \beta_c \cdot (CmDelta + CqTheta) / 2 + (h \cdot \beta_c)^2 / 2 \cdot CqDelta)$$

$$= 6(2666 + 50 / \cos(0)) \cdot 23048842 / 500^3 \cdot (1 + 500 \cdot 0.0049 \cdot (1 + 1) / 2 + (500 \cdot 0.0049)^2 / 2 \cdot 1)$$

$$= 19556.1 \text{ N/mm}^2$$

Channel coefficient [δ_c]:

$$= D_c^2 / (4 \cdot E_c \cdot t_c) \cdot (1 - D_c \cdot v_c / (2(D_c + t_c)))$$

$$= 2666^2 / (4 \cdot 204092 \cdot 50) \cdot (1 - 2666 \cdot 0.3 / (2(2666 + 50)))$$

$$= 0.14850\text{E-}03 \text{ mm/kPa}$$

Channel coefficient [zetac]:

$$zetac = 0$$

Parameter [ω_c]:

$$= p_c \cdot k_c [\delta_c (CmTheta \cdot \beta_c + CqDelta \cdot h \cdot \beta_c^2) - \zeta_c (CmTheta + h \cdot CqTheta \cdot h \cdot \beta_c / 2)]$$

$$= 1.04 \cdot 23048842 [0.000148497 (1 + 0.0049 \cdot 1 \cdot 500) - 0.000 (1/500 + 1 \cdot 0.0049/2)]$$

$$= 60688.4023 \text{ mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 5:

Diameter ratio [K]:

$$= A / D_o$$

$$= 2980 / 2571$$

$$= 1.1591$$

Coefficient [F]:

$$= (1 - v^*) / E^* \cdot (\lambda_s + \lambda_c + E \cdot \ln(K))$$

$$= (1 - 0.42) / 39041 \cdot (144329 + 19556 + 204092 \cdot \ln(1.16))$$

$$= 2.8873$$

ASME Section VIII-2 Part 4.18.7.4 Step 6:

Moment on tubesheet due to pressures (Ps, Pt) [Mts]:

$$= D_o^2 / 16 \cdot [(p_s - 1) \cdot (p_s^2 + 1) \cdot P_s - (p_c - 1) \cdot (p_c^2 + 1) \cdot P_t]$$

$$= 2571^2 / 16 \cdot [(1.03 - 1) \cdot (1.03^2 + 1) \cdot 18.6 - (1.04 - 1) \cdot (1.04^2 + 1) \cdot 0]$$

$$= 550200.8125 \text{ N/mm}^2 \cdot \text{mm}^2$$

Moment acting on unperforated tubesheet rim [M*]:

$$= Mts + \omega_c \cdot P_t - \omega_s \cdot P_s$$

$$= 550201 + 60688 \cdot 0 - 117511 \cdot 18.6$$

$$= -1631401.5000 \text{ N/mm}^2 \cdot \text{mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 7:

Maximum bending moment acting on periphery of tubesheet [Mp]:



PV Elite 28 Licensee: SL User

FileName : 103-C2

Page 44 of 192

ASME TS Calc: Pressure Test

Case: 1 3:14p Apr 10, 2026

$$\begin{aligned}
 &= ((M \cdot) - Do^2/32 \cdot F(Ps - Pt)) / (1 + F) \\
 &= ((-1631401) - 2571^2/32 \cdot 2.89 (18.6 - 0)) / (1 + 2.89) \\
 &= -3267854.7500 \text{ N/mm}^2 \cdot \text{mm}^2
 \end{aligned}$$

Maximum bending moment acting on center of tubesheet [Mo]:

$$\begin{aligned}
 &= Mp + Do^2/64 \cdot (3 + \nu \cdot) \cdot (Ps - Pt) \\
 &= -3267854 + 2571^2/64 \cdot (3 + 0.42) \cdot (18.6 - 0) \\
 &= 3287476.5000 \text{ N/mm}^2 \cdot \text{mm}^2
 \end{aligned}$$

Maximum bending moment acting on tubesheet [M]:

$$\begin{aligned}
 &= \max(| Mp |; | Mo |) \\
 &= \max(| -3267854 |; | 3287476 |) \\
 &= 3287476.5000 \text{ N/mm}^2 \cdot \text{mm}^2
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 8:

Tubesheet Bending Stress at original thickness:

$$\begin{aligned}
 &= 6 \cdot M / (\mu \cdot (h - hg')^2) \\
 &= 6 \cdot 3287476 / (0.23 (500 - 5)^2) \\
 &= 350357.3750 \text{ kPa}
 \end{aligned}$$

Allowable Tubesheet Bending Stress [σ_{All}]:

$$\begin{aligned}
 &= 2 \cdot \beta T \cdot Sy \\
 &= 2 \cdot 0.95 \cdot 309976 \\
 &= 588955.12 \text{ kPa}
 \end{aligned}$$

Tubesheet bending stress at final thickness [σ]:

$$\begin{aligned}
 &= 6 \cdot M / (\mu \cdot (h - hg')^2) \\
 &= 6 \cdot 3376321 / (0.23 (389 - 5)^2) \\
 &= 588506.8125 \text{ kPa}
 \end{aligned}$$

Required tubesheet thickness for Bending Stress [HreqB]:

$$\begin{aligned}
 &= h + Cats + Catc \\
 &= 389 + 0 + 0 \\
 &= 388.7298 \text{ mm}
 \end{aligned}$$

Required tubesheet thickness for given loadings (includes CA) [Hreq]:

$$\begin{aligned}
 &= \max(HreqB; HreqS) \\
 &= \max(389 ; 237) \\
 &= 388.7298 \text{ mm}
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 9:

Absolute pressure differential [$| Ps - Pt |$]:

$$\begin{aligned}
 &= | Ps - Pt | \\
 &= | 18.6 - 0 | \\
 &= 18.565 \text{ N/mm}^2
 \end{aligned}$$

Shear Stress check [τ_{limit}]:

$$\begin{aligned}
 &= 4 \cdot \mu \cdot h/Do \cdot 0.8 \cdot \beta T \cdot Sy \\
 &= 4 \cdot 0.2 \cdot 500/2571 \cdot 0.8 \cdot 0.95 \cdot 309976 \\
 &= 36653.348 \text{ kPa}
 \end{aligned}$$

$| Ps - Pt |$ is $\leq$ the Shear Stress limit (τ_{limit}). Therefore, the required thickness for shear is for information only.

Minimum required thickness for Shear [HreqS]:

$$\begin{aligned}
 &= 1 / (4 \cdot \mu \cdot (0.8 \cdot \beta T \cdot Sy)) \cdot (4 \cdot Ap/Cp) \cdot | Ps - Pt | + Cats + Catc \\
 &= 1 / (4 \cdot 0.2 \cdot 235582) \cdot (4 \cdot 4.95/8219) \cdot | 18.6 - 0 | + 0 \\
 &= 237.4991 \text{ mm}
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 10:

Note: For a given Shell thickness of: 142.000mm

Min. Shell len. adjacent to the tubesheet is: 1106.261 mm



PV Elite 28 Licensee: SL User

FileName : 103-C2

Page 45 of 192

ASME TS Calc: Pressure Test

Case: 1 3:14p Apr 10,2026

Note: For a given Channel thickness of: 50.000 mm
Min. Channel len. adjacent to the tubesheet is: 657.185mm

Tube Metal Area [At]:
= $\pi (d_o - t) t$
= $\pi (25.4 - 3.05) 3.05$
= 0.000 m²

Tube Weld Size Results per UW-20.3:

Tube Strength [Ft]:
= $\pi \cdot t (d_o - t) S$
= $\pi \cdot 3.05 (25.4 - 3.05) 161488$
= 34562.801 N

Groove Weld Strength [Fg]:
= $0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w$ (but not > Ft)
= $0.85 \cdot \pi \cdot 3 (25.4 + 0.67 \cdot 3) 161488$
= 34562.8008 N

Weld Strength Factor [fw]:
= S / S_w where $S_w = \min(S \text{ and } S_t)$
= 161488/161488
= 1.000

Design Strength Ratio [fd]:
= 1 for full strength welds

Fillet Weld Strength [Ff]:
= $0.55 \cdot \pi \cdot a_f (d_o + 0.67 \cdot a_f) S_w$ (but not > Ft)
= $0.55 \cdot \pi \cdot 3 (25.4 + 0.67 \cdot 3) 161488$
= 22944.0781 N

Minimum Length of Weld [ar]:
= $2 \left(\sqrt{((0.75 \cdot d_o)^2 + 1.07 \cdot t \cdot (d_o - t) f_w \cdot f_d)} - 0.75 \cdot d_o \right)$
= $2 \left(\sqrt{((0.75 \cdot 25.4)^2 + 1.07 \cdot 3.05 \cdot (25.4 - 3.05) 1 \cdot 1)} - 0.75 \cdot 25.4 \right)$
= 3.652 mm

Minimum Required Fillet Weld Leg [afr]: 1.8288 mm
Minimum Required Groove Weld Leg [agr]: 1.8288 mm

Tube-Tubesheet Jt allowable, 34563 is >= tube strength 34563 N

Note: This tube-tubesheet joint is a Full Strength joint

Stress/Force summary for loadcase D2 un-corr. (Psd,max + Ptd,min):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet bend. stress	350357.4 <=	588955.1 kPa	Ok
Tubesheet shear stress	111901.0 <=	235582.0 kPa	Ok

Thickness results for loadcase D2 un-corr. (Psd,max + Ptd,min):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	388.730	500.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.829	3.000	Ok
Tube-Tubesheet Groove Weld Leg :	1.829	3.000	Ok

Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)



PV Elite 28 Licensee: SL User

FileName : 103-C2

Page 46 of 192

ASME TS Calc: Pressure Test

Case: 1 3:14p Apr 10, 2026

Results for ASME U-tube Tubesheet Calculations for Configuration a:

Per Edition 2025, Original Thickness :

ASME Section VIII-2 Part 4.18.7.4 Step 1:

Equivalent outer tube limit circle diameter [Do]:

$$\begin{aligned} &= 2 \cdot r_o + dt \\ &= 2 \cdot 1273 + 25.4 \\ &= 2570.920 \text{ mm} \end{aligned}$$

Basic ligament efficiency for shear [μ]:

$$\begin{aligned} &= (p - dt) / p \\ &= (31.8 - 25.4) / 31.8 \\ &= 0.200 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 2:

Ratio [ps]:

$$\begin{aligned} &= D_s / D_o \text{ (Configurations a, b, c)} \\ &= 2660 / 2571 \\ &= 1.0346 \end{aligned}$$

Ratio [pc]:

$$\begin{aligned} &= D_c / D_o \text{ (Configurations a, e, f)} \\ &= 2666 / 2571 \\ &= 1.0370 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 3:

Ratio [p]:

$$\begin{aligned} &= l_{tx} / h \\ &= 50 / 500 \\ &= 0.100 \text{ (must be } 0 \leq \rho \leq 1 \text{)} \end{aligned}$$

Effective tube hole Diameter [d*]:

$$\begin{aligned} &= \max(dt - 2t_t (E_t T / E) \cdot (S_t T / S) \cdot (\rho); dt - 2t_t) \\ &= \max(25.4 - 2 \cdot 3.05 (195128 / 204092) \cdot (114991 / 147989) \cdot (0.1); 25.4 - 2 \cdot 3.05) \\ &= 24.9471 \text{ mm} \end{aligned}$$

Effective tube pitch [p*]:

$$\begin{aligned} &= p / \sqrt{ (1 - 4 \cdot \min(A_l; 4 \cdot D_o \cdot p) / (\pi \cdot D_o^2)) } \\ &= 31.8 / \sqrt{ (1 - 4 \cdot \min(0.2; 4 \cdot 2571 \cdot 31.8) / (\pi \cdot 2571^2)) } \\ &= 32.3886 \text{ mm} \end{aligned}$$

Effective ligament efficiency for bending [μ^*]:

$$\begin{aligned} &= (p^* - d^*) / p^* \\ &= (32.4 - 24.9) / 32.4 \\ &= 0.2298 \end{aligned}$$

$h/p = 15.748032$; $\mu^* = 0.229756$; $E^* = 39041 \text{ N/mm}^2$

$E^*/E = 0.191293$; $\nu^* = 0.419013$

Note: As h/p (15.748) is > 2 , data values for $h/p = 2$ were used.

ASME Section VIII-2 Part 4.18.7.4 Step 4:

Shell coefficient [β_s]:

$$\begin{aligned} &= ((12 (1 - \nu_s^2))^{1/4}) / \sqrt{ ((D_s + t_s) t_s) } \\ &= ((12 (1 - 0.3^2))^{1/4}) / \sqrt{ ((2660 + 142) 142) } \\ &= 0.0029 \text{ 1/mm} \end{aligned}$$



PV Elite 28 Licensee: SL User

FileName : 103-C2

Page 47 of 192

ASME TS Calc: Pressure Test

Case: 1 3:14p Apr 10, 2026

Shell coefficient [ks]:

$$= \beta_s \cdot E_s \cdot t_s^3 / (6(1 - \nu_s^2))$$

$$= 0.0029 \cdot 204092 \cdot 142^3 / (6(1 - 0.3^2))$$

$$= 308443584.0000 \text{ N/mm}^2 \cdot \text{mm}^2$$

Shell coefficient [λs]:

$$= 6(D_s + t_s) \cdot k_s / h^3 \cdot (1 + h \cdot \beta_s + (h \cdot \beta_s)^2 / 2)$$

$$= 6(2660 + 142) \cdot 308443584 / 500^3 \cdot (1 + 500 \cdot 0.0029 + (500 \cdot 0.0029)^2 / 2)$$

$$= 144328.7 \text{ N/mm}^2$$

Shell coefficient [δs]:

$$= D_s^2 / (4 \cdot E_s \cdot t_s) \cdot (1 - D_s \cdot \nu_s / (2(D_s + t_s)))$$

$$= 2660^2 / (4 \cdot 204092 \cdot 142) \cdot (1 - 2660 \cdot 0.3 / (2(2660 + 142)))$$

$$= 0.52348\text{E-}04 \text{ mm/kPa}$$

Parameter [ws]:

$$= \rho_s \cdot k_s \cdot \beta_s \cdot \delta_s (1 + h \cdot \beta_s)$$

$$= 1.03 \cdot 308443584 \cdot 0.0029 \cdot 0.52348\text{E-}04 (1 + 500 \cdot 0.0029)$$

$$= 117511.0781 \text{ mm}^2$$

Channel coefficient [βc]:

$$= ((12(1 - \nu_c^2))^{1/4}) / \sqrt{(D_c + t_c)t_c}$$

$$= ((12(1 - 0.3^2))^{1/4}) / \sqrt{(2666 + 50)50}$$

$$= 0.0049 \text{ 1/mm}$$

Channel coefficient [kc]:

$$= \beta_c \cdot E_c \cdot t_c^3 / (6(1 - \nu_s^2))$$

$$= 0.0049 \cdot 204092 \cdot 50^3 / (6(1 - 0.3^2))$$

$$= 23048842.0000 \text{ N/mm}^2 \cdot \text{mm}^2$$

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:

$$CqDelta = CqTheta = CmTheta = CmDelta = 1$$

Channel coefficient [λc]:

$$= 6(D_c + t_c / \cos(\theta)) \cdot k_c / h^3 \cdot (CmTheta + h \cdot \beta_c \cdot (Cm\Delta + CqTheta) / 2 + (h \cdot \beta_c)^2 / 2 \cdot Cq\Delta)$$

$$= 6(2666 + 50 / \cos(0)) \cdot 23048842 / 500^3 \cdot (1 + 500 \cdot 0.0049 \cdot (1 + 1) / 2 + (500 \cdot 0.0049)^2 / 2 \cdot 1)$$

$$= 19556.1 \text{ N/mm}^2$$

Channel coefficient [δc]:

$$= D_c^2 / (4 \cdot E_c \cdot t_c) \cdot (1 - D_c \cdot \nu_c / (2(D_c + t_c)))$$

$$= 2666^2 / (4 \cdot 204092 \cdot 50) \cdot (1 - 2666 \cdot 0.3 / (2(2666 + 50)))$$

$$= 0.14850\text{E-}03 \text{ mm/kPa}$$

Channel coefficient [zetac]:

$$zetac = 0$$

Parameter [wc]:

$$= \rho_c \cdot k_c [\delta_c (CmTheta \cdot \beta_c + Cq\Delta \cdot h \cdot \beta_c^2) - \zeta_c (CmTheta + h \cdot CqTheta \cdot h \cdot \beta_c / 2)]$$

$$= 1.04 \cdot 23048842 [0.000148497 (1 + 0.0049 \cdot 1 \cdot 500) - 0.000 (1/500 + 1 \cdot 0.0049/2)]$$

$$= 60688.4023 \text{ mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 5:

Diameter ratio [K]:

$$= A / D_o$$

$$= 2980 / 2571$$

$$= 1.1591$$

Coefficient [F]:

$$= (1 - \nu^*) / E^* \cdot (\lambda_s + \lambda_c + E \cdot \ln(K))$$



PV Elite 28 Licensee: SL User

FileName : 103-C2

Page 48 of 192

ASME TS Calc: Pressure Test

Case: 1 3:14p Apr 10, 2026

$$= (1 - 0.42) / 39041 \cdot (144329 + 19556 + 204092 \cdot \ln(1.16))$$

$$= 2.8873$$

ASME Section VIII-2 Part 4.18.7.4 Step 6:

Moment on tubesheet due to pressures (Ps, Pt) [Mts]:

$$= Do^2 / 16 \cdot [(ps - 1) \cdot (ps^2 + 1) \cdot Ps - (pc - 1) \cdot (pc^2 + 1) \cdot Pt]$$

$$= 2571^2 / 16 \cdot [(1.03 - 1) \cdot (1.03^2 + 1) \cdot 18.6 - (1.04 - 1) \cdot (1.04^2 + 1) \cdot 6.44]$$

$$= 346158.8125 \text{ N/mm}^2 \cdot \text{mm}^2$$

Moment acting on unperforated tubesheet rim [M*]:

$$= Mts + \omega c \cdot Pt - \omega s \cdot Ps$$

$$= 346159 + 60688 \cdot 6.44 - 117511 \cdot 18.6$$

$$= -1444891.0000 \text{ N/mm}^2 \cdot \text{mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 7:

Maximum bending moment acting on periphery of tubesheet [Mp]:

$$= ((M*) - Do^2 / 32 \cdot F(Ps - Pt)) / (1 + F)$$

$$= ((-1444891) - 2571^2 / 32 \cdot 2.89(18.6 - 6.44)) / (1 + 2.89)$$

$$= -2232586.2500 \text{ N/mm}^2 \cdot \text{mm}^2$$

Maximum bending moment acting on center of tubesheet [Mo]:

$$= Mp + Do^2 / 64 \cdot (3 + \nu) \cdot (Ps - Pt)$$

$$= -2232586 + 2571^2 / 64 \cdot (3 + 0.42) \cdot (18.6 - 6.44)$$

$$= 2050413.8750 \text{ N/mm}^2 \cdot \text{mm}^2$$

Maximum bending moment acting on tubesheet [M]:

$$= \max(|Mp|; |Mo|)$$

$$= \max(|-2232586|; |2050413|)$$

$$= 2232586.2500 \text{ N/mm}^2 \cdot \text{mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 8:

Tubesheet Bending Stress at original thickness:

$$= 6 \cdot M / (\mu \cdot (h - hg')^2)$$

$$= 6 \cdot 2232586 / (0.23(500 - 5)^2)$$

$$= 237934.1719 \text{ kPa}$$

Allowable Tubesheet Bending Stress [σ_{All}]:

$$= 2 \cdot \beta T \cdot Sy$$

$$= 2 \cdot 0.95 \cdot 309976$$

$$= 588955.12 \text{ kPa}$$

Tubesheet bending stress at final thickness [σ]:

$$= 6 \cdot M / (\mu \cdot (h - hg')^2)$$

$$= 6 \cdot 2313161 / (0.24(320 - 5)^2)$$

$$= 588782.5625 \text{ kPa}$$

Required tubesheet thickness for Bending Stress [HreqB]:

$$= h + Cats + Catc$$

$$= 320 + 0 + 0$$

$$= 319.9509 \text{ mm}$$

Required tubesheet thickness for given loadings (includes CA) [Hreq]:

$$= \max(HreqB; HreqS)$$

$$= \max(320; 155)$$

$$= 319.9509 \text{ mm}$$

ASME Section VIII-2 Part 4.18.7.4 Step 9:

Absolute pressure differential [|Ps - Pt|]:

$$= |Ps - Pt|$$



PV Elite 28 Licensee: SL User

FileName : 103-C2

Page 49 of 192

ASME TS Calc: Pressure Test

Case: 1 3:14p Apr 10, 2026

$$= | 18.6 - 6.44 |$$

$$= 12.130 \text{ N/mm}^2$$

Shear Stress check [τ_{limit}]:

$$= 4 \cdot \mu \cdot h/Do \cdot 0.8 \cdot \beta T \cdot Sy$$

$$= 4 \cdot 0.2 \cdot 500/2571 \cdot 0.8 \cdot 0.95 \cdot 309976$$

$$= 36653.348 \text{ kPa}$$

$| P_s - P_t |$ is $\leq$ the Shear Stress limit (τ_{limit}). Therefore, the required thickness for shear is for information only.

Minimum required thickness for Shear [H_{reqS}]:

$$= 1/(4 \cdot \mu \cdot (0.8 \cdot \beta T \cdot Sy)) \cdot (4 \cdot Ap/Cp) \cdot | P_s - P_t | + Cats + Catc$$

$$= 1/(4 \cdot 0.2 \cdot 235582) \cdot (4 \cdot 4.95/8219) \cdot | 18.6 - 6.44 | + 0$$

$$= 155.1727 \text{ mm}$$

ASME Section VIII-2 Part 4.18.7.4 Step 10:

Note: For a given Shell thickness of: 142.000mm

Min. Shell len. adjacent to the tubesheet is: 1106.261 mm

Note: For a given Channel thickness of: 50.000 mm

Min. Channel len. adjacent to the tubesheet is: 657.185mm

Tube Metal Area [A_t]:

$$= \pi (d_o - t) t$$

$$= \pi (25.4 - 3.05) 3.05$$

$$= 0.000 \text{ m}^2$$

Tube Weld Size Results per UW-20.3:

Tube Strength [F_t]:

$$= \pi \cdot t (d_o - t) S$$

$$= \pi \cdot 3.05 (25.4 - 3.05) 161488$$

$$= 34562.801 \text{ N}$$

Groove Weld Strength [F_g]:

$$= 0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w \text{ (but not } > F_t)$$

$$= 0.85 \cdot \pi \cdot 3 (25.4 + 0.67 \cdot 3) 161488$$

$$= 34562.8008 \text{ N}$$

Weld Strength Factor [f_w]:

$$= S / S_w \text{ where } S_w = \min(S \text{ and } S_t)$$

$$= 161488/161488$$

$$= 1.000$$

Design Strength Ratio [f_d]:

$$= 1 \text{ for full strength welds}$$

Fillet Weld Strength [F_f]:

$$= 0.55 \cdot \pi \cdot a_f (d_o + 0.67 \cdot a_f) S_w \text{ (but not } > F_t)$$

$$= 0.55 \cdot \pi \cdot 3 (25.4 + 0.67 \cdot 3) 161488$$

$$= 22944.0781 \text{ N}$$

Minimum Length of Weld [a_r]:

$$= 2 \left(\sqrt{(0.75 \cdot d_o)^2 + 1.07 \cdot t \cdot (d_o - t) f_w \cdot f_d} - 0.75 \cdot d_o \right)$$

$$= 2 \left(\sqrt{(0.75 \cdot 25.4)^2 + 1.07 \cdot 3.05 \cdot (25.4 - 3.05) 1 \cdot 1} - 0.75 \cdot 25.4 \right)$$

$$= 3.652 \text{ mm}$$

Minimum Required Fillet Weld Leg [a_{fr}]: 1.8288 mm

Minimum Required Groove Weld Leg [a_{gr}]: 1.8288 mm

Tube-Tubesheet Jt allowable, 34563 is $\geq$ tube strength 34563 N



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FileName : 103-C2

Page 50 of 192

ASME TS Calc: Pressure Test

Case: 1 3:14p Apr 10,2026

Note: This tube-tubesheet joint is a Full Strength joint

Stress/Force summary for loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet bend. stress	237934.2	<= 588955.1 kPa	Ok
Tubesheet shear stress	73111.8	<= 235582.0 kPa	Ok

Thickness results for loadcase D3 un-corr. (Psd,max + Ptd,max):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	319.951	500.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.829	3.000	Ok
Tube-Tubesheet Groove Weld Leg :	1.829	3.000	Ok

U-Tube Tubesheet results per ASME Div. 2 Part 4.18.7 2025

Results for 3 Load Cases:

Case#	--Reqd. Thk. + CA Tbsht Extnsn	----	Tubesheet Bend Allwd	Stresses Shear Allwd	Case Type	Pass/ Fail
D1uc	231.072 ...	131838	588955	38789 235582	Ps+Pt D1	Ok
D2uc	388.730 ...	350357	588955	111901 235582	Ps+Pt D2	Ok
D3uc	319.951 ...	237934	588955	73112 235582	Ps+Pt D3	Ok
Max:	388.7298 ... mm	0.595	0.475	(Str. Ratio)		

Load Case Definitions:

[Ps & Pt]:

Shell-side and Tube-side Design or Operating Pressures
derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the
Load Case Tables

[c]:

With or Without Corrosion Allowance

[D1, D2, D3]:

Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:

Design Load Case using the Minimum (Vacuum) Pressures (if specified)

Shell and Channel Stress Summary:

Case#	-- Integral Cyl. Req'd. Thk. Shell Channel	Shell Stress	Stress Allwd	Channel Stress	Stress Allwd	Pass Fail
D1uc		...	441716	...	441716	Ok
D2uc		...	441716	...	441716	Ok
D3uc		...	441716	...	441716	Ok
Max		...	...	...	...	

Note:

Because there was no net pressure differential across the tubesheet due to
the specified pressures psd,min and ptd,min; load case D4 was not processed as
the resulting tubesheet bending stress would be zero.

Summary of Thickness Comparisons for 3 Load Cases:



PV Elite 28 Licensee: SL User

FileName : 103-C2

Page 51 of 192

ASME TS Calc: Pressure Test

Case: 1 3:14p Apr 10,2026

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	388.730	500.000	Ok
Tube thickness :	2.312	3.048	Ok
Tube-tubesheet fillet weld leg :	1.829	3.000	Ok
Tube-tubesheet groove weld leg :	1.829	3.000	Ok

Given Shell Thickness: 142.0000 mm

Given Channel Thickness: 50.0000 mm

Min Shell length of thk, (142.000) adj. to tubesheet: 1106.261 mm

Min Channel length of thk, (50.000) adj. to tubesheet: 657.185 mm

Note: This is a full strength Tube to Tubesheet Joint.

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FileName : 103-C2

Page 52 of 192

External Pressure Calculations: Step: 10 3:14pm Apr 10,2026

External Pressure Calculation Results Summary:

ASME Code, Section VIII Division 1, 2025

External Pressure Calculations:

From	To	Section Length m	Outside Diameter mm	Corroded Thickness mm	Factor A	Factor B kPa
10	20	No Calc	2722	26	No Calc	No Calc
20	Tsht	2.9443	2766	48	No Calc	No Calc
Tsht	40	No Calc	...	48	No Calc	No Calc
40	50	6.24336	2944	139	No Calc	No Calc
50	60	No Calc	2800	67	No Calc	No Calc

External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	External Design Pressure N/mm ²	External M.A.W.P. N/mm ²
10	20	28	No Calc	...	3.77
20	Tsht	50	No Calc	...	5.01
Tsht	40	50	No Calc	...	No Calc
40	50	142	15.1577	0.10131	14.8
50	60	70	6.70047	0.10131	12.1

Minimum

3.77

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners m	Allowable Length Bet. Stiffeners m	Ring Inertia Required mm ⁴	Ring Inertia Available mm ⁴
10	20	No Calc	No Calc	No Calc	No Calc
20	Tsht	2.9443	No Calc	No Calc	No Calc
Tsht	40	No Calc	No Calc	No Calc	No Calc
40	50	6.24336	25400	No Calc	No Calc
50	60	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

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FileName : 103-C2

Page 53 of 192

Element and Detail Weights:

Step: 11

3:14pm Apr 10,2026

Element and Detail Weights:

From	To	Element Metal Wgt. kgm	Element ID Volume m ³	Corroded Metal Wgt. kgm	Corroded ID Volume m ³	Extra due Misc % kgm
10	20	3017.72	4.8436	2844.4	4.86543	301.772
20	40	8266.46	13.626	7941.64	13.6669	826.646
40	50	56190.7	12.0466	55062.5	12.1887	5619.07
50	60	7508.07	4.81097	7249.06	4.8436	750.807
Total		74982	35.33	73097	35.56	7498



For elements specified as shell side elements, the volume(s) shown above for those elements, reflects the displacement of the tubes.

Weight of Details:

From	Type	Weight of Detail kgm	X Offset, Dtl. Cent. m	Y Offset, Dtl. Cent. m	Z Offset, Dtl. Cent. m	Description
10	Liqd	91.7347	-0.44433	...	...	Liquid: 10
10	Lini	267.659	-0.6665	...	...	316SS weld over
10	Nozl	2403.79	-1.333	...	...	Manhole
10	Wght	500	-1.333	...	...	Davit
20	Liqd	258.068	1.25	...	...	Liquid: 20
20	Lini	502.54	1.25	...	...	316SS Weld over
20	Nozl	1707.44	1.1	1.7902	...	N1- Tube inlet
20	Nozl	1707.45	1.1	-1.7902	...	N2- Tube outlet
20	Wght	2200	1.25	...	...	PP gurad
40	Sadl	2980.72	1.15	1.729	...	Left Saddle
40	Sadl	2980.72	4.55	1.729	...	Right Saddle
40	Liqd	12332.7	2.9	...	...	Liquid: 40
40	Nozl	464.161	1.2	1.457	...	Shell outlet-1
40	Nozl	495.491	2.5	-1.457	...	Shell inlet-1
40	Wght	85418	1.15	4.99	...	103-C1 WEI (Ope
40	Wght	85418	4.55	4.99	...	103-C1 WEI (Ope
40	Wght	100483	1.15	4.99	...	103-C1 WEI (Hyd
40	Wght	100483	4.55	4.99	...	103-C1 WEI (Hyd
40	Wght	75000	1.15	4.99	...	103-C1 WEI (Emp
40	Wght	75000	4.55	4.99	...	103-C1 WEI (EMP
40	Wght	400	1.15	-2.3	...	Fireproofing 1
40	Wght	400	4.55	-2.3	...	Fireproofing 2
40	Forc	...	1.15	...	...	Left saddle- wi
40	Forc	...	4.55	...	...	Right saddle-wi
40	Forc	...	1.15	...	...	Left sadd.-seis
40	Forc	...	4.55	...	...	Right sadd. sei
50	Liqd	4925.23	0.44333	...	...	Liquid: 50
20	FTsh	17752.1	2.75	...	...	Tubesheet
20	Tube	73784.8	5.25	...	...	Tubes
40	Baf1	512.345	0.6875	...	...	Baffles
40	Baf1	512.345	1.375	...	...	Baffles
40	Baf1	512.345	2.0625	...	...	Baffles
40	Baf1	512.345	2.75	...	...	Baffles
40	Baf1	512.345	3.4375	...	...	Baffles
40	Baf1	512.345	4.125	...	...	Baffles
40	Baf1	513.406	4.8125	...	...	Baffles

Total Weight of Each Detail Type:



PV Elite 28 Licensee: SL User

FileName : 103-C2

Page 54 of 192

Element and Detail Weights: Step: 11 3:14pm Apr 10,2026

Saddles	5961.4
Liquid	17607.7
Cladding & Overlay	770.2
Nozzles	6778.3
Weights	525302.1
Exchanger Components	95124.4
Liquid in Tubes	229.7

Sum of the Detail Weights 651773.9 kgm

Weight Summation Results: (kgm)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	82481.3	82481.3	82481.3	82481.3	82481.3	82481.3
Saddles	5961.4	5961.4	5961.4	5961.4	5961.4	5961.4
Cladding	770.2	770.2	770.2	770.2	770.2	770.2
Nozzles	6778.3	6778.3	6778.3	6778.3	6778.3	6778.3
Wld Weights	2700.0	2700.0	2700.0	2700.0	2700.0	2700.0
Exchanger	95124.4	95124.4	95124.4	95124.4	95124.4	95124.4
Empty Weights	...	...	...	150800.0	150800.0	...
Test Weights	...	201766.0	...	...	...	...
Ope Weights	...	...	...	...	...	171636.0
Ope. Liquid	...	...	...	...	...	17607.7
Tube Ope Lqd	...	...	...	...	...	229.7
Test Liquid	...	36070.1	...	...	...	...
Tube Tst Lqd	...	12296.8	...	...	...	...
Totals	193815.6	443948.5	193815.6	344615.6	344615.6	383289.0

Field Installation Options:

Miscellaneous Weight Percent: 10.0 %

Note that the above value for the miscellaneous weight percent has been applied to the shells/heads/flange/tubesheets/tubes etc. in the weight calculations for metallic components.

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	193815.6 kgm
Shop Test Wt.	- Fabricated weight + water (full)	443948.5 kgm
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	193815.6 kgm
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	344615.6 kgm
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	344615.6 kgm
Operating Wt.	- Empty weight + operating liq. uncorroded	383289.0 kgm
Oper. Wt. + CA	- Corr wt. + operating liquid	381215.1 kgm
Field Test Wt.	- Empty weight + test medium (full)	443948.5 kgm

Exchanger Tube Data

Volume of Exchanger tubes :	12.0 m ³
Weight of Ope Liq in tubes :	229.7 kgm
Weight of Water in tubes :	12296.8 kgm

Surface Areas of Elements:

From	To	Outside Surface Area m ²	Inside Surface Area m ²
10	20	11.7384	11.1618
20	40	21.7187	20.9335
40	50	53.63	48.4565
50	60	12.524	11.1116



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FileName : 103-C2

Page 55 of 192

Element and Detail Weights: Step: 11 3:14pm Apr 10,2026

Total	99.6	91.7 m ²
Total	1072.5	986.9 ft ²

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FileName : 103-C2

Page 56 of 192

Nozzle Flange MAWP:

Step: 12 3:14pm Apr 10,2026

Nozzle Flange MAWP Results: (N/mm² & °C)

Nozzle Description	Flange Rating		Des. Temp	Class	Grade/Group	Equiv. Press	Max Pressure		
	Ope.	Ambient					UG-44	50%	DNV
Manhole	11.38	15.31	343	900	GR 1.1	...	...	...	...

Shellside Flange Rating**Channelside Flange Rating**Lowest Flange Pressure Rating was (Ope)[TubeSide]: 11.378 N/mm²Lowest Flange Pressure Rating was (Amb)[TubeSide]: 15.311 N/mm²

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

Large Diameter Flange Pressure Ratings are per ASME B16.47 2025 Edition

Warning:

There are nozzles in this model, but no flange MAWPs were de-rated. Be sure this is what you intended. There is a check box in the nozzle dialog that instructs PV Elite to perform the de-rating for each nozzle flange. See ASME VIII-1, UG-44(b) for more information.

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PV Elite 28 Licensee: SL User

FileName : 103-C2

Page 57 of 192

Wind Load Calculation:

Step: 13 3:14pm Apr 10,2026

Wind Load Calculation Results:

Input Values:

Wind Design Code	ASCE/SEI 7-16
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed [V]	27 m/sec
Surface Roughness Category	C: Open Terrain
Type of Surface	Moderately Smooth
Base Elevation	0.3 m
Percent Wind for Hydrotest	50.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 m
Distance Upwind of Crest Lh	0 m
Distance from Crest to the Vessel x	0 m
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 m

Wind Analysis Results

The importance factor is not used in the calculation of wind pressure vs. elevation for the selected wind code.

Static Gust-Effect Factor, Operating Case [G]:

$$= \min(0.85, 0.925((1 + 1.7 \cdot gQ \cdot I_{zbar} \cdot Q) / (1 + 1.7 \cdot gV \cdot I_{zbar})))$$

$$= \min(0.85, 0.925((1 + 1.7 \cdot 3.4 \cdot 0.23 \cdot 0.97) / (1 + 1.7 \cdot 3.4 \cdot 0.23)))$$

$$= \min(0.85, 0.91)$$

$$= 0.850$$

Natural Frequency of Vessel (Operating)	33.000 Hz
Natural Frequency of Vessel (Empty)	33.000 Hz
Natural Frequency of Vessel (Test)	33.000 Hz

User Entered Importance Factor is	1.000
Force Coefficient [Cf]	0.547
Structure Height to Diameter ratio	3.843

This is classified as a rigid structure. Static analysis performed.

Sample Calculation for the First Element:

The ASCE code performs calculations in Imperial Units. The wind pressure is therefore computed in these units.

Value of $[\alpha]$ and $[Z_g]$:

Exposure Category: C from Table 26.11-1
 $\alpha = 9.5 : Z_g = 274 \text{ m}$

Effective Height $[z]$:

= Centroid Height + Vessel Base Elevation
 $= 2.3 + 0.3 = 2.6 \text{ m}$
 $= 8.53 \text{ ft. Imperial Units}$

Velocity Pressure coefficient evaluated at height z $[K_z]$:

$$\text{Because } z (8.53 \text{ ft.}) < 15 \text{ ft.}$$

$$= 2.01(15 / Z_g)^{2/\alpha}$$

$$= 2.01(15/900)^{2/9.5}$$

$$= 0.85$$

Type of Hill: No Hill



PV Elite 28 Licensee: SL User

FileName : 103-C2

Page 58 of 192

Wind Load Calculation:

Step: 13 3:14pm Apr 10, 2026

Wind Directionality Factor [Kd]:

= 1, per Table 26.6-1

As there is No Hill Present: [Kzt]:

K1 = 0, K2 = 0, K3 = 0

Topographical Factor [Kzt]:

= (1 + K1 • K2 • K3)²

= (1 + 0 • 0 • 0)²

= 1

Velocity Pressure evaluated at height z, Imperial Units [qz]:

= max(16, 0.00256 • Kz • Kzt • Kd • Ke • V(mph)²)

= max(16, 0.00256 • 0.85 • 1 • 1 • 1 • 60.4²)

= 16 psf [0.77] kPa

Force on the first element [F]:

= qz • G • Cf • WindArea

= 16 • 0.85 • 0.55 • 31.3

= 233 lbs. [1037] N

Element	Hgt (z) m	K1	K2	K3	Kz	Kzt	qz kPa
Channel head	2.6	0.000	0.000	0.000	0.849	1.000	0.766
Channel shell	2.6	0.000	0.000	0.000	0.849	1.000	0.766
Shell	2.6	0.000	0.000	0.000	0.849	1.000	0.766
Shell head	2.6	0.000	0.000	0.000	0.849	1.000	0.766

Wind Loads on Masses/Equipment/Piping

ID	Wind Area m ²	Elevation m	Pressure kPa	Force N
Davit	0.00	2.60	0.77	0.00
PP gurad	0.00	2.60	0.77	0.00
103-C1 WEI (O	9.05	7.59	0.77	6935.63
103-C1 WEI (O	9.05	7.59	0.77	6935.63
103-C1 WEI (H	0.00	7.59	0.77	0.00
103-C1 WEI (H	0.00	7.59	0.77	0.00
103-C1 WEI (E	0.00	7.59	0.77	0.00
103-C1 WEI (E	0.00	7.59	0.77	0.00
Fireproofing	0.00	0.30	0.77	0.00
Fireproofing	0.00	0.30	0.77	0.00

Wind Load Calculation:

From	To	Wind Height m	Wind Diameter m	Wind Area m ²	Wind Pressure kPa	Element Wind Load N
10	20	2.6	3.2664	2.909	0.76606	1140.82
20	40	2.6	3.3192	8.29594	0.76606	3253.4
40	50	2.6	3.5328	20.4852	0.76606	23292
50	60	2.6	3.36	3.0781	0.76606	1207.13

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.

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PV Elite 28 Licensee: SL User

FileName : 103-C2

Page 59 of 192

Earthquake Load Calculation: Step: 14 3:14pm Apr 10,2026

Earthquake Load Calculation Results:

Earthquake Load Calculation:

Input Values:

Seismic Design Code		ASCE/SEI 7-16
Seismic Load Reduction Scale Factor		1.000
Importance Factor		1.250
Table Value Fa		1.600
Table Value Fv		2.400
Max. Mapped Res. Acceleration	[Ss]	0.147
Max. Eff. Ground Acceleration	[S]	0.065
Force Modification Factor R		3.000
Site Class		D
Component Elevation Ratio	z/h	0.000
Amplification Factor	Ap	1.000
Force Factor		0.000
Consider Vertical Acceleration		Yes
Minimum Acceleration Multiplier		0.000
User Value of Sds (used if > 0)		0.000
User Value of Sd1 (used if > 0)		0.000
Moment Reduction Factor Tau		1.000

Seismic Analysis Results:

$$\begin{aligned} SMS &= Fa * Ss = 1.6 * 0.15 = 0.24 \\ SM1 &= Fv * S1 = 2.4 * 0.065 = 0.16 \\ SDS &= 2/3 * Sms = 2/3 * 0.24 = 0.16 \\ SD1 &= 2/3 * Sm1 = 2/3 * 0.16 = 0.1 \end{aligned}$$

Check Approximate Fundamental Period from 12.8-7 [Ta]:
 $= Ct \cdot h_{nx}$ where $Ct = 0.020$, $x = 0.75$ and h_n = Structural Height (ft.)
 $= 0.020(11.90.75)$
 $= 0.128$ seconds

The Coefficient C_u from Table 12.8-1 is : 1.700

Fundamental Period (1/Frequency) [T]:
 $= (1/\text{Natural Frequency}) = (1/33)$
 $= 0.030$

Check the Value of T which is the smaller of $C_u \cdot T_a$ and T:
 $= \text{Minimum Value of } (1.7 \cdot 0.13, 0.03) \text{ per 12.8.2}$
 $= 0.030$
 As the time period is < 0.06 second, use section 15.4.2.

Base Shear per equation 15.4-5, [V]:
 $= 0.3 \cdot Sds \cdot W \cdot I$
 $= 0.3 \cdot 0.16 \cdot 3700073 \cdot 1.25$
 $= 217564.328 \text{ N}$

Vertical load per 12.4-4, [Ev]:
 $= 0.2 \cdot Sds \cdot W$
 $= 0.2 \cdot 0.16 \cdot 3700073$
 $= 116034.3 \text{ N}$

Earthquake Load Calculation:

From	To	Earthquake Height m	Earthquake Weight N	Element Ope Load N
10	20	1.333	616679	36315.2



PV Elite 28 Licensee: SL User

FileName : 103-C2

Page 60 of 192

Earthquake Load Calculation: Step: 14 3:14pm Apr 10,2026

20	40	1.333	616679	36315.2
40	Sad1	1.33	616679	36233.5
Sad1	50	1.33	616679	36233.5
40	50	1.33	616679	36233.5
50	60	1.33	616679	36233.5

The Earthquake Weight does not include the weight of the saddles.

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PV Elite 28 Licensee: SL User

FileName : 103-C2

Page 61 of 192

Center of Gravity Calculation: Step: 15 3:14pm Apr 10,2026

Shop/Field Installation Options :

Lining is installed in the Shop.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Saddles	5.850 m
Center of Gravity of the Liquid	6.734 m
Center of Gravity of the Lining	0.584 m
Center of Gravity of the Nozzles	0.771 m
Center of Gravity of the Added Weights (Operating)	5.771 m
Center of Gravity of the Added Weights (Empty)	5.761 m
Center of Gravity of the Tubesheet(s)	2.750 m
Center of Gravity of the Tubes	5.250 m
Center of Gravity of the Baffles	5.751 m
Center of Gravity of Bare Shell New and Cold	5.482 m
Center of Gravity of Bare Shell Corroded	5.494 m
Vessel CG in the Operating Condition	5.417 m
Vessel CG in the Fabricated (Shop/Empty) Condition	5.321 m
Vessel CG in the Test Condition	5.169 m

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PV Elite 28 Licensee: SL User

FileName : 103-C2

Page 62 of 192

Saddle Calcs: Operating Case: Step: 16 3:14pm Apr 10,2026

ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Operating Case

Wear Pad Bearing angle is less than saddle angle + saddle/12 degrees. The wear plate will be ignored.

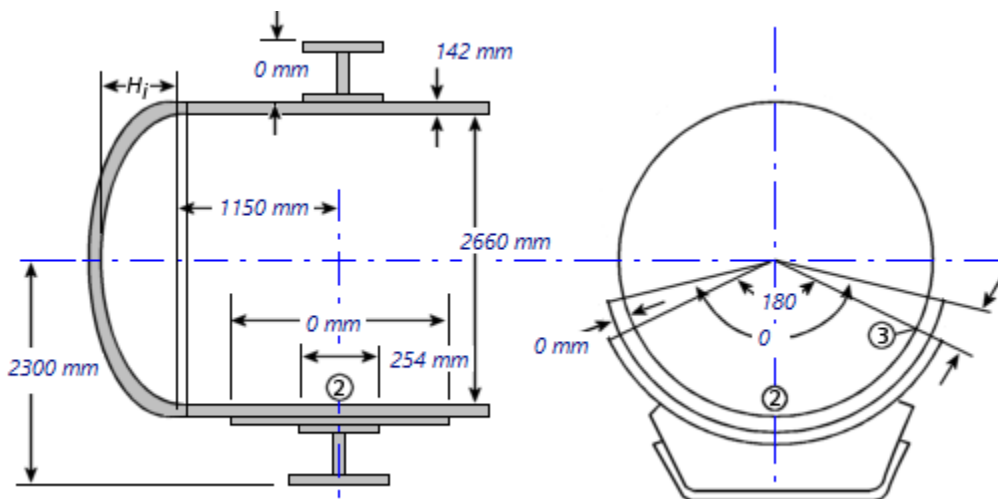
Minimum Wear Plate Angle [θ1]:
= 192.0 degrees

Note:
Wear Pad Width (.00) is less than $1.56 \cdot \sqrt{r_m \cdot t}$
and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:
= $\min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$
= $\min(254 + 1.56 \cdot \sqrt{1402 \cdot 139}, 2 \cdot 1150)$
= 942.7846 mm

Input and Calculated Values:

Vessel Mean Radius	Rm	1402.50	mm
Shell Thickness used in this Case	t	142.000	mm
Stiffened Vessel Length per 4.15.6	L	5.80	m
Distance from Saddle to Vessel tangent	a1 or a	1150.00	mm
Saddle Width	b1 or b	254.00	mm
Saddle Bearing Angle	delta or theta	180.00	degrees
Inside or Mean Depth of Head	Hi or hm	1.35	m
Shell Allowable Stress used in Calculation		147988.73	kPa
Head Allowable Stress used in Calculation		137996.80	kPa
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	2300.00	mm
Coefficient of Friction	mu	0.00	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case		3176314.75	N
Pressure used in Analysis	P	14.294	N/mm ²



PV Elite 28 Licensee: SL User

FileName : 103-C2

Page 63 of 192

Saddle Calcs: Operating Case: Step: 16 3:14pm Apr 10,2026

Horizontal Vessel Analysis Results:	Actual kPa	Allowable kPa
Long. Stress at Top of Midspan	72230.48	147988.73
Long. Stress at Bottom of Midspan	71985.23	147988.73
Long. Stress at Top of Saddles	79165.88	147988.73
Long. Stress at Bottom of Saddles	67860.66	147988.73
Tangential Shear in Shell	4331.66	118390.98
Circ. Stress at Horn of Saddle	12131.26	184985.91
Circ. Compressive Stress in Shell	1513.14	147988.73

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= \text{Ftr} (\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3 (28893/2 + 176413) \cdot 2300/2500$$

$$= 526773.3 \text{ N}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(\text{Fl,wind}; \text{Sum of X Forces}) \cdot B / \text{Ls}$$

$$= \max(8260, 352826) \cdot 2300/3400$$

$$= 5587.6 \text{ N}$$

Saddle Reaction Force due to Earthquake Fl [Fsl]:

$$= \max(\text{Fl,eq}; \text{Sum of X Forces}) \cdot B / \text{Ls}$$

$$= \max(217564, 352826) \cdot 2300/3400$$

$$= 238676.7 \text{ N}$$

Saddle Reaction Force due to Earthquake Ft [Fst]:

$$= \text{Ftr} (\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3 (217564/2 + 176413) \cdot 2300/2500$$

$$= 787139.2 \text{ N}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$$

$$= 2389175 + \max(5588, 526773, 238677, 787139)$$

$$= 3176314.8 \text{ N}$$

Longitudinal Wind Force [Fl]:

$$= \text{WindScalar} \cdot \text{WindPress} (\text{Platform Area} + (\pi/4 (\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 766 (0 + \pi/4 (2.94 \cdot 1.2)^2)$$

$$= 8260.342 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	3205543.75	N
Transverse Shear Load	285195.38	N
Longitudinal Shear Load	352826.41	N

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

$$= \text{Baseplate Length}$$

$$= 2500.000 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.2286	K2 = 0.5771	K3 = 0.2597	K4 = 0.2157
K5 = 0.6243	K6 = 0.0174	K7 = 0.0127	K8 = 0.2500
K9 = 0.1396	K10 = 0.0146	K1* = 0.3799	

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:



PV Elite 28 Licensee: SL User

FileName : 103-C2

Page 64 of 192

Saddle Calcs: Operating Case: Step: 16 3:14pm Apr 10, 2026

$$\begin{aligned}
 &= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L))/(1 + (4hm)/3L)] \\
 &= -3176314 \cdot 1.15 [1 - (1 - 1.15/5.8 + (1.4^2 - 1.35^2)/(2 \cdot 1.15 \cdot 5.8))/(1 + (4 \cdot 1.35)/(3 \cdot 5.8))] \\
 &= -1386284.6 \text{ N-m}
 \end{aligned}$$

Moment per Equation 4.15.2 [M2]:

$$\begin{aligned}
 &= Q \cdot L/4 ((1 + 2(Rm^2 - hm^2)/(L^2))/(1 + (4hm^2)/(3L)) - 4a/L) \\
 &= 3176314 \cdot 5.8/4 ((1 + 2(1.4^2 - 1.35^2)/(5.8^2))/(1 + (4 \cdot 1.35)/(3 \cdot 5.8)) - 4 \cdot 1.15/5.8) \\
 &= -105369.4 \text{ N-m}
 \end{aligned}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$\begin{aligned}
 &= P \cdot Rm/(2t) - M2/(\pi \cdot Rm^2 \cdot t) \\
 &= 14.3 \cdot 1402/(2 \cdot 139) - 105369/(\pi \cdot 1402^2 \cdot 139) \\
 &= 72230.48 \text{ kPa}
 \end{aligned}$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$\begin{aligned}
 &= P \cdot Rm/(2t) + M2/(\pi \cdot Rm^2 \cdot t) \\
 &= 14.3 \cdot 1402/(2 \cdot 139) + 105369/(\pi \cdot 1402^2 \cdot 139) \\
 &= 71985.23 \text{ kPa}
 \end{aligned}$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_3]:

$$\begin{aligned}
 &= P \cdot Rm/(2t) - M1/(K1 \cdot \pi \cdot Rm^2 \cdot t) \\
 &= 14.3 \cdot 1402/(2 \cdot 139) - 1386284/(0.23 \cdot \pi \cdot 1402^2 \cdot 139) \\
 &= 79165.88 \text{ kPa}
 \end{aligned}$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_4]:

$$\begin{aligned}
 &= P \cdot Rm/(2t) + M1/(K1 \cdot \pi \cdot Rm^2 \cdot t) \\
 &= 14.3 \cdot 1402/(2 \cdot 139) + 1386284/(0.38 \cdot \pi \cdot 1402^2 \cdot 139) \\
 &= 67860.66 \text{ kPa}
 \end{aligned}$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$\begin{aligned}
 &= Q(L - 2a)/(L + (4 \cdot hm/3)) \\
 &= 3176314 (5.8 - 2 \cdot 1.15)/(5.8 + (4 \cdot 1.35/3)) \\
 &= 1463296.8 \text{ N}
 \end{aligned}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$\begin{aligned}
 &= K2 \cdot T / (\pi \cdot Rm \cdot t) \\
 &= 0.58 \cdot 1463296/(1402 \cdot 139) \\
 &= 4331.66 \text{ kPa}
 \end{aligned}$$

Decay Length (4.15.20) [x_1, x_2]:

$$\begin{aligned}
 &= 0.78 \cdot \sqrt{(Rm \cdot t)} \\
 &= 0.78 \cdot \sqrt{(1402 \cdot 139)} \\
 &= 344.392 \text{ mm}
 \end{aligned}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$\begin{aligned}
 &= -K5 \cdot Q \cdot k / (t(b + X1 + X2)) \\
 &= -0.62 \cdot 3176314 \cdot 0.1/(139(254 + 344 + 344)) \\
 &= -1513.14 \text{ kPa}
 \end{aligned}$$

Circ. Comp. Stress at Horn of Saddle, $L < 8Rm$ (4.15.23) [σ_7]:

$$\begin{aligned}
 &= -Q/(4 \cdot t(b + X1 + X2)) - 12 \cdot K7 \cdot Q \cdot Rm/(L \cdot t^2) \\
 &= -3176314/(4 \cdot 139(254 + 344 + 344)) - 12 \cdot 0.013 \cdot 3176314 \cdot 1402/(5.8 \cdot 139^2) \\
 &= -12131.26 \text{ kPa}
 \end{aligned}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$\begin{aligned}
 &= \min(b + 1.56 \cdot \sqrt{(Rm \cdot t)}, 2a) \\
 &= \min(254 + 1.56 \cdot \sqrt{(1402 \cdot 139)}, 2 \cdot 1.15) \\
 &= 942.78 \text{ mm}
 \end{aligned}$$

Distance between Saddle Supports [Ls]:

$$= 3400.0 \text{ mm}$$



PV Elite 28 Licensee: SL User

FileName : 103-C2

Page 65 of 192

Saddle Calcs: Operating Case: Step: 16 3:14pm Apr 10,2026

Free Un-Restrained Thermal Expansion between the Saddles [Exp]:

$$= \alpha \cdot Ls(\text{Design Temperature} - \text{Ambient Temperature})$$

$$= 0.13497\text{E-}04 \cdot 3400 (343 - 21.1)$$

$$= 14.771 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2500.0000	mm
Baseplate Thickness	Bpthk	100.0000	mm
Baseplate Width	Bpwid	552.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	35.0000	mm
Web Thickness	Webtk	35.0000	mm
Web Location	Webloc	Side	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	194672.1	kPa
Height of Web at Center	Hw,c	306.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3)/12$$

$$= ((1.56 \cdot \sqrt{(1333 \cdot 139)} + 304)139^3)/12$$

$$= 218318624.000 \text{ mm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2$$

$$= 218318624 + 0.14 (330 - 69.5)^2$$

$$= 9441132544.000 \text{ mm}^4$$

	B mm	D mm	Y mm	A m ²	AY mm ³	I + AD ⁽²⁾ mm ⁴
Shell	975.5	139.0	69.5	0.1	9423834.0	0.944E+10
Wearplate	304.0	30.0	154.0	0.0	1404480.0	0.284E+09
Web	35.0	698.0	518.0	0.0	12654744.0	0.185E+10
BasePlate	552.0	100.0	917.0	0.1	50618408.0	0.190E+11
Totals	...	...	...	0.2	74101464.0	0.306E+11

Distance to Centroid [C1]:

$$= AY / A$$

$$= 74101464/0.22$$

$$= 330.302 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 180/2$$

$$= 90.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(90) - 0.5 \cdot \sin(90)^2) / (\pi - 1.57 + \sin(90) \cos(90))$$

$$= 0.3183$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.32 \cdot 3176314$$

$$= 1011052.3125 \text{ N}$$

$$\text{Tension Stress, } St = (Fh/As) = 11392.4434 \text{ kPa}$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 116803.2656 \text{ kPa}$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\text{theta}/2) / (\text{theta}/2 \text{ in radians})$$

$$= 2300 - 1333 \cdot \sin(180/2)/1.57$$



PV Elite 28 Licensee: SL User

FileName : 103-C2

Page 66 of 192

Saddle Calcs: Operating Case: Step: 16 3:14pm Apr 10,2026

= 1451.386 mm

Bending Moment, $M = F_h \cdot d = 1468022.0000 \text{ N-m}$

Bending Stress, $S_b = (M \cdot C_1 / I) = 15827.4062 \text{ kPa}$

Allowed Stress, $S_a = 2/3 \cdot \text{Yield Str} = 129781.4062 \text{ kPa}$

Minimum Thickness of Baseplate per Moss:

The maximum of $2/3 \cdot \text{Yield}$ and the Saddle Allowable Stress value was used to determine the baseplate thickness.

$$= \sqrt{\frac{3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}}{(2 \cdot \text{BasePlateLength} \cdot \text{AllStress})}}$$

$$= \sqrt{\frac{3(3176314 + 29229) \cdot 552}{(2 \cdot 2500 \cdot 147989)}}$$

$$= 84.701 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 180/2) (1330 + 142 + 30)$$

$$= 3004.000 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 3004 / (5 - 1)$$

$$= 751.000 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 751 \cdot 552 / 2$$

$$= 0.207 \text{ m}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 3205544 / (2500 \cdot 552)$$

$$= 2323433.750 \text{ N/m}^2$$

Axial Load [P]:

$$= A_p \cdot B_p$$

$$= 0.21 \cdot 2323433$$

$$= 481472.719 \text{ N}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 0.0077 + 0.013$$

$$= 0.021 \text{ m}^2$$

Compressive Stress [Sc]:

$$= P / A_r$$

$$= 481473 / 0.021$$

$$= 23140.004 \text{ kPa}$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib	35.0	219.0	144.5	0.0	1107592.5	0.800E+08
Web	375.5	35.0	17.5	0.0	229993.8	0.301E+08
Totals	...	...	...	...	1337586.2	0.110E+09

Rib dimension [D]:

= Saddle Width - Web Thickness



PV Elite 28 Licensee: SL User

FileName : 103-C2

Page 67 of 192

Saddle Calcs: Operating Case: Step: 16 3:14pm Apr 10, 2026

= 254 - 35
= 219.000 mm

Distance to Centroid from Datum [ytot]:

= AY / A
= 1337586/0.021
= 64.284 mm

Distance to Centroid [C1]:

= max(ytot, Saddle Width - ytot)
= max(64.3, 254 - 64.3)
= 189.716 mm

Radius of Gyration [r]:

= $\sqrt{\text{Total Inertia} / \text{Total Area}}$
= $\sqrt{110063424/0.021}$
= 72.730 mm

Length of Outer Rib [L]:

= Saddle Height - cos(theta/2)(radius + shlthk + wpdthk) - bpthk
= 2300 - cos(180/2)(1330 + 142 + 30) - 100
= 2200.000 mm

Intermediate Term [Cc]:

= $\sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$
= $\sqrt{2 \cdot \pi^2 \cdot 199955/194672}$
= 142.386

Slenderness ratio [KL/r]:

= KL/r
= 1 • 2200/72.7
= 30.249

Bending Moment [Rm]:

= FI / (2 • Bplen) • e • L / 2
= 352826 / (2 • 2500) • 751 • 2200/2
= 58317.621 N-m

Compressive Allowable, KL/r < Cc (30.2 < 142) per AISC E2-1 [Sca]:

= (1 - (KL/r)² / (2 • Cc²)) Fy / (5/3 + 3 • (KL/r) / (8 • Cc) - (KL/r)³ / (8 • Cc³))
= (1 - (30.2)² / (2 • 142²)) 194672 /
(5/3 + 3 • (30.2) / (8 • 142) - (30.2³) / (8 • 142³))
= 109034 kPa

AISC Unity Check of Outside Ribs (must be ≤ 1)

= Sc/Sca + (Rm • C1 / I) / Sba
= 23140/109034 + (58318 • 190/110063536) / 129781
= 0.986

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	35.0	219.0	144.5	0.0	1107592.5	0.105E+09
Web	751.0	35.0	17.5	0.0	459987.5	0.243E+08
Totals	...	...	...	...	1567580.0	0.129E+09

Distance to Centroid from Datum [ytot]:

= AY / A
= 1567580/0.034
= 46.173 mm

Distance to Centroid [C1]:

= max(ytot, abs(Saddle Width - ytot))



PV Elite 28 Licensee: SL User

FileName : 103-C2

Page 68 of 192

Saddle Calcs: Operating Case: Step: 16 3:14pm Apr 10, 2026

$$= \max(46.2, \text{abs}(254 - 46.2))$$

$$= 207.827 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$$

$$= 2300 - \cos(1502/2)(100 + 0 + 0) - 0$$

$$= 698.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{129034936 / 0.034}$$

$$= 61.650 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 698 / 61.7$$

$$= 11.322$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 352826 / (2 \cdot 2500)$$

$$= 70.565 \text{ N/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 70.6 \cdot 751 \cdot 698$$

$$= 37005.176 \text{ N-m}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 37005 / 620878$$

$$= 59578.828 \text{ kPa}$$

Compressive Allowable, $KL/r < C_c$ ($11.3 < 142$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (11.3)^2 / (2 \cdot 142^2)) 194672 / (5/3 + 3 \cdot (11.3) / (8 \cdot 142) - (11.3)^3 / (8 \cdot 142^3))$$

$$= 114392 \text{ kPa}$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C1/I) / S_{ba}$$

$$= 22142 / 114392 + (37005 \cdot 208 / 129034936) / 129781$$

$$= 0.653$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	8
Total Number of Bolts in Tension/Baseplate	Nbt	4
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	283984.00 kPa
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	70.0 mm
User Defined Area in a Single Bolt	UserBoltArea	0.0012 m²
BasePlate Allowable Stress	S	129114.48 kPa
Area Available in a Single Bolt	BltArea	0.0010 m²
Saddle Load Q0 (Weight)	Q0	2418404.5 N
Saddle Load QL (Wind/Seismic contribution)	QL	238676.7 N
Maximum Transverse Force	Ft	285195.4 N
Maximum Longitudinal Force	F1	352826.4 N
Saddle Bolted to Steel Foundation		No

Friction force due to thermal growth, [Ff]:

$$= 582464.4 \text{ N}$$



PV Elite 28 Licensee: SL User

FileName : 103-C2

Page 69 of 192

Saddle Calcs: Operating Case: Step: 16 3:14pm Apr 10, 2026

Shear Stress in a Single Bolt, Longitudinal Direction [rb,l]:
 $= \max(F_f, F_l) / (\text{Bolt Area} \cdot \text{Number of Bolts})$
 $= 582464 / (0.00098 \cdot 8)$
 $= 74334.2 \text{ kPa}$. Must be less than 169990.0 kPa.

Shear Stress in a Single Bolt, Transverse Direction [rb,t]:
 $= F_t / (\text{Bolt Area} \cdot \text{Number of Bolts})$
 $= 285195 / (0.00098 \cdot 8)$
 $= 36396.7 \text{ kPa}$. Must be less than 169990.0 kPa.

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
 $= 0.0$ (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
 $= F_l / (\text{BoltShearAllowable} \cdot \text{Nbolts})$
 $= 352826 / (169990 \cdot 8)$
 $= 0.0003 \text{ m}^2$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
 $= B \cdot F_t + \text{Sum of X Moments}$
 $= 2300 \cdot 285195 + 0$
 $= 656215.31 \text{ N-m}$

Eccentricity (e):
 $= Rmom / QO$
 $= 656215 / 2418404$
 $= 271.23 \text{ mm} < Bplen/6$ --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
 $= 0$ (No Uplift)

Required Area of a Single Bolt [Bltarear]:
 $= \max[Bltarearl, Bltarears, Bltareart]$
 $= \max[0, 0.00026, 0]$
 $= 0.0003 \text{ m}^2$

MDMT Calculations for: Left Saddle, Saddle to Wearplate Junction, Curve: A

Govrn. thk, tg = 30, c = 3 mm, E* = 1
 Thickness Ratio = $tr \cdot E^* / (tg - c) = -1$, Temp. Reduction = 0 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve A	24 °C
Min Metal Temp per UCS-66 and UCS-68(c), PWHT credit	7 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Wear Pad Bearing angle is less than saddle angle + saddle/12 degrees. The wear plate will be ignored.

Minimum Wear Plate Angle [01]:
 $= 192.0 \text{ degrees}$

Note:
 Wear Pad Width (.00) is less than $1.56 \cdot \sqrt{rm \cdot t}$
 and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:
 $= \min(b + 1.56 \cdot \sqrt{Rm \cdot t}, 2a)$



PV Elite 28 Licensee: SL User

FileName : 103-C2

Page 70 of 192

Saddle Calcs: Operating Case: Step: 16 3:14pm Apr 10,2026

$$= \min(254 + 1.56\sqrt{(1402 \cdot 139)}, 2 \cdot 1150)$$

$$= 942.7846 \text{ mm}$$

Input and Calculated Values:

Vessel Mean Radius	Rm	1402.50	mm
Shell Thickness used in this Case	t	142.000	mm
Stiffened Vessel Length per 4.15.6	L	5.80	m
Distance from Saddle to Vessel tangent	a1 or a	1150.00	mm
Saddle Width	b1 or b	254.00	mm
Saddle Bearing Angle	delta or theta	180.00	degrees
Inside or Mean Depth of Head	Hi or hm	1.37	m
Shell Allowable Stress used in Calculation		147988.73	kPa
Head Allowable Stress used in Calculation		147988.73	kPa
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	2300.00	mm
Coefficient of Friction	mu	0.40	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case		2214071.25	N
Pressure used in Analysis	P	14.294	N/mm ²

Horizontal Vessel Analysis Results:		Actual kPa	Allowable kPa
Long. Stress at Top of Midspan		72211.14	147988.73
Long. Stress at Bottom of Midspan		72004.58	147988.73
Long. Stress at Top of Saddles		77090.90	147988.73
Long. Stress at Bottom of Saddles		69109.29	147988.73
Tangential Shear in Shell		3009.64	118390.98
Circ. Stress at Horn of Saddle		8456.17	184985.91
Circ. Compressive Stress in Shell		1054.75	147988.73

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3 (28893/2 + 176413) \cdot 2300/2500$$

$$= 526773.3 \text{ N}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(\text{Fl,wind}; \text{Sum of X Forces}) \cdot B / \text{Ls}$$

$$= \max(8260, 352826) \cdot 2300/3400$$

$$= 5587.6 \text{ N}$$

Saddle Reaction Force due to Earthquake Fl [Fsl]:

$$= \max(\text{Fl,eq}; \text{Sum of X Forces}) \cdot B / \text{Ls}$$

$$= \max(217564, 352826) \cdot 2300/3400$$

$$= 238676.7 \text{ N}$$

Saddle Reaction Force due to Earthquake Ft [Fst]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3 (217564/2 + 176413) \cdot 2300/2500$$

$$= 787139.2 \text{ N}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$$



PV Elite 28 Licensee: SL User

FileName : 103-C2

Page 71 of 192

Saddle Calcs: Operating Case: Step: 16 3:14pm Apr 10, 2026

$$= 1426932 + \max(5588, 526773, 238677, 787139)$$

$$= 2214071.2 \text{ N}$$

Longitudinal Wind Force [FI]:

$$= \text{WindScalar} \cdot \text{WindPress} (\text{Platform Area} + (\pi/4 (\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 766 (0 + \pi/4 (2.94 \cdot 1.2)^2)$$

$$= 8260.342 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	2243300.25	N
Transverse Shear Load	285195.38	N
Longitudinal Shear Load	352826.41	N

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

$$= \text{Baseplate Length}$$

$$= 2500.000 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.2286	K2 = 0.5771	K3 = 0.2597	K4 = 0.2157
K5 = 0.6243	K6 = 0.0174	K7 = 0.0127	K8 = 0.2500
K9 = 0.1396	K10 = 0.0146	K1* = 0.3799	

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h \cdot m^2)/(2a \cdot L))/(1 + (4h \cdot m)/(3L))]$$

$$= -2214071 \cdot 1.15 [1 - (1 - 1.15/5.8 + (1.4^2 - 1.37^2)/(2 \cdot 1.15 \cdot 5.8))/(1 + (4 \cdot 1.37)/(3 \cdot 5.8))]$$

$$= -978731.7 \text{ N-m}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4 ((1 + 2(R_m^2 - h \cdot m^2)/(L^2))/(1 + (4h \cdot m)/(3L)) - 4a/L)$$

$$= 2214071 \cdot 5.8/4 ((1 + 2(1.4^2 - 1.37^2)/(5.8^2))/(1 + (4 \cdot 1.37)/(3 \cdot 5.8)) - 4 \cdot 1.15/5.8)$$

$$= -88751.3 \text{ N-m}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m/(2t) - M_2/(\pi \cdot R_m^2 \cdot t)$$

$$= 14.3 \cdot 1402/(2 \cdot 139) - 88751/(\pi \cdot 1402^2 \cdot 139)$$

$$= 72211.14 \text{ kPa}$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m/(2t) + M_2/(\pi \cdot R_m^2 \cdot t)$$

$$= 14.3 \cdot 1402/(2 \cdot 139) + 88751/(\pi \cdot 1402^2 \cdot 139)$$

$$= 72004.58 \text{ kPa}$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot R_m/(2t) - M_1/(K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 14.3 \cdot 1402/(2 \cdot 139) - 978732/(0.23 \cdot \pi \cdot 1402^2 \cdot 139)$$

$$= 77090.90 \text{ kPa}$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$= P \cdot R_m/(2t) + M_1/(K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 14.3 \cdot 1402/(2 \cdot 139) + 978732/(0.38 \cdot \pi \cdot 1402^2 \cdot 139)$$

$$= 69109.29 \text{ kPa}$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a)/(L + (4 \cdot h \cdot m/3))$$

$$= 2214071 (5.8 - 2 \cdot 1.15)/(5.8 + (4 \cdot 1.37/3))$$

$$= 1016699.6 \text{ N}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$



PV Elite 28 Licensee: SL User

FileName : 103-C2

Page 72 of 192

Saddle Calcs: Operating Case: Step: 16 3:14pm Apr 10,2026

$$= 0.58 \cdot 1016699 / (1402 \cdot 139)$$

$$= 3009.64 \text{ kPa}$$

Decay Length (4.15.20) [x1,x2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(1402 \cdot 139)}$$

$$= 344.392 \text{ mm}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K5 \cdot Q \cdot k / (t(b + X1 + X2))$$

$$= -0.62 \cdot 2214071 \cdot 0.1 / (139(254 + 344 + 344))$$

$$= -1054.75 \text{ kPa}$$

Circ. Comp. Stress at Horn of Saddle, L<8Rm (4.15.23) [σ_7^*]:

$$= -Q / (4 \cdot t(b + X1 + X2)) - 12 \cdot K7 \cdot Q \cdot R_m / (L \cdot t^2)$$

$$= -2214071 / (4 \cdot 139(254 + 344 + 344)) -$$

$$12 \cdot 0.013 \cdot 2214071 \cdot 1402 / (5.8 \cdot 139^2)$$

$$= -8456.17 \text{ kPa}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(254 + 1.56 \cdot \sqrt{(1402 \cdot 139)}, 2 \cdot 1.15)$$

$$= 942.78 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2500.0000	mm
Baseplate Thickness	Bpthk	100.0000	mm
Baseplate Width	Bpwid	552.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	35.0000	mm
Web Thickness	Webtk	35.0000	mm
Web Location	Webloc	Side	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	194672.1	kPa
Height of Web at Center	Hw,c	306.0	mm
Friction Coefficient	mu	0.400	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(1333 \cdot 139)} + 304)^3) / 12$$

$$= 218318624.000 \text{ mm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 218318624 + 0.14(330 - 69.5)^2$$

$$= 9441132544.000 \text{ mm}^4$$

	B mm	D mm	Y mm	A m ²	AY mm ³	I + AD ² mm ⁴
Shell	975.5	139.0	69.5	0.1	9423834.0	0.944E+10
Wearplate	304.0	30.0	154.0	0.0	1404480.0	0.284E+09
Web	35.0	698.0	518.0	0.0	12654744.0	0.185E+10
BasePlate	552.0	100.0	917.0	0.1	50618408.0	0.190E+11
Totals	...	...	...	0.2	74101464.0	0.306E+11

Distance to Centroid [C1]:

$$= AY / A$$

$$= 74101464 / 0.22$$

$$= 330.302 \text{ mm}$$

Angle [β]:



PV Elite 28 Licensee: SL User

FileName : 103-C2

Page 73 of 192

Saddle Calcs: Operating Case: Step: 16 3:14pm Apr 10, 2026

$$\begin{aligned}
 &= 180 - \text{Saddle Angle}/2 \\
 &= 180 - 180/2 \\
 &= 90.0
 \end{aligned}$$

Saddle Splitting Coefficient [K1]:

$$\begin{aligned}
 &= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta)) \\
 &= (1 + \cos(90) - 0.5 \cdot \sin(90)^2) / (\pi - 1.57 + \sin(90) \cos(90)) \\
 &= 0.3183
 \end{aligned}$$

Saddle Splitting Force [Fh]:

$$\begin{aligned}
 &= K1 \cdot Q \\
 &= 0.32 \cdot 2214071 \\
 &= 704760.6250 \text{ N}
 \end{aligned}$$

$$\begin{aligned}
 \text{Tension Stress, } St &= (Fh / As) = 7941.1777 \text{ kPa} \\
 \text{Allowed Stress, } Sa &= 0.6 \cdot \text{Yield Str} = 116803.2656 \text{ kPa}
 \end{aligned}$$

Saddle Splitting Dimension [d]:

$$\begin{aligned}
 &= B - R \cdot \sin(\text{theta}/2) / (\text{theta}/2 \text{ in radians}) \\
 &= 2300 - 1333 \cdot \sin(180/2) / 1.57 \\
 &= 1451.386 \text{ mm}
 \end{aligned}$$

$$\text{Bending Moment, } M = Fh \cdot d = 1023294.3750 \text{ N-m}$$

$$\begin{aligned}
 \text{Bending Stress, } Sb &= (M \cdot C1 / I) = 11032.5977 \text{ kPa} \\
 \text{Allowed Stress, } Sa &= 2/3 \cdot \text{Yield Str} = 129781.4062 \text{ kPa}
 \end{aligned}$$

Minimum Thickness of Baseplate per Moss:

The maximum of 2/3*Yield and the Saddle Allowable Stress value was used to determine the baseplate thickness.

$$\begin{aligned}
 &= \sqrt{3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (2 \cdot \text{BasePlateLength} \cdot \text{AllStress})} \\
 &= \sqrt{3(2214071 + 29229)552 / (2 \cdot 2500 \cdot 147989)} \\
 &= 70.857 \text{ mm}
 \end{aligned}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$\begin{aligned}
 &= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk}) \\
 &= 2 \cdot \cos(90 - 180/2) (1330 + 142 + 30) \\
 &= 3004.000 \text{ mm}
 \end{aligned}$$

Distance between Ribs [e]:

$$\begin{aligned}
 &= \text{Web Length} / (\text{Nribs} - 1) \\
 &= 3004 / (5 - 1) \\
 &= 751.000 \text{ mm}
 \end{aligned}$$

Baseplate Pressure Area [Ap]:

$$\begin{aligned}
 &= e \cdot \text{Bpwid} / 2 \\
 &= 751 \cdot 552 / 2 \\
 &= 0.207 \text{ m}^2
 \end{aligned}$$

Bearing Pressure [Bp]:

$$\begin{aligned}
 &= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth}) \\
 &= 2243300 / (2500 \cdot 552) \\
 &= 1625983.000 \text{ N/m}^2
 \end{aligned}$$

Axial Load [P]:

$$\begin{aligned}
 &= Ap \cdot Bp \\
 &= 0.21 \cdot 1625983 \\
 &= 336943.688 \text{ N}
 \end{aligned}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$



PV Elite 28 Licensee: SL User

FileName : 103-C2

Page 74 of 192

Saddle Calcs: Operating Case: Step: 16 3:14pm Apr 10,2026

$$= 0.0077 + 0.013$$

$$= 0.021 \text{ m}^2$$

Compressive Stress [Sc]:

$$= P/Ar$$

$$= 336944/0.021$$

$$= 16193.812 \text{ kPa}$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib	35.0	219.0	144.5	0.0	1107592.5	0.800E+08
Web	375.5	35.0	17.5	0.0	229993.8	0.301E+08
Totals	...	...	...	...	1337586.2	0.110E+09

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 254 - 35$$

$$= 219.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 1337586/0.021$$

$$= 64.284 \text{ mm}$$

Distance to Centroid [C1]:

$$= \max(y_{tot}, \text{Saddle Width} - y_{tot})$$

$$= \max(64.3, 254 - 64.3)$$

$$= 189.716 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ \text{Total Inertia} / \text{Total Area} }$$

$$= \sqrt{ 110063424/0.021 }$$

$$= 72.730 \text{ mm}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\theta/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 2300 - \cos(180/2) (1330 + 142 + 30) - 100$$

$$= 2200.000 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{ (2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}) }$$

$$= \sqrt{ (2 \cdot \pi^2 \cdot 199955/194672) }$$

$$= 142.386$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 2200/72.7$$

$$= 30.249$$

Bending Moment [Rm]:

$$= FI / (2 \cdot B_{plen}) \cdot e \cdot L / 2$$

$$= 352826 / (2 \cdot 2500) \cdot 751 \cdot 2200/2$$

$$= 58317.621 \text{ N-m}$$

Compressive Allowable, $KL/r < C_c$ ($30.2 < 142$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (30.2)^2 / (2 \cdot 142^2)) 194672 / (5/3 + 3 \cdot (30.2) / (8 \cdot 142) - (30.2)^3 / (8 \cdot 142^3))$$

$$= 109034 \text{ kPa}$$

AISC Unity Check of Outside Ribs (must be ≤ 1)



PV Elite 28 Licensee: SL User

FileName : 103-C2

Page 75 of 192

Saddle Calcs: Operating Case: Step: 16 3:14pm Apr 10,2026

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 16194/109034 + (58318 \cdot 190/110063536) / 129781$$

$$= 0.923$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	35.0	219.0	144.5	0.0	1107592.5	0.105E+09
Web	751.0	35.0	17.5	0.0	459987.5	0.243E+08
Totals	...	...	...	...	1567580.0	0.129E+09

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 1567580/0.034$$

$$= 46.173 \text{ mm}$$

Distance to Centroid [C1]:

$$= \max(ytot, \text{abs}(\text{Saddle Width} - ytot))$$

$$= \max(46.2, \text{abs}(254 - 46.2))$$

$$= 207.827 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - Bpthk$$

$$= 2300 - \cos(1502/2) (100 + 0 + 0) - 0$$

$$= 698.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ (\text{Total Inertia} / \text{Total Area}) }$$

$$= \sqrt{ (129034936/0.034) }$$

$$= 61.650 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 698/61.7$$

$$= 11.322$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 352826/(2 \cdot 2500)$$

$$= 70.565 \text{ N/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 70.6 \cdot 751 \cdot 698$$

$$= 37005.176 \text{ N-m}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 37005/620878$$

$$= 59578.828 \text{ kPa}$$

Compressive Allowable, $KL/r < Cc$ ($11.3 < 142$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (11.3)^2 / (2 \cdot 142^2)) 194672 / (5/3 + 3 \cdot (11.3) / (8 \cdot 142) - (11.3)^3 / (8 \cdot 142^3))$$

$$= 114392 \text{ kPa}$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (Mbase,c \cdot C1/I) / Sba$$

$$= 15495/114392 + (37005 \cdot 208/129034936) / 129781$$

$$= 0.595$$

Input Data for Base Plate Bolting Calculations:



PV Elite 28 Licensee: SL User

FileName : 103-C2

Page 76 of 192

Saddle Calcs: Operating Case: Step: 16 3:14pm Apr 10,2026

Total Number of Bolts per BasePlate	Nbolts	8
Total Number of Bolts in Tension/Baseplate	Nbt	4
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	283984.00 kPa
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	70.0 mm
User Defined Area in a Single Bolt	UserBoltArea	0.0012 m²
BasePlate Allowable Stress	S	129114.48 kPa
Area Available in a Single Bolt	BltArea	0.0010 m²
Saddle Load Q0 (Weight)	Q0	1456160.9 N
Saddle Load QL (Wind/Seismic contribution)	QL	238676.7 N
Maximum Transverse Force	Ft	285195.4 N
Maximum Longitudinal Force	Fl	352826.4 N
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [tb,l]:

= Fl / (Bolt Area • Number of Bolts)

= 352826/(0.00098 • 8)

= 45027.8 kPa. Must be less than 169990.0 kPa.

Shear Stress in a Single Bolt, Transverse Direction [tb,t]:

= Ft / (Bolt Area • Number of Bolts)

= 285195/(0.00098 • 8)

= 36396.7 kPa. Must be less than 169990.0 kPa.

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

= 0.0 (Q0 > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:

= Fl / (BoltShearAllowable • Nbolts)

= 352826/(169990 • 8)

= 0.0003 m²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

= B • Ft + Sum of X Moments

= 2300 • 285195 + 0

= 656215.31 N-m

Eccentricity (e):

= Rmom / Q0

= 656215/1456160

= 450.46 mm > Bplen/6 --> Uplift in Transverse direction

f = Bplen / 2 - Edgedis

= 2500/2 - 70

= 1180.00 mm

Modular Ratio Of Steel/Concrete (n1):

= ES / EC

= 203390704/21525068

= 9.45

K1 = 3 (e - 0.5 • Bplen)

= 3 (450 - 0.5•2500)

= -2398.61 mm

K2 = 6 • n1 • At / Bpwid • (f + e)

= 6 • 9.45 • 0.0039/552 • (1180 + 450)

= 656103.87 mm²



PV Elite 28 Licensee: SL User

FileName : 103-C2

Page 77 of 192

Saddle Calcs: Operating Case: Step: 16 3:14pm Apr 10, 2026

$$\begin{aligned} K3 &= -K2 \cdot (0.5 \cdot Bplen + f) \\ &= -656104 \cdot (0.5 \cdot 2500 + 1180) \\ &= -1594332405.01 \text{ mm}^3 \end{aligned}$$

Iteratively Solving for the Effective Bearing Length:

$$\begin{aligned} Y^3 + K1 \cdot Y^2 + K2 \cdot Y + K3 &= 0 \\ Y^3 + -2399 \cdot Y^2 + 656104 \cdot Y + -1594332416 &= 0 \\ Y &= 2401.81 \text{ mm} \end{aligned}$$

$$\begin{aligned} Num &= (Bplen / 2 - Y / 3 - e) \\ &= (2500/2 - 2402/3 - 450) \\ &= -1.07 \end{aligned}$$

$$\begin{aligned} Denom &= (Bplen / 2 - Y / 3 + f) \\ &= (2500/2 - 2402/3 + 1180) \\ &= 1629.40 \end{aligned}$$

$$\begin{aligned} \text{Total Bolt Tension Force [Tforce]:} \\ &= -QO \cdot Num / Denom \\ &= -1456160 \cdot -1.07/1629 \\ &= 955.06 \text{ N} \end{aligned}$$

$$\begin{aligned} \text{Bolt Area Required due to Transverse Load [Bltareart]:} \\ &= Tforce / (Stba \cdot Nbt) \\ &= 955 / (283984 \cdot 4) \\ &= 0.0000 \text{ m}^2 \end{aligned}$$

$$\begin{aligned} \text{Required Area of a Single Bolt [Bltarear]:} \\ &= \max[Bltarearl, Bltarears, Bltareart] \\ &= \max[0, 0.00026, 0.84059E-06] \\ &= 0.0003 \text{ m}^2 \end{aligned}$$

Baseplate Thickness Calculation per D. Moss:

$$\begin{aligned} \text{Bearing Pressure (fc)} \\ &= 2(QO + Tforce) / (Y \cdot Bpwid) \\ &= 2(1456160 + 955) / (2402 \cdot 552) \\ &= 2.20 \text{ N/mm}^2 \end{aligned}$$

$$\begin{aligned} \text{Distance from Baseplate Edge to the Web [ADIST]:} \\ &= (Bplen - WebIngh) / 2 \\ &= (2500 - 2449) / 2 \\ &= 25.4000 \text{ mm} \end{aligned}$$

$$\begin{aligned} \text{Overturning Moment due To Bolt Tension [Mt]:} \\ &= Tforce \cdot Adist \\ &= 955 \cdot 0.025 \\ &= 24.27 \text{ N-m} \end{aligned}$$

$$\begin{aligned} \text{Equivalent Bearing Pressure (f1):} \\ &= fc \cdot (Y - Adist) / Y \\ &= 2.2 \cdot (2402 - 25.4) / 2402 \\ &= 2.18 \text{ N/mm}^2 \end{aligned}$$

$$\begin{aligned} \text{Overturning Moment due to Bearing Pressure [Mc]:} \\ &= (Adist^2 \cdot Bpwid / 6) \cdot (f1 + 2 \cdot fc) \\ &= (25.4^2 \cdot 552/6) \cdot (2.18 + 2 \cdot 2.2) \\ &= 390.18 \text{ N-m} \end{aligned}$$

$$\begin{aligned} \text{Baseplate Required Thickness [Treq]:} \\ &= \sqrt{(6 \cdot \max(Mt, Mc) / (Bpwid \cdot Sba))} \\ &= \sqrt{(6 \cdot \max(24.3, 390) / (552 \cdot 193672))} \\ &= 4.6787 \text{ mm} \end{aligned}$$



PV Elite 28 Licensee: SL User

FileName : 103-C2

Page 78 of 192

Saddle Calcs: Operating Case: Step: 16 3:14pm Apr 10, 2026

MDMT Calculations for: Right Saddle, Saddle to Wearplate Junction, Curve: A

Govrn. thk, tg = 30, c = 3 mm, $E^* = 1$

Thickness Ratio = $tr * E^* / (tg - c) = -1$, Temp. Reduction = 0 °C

Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve A	24 °C
--	-------

Min Metal Temp per UCS-66 and UCS-68(c), PWHT credit	7 °C
--	------

Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
--	---------

Review notes about nozzle loadings and supports in the Warnings report.

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FileName : 103-C2

Page 79 of 192

Saddle Calcs: Test Case:

Step: 17 3:14pm Apr 10,2026

ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Test Case

Wear Pad Bearing angle is less than saddle angle + saddle/12 degrees. The wear plate will be ignored.

Minimum Wear Plate Angle [θ1]:

= 192.0 degrees

Note:

Wear Pad Width (.00) is less than $1.56 \cdot \sqrt{R_m \cdot t}$ and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

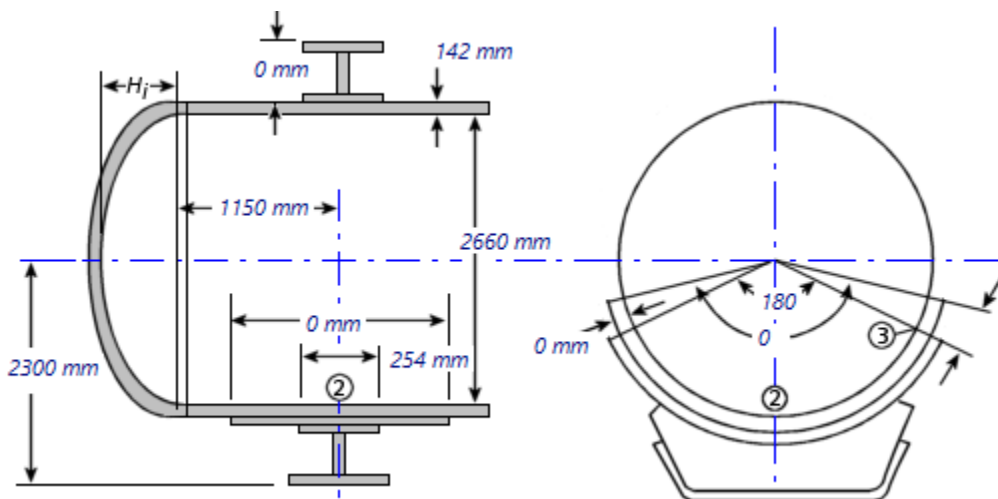
= $\min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$

= $\min(254 + 1.56 \cdot \sqrt{1401 \cdot 142}, 2 \cdot 1150)$

= 949.8055 mm

Input and Calculated Values:

Vessel Mean Radius	Rm	1401.00	mm
Shell Thickness used in this Case	t	142.000	mm
Stiffened Vessel Length per 4.15.6	L	5.80	m
Distance from Saddle to Vessel tangent	a1 or a	1150.00	mm
Saddle Width	b1 or b	254.00	mm
Saddle Bearing Angle	delta or theta	180.00	degrees
Inside or Mean Depth of Head	Hi or hm	1.35	m
Shell Allowable Stress used in Calculation		278978.75	kPa
Head Allowable Stress used in Calculation		278978.75	kPa
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	2300.00	mm
Coefficient of Friction	mu	0.00	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces		3675187.50	N
Pressure used in Analysis	P	18.579	N/mm ²



PV Elite 28 Licensee: SL User

FileName : 103-C2

Page 80 of 192

Saddle Calcs: Test Case:

Step: 17 3:14pm Apr 10, 2026

Horizontal Vessel Analysis Results:	Actual kPa	Allowable kPa
Long. Stress at Top of Midspan	91784.55	278978.75
Long. Stress at Bottom of Midspan	91506.99	278978.75
Long. Stress at Top of Saddles	99656.45	278978.75
Long. Stress at Bottom of Saddles	86825.31	278978.75
Tangential Shear in Shell	4912.22	165217.41
Circ. Stress at Horn of Saddle	13548.96	418468.09
Circ. Compressive Stress in Shell	1701.14	278978.75

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3 (14447/2 + 0) \cdot 2300/2500$$

$$= 19936.4 \text{ N}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(\text{Fl,wind}; \text{Sum of X Forces}) \cdot B / \text{Ls}$$

$$= \max(4130, 0) \cdot 2300/3400$$

$$= 2793.8 \text{ N}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$$

$$= 3655251 + \max(2794, 19936, 0, 0)$$

$$= 3675187.5 \text{ N}$$

Longitudinal Wind Force [Fl]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 766 (0 + \pi/4(2.94 \cdot 1.2)^2)$$

$$= 8260.342 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	3704416.50	N
Transverse Shear Load	7223.34	N
Longitudinal Shear Load	4129.96	N

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

$$= \text{Baseplate Length}$$

$$= 2500.000 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.2286	K2 = 0.5771	K3 = 0.2597	K4 = 0.2157
K5 = 0.6243	K6 = 0.0174	K7 = 0.0128	K8 = 0.2500
K9 = 0.1396	K10 = 0.0146	K1* = 0.3799	

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L))/(1 + (4hm)/3L)]$$

$$= -3675187 \cdot 1.15 [1 - (1 - 1.15/5.8 + (1.4^2 - 1.35^2)/(2 \cdot 1.15 \cdot 5.8))/(1 + (4 \cdot 1.35)/(3 \cdot 5.8))]$$

$$= -1603919.5 \text{ N-m}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4((1 + 2(Rm^2 - hm^2)/(L^2))/(1 + (4hm^2)/(3L)) - 4a/L)$$

$$= 3675187 \cdot 5.8/4((1 + 2(1.4^2 - 1.35^2)/(5.8^2))/(1 + (4 \cdot 1.35)/(3 \cdot 5.8)) - 4 \cdot 1.15/5.8)$$

$$= -121563.5 \text{ N-m}$$



PV Elite 28 Licensee: SL User

FileName : 103-C2

Page 81 of 192

Saddle Calcs: Test Case: Step: 17 3:14pm Apr 10, 2026

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m / (2t) - M_2 / (\pi \cdot R_m^2 \cdot t)$$

$$= 18.6 \cdot 1401 / (2 \cdot 142) - 121563 / (\pi \cdot 1401^2 \cdot 142)$$

$$= 91784.55 \text{ kPa}$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t)$$

$$= 18.6 \cdot 1401 / (2 \cdot 142) + 121563 / (\pi \cdot 1401^2 \cdot 142)$$

$$= 91506.99 \text{ kPa}$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot R_m / (2t) - M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 18.6 \cdot 1401 / (2 \cdot 142) - 1603919 / (0.23 \cdot \pi \cdot 1401^2 \cdot 142)$$

$$= 99656.45 \text{ kPa}$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$= P \cdot R_m / (2t) + M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 18.6 \cdot 1401 / (2 \cdot 142) + 1603919 / (0.38 \cdot \pi \cdot 1401^2 \cdot 142)$$

$$= 86825.31 \text{ kPa}$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot h_m / 3))$$

$$= 3675187 \cdot (5.8 - 2 \cdot 1.15) / (5.8 + (4 \cdot 1.35 / 3))$$

$$= 1693419.5 \text{ N}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 0.58 \cdot 1693419 / (1401 \cdot 142)$$

$$= 4912.22 \text{ kPa}$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{R_m \cdot t}$$

$$= 0.78 \cdot \sqrt{1401 \cdot 142}$$

$$= 347.903 \text{ mm}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K_5 \cdot Q \cdot k / (t(b + X_1 + X_2))$$

$$= -0.62 \cdot 3675187 \cdot 0.1 / (142(254 + 348 + 348))$$

$$= -1701.14 \text{ kPa}$$

Circ. Comp. Stress at Horn of Saddle, $L < 8R_m$ (4.15.23) [σ_{*7}]:

$$= -Q / (4 \cdot t(b + X_1 + X_2)) - 12 \cdot K_7 \cdot Q \cdot R_m / (L \cdot t^2)$$

$$= -3675187 / (4 \cdot 142(254 + 348 + 348)) - 12 \cdot 0.013 \cdot 3675187 \cdot 1401 / (5.8 \cdot 142^2)$$

$$= -13548.96 \text{ kPa}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$$

$$= \min(254 + 1.56 \cdot \sqrt{1401 \cdot 142}, 2 \cdot 1.15)$$

$$= 949.81 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2500.0000	mm
Baseplate Thickness	Bpthk	100.0000	mm
Baseplate Width	Bpwid	552.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	35.0000	mm
Web Thickness	Webtk	35.0000	mm
Web Location	Webloc	Side	
Saddle Yield Stress	Sy	259980.2	kPa
Height of Web at Center	Hw,c	306.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)



PV Elite 28 Licensee: SL User

FileName : 103-C2

Page 82 of 192

Saddle Calcs: Test Case: Step: 17 3:14pm Apr 10,2026

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t) + \text{WearPlateWidth}})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(1330 \cdot 142) + 304})^3) / 12$$

$$= 234299104.000 \text{ mm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2$$

$$= 234299104 + 0.14 (328 - 71)^2$$

$$= 9443524608.000 \text{ mm}^4$$

	B mm	D mm	Y mm	A m ²	AY mm ³	I + AD ⁽²⁾ mm ⁴
Shell	981.9	142.0	71.0	0.1	9899971.0	0.944E+10
Wearplate	304.0	30.0	157.0	0.0	1431840.0	0.267E+09
Web	35.0	698.0	521.0	0.0	12728034.0	0.190E+10
BasePlate	552.0	100.0	920.0	0.1	50784000.0	0.194E+11
Totals	...	...	...	0.2	74843840.0	0.310E+11

Distance to Centroid [C1]:

$$= AY / A$$

$$= 74843840 / 0.23$$

$$= 327.995 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 180 / 2$$

$$= 90.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(90) - 0.5 \cdot \sin(90)^2) / (\pi - 1.57 + \sin(90) \cos(90))$$

$$= 0.3183$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.32 \cdot 3675187$$

$$= 1169848.3750 \text{ N}$$

$$\text{Tension Stress, } St = (Fh / As) = 13181.7441 \text{ kPa}$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 155988.1250 \text{ kPa}$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 2300 - 1330 \cdot \sin(180/2) / 1.57$$

$$= 1453.296 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 1700824.7500 \text{ N-m}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 17985.9570 \text{ kPa}$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 173320.1406 \text{ kPa}$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (2 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3(3675187 + 29229) \cdot 552) / (2 \cdot 2500 \cdot 173320)}$$

$$= 84.137 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle} / 2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 180/2) (1330 + 142 + 30)$$

$$= 3004.000 \text{ mm}$$



PV Elite 28 Licensee: SL User

FileName : 103-C2

Page 83 of 192

Saddle Calcs: Test Case: Step: 17 3:14pm Apr 10,2026

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nribs} - 1)$$

$$= 3004 / (5 - 1)$$

$$= 751.000 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 751 \cdot 552 / 2$$

$$= 0.207 \text{ m}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 3704416 / (2500 \cdot 552)$$

$$= 2685025.500 \text{ N/m}^2$$

Axial Load [P]:

$$= \text{Ap} \cdot \text{Bp}$$

$$= 0.21 \cdot 2685025$$

$$= 556403.375 \text{ N}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 0.0077 + 0.013$$

$$= 0.021 \text{ m}^2$$

Compressive Stress [Sc]:

$$= P / \text{Ar}$$

$$= 556403 / 0.021$$

$$= 26741.240 \text{ kPa}$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib	35.0	219.0	144.5	0.0	1107592.5	0.800E+08
Web	375.5	35.0	17.5	0.0	229993.8	0.301E+08
Totals	...	...	...	...	1337586.2	0.110E+09

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 254 - 35$$

$$= 219.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= \text{AY} / \text{A}$$

$$= 1337586 / 0.021$$

$$= 64.284 \text{ mm}$$

Distance to Centroid [C1]:

$$= \max(\text{ytot}, \text{Saddle Width} - \text{ytot})$$

$$= \max(64.3, 254 - 64.3)$$

$$= 189.716 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(110063424 / 0.021)}$$

$$= 72.730 \text{ mm}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\text{theta}/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 2300 - \cos(180/2) (1330 + 142 + 30) - 100$$

$$= 2200.000 \text{ mm}$$



PV Elite 28 Licensee: SL User

FileName : 103-C2

Page 84 of 192

Saddle Calcs: Test Case: Step: 17 3:14pm Apr 10, 2026

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 199955/259980)}$$

$$= 123.211$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 2200/72.7$$

$$= 30.249$$

Bending Moment [Rm]:

$$= FI / (2 \cdot B_{\text{plen}}) \cdot e \cdot L / 2$$

$$= 4130 / (2 \cdot 2500) \cdot 751 \cdot 2200 / 2$$

$$= 682.628 \text{ N-m}$$

Compressive Allowable, $KL/r < C_c$ ($30.2 < 123$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (30.2)^2 / (2 \cdot 123^2)) 259980 / (5/3 + 3 \cdot (30.2) / (8 \cdot 123) - (30.2)^3 / (8 \cdot 123^3))$$

$$= 143519 \text{ kPa}$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 26741/143519 + (683 \cdot 190/110063536) / 173320$$

$$= 0.193$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	35.0	219.0	144.5	0.0	1107592.5	0.105E+09
Web	751.0	35.0	17.5	0.0	459987.5	0.243E+08
Totals	...	...	...	...	1567580.0	0.129E+09

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 1567580/0.034$$

$$= 46.173 \text{ mm}$$

Distance to Centroid [C1]:

$$= \max(y_{\text{tot}}, \text{abs}(\text{Saddle Width} - y_{\text{tot}}))$$

$$= \max(46.2, \text{abs}(254 - 46.2))$$

$$= 207.827 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - B_{\text{pthk}}$$

$$= 2300 - \cos(150/2) (100 + 0 + 0) - 0$$

$$= 698.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(129034936/0.034)}$$

$$= 61.650 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 698/61.7$$

$$= 11.322$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 4130 / (2 \cdot 2500)$$

$$= 0.826 \text{ N/mm}$$



PV Elite 28 Licensee: SL User

FileName : 103-C2

Page 85 of 192

Saddle Calcs: Test Case: Step: 17 3:14pm Apr 10, 2026

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.83 \cdot 751 \cdot 698$$

$$= 433.159 \text{ N-m}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 433/620878$$

$$= 697.391 \text{ kPa}$$

Compressive Allowable, $KL/r < C_c$ ($11.3 < 123$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (11.3)^2 / (2 \cdot 123^2)) 259980 /$$

$$(5/3 + 3 \cdot (11.3) / (8 \cdot 123) - (11.3^3) / (8 \cdot 123^3))$$

$$= 152192 \text{ kPa}$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C1/I) / S_{ba}$$

$$= 25588/152192 + (433 \cdot 208/129034936) / 173320$$

$$= 0.172$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	8
Total Number of Bolts in Tension/Baseplate	Nbt	4
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	171986.91 kPa
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	70.0 mm
User Defined Area in a Single Bolt	UserBoltArea	0.0012 m ²
BasePlate Allowable Stress	S	137989.50 kPa
Area Available in a Single Bolt	BltArea	0.0010 m ²
Saddle Load Q0 (Weight)	Q0	3684480.0 N
Saddle Load QL (Wind/Seismic contribution)	QL	2793.8 N
Maximum Transverse Force	Ft	7223.3 N
Maximum Longitudinal Force	Fl	4130.0 N
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [τ_b ,l]:

$$= F_l / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 4130 / (0.00098 \cdot 8)$$

$$= 527.1 \text{ kPa. Must be less than } 103192.1 \text{ kPa.}$$

Shear Stress in a Single Bolt, Transverse Direction [τ_b ,t]:

$$= F_t / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 7223 / (0.00098 \cdot 8)$$

$$= 921.8 \text{ kPa. Must be less than } 103192.1 \text{ kPa.}$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (} Q_0 > Q_L \text{ --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= F_l / (\text{BoltShearAllowable} \cdot \text{Nbolts})$$

$$= 4130 / (103192 \cdot 8)$$

$$= 0.0000 \text{ m}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot F_t + \text{Sum of X Moments}$$

$$= 2300 \cdot 7223 + 0$$

$$= 16620.42 \text{ N-m}$$



PV Elite 28 Licensee: SL User

FileName : 103-C2

Page 86 of 192

Saddle Calcs: Test Case: Step: 17 3:14pm Apr 10, 2026

Eccentricity (e):
 = Rmom / QO
 = 16620/3684480
 = 4.51 mm < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
 = 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:
 = max[Bltarearl, Bltarears, Bltareart]
 = max[0, 0.50016E-05, 0]
 = 0.0000 m²

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Wear Pad Bearing angle is less than saddle angle + saddle/12 degrees. The wear plate will be ignored.

Minimum Wear Plate Angle [Ø1]:
 = 192.0 degrees

Note:
 Wear Pad Width (.00) is less than 1.56*sqrt(rm*t)
 and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:
 = min(b + 1.56*sqrt(Rm * t), 2a)
 = min(254 + 1.56*sqrt(1401 * 142), 2 * 1150)
 = 949.8055 mm

Input and Calculated Values:

Vessel Mean Radius	Rm	1401.00	mm
Shell Thickness used in this Case	t	142.000	mm
Stiffened Vessel Length per 4.15.6	L	5.80	m
Distance from Saddle to Vessel tangent	a1 or a	1150.00	mm
Saddle Width	b1 or b	254.00	mm
Saddle Bearing Angle	delta or theta	180.00	degrees
Inside or Mean Depth of Head	Hi or hm	1.37	m
Shell Allowable Stress used in Calculation		278978.75	kPa
Head Allowable Stress used in Calculation		278978.75	kPa
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	2300.00	mm
Coefficient of Friction	mu	0.40	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces		2131426.00	N
Pressure used in Analysis	P	18.579	N/mm ²

Horizontal Vessel Analysis Results:		Actual kPa	Allowable kPa
Long. Stress at Top of Midspan		91742.62	278978.75
Long. Stress at Bottom of Midspan		91548.93	278978.75
Long. Stress at Top of Saddles		96349.62	278978.75
Long. Stress at Bottom of Saddles		88815.21	278978.75
Tangential Shear in Shell		2839.87	165217.41



PV Elite 28 Licensee: SL User

FileName : 103-C2

Page 87 of 192

Saddle Calcs: Test Case: Step: 17 3:14pm Apr 10, 2026

Circ. Stress at Horn of Saddle	7857.73	418468.09
Circ. Compressive Stress in Shell	986.58	278978.75

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$$

$$= 3 (14447/2 + 0) \cdot 2300/2500$$

$$= 19936.4 \text{ N}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(F_{l, \text{wind}}, \text{Sum of X Forces}) \cdot B / L_s$$

$$= \max(4130, 0) \cdot 2300/3400$$

$$= 2793.8 \text{ N}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$$

$$= 2111489 + \max(2794, 19936, 0, 0)$$

$$= 2131426.0 \text{ N}$$

Longitudinal Wind Force [Fl]:

$$= \text{WindScalar} \cdot \text{WindPress} (\text{Platform Area} + (\pi/4 (\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 766 (0 + \pi/4 (2.94 \cdot 1.2)^2)$$

$$= 8260.342 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	2160655.00	N
Transverse Shear Load	7223.34	N
Longitudinal Shear Load	4129.96	N

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

$$= \text{Baseplate Length}$$

$$= 2500.000 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.2286	K2 = 0.5771	K3 = 0.2597	K4 = 0.2157
K5 = 0.6243	K6 = 0.0174	K7 = 0.0128	K8 = 0.2500
K9 = 0.1396	K10 = 0.0146	K1* = 0.3799	

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L)) / (1 + (4h_m)/3L)]$$

$$= -2131426 \cdot 1.15 [1 - (1 - 1.15/5.8 + (1.4^2 - 1.37^2)/(2 \cdot 1.15 \cdot 5.8)) / (1 + (4 \cdot 1.37)/(3 \cdot 5.8))]$$

$$= -941817.4 \text{ N-m}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4 ((1 + 2(R_m^2 - h_m^2)/(L^2)) / (1 + (4h_m^2)/(3L)) - 4a/L)$$

$$= 2131426 \cdot 5.8/4 ((1 + 2(1.4^2 - 1.37^2)/(5.8^2)) / (1 + (4 \cdot 1.37)/(3 \cdot 5.8)) - 4 \cdot 1.15/5.8)$$

$$= -84832.3 \text{ N-m}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m / (2t) - M_2 / (\pi \cdot R_m^2 t)$$

$$= 18.6 \cdot 1401 / (2 \cdot 142) - -84832 / (\pi \cdot 1401^2 \cdot 142)$$

$$= 91742.62 \text{ kPa}$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t)$$

$$= 18.6 \cdot 1401 / (2 \cdot 142) + -84832 / (\pi \cdot 1401^2 \cdot 142)$$

$$= 91548.93 \text{ kPa}$$



PV Elite 28 Licensee: SL User

FileName : 103-C2

Page 88 of 192

Saddle Calcs: Test Case: Step: 17 3:14pm Apr 10, 2026

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$= P \cdot R_m / (2t) - M_1 / (K_1 \cdot \pi \cdot R_m^2 t)$$

$$= 18.6 \cdot 1401 / (2 \cdot 142) - 941817 / (0.23 \cdot \pi \cdot 1401^2 \cdot 142)$$

$$= 96349.62 \text{ kPa}$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot R_m / (2t) + M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 18.6 \cdot 1401 / (2 \cdot 142) + 941817 / (0.38 \cdot \pi \cdot 1401^2 \cdot 142)$$

$$= 88815.21 \text{ kPa}$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a) / (L + (4 \cdot h_m / 3))$$

$$= 2131426 (5.8 - 2 \cdot 1.15) / (5.8 + (4 \cdot 1.37 / 3))$$

$$= 979006.0 \text{ N}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 0.58 \cdot 979006 / (1401 \cdot 142)$$

$$= 2839.87 \text{ kPa}$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{R_m \cdot t}$$

$$= 0.78 \cdot \sqrt{1401 \cdot 142}$$

$$= 347.903 \text{ mm}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K_5 \cdot Q \cdot k / (t(b + X_1 + X_2))$$

$$= -0.62 \cdot 2131426 \cdot 0.1 / (142(254 + 348 + 348))$$

$$= -986.58 \text{ kPa}$$

Circ. Comp. Stress at Horn of Saddle, $L < 8R_m$ (4.15.23) [σ^*7]:

$$= -Q / (4 \cdot t(b + X_1 + X_2)) - 12 \cdot K_7 \cdot Q \cdot R_m / (L \cdot t^2)$$

$$= -2131426 / (4 \cdot 142(254 + 348 + 348)) - 12 \cdot 0.013 \cdot 2131426 \cdot 1401 / (5.8 \cdot 142^2)$$

$$= -7857.73 \text{ kPa}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$$

$$= \min(254 + 1.56 \cdot \sqrt{1401 \cdot 142}, 2 \cdot 1.15)$$

$$= 949.81 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2500.0000	mm
Baseplate Thickness	Bpthk	100.0000	mm
Baseplate Width	Bpwid	552.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	35.0000	mm
Web Thickness	Webtk	35.0000	mm
Web Location	Webloc	Side	
Saddle Yield Stress	Sy	259980.2	kPa
Height of Web at Center	Hw, c	306.0	mm
Friction Coefficient	mu	0.400	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{R \cdot t} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{1330 \cdot 142}) + 304)^3 / 12$$

$$= 234299104.000 \text{ mm}^4$$

Resolved Inertia for the Shell [I, shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C_1 - Y)^2$$

$$= 234299104 + 0.14(328 - 71)^2$$



PV Elite 28 Licensee: SL User

FileName : 103-C2

Page 89 of 192

Saddle Calcs: Test Case: Step: 17 3:14pm Apr 10,2026

= 9443524608.000 mm⁴

	B mm	D mm	Y mm	A m ²	AY mm ³	I + AD ² (2) mm ⁴
Shell	981.9	142.0	71.0	0.1	9899971.0	0.944E+10
Wearplate	304.0	30.0	157.0	0.0	1431840.0	0.267E+09
Web	35.0	698.0	521.0	0.0	12728034.0	0.190E+10
BasePlate	552.0	100.0	920.0	0.1	50784000.0	0.194E+11
Totals	...	...	...	0.2	74843840.0	0.310E+11

Distance to Centroid [C1]:

= AY / A

= 74843840/0.23

= 327.995 mm

Angle [β]:

= 180 - Saddle Angle/2

= 180 - 180/2

= 90.0

Saddle Splitting Coefficient [K1]:

= (1 + cos(β) - 0.5•sin(β)²)/(π - β + sin(β)cos(β))

= (1 + cos(90) - 0.5•sin(90)²)/(π - 1.57 + sin(90)cos(90))

= 0.3183

Saddle Splitting Force [Fh]:

= K1 • Q

= 0.32 • 2131426

= 678453.9375 N

Tension Stress, St = (Fh/As) = 7644.7573 kPa

Allowed Stress, Sa = 0.6 • Yield Str = 155988.1250 kPa

Saddle Splitting Dimension [d]:

= B - R • sin(theta/2)/(theta/2 in radians)

= 2300 - 1330 • sin(180/2)/1.57

= 1453.296 mm

Bending Moment, M = Fh • d = 986393.9375 N-m

Bending Stress, Sb = (M • C1 / I) = 10430.9619 kPa

Allowed Stress, Sa = 2/3 • Yield Str = 173320.1406 kPa

Minimum Thickness of Baseplate per Moss:

= √(3(Q + SaddleWt)BasePlateWidth / (2 • BasePlateLength • AllStress))

= √(3(2131426 + 29229)552/(2 • 2500 • 173320))

= 64.257 mm

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

= 2 • cos(90 - Saddle Angle/2) (Inside Radius + Shell Thk + Wear Plate Thk)

= 2 • cos(90 - 180/2) (1330 + 142 + 30)

= 3004.000 mm

Distance between Ribs [e]:

= Web Length / (Nribs - 1)

= 3004/(5 - 1)

= 751.000 mm

Baseplate Pressure Area [Ap]:

= e • Bpwid / 2

= 751 • 552/2

= 0.207 m²



PV Elite 28 Licensee: SL User

FileName : 103-C2

Page 90 of 192

Saddle Calcs: Test Case: Step: 17 3:14pm Apr 10, 2026

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 2160655 / (2500 \cdot 552)$$

$$= 1566080.250 \text{ N/m}^2$$

Axial Load [P]:

$$= A_p \cdot B_p$$

$$= 0.21 \cdot 1566080$$

$$= 324530.375 \text{ N}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 0.0077 + 0.013$$

$$= 0.021 \text{ m}^2$$

Compressive Stress [Sc]:

$$= P / A_r$$

$$= 324530 / 0.021$$

$$= 15597.218 \text{ kPa}$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib	35.0	219.0	144.5	0.0	1107592.5	0.800E+08
Web	375.5	35.0	17.5	0.0	229993.8	0.301E+08
Totals	...	...	...	...	1337586.2	0.110E+09

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 254 - 35$$

$$= 219.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 1337586 / 0.021$$

$$= 64.284 \text{ mm}$$

Distance to Centroid [C1]:

$$= \max(\text{ytot}, \text{Saddle Width} - \text{ytot})$$

$$= \max(64.3, 254 - 64.3)$$

$$= 189.716 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(110063424 / 0.021)}$$

$$= 72.730 \text{ mm}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\text{theta}/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 2300 - \cos(180/2) (1330 + 142 + 30) - 100$$

$$= 2200.000 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 199955 / 259980)}$$

$$= 123.211$$

Slenderness ratio [KL/r]:

$$= KL / r$$

$$= 1 \cdot 2200 / 72.7$$

$$= 30.249$$



PV Elite 28 Licensee: SL User

FileName : 103-C2

Page 91 of 192

Saddle Calcs: Test Case: Step: 17 3:14pm Apr 10,2026

Bending Moment [Rm]:

$$= FI / (2 \cdot B_{plen}) \cdot e \cdot L / 2$$

$$= 4130 / (2 \cdot 2500) \cdot 751 \cdot 2200 / 2$$

$$= 682.628 \text{ N-m}$$

Compressive Allowable, $KL/r < C_c$ ($30.2 < 123$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (30.2)^2 / (2 \cdot 123^2)) 259980 /$$

$$(5/3 + 3 \cdot (30.2) / (8 \cdot 123) - (30.2^3) / (8 \cdot 123^3))$$

$$= 143519 \text{ kPa}$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 15597 / 143519 + (683 \cdot 190 / 110063536) / 173320$$

$$= 0.115$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	35.0	219.0	144.5	0.0	1107592.5	0.105E+09
Web	751.0	35.0	17.5	0.0	459987.5	0.243E+08
Totals	...	...	...	...	1567580.0	0.129E+09

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 1567580 / 0.034$$

$$= 46.173 \text{ mm}$$

Distance to Centroid [C1]:

$$= \max(y_{tot}, \text{abs}(\text{Saddle Width} - y_{tot}))$$

$$= \max(46.2, \text{abs}(254 - 46.2))$$

$$= 207.827 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - B_{pthk}$$

$$= 2300 - \cos(1502/2) (100 + 0 + 0) - 0$$

$$= 698.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ (\text{Total Inertia} / \text{Total Area}) }$$

$$= \sqrt{ (129034936 / 0.034) }$$

$$= 61.650 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 698 / 61.7$$

$$= 11.322$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 4130 / (2 \cdot 2500)$$

$$= 0.826 \text{ N/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.83 \cdot 751 \cdot 698$$

$$= 433.159 \text{ N-m}$$

Bending Stress due to Transverse Force and Weight Load [$\sigma_{B,base,c}$]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 433 / 620878$$

$$= 697.391 \text{ kPa}$$



PV Elite 28 Licensee: SL User

FileName : 103-C2

Page 92 of 192

Saddle Calcs: Test Case: Step: 17 3:14pm Apr 10,2026

Compressive Allowable, $KL/r < C_c$ ($11.3 < 123$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (11.3)^2 / (2 \cdot 123^2)) 259980 / (5/3 + 3 \cdot (11.3) / (8 \cdot 123) - (11.3^3) / (8 \cdot 123^3))$$

$$= 152192 \text{ kPa}$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C_1 / I) / S_{ba}$$

$$= 14924 / 152192 + (433 \cdot 208 / 129034936) / 173320$$

$$= 0.102$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	8
Total Number of Bolts in Tension/Baseplate	Nbt	4
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	129990.10 kPa
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	70.0 mm
User Defined Area in a Single Bolt	UserBoltArea	0.0012 m ²
BasePlate Allowable Stress	S	137989.50 kPa
Area Available in a Single Bolt	BlArea	0.0010 m ²
Saddle Load Q0 (Weight)	Q0	2140718.8 N
Saddle Load QL (Wind/Seismic contribution)	QL	2793.8 N
Maximum Transverse Force	Ft	7223.3 N
Maximum Longitudinal Force	Fl	4130.0 N
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [tb,l]:

$$= F_l / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 4130 / (0.00098 \cdot 8)$$

$$= 527.1 \text{ kPa. Must be less than } 77994.1 \text{ kPa.}$$

Shear Stress in a Single Bolt, Transverse Direction [tb,t]:

$$= F_t / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 7223 / (0.00098 \cdot 8)$$

$$= 921.8 \text{ kPa. Must be less than } 77994.1 \text{ kPa.}$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (Q0 > QL --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= F_l / (\text{BoltShearAllowable} \cdot \text{Nbolts})$$

$$= 4130 / (77994 \cdot 8)$$

$$= 0.0000 \text{ m}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot F_t + \text{Sum of X Moments}$$

$$= 2300 \cdot 7223 + 0$$

$$= 16620.42 \text{ N-m}$$

Eccentricity (e):

$$= R_{mom} / Q_0$$

$$= 16620 / 2140718$$

$$= 7.76 \text{ mm} < B_{plen} / 6 \text{ --> No Uplift in Transverse direction}$$

Bolt Area due to Transverse Load [Bltareart]:

$$= 0 \text{ (No Uplift)}$$

Required Area of a Single Bolt [Bltarear]:



PV Elite 28 Licensee: SL User

FileName : 103-C2

Page 93 of 192

Saddle Calcs: Test Case:

Step: 17 3:14pm Apr 10, 2026

= max[Bltarearl, Bltarears, Bltareart]

= max[0, 0.66176E-05, 0]

= 0.0000 m²

Review notes about nozzle loadings and supports in the Warnings report.

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PV Elite 28 Licensee: SL User

FileName : 103-C2

Page 94 of 192

Saddle Calcs: Ext. Pressure: Step: 18 3:14pm Apr 10,2026

ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : External Pressure

Wear Pad Bearing angle is less than saddle angle + saddle/12 degrees. The wear plate will be ignored.

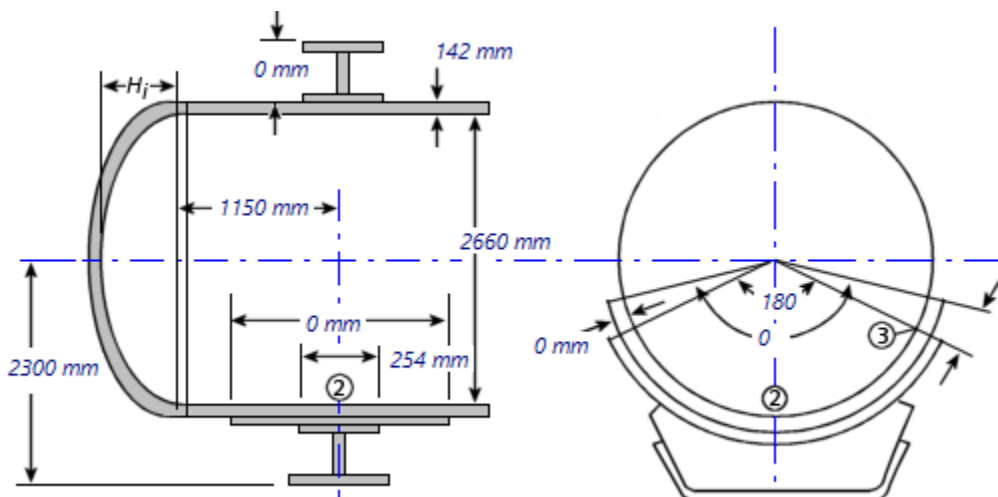
Minimum Wear Plate Angle [θ1]:
= 192.0 degrees

Note:
Wear Pad Width (.00) is less than $1.56 \cdot \sqrt{r_m \cdot t}$
and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:
= $\min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$
= $\min(254 + 1.56 \cdot \sqrt{1402 \cdot 139}, 2 \cdot 1150)$
= 942.7846 mm

Input and Calculated Values:

Vessel Mean Radius	Rm	1402.50	mm
Shell Thickness used in this Case	t	142.000	mm
Stiffened Vessel Length per 4.15.6	L	5.80	m
Distance from Saddle to Vessel tangent	a1 or a	1150.00	mm
Saddle Width	b1 or b	254.00	mm
Saddle Bearing Angle	delta or theta	180.00	degrees
Inside or Mean Depth of Head	Hi or hm	1.35	m
Shell Allowable Stress used in Calculation		147988.73	kPa
Head Allowable Stress used in Calculation		137996.80	kPa
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	2300.00	mm
Coefficient of Friction	mu	0.00	



Results for Horizontal Vessel Analysis:

Saddle Force Q, External Pressure Case		3176314.75	N
Pressure used in Analysis	P	-0.101	N/mm ²



PV Elite 28 Licensee: SL User

FileName : 103-C2

Page 95 of 192

Saddle Calcs: Ext. Pressure: Step: 18 3:14pm Apr 10,2026

Horizontal Vessel Analysis Results:	Actual kPa	Allowable kPa
Long. Stress at Top of Midspan	-388.43	-215195.78
Long. Stress at Top of Midspan	388.43	147988.73
Long. Stress at Bottom of Midspan	-633.68	-215195.78
Long. Stress at Bottom of Midspan	633.68	147988.73
Long. Stress at Top of Saddles	6546.97	147988.73
Long. Stress at Bottom of Saddles	-4758.26	-215195.78
Long. Stress at Bottom of Saddles	4758.26	147988.73
Tangential Shear in Shell	4331.66	118390.98
Circ. Stress at Horn of Saddle	12131.26	184985.91
Circ. Compressive Stress in Shell	1513.14	147988.73

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= Ftr(Ft/Num of Saddles + Z Force Load) \cdot B / E$$

$$= 3 (28893/2 + 176413) \cdot 2300/2500$$

$$= 526773.3 \text{ N}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(Fl,wind; \text{Sum of X Forces}) \cdot B / Ls$$

$$= \max(8260, 352826) \cdot 2300/3400$$

$$= 5587.6 \text{ N}$$

Saddle Reaction Force due to Earthquake Fl [Fsl]:

$$= \max(Fl,eq; \text{Sum of X Forces}) \cdot B / Ls$$

$$= \max(217564, 352826) \cdot 2300/3400$$

$$= 238676.7 \text{ N}$$

Saddle Reaction Force due to Earthquake Ft [Fst]:

$$= Ftr(Ft/Num of Saddles + Z Force Load) \cdot B / E$$

$$= 3 (217564/2 + 176413) \cdot 2300/2500$$

$$= 787139.2 \text{ N}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(Fwl, Fwt, Fsl, Fst)$$

$$= 2389175 + \max(5588, 526773, 238677, 787139)$$

$$= 3176314.8 \text{ N}$$

Longitudinal Wind Force [Fl]:

$$= \text{WindScalar} \cdot \text{WindPress} (\text{Platform Area} + (\pi/4 (OD \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 766 (0 + \pi/4 (2.94 \cdot 1.2)^2)$$

$$= 8260.342 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight) 3205543.75 N

Transverse Shear Load 285195.38 N

Longitudinal Shear Load 352826.41 N

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

$$= \text{Baseplate Length}$$

$$= 2500.000 \text{ mm}$$

The Computed K values from Table 4.15.1:

$$K1 = 0.2286 \quad K2 = 0.5771 \quad K3 = 0.2597 \quad K4 = 0.2157$$

$$K5 = 0.6243 \quad K6 = 0.0174 \quad K7 = 0.0127 \quad K8 = 0.2500$$

$$K9 = 0.1396 \quad K10 = 0.0146 \quad K1* = 0.3799$$



PV Elite 28 Licensee: SL User

FileName : 103-C2

Page 96 of 192

Saddle Calcs: Ext. Pressure: Step: 18 3:14pm Apr 10, 2026

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a \left[1 - \left(1 - \frac{a}{L} + \frac{(Rm^2 - hm^2)}{(2a \cdot L)} \right) / \left(1 + \frac{(4hm)}{3L} \right) \right]$$

$$= -3176314 \cdot 1.15 \left[1 - \left(1 - \frac{1.15}{5.8} + \frac{(1.4^2 - 1.35^2)}{(2 \cdot 1.15 \cdot 5.8)} \right) / \left(1 + \frac{(4 \cdot 1.35)}{(3 \cdot 5.8)} \right) \right]$$

$$= -1386284.6 \text{ N-m}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L / 4 \left(\frac{(1 + 2(Rm^2 - hm^2)/(L^2))}{(1 + (4hm^2)/(3L))} - 4a/L \right)$$

$$= 3176314 \cdot 5.8 / 4 \left(\frac{(1 + 2(1.4^2 - 1.35^2)/(5.8^2))}{(1 + (4 \cdot 1.35)/(3 \cdot 5.8))} - 4 \cdot 1.15 / 5.8 \right)$$

$$= -105369.4 \text{ N-m}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot Rm / (2t) - M2 / (\pi \cdot Rm^2 t)$$

$$= -0.1 \cdot 1402 / (2 \cdot 139) - -105369 / (\pi \cdot 1402^2 \cdot 139)$$

$$= -388.43 \text{ kPa}$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot Rm / (2t) + M2 / (\pi \cdot Rm^2 \cdot t)$$

$$= -0.1 \cdot 1402 / (2 \cdot 139) + -105369 / (\pi \cdot 1402^2 \cdot 139)$$

$$= -633.68 \text{ kPa}$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot Rm / (2t) - M1 / (K1 \cdot \pi \cdot Rm^2 t)$$

$$= -0.1 \cdot 1402 / (2 \cdot 139) - -1386284 / (0.23 \cdot \pi \cdot 1402^2 \cdot 139)$$

$$= 6546.97 \text{ kPa}$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$= P \cdot Rm / (2t) + M1 / (K1 \cdot \pi \cdot Rm^2 \cdot t)$$

$$= -0.1 \cdot 1402 / (2 \cdot 139) + -1386284 / (0.38 \cdot \pi \cdot 1402^2 \cdot 139)$$

$$= -4758.26 \text{ kPa}$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a) / (L + (4 \cdot hm / 3))$$

$$= 3176314 \cdot (5.8 - 2 \cdot 1.15) / (5.8 + (4 \cdot 1.35 / 3))$$

$$= 1463296.8 \text{ N}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (Rm \cdot t)$$

$$= 0.58 \cdot 1463296 / (1402 \cdot 139)$$

$$= 4331.66 \text{ kPa}$$

Decay Length (4.15.20) [x1, x2]:

$$= 0.78 \cdot \sqrt{(Rm \cdot t)}$$

$$= 0.78 \cdot \sqrt{(1402 \cdot 139)}$$

$$= 344.392 \text{ mm}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K5 \cdot Q \cdot k / (t(b + X1 + X2))$$

$$= -0.62 \cdot 3176314 \cdot 0.1 / (139(254 + 344 + 344))$$

$$= -1513.14 \text{ kPa}$$

Circ. Comp. Stress at Horn of Saddle, L < 8Rm (4.15.23) [σ_{7*}]:

$$= -Q / (4 \cdot t(b + X1 + X2)) - 12 \cdot K7 \cdot Q \cdot Rm / (L \cdot t^2)$$

$$= -3176314 / (4 \cdot 139(254 + 344 + 344)) - 12 \cdot 0.013 \cdot 3176314 \cdot 1402 / (5.8 \cdot 139^2)$$

$$= -12131.26 \text{ kPa}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(Rm \cdot t)}, 2a)$$

$$= \min(254 + 1.56 \cdot \sqrt{(1402 \cdot 139)}, 2 \cdot 1.15)$$

$$= 942.78 \text{ mm}$$



PV Elite 28 Licensee: SL User

FileName : 103-C2

Page 97 of 192

Saddle Calcs: Ext. Pressure: Step: 18 3:14pm Apr 10,2026

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2500.0000	mm
Baseplate Thickness	Bpthk	100.0000	mm
Baseplate Width	Bpwid	552.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	35.0000	mm
Web Thickness	Webtk	35.0000	mm
Web Location	Webloc	Side	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	194672.1	kPa
Height of Web at Center	Hw,c	306.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3)/12$$

$$= ((1.56 \cdot \sqrt{(1333 \cdot 139)} + 304)139^3)/12$$

$$= 218318624.000 \text{ mm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 218318624 + 0.14(330 - 69.5)^2$$

$$= 9441132544.000 \text{ mm}^4$$

	B mm	D mm	Y mm	A m ²	AY mm ³	I + AD ² (2) mm ⁴
Shell	975.5	139.0	69.5	0.1	9423834.0	0.944E+10
Wearplate	304.0	30.0	154.0	0.0	1404480.0	0.284E+09
Web	35.0	698.0	518.0	0.0	12654744.0	0.185E+10
BasePlate	552.0	100.0	917.0	0.1	50618408.0	0.190E+11
Totals	...	...	...	0.2	74101464.0	0.306E+11

Distance to Centroid [C1]:

$$= AY / A$$

$$= 74101464/0.22$$

$$= 330.302 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 180/2$$

$$= 90.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2)/(\pi - \beta + \sin(\beta)\cos(\beta))$$

$$= (1 + \cos(90) - 0.5 \cdot \sin(90)^2)/(\pi - 1.57 + \sin(90)\cos(90))$$

$$= 0.3183$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.32 \cdot 3176314$$

$$= 1011052.3125 \text{ N}$$

$$\text{Tension Stress, } St = (Fh/As) = 11392.4434 \text{ kPa}$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 116803.2656 \text{ kPa}$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\text{theta}/2)/(\text{theta}/2 \text{ in radians})$$

$$= 2300 - 1333 \cdot \sin(180/2)/1.57$$

$$= 1451.386 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 1468022.0000 \text{ N-m}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 15827.4062 \text{ kPa}$$



PV Elite 28 Licensee: SL User

FileName : 103-C2

Page 98 of 192

Saddle Calcs: Ext. Pressure: Step: 18 3:14pm Apr 10,2026

Allowed Stress, $S_a = 2/3 \cdot \text{Yield Str} = 129781.4062 \text{ kPa}$

Minimum Thickness of Baseplate per Moss:

The maximum of $2/3 \cdot \text{Yield}$ and the Saddle Allowable Stress value was used to determine the baseplate thickness.

$$= \sqrt{\frac{3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}}{2 \cdot \text{BasePlateLength} \cdot \text{AllStress}}}$$

$$= \sqrt{\frac{3(3176314 + 29229) \cdot 552}{2 \cdot 2500 \cdot 147989}}$$

$$= 84.701 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) \cdot (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 180/2) \cdot (1330 + 142 + 30)$$

$$= 3004.000 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 3004 / (5 - 1)$$

$$= 751.000 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 751 \cdot 552 / 2$$

$$= 0.207 \text{ m}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 3205544 / (2500 \cdot 552)$$

$$= 2323433.750 \text{ N/m}^2$$

Axial Load [P]:

$$= A_p \cdot B_p$$

$$= 0.21 \cdot 2323433$$

$$= 481472.719 \text{ N}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 0.0077 + 0.013$$

$$= 0.021 \text{ m}^2$$

Compressive Stress [Sc]:

$$= P / A_r$$

$$= 481473 / 0.021$$

$$= 23140.004 \text{ kPa}$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib	35.0	219.0	144.5	0.0	1107592.5	0.800E+08
Web	375.5	35.0	17.5	0.0	229993.8	0.301E+08
Totals	...	...	...	...	1337586.2	0.110E+09

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 254 - 35$$

$$= 219.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$



PV Elite 28 Licensee: SL User

FileName : 103-C2

Page 99 of 192

Saddle Calcs: Ext. Pressure: Step: 18 3:14pm Apr 10,2026

$$= 1337586/0.021$$

$$= 64.284 \text{ mm}$$

Distance to Centroid [C1]:

$$= \max(y_{tot}, \text{Saddle Width} - y_{tot})$$

$$= \max(64.3, 254 - 64.3)$$

$$= 189.716 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ \text{Total Inertia} / \text{Total Area} }$$

$$= \sqrt{ 110063424/0.021 }$$

$$= 72.730 \text{ mm}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\theta/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 2300 - \cos(180/2) (1330 + 142 + 30) - 100$$

$$= 2200.000 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{ (2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}) }$$

$$= \sqrt{ (2 \cdot \pi^2 \cdot 199955/194672) }$$

$$= 142.386$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 2200/72.7$$

$$= 30.249$$

Bending Moment [Rm]:

$$= FI / (2 \cdot \text{Bplen}) \cdot e \cdot L / 2$$

$$= 352826 / (2 \cdot 2500) \cdot 751 \cdot 2200/2$$

$$= 58317.621 \text{ N-m}$$

Compressive Allowable, $KL/r < Cc$ ($30.2 < 142$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (30.2)^2 / (2 \cdot 142^2)) 194672 / (5/3 + 3 \cdot (30.2) / (8 \cdot 142) - (30.2^3) / (8 \cdot 142^3))$$

$$= 109034 \text{ kPa}$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 23140/109034 + (58318 \cdot 190/110063536) / 129781$$

$$= 0.986$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	35.0	219.0	144.5	0.0	1107592.5	0.105E+09
Web	751.0	35.0	17.5	0.0	459987.5	0.243E+08
Totals	...	...	...	...	1567580.0	0.129E+09

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 1567580/0.034$$

$$= 46.173 \text{ mm}$$

Distance to Centroid [C1]:

$$= \max(y_{tot}, \text{abs}(\text{Saddle Width} - y_{tot}))$$

$$= \max(46.2, \text{abs}(254 - 46.2))$$

$$= 207.827 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$$



PV Elite 28 Licensee: SL User

FileName : 103-C2

Page 100 of 192

Saddle Calcs: Ext. Pressure: Step: 18 3:14pm Apr 10, 2026

$$= 2300 - \cos(1502/2)(100 + 0 + 0) - 0$$

$$= 698.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{129034936 / 0.034}$$

$$= 61.650 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 698 / 61.7$$

$$= 11.322$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 352826 / (2 \cdot 2500)$$

$$= 70.565 \text{ N/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 70.6 \cdot 751 \cdot 698$$

$$= 37005.176 \text{ N-m}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 37005 / 620878$$

$$= 59578.828 \text{ kPa}$$

Compressive Allowable, $KL/r < C_c$ ($11.3 < 142$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (11.3)^2 / (2 \cdot 142^2)) 194672 / (5/3 + 3 \cdot (11.3) / (8 \cdot 142) - (11.3)^3 / (8 \cdot 142^3))$$

$$= 114392 \text{ kPa}$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C_{1/I}) / S_{ba}$$

$$= 22142 / 114392 + (37005 \cdot 208 / 129034936) / 129781$$

$$= 0.653$$

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Wear Pad Bearing angle is less than saddle angle + saddle/12 degrees. The wear plate will be ignored.

Minimum Wear Plate Angle [θ_1]:

$$= 192.0 \text{ degrees}$$

Note:
Wear Pad Width (.00) is less than $1.56 \cdot \sqrt{r_m \cdot t}$
and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$$

$$= \min(254 + 1.56 \cdot \sqrt{1402 \cdot 139}, 2 \cdot 1150)$$

$$= 942.7846 \text{ mm}$$

Input and Calculated Values:

Vessel Mean Radius	Rm	1402.50	mm
Shell Thickness used in this Case	t	142.000	mm
Stiffened Vessel Length per 4.15.6	L	5.80	m
Distance from Saddle to Vessel tangent	a1 or a	1150.00	mm
Saddle Width	b1 or b	254.00	mm
Saddle Bearing Angle	delta or theta	180.00	degrees



PV Elite 28 Licensee: SL User

FileName : 103-C2

Page 101 of 192

Saddle Calcs: Ext. Pressure: Step: 18 3:14pm Apr 10,2026

Inside or Mean Depth of Head	Hi or hm	1.37 m
Shell Allowable Stress used in Calculation		147988.73 kPa
Head Allowable Stress used in Calculation		147988.73 kPa
Circumferential Efficiency in Plane of Saddle		1.00
Circumferential Efficiency at Mid-Span		1.00
Distance from Saddle Base to Centerline	B	2300.00 mm
Coefficient of Friction	mu	0.40

Results for Horizontal Vessel Analysis:

Saddle Force Q, External Pressure Case		2214071.25 N
Pressure used in Analysis	P	-0.101 N/mm ²

Horizontal Vessel Analysis Results:	Actual kPa	Allowable kPa
Long. Stress at Top of Midspan	-407.77	-215195.78
Long. Stress at Top of Midspan	407.77	147988.73
Long. Stress at Bottom of Midspan	-614.34	-215195.78
Long. Stress at Bottom of Midspan	614.34	147988.73
Long. Stress at Top of Saddles	4471.98	147988.73
Long. Stress at Bottom of Saddles	-3509.62	-215195.78
Long. Stress at Bottom of Saddles	3509.62	147988.73
Tangential Shear in Shell	3009.64	118390.98
Circ. Stress at Horn of Saddle	8456.17	184985.91
Circ. Compressive Stress in Shell	1054.75	147988.73

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= Ftr(Ft/Num of Saddles + Z Force Load) \cdot B / E$$

$$= 3 (28893/2 + 176413) \cdot 2300/2500$$

$$= 526773.3 \text{ N}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(Fl,wind; \text{Sum of X Forces}) \cdot B / Ls$$

$$= \max(8260, 352826) \cdot 2300/3400$$

$$= 5587.6 \text{ N}$$

Saddle Reaction Force due to Earthquake Fl [Fsl]:

$$= \max(Fl,eq; \text{Sum of X Forces}) \cdot B / Ls$$

$$= \max(217564, 352826) \cdot 2300/3400$$

$$= 238676.7 \text{ N}$$

Saddle Reaction Force due to Earthquake Ft [Fst]:

$$= Ftr(Ft/Num of Saddles + Z Force Load) \cdot B / E$$

$$= 3 (217564/2 + 176413) \cdot 2300/2500$$

$$= 787139.2 \text{ N}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(Fwl, Fwt, Fsl, Fst)$$

$$= 1426932 + \max(5588, 526773, 238677, 787139)$$

$$= 2214071.2 \text{ N}$$

Longitudinal Wind Force [Fl]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4 (OD \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 766 (0 + \pi/4 (2.94 \cdot 1.2)^2)$$

$$= 8260.342 \text{ N}$$

Summary of Loads at the Base of this Saddle:



PV Elite 28 Licensee: SL User

FileName : 103-C2

Page 102 of 192

Saddle Calcs: Ext. Pressure: Step: 18 3:14pm Apr 10, 2026

Vertical Load (including saddle weight)	2243300.25	N
Transverse Shear Load	285195.38	N
Longitudinal Shear Load	352826.41	N

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

= Baseplate Length
= 2500.000 mm

The Computed K values from Table 4.15.1:

K1 = 0.2286	K2 = 0.5771	K3 = 0.2597	K4 = 0.2157
K5 = 0.6243	K6 = 0.0174	K7 = 0.0127	K8 = 0.2500
K9 = 0.1396	K10 = 0.0146	K1* = 0.3799	

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a \left[1 - \left(1 - \frac{a}{L} + \frac{(R_m^2 - h_m^2)}{(2a \cdot L)} \right) / (1 + (4h_m)/3L) \right]$$

$$= -2214071 \cdot 1.15 \left[1 - \left(1 - \frac{1.15}{5.8} + \frac{(1.4^2 - 1.37^2)}{(2 \cdot 1.15 \cdot 5.8)} \right) / (1 + (4 \cdot 1.37)/(3 \cdot 5.8)) \right]$$

$$= -978731.7 \text{ N-m}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L / 4 \left(\frac{1 + 2(R_m^2 - h_m^2)/(L^2)}{(1 + (4h_m^2)/(3L))} - 4a/L \right)$$

$$= 2214071 \cdot 5.8 / 4 \left(\frac{1 + 2(1.4^2 - 1.37^2)/(5.8^2)}{(1 + (4 \cdot 1.37)/(3 \cdot 5.8))} - 4 \cdot 1.15/5.8 \right)$$

$$= -88751.3 \text{ N-m}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m / (2t) - M2 / (\pi \cdot R_m^2 t)$$

$$= -0.1 \cdot 1402 / (2 \cdot 139) - 88751 / (\pi \cdot 1402^2 \cdot 139)$$

$$= -407.77 \text{ kPa}$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m / (2t) + M2 / (\pi \cdot R_m^2 \cdot t)$$

$$= -0.1 \cdot 1402 / (2 \cdot 139) + 88751 / (\pi \cdot 1402^2 \cdot 139)$$

$$= -614.34 \text{ kPa}$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot R_m / (2t) - M1 / (K1 \cdot \pi \cdot R_m^2 t)$$

$$= -0.1 \cdot 1402 / (2 \cdot 139) - 978732 / (0.23 \cdot \pi \cdot 1402^2 \cdot 139)$$

$$= 4471.98 \text{ kPa}$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$= P \cdot R_m / (2t) + M1 / (K1^* \cdot \pi \cdot R_m^2 \cdot t)$$

$$= -0.1 \cdot 1402 / (2 \cdot 139) + 978732 / (0.38 \cdot \pi \cdot 1402^2 \cdot 139)$$

$$= -3509.62 \text{ kPa}$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a) / (L + (4 \cdot h_m/3))$$

$$= 2214071 (5.8 - 2 \cdot 1.15) / (5.8 + (4 \cdot 1.37/3))$$

$$= 1016699.6 \text{ N}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (R_m \cdot t)$$

$$= 0.58 \cdot 1016699 / (1402 \cdot 139)$$

$$= 3009.64 \text{ kPa}$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(1402 \cdot 139)}$$

$$= 344.392 \text{ mm}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:



PV Elite 28 Licensee: SL User

FileName : 103-C2

Page 103 of 192

Saddle Calcs: Ext. Pressure: Step: 18 3:14pm Apr 10, 2026

$$= -K5 \cdot Q \cdot k / (t(b + X1 + X2))$$

$$= -0.62 \cdot 2214071 \cdot 0.1 / (139(254 + 344 + 344))$$

$$= -1054.75 \text{ kPa}$$

Circ. Comp. Stress at Horn of Saddle, $L < 8R_m$ (4.15.23) $[\sigma_7^*]$:

$$= -Q / (4 \cdot t(b + X1 + X2)) - 12 \cdot K7 \cdot Q \cdot R_m / (L \cdot t^2)$$

$$= -2214071 / (4 \cdot 139(254 + 344 + 344)) -$$

$$12 \cdot 0.013 \cdot 2214071 \cdot 1402 / (5.8 \cdot 139^2)$$

$$= -8456.17 \text{ kPa}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(254 + 1.56 \cdot \sqrt{(1402 \cdot 139)}, 2 \cdot 1.15)$$

$$= 942.78 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2500.0000	mm
Baseplate Thickness	Bpthk	100.0000	mm
Baseplate Width	Bpwid	552.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	35.0000	mm
Web Thickness	Webtk	35.0000	mm
Web Location	Webloc	Side	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	194672.1	kPa
Height of Web at Center	Hw,c	306.0	mm
Friction Coefficient	mu	0.400	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(1333 \cdot 139)} + 304)^3) / 12$$

$$= 218318624.000 \text{ mm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 218318624 + 0.14(330 - 69.5)^2$$

$$= 9441132544.000 \text{ mm}^4$$

	B mm	D mm	Y mm	A m ²	AY mm ³	I + AD ⁽²⁾ mm ⁴
Shell	975.5	139.0	69.5	0.1	9423834.0	0.944E+10
Wearplate	304.0	30.0	154.0	0.0	1404480.0	0.284E+09
Web	35.0	698.0	518.0	0.0	12654744.0	0.185E+10
BasePlate	552.0	100.0	917.0	0.1	50618408.0	0.190E+11
Totals	...	...	...	0.2	74101464.0	0.306E+11

Distance to Centroid [C1]:

$$= AY / A$$

$$= 74101464 / 0.22$$

$$= 330.302 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 180 / 2$$

$$= 90.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(90) - 0.5 \cdot \sin(90)^2) / (\pi - 1.57 + \sin(90) \cos(90))$$

$$= 0.3183$$



PV Elite 28 Licensee: SL User

FileName : 103-C2

Page 104 of 192

Saddle Calcs: Ext. Pressure: Step: 18 3:14pm Apr 10, 2026

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.32 \cdot 2214071$$

$$= 704760.6250 \text{ N}$$

$$\text{Tension Stress, } St = (Fh / As) = 7941.1777 \text{ kPa}$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 116803.2656 \text{ kPa}$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 2300 - 1333 \cdot \sin(180/2) / 1.57$$

$$= 1451.386 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 1023294.3750 \text{ N-m}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 11032.5977 \text{ kPa}$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 129781.4062 \text{ kPa}$$

Minimum Thickness of Baseplate per Moss:

The maximum of $2/3 \cdot \text{Yield}$ and the Saddle Allowable Stress value was used to determine the baseplate thickness.

$$= \sqrt{3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (2 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{3(2214071 + 29229)552 / (2 \cdot 2500 \cdot 147989)}$$

$$= 70.857 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 180/2) (1330 + 142 + 30)$$

$$= 3004.000 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nribs} - 1)$$

$$= 3004 / (5 - 1)$$

$$= 751.000 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 751 \cdot 552 / 2$$

$$= 0.207 \text{ m}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 2243300 / (2500 \cdot 552)$$

$$= 1625983.000 \text{ N/m}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 0.21 \cdot 1625983$$

$$= 336943.688 \text{ N}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 0.0077 + 0.013$$

$$= 0.021 \text{ m}^2$$

Compressive Stress [Sc]:

$$= P / Ar$$

$$= 336944 / 0.021$$

$$= 16193.812 \text{ kPa}$$

Check of Outside Ribs:



PV Elite 28 Licensee: SL User

FileName : 103-C2

Page 105 of 192

Saddle Calcs: Ext. Pressure: Step: 18 3:14pm Apr 10, 2026

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib	35.0	219.0	144.5	0.0	1107592.5	0.800E+08
Web	375.5	35.0	17.5	0.0	229993.8	0.301E+08
Totals	...	...	...	...	1337586.2	0.110E+09

Rib dimension [D]:

= Saddle Width - Web Thickness

= 254 - 35

= 219.000 mm

Distance to Centroid from Datum [ytot]:

= AY / A

= 1337586/0.021

= 64.284 mm

Distance to Centroid [C1]:

= max(ytot, Saddle Width - ytot)

= max(64.3, 254 - 64.3)

= 189.716 mm

Radius of Gyration [r]:

= $\sqrt{\text{Total Inertia} / \text{Total Area}}$

= $\sqrt{110063424/0.021}$

= 72.730 mm

Length of Outer Rib [L]:

= Saddle Height - cos(theta/2)(radius + shlthk + wpdthk) - bpthk

= 2300 - cos(180/2)(1330 + 142 + 30) - 100

= 2200.000 mm

Intermediate Term [Cc]:

= $\sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$

= $\sqrt{2 \cdot \pi^2 \cdot 199955/194672}$

= 142.386

Slenderness ratio [KL/r]:

= KL/r

= 1 • 2200/72.7

= 30.249

Bending Moment [Rm]:

= FI / (2 • Bplen) • e • L / 2

= 352826 / (2 • 2500) • 751 • 2200/2

= 58317.621 N-m

Compressive Allowable, KL/r < Cc (30.2 < 142) per AISC E2-1 [Sca]:

= $(1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$

= $(1 - (30.2)^2 / (2 \cdot 142^2)) 194672 /$

$(5/3 + 3 \cdot (30.2) / (8 \cdot 142) - (30.2^3) / (8 \cdot 142^3))$

= 109034 kPa

AISC Unity Check of Outside Ribs (must be ≤ 1)

= Sc/Sca + (Rm • C1 / I) / Sba

= 16194/109034 + (58318 • 190/110063536) / 129781

= 0.923

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
--	---	---	---	---	----	----



PV Elite 28 Licensee: SL User

FileName : 103-C2

Page 106 of 192

Saddle Calcs: Ext. Pressure: Step: 18 3:14pm Apr 10, 2026

Rib	35.0	219.0	144.5	0.0	1107592.5	0.105E+09
Web	751.0	35.0	17.5	0.0	459987.5	0.243E+08
Totals	...	...	...	...	1567580.0	0.129E+09

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 1567580 / 0.034$$

$$= 46.173 \text{ mm}$$

Distance to Centroid [C1]:

$$= \max(ytot, \text{abs}(\text{Saddle Width} - ytot))$$

$$= \max(46.2, \text{abs}(254 - 46.2))$$

$$= 207.827 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$$

$$= 2300 - \cos(1502/2) (100 + 0 + 0) - 0$$

$$= 698.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ (\text{Total Inertia} / \text{Total Area}) }$$

$$= \sqrt{ (129034936 / 0.034) }$$

$$= 61.650 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 698 / 61.7$$

$$= 11.322$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 352826 / (2 \cdot 2500)$$

$$= 70.565 \text{ N/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 70.6 \cdot 751 \cdot 698$$

$$= 37005.176 \text{ N-m}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 37005 / 620878$$

$$= 59578.828 \text{ kPa}$$

Compressive Allowable, $KL/r < C_c$ ($11.3 < 142$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (11.3)^2 / (2 \cdot 142^2)) 194672 /$$

$$(5/3 + 3 \cdot (11.3) / (8 \cdot 142) - (11.3^3) / (8 \cdot 142^3))$$

$$= 114392 \text{ kPa}$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (M_{base,c} \cdot C_{1/I}) / S_{ba}$$

$$= 15495 / 114392 + (37005 \cdot 208 / 129034936) / 129781$$

$$= 0.595$$

Review notes about nozzle loadings and supports in the Warnings report.

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FileName : 103-C2

Page 107 of 192

Nozzle Calcs.: Manhole

Nozl: 11 3:14pm Apr 10,2026

Input, Nozzle Desc: Manhole

From: 10

Pressure for Reinforcement Calculations	P	4.9502	N/mm ²
Temperature for Internal Pressure	Temp	343	°C
Parent Material		SA-387 11	
Parent Allowable Stress at Temperature	Sv	137996.80	kPa
Parent Allowable Stress At Ambient	Sva	137996.80	kPa
Inside Diameter of Hemispherical Head	D	2666.00	mm
Head Finished (Minimum) Thickness	t	28.0000	mm
Head Internal Corrosion Allowance	c	2.0000	mm
Head External Corrosion Allowance	co	0.0000	mm
Distance from Head Centerline	L1	0.0000	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-336 F11 +IT	
Material UNS Number		K11572	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	147991.42	kPa
Allowable Stress At Ambient	Sna	147991.42	kPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		24.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	24.6100	mm
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	75.0000	mm
Height of Beveled Transition	L'	75.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		150.0000	mm
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	396.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		50.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		900	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



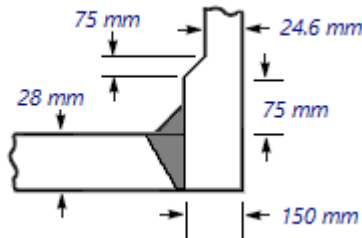
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FileName : 103-C2

Page 108 of 192

Nozzle Calcs.: Manhole

Nozl: 11 3:14pm Apr 10, 2026



Reinforcement CALCULATION, Description: Manhole

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	24.000 in.
Actual Thickness Used in Calculation	0.969 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Spherical Head, tr [Int. Press]
 $= P \cdot R / (2 \cdot S_v \cdot E - 0.2 \cdot P)$ per UG-27 (d)
 $= 4.95 \cdot 1335 / (2 \cdot 137997 \cdot 1 - 0.2 \cdot 4.95)$
 $= 24.0295 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 4.95 \cdot 308 / (147991 \cdot 1 - 0.6 \cdot 4.95)$
 $= 10.5060 \text{ mm}$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 125 / 0.5773$
 $= 217.1819 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1231.2001	mm
Parallel to Vessel Wall, opening length	d	615.6000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	65.0000	mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 147991 / 137997)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 147991 / 137997)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1, 1)$
 $= 1.000$



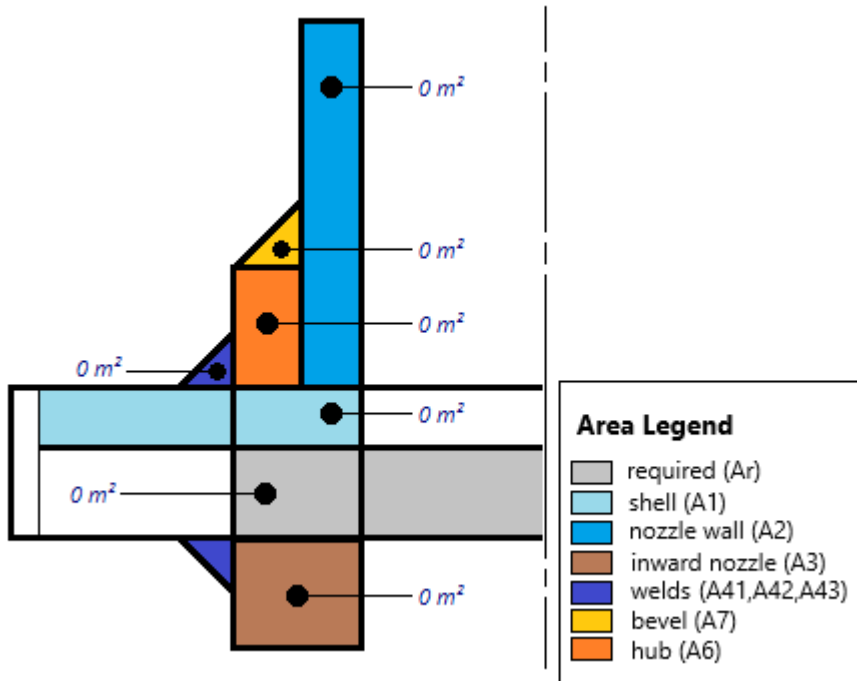
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FileName : 103-C2

Page 109 of 192

Nozzle Calcs.: Manhole

Nozl: 11 3:14pm Apr 10, 2026



Results of Nozzle Reinforcement Area Calculations: (m²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	0.015	NA	NA
Area in Shell	A1	0.001	NA	NA
Area in Nozzle Wall	A2	0.001	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	0.000	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub	A6	0.016	NA	NA
TOTAL AREA AVAILABLE	Atot	0.019	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (616 \cdot 24 \cdot 1 + 2 \cdot 147 \cdot 24 \cdot 1 \cdot (1 - 1))$$

$$= 0.015 \text{ m}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 616(1 \cdot 26 - 1 \cdot 24) - 2 \cdot 147$$

$$(1 \cdot 26 - 1 \cdot 24) \cdot (1 - 1)$$

$$= 0.001 \text{ m}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tnp)(tn - trn)fr2$$

$$= (2 \cdot 65)(21.6 - 10.5)1$$

$$= 0.001 \text{ m}^2$$



PV Elite 28 Licensee: SL User

FileName : 103-C2

Page 110 of 192

Nozzle Calcs.: Manhole

Nozl: 11 3:14pm Apr 10, 2026

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot fr2 + (W_i - can / 0.707)^2 \cdot fr2$$

$$= 10^2 \cdot 1 + (0)^2 \cdot 1$$

$$= 0.000 \text{ m}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lnp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot fr2$$

$$= (2 \cdot \min(65, 396, 75)) \cdot (150 - 24.6) \cdot 1$$

$$= 0.016 \text{ m}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 13.5060 mm
Wall Thickness per UG16(b),	tr16b = 5.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 26.0295 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 26.0295 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 5.5000 mm
Wall Thickness per table UG-45	tb3 = 11.3312 mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[11.3, \max(26, 5.5)]$$

$$= 11.3312 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(13.5, 11.3)$$

$$= 13.5060 \text{ mm}$$

Available Nozzle Neck Thickness = 24.6100 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

Govrn. thk, tg = 24.6, tr = 10.5, c = 3 mm, E* = 1

Thickness Ratio = tr * E* / (tg - c) = 0.49, Temp. Reduction = 34 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	18 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-16 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: A

Govrn. thk, tg = 28, tr = 24, c = 2 mm, E* = 1

Thickness Ratio = tr * E* / (tg - c) = 0.92, Temp. Reduction = 4 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	22 °C
Min Metal Temp. at Required thickness (UCS 66.1)	18 °C

Gov. MDMT of the nozzle to shell joint welded assembly : 18 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :

$$\text{Design Pressure/Ambient Rating} = 4.95/15.31 = 0.323$$

Weld Size Calculations, Description: Manhole

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

Required Thickness	Actual Thickness
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FileName : 103-C2

Page 111 of 192

Nozzle Calcs.: Manhole

Nozl: 11 3:14pm Apr 10, 2026

Nozzle Weld 6.0000 = Min per Code 7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A - A1 + 2 \cdot tn \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv)$$

$$= \max(0, (0.015 - 0.0012 + 2 \cdot 147 \cdot 1 \cdot (1 \cdot 26 - 24)) \cdot 137997)$$

$$= 1953816.75 \text{ N}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2 + A4 - (Wi - Can / .707)^2 \cdot fr2) \cdot Sv$$

$$= (0.018 + 0.99975E-04 - 0 \cdot 1) \cdot 137997$$

$$= 2462378.75 \text{ N}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$$

$$= (0.018 + 0 + 0.99975E-04 + (0.0076)) \cdot 137997$$

$$= 3517197.50 \text{ N}$$

Weld Load [W3]:

$$= (A2 + A3 + A4 + A5 + (2 \cdot tn \cdot t \cdot fr1)) \cdot S$$

$$= (0.018 + 0 + 0.99975E-04 + 0 + (0.0076)) \cdot 137997$$

$$= 3517197.50 \text{ N}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$$

$$= (3.14/2.0) \cdot 910 \cdot 10 \cdot 0.49 \cdot 137997$$

$$= 966104. \text{ N}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.14 \cdot 381) \cdot (150 - 3) \cdot 0.7 \cdot 147991$$

$$= 18241346. \text{ N}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$$

$$= (3.14/2.0) \cdot 910 \cdot (28 - 2) \cdot 0.74 \cdot 147991$$

$$= 4068181. \text{ N}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (966104 + 18241346) = 19207450 \text{ N}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (966104 + 0 + 4068181 + 0) = 5034284 \text{ N}$$

Failure Path [Path 3-3]:

$$= (Sonw + Tngw + Sinw)$$

$$= (966104 + 4068181 + 0) = 5034284 \text{ N}$$

Summary of Failure Path Calculations:

Path 1-1 = 19207450 N	, must exceed W = 1953816 N	or W1 = 2462378 N
Path 2-2 = 5034284 N	, must exceed W = 1953816 N	or W2 = 3517197 N
Path 3-3 = 5034284 N	, must exceed W = 1953816 N	or W3 = 3517197 N



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FileName : 103-C2

Page 112 of 192

Nozzle Calcs.: Manhole

Nozl: 11 3:14pm Apr 10, 2026

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 4.95 N/mm²

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 79.9852 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 503.9853 mm

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FileName : 103-C2

Page 113 of 192

Nozzle Calcs.: N1- Tube inlet

Nozl: 12 3:14pm Apr 10,2026

Input, Nozzle Desc: N1- Tube inlet

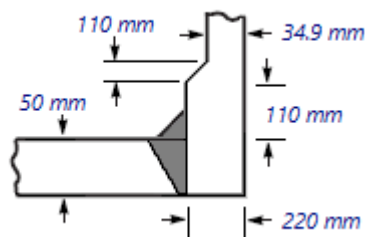
From: 20

Pressure for Reinforcement Calculations	P	4.9500	N/mm ²
Temperature for Internal Pressure	Temp	343	°C
Parent Material		SA-387 11	+IT
Parent Allowable Stress at Temperature	Sv	147988.41	kPa
Parent Allowable Stress At Ambient	Sva	147988.41	kPa
Inside Diameter of Cylindrical Shell	D	2666.00	mm
Shell Finished (Minimum) Thickness	t	50.0000	mm
Shell Internal Corrosion Allowance	c	2.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		1.10	m
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-336 F11	+IT
Material UNS Number		K11572	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	147988.41	kPa
Allowable Stress At Ambient	Sna	147988.41	kPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		36.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	34.9240	mm
Hub Height of Integral Nozzle	h	110.0000	mm
Height of Beveled Transition	L'	110.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		220.0000	mm
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	432.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	15.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	50.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: N1- Tube inlet



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FileName : 103-C2

Page 114 of 192

Nozzle Calcs.: N1- Tube inlet Nozl: 12 3:14pm Apr 10,2026

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 36.000 in.
Actual Thickness Used in Calculation 1.375 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $4.95 \cdot 1335 / (147988 \cdot 1 - 0.6 \cdot 4.95)$
= 45.5656 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $4.95 \cdot 460 / (147988 \cdot 1 - 0.6 \cdot 4.95)$
= 15.7073 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
= (Hub Thickness - Neck Thickness) / $\tan(30)$
= $185 / 0.5773$
= 320.5612 mm

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40 Modified by Appendix 1-7, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1840.8000	mm
Parallel to Vessel Wall, opening length	d	920.4000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	120.0000	mm

Weld Strength Reduction Factor [fr1]:
= $\min(1, S_n / S_v)$
= $\min(1, 147988 / 147988)$
= 1.000

Weld Strength Reduction Factor [fr2]:
= $\min(1, S_n / S_v)$
= $\min(1, 147988 / 147988)$
= 1.000

Weld Strength Reduction Factor [fr3]:
= $\min(fr2, fr4)$
= $\min(1, 1)$
= 1.000



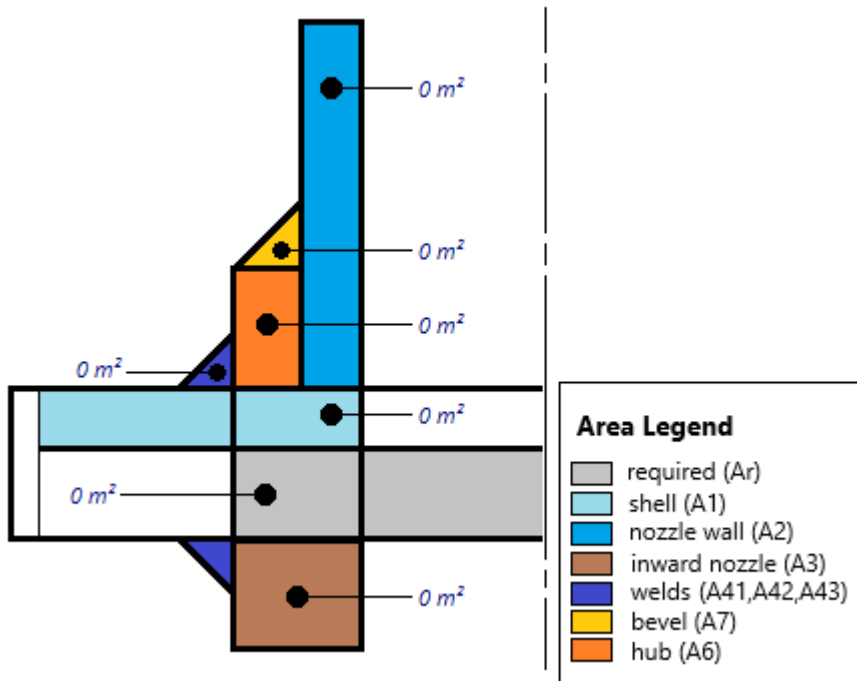
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FileName : 103-C2

Page 115 of 192

Nozzle Calcs.: N1- Tube inlet

Nozl: 12 3:14pm Apr 10, 2026



Results of Nozzle Reinforcement Area Calculations: (m²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	0.042	NA	NA
Area in Shell	A1	0.002	NA	NA
Area in Nozzle Wall	A2	0.004	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	0.000	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub + Bevel	A6	0.044	NA	NA
TOTAL AREA AVAILABLE	Atot	0.051	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (920 \cdot 45.6 \cdot 1 + 2 \cdot 217 \cdot 45.6 \cdot 1 \cdot (1 - 1))$$

$$= 0.042 \text{ m}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 920(1 \cdot 48 - 1 \cdot 45.6) - 2 \cdot 217$$

$$(1 \cdot 48 - 1 \cdot 45.6) \cdot (1 - 1)$$

$$= 0.002 \text{ m}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tnp)(tn - trn)fr2$$

$$= (2 \cdot 120)(31.9 - 15.7)1$$

$$= 0.004 \text{ m}^2$$



PV Elite 28 Licensee: SL User

FileName : 103-C2

Page 116 of 192

Nozzle Calcs.: N1- Tube inlet Nozl: 12 3:14pm Apr 10, 2026

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot fr_2 + (W_i - can / 0.707)^2 \cdot fr_2$$

$$= 15^2 \cdot 1 + (0)^2 \cdot 1$$

$$= 0.000 \text{ m}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lnp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot fr_2$$

$$= (2 \cdot \min(120, 432, 110)) \cdot (220 - 34.9) \cdot 1$$

$$= 0.044 \text{ m}^2 \text{ Includes Bevel Area of } 0.004 \text{ m}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 18.7073 mm
Wall Thickness per UG16(b),	tr16b = 5.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 47.5656 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 47.5656 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 5.5000 mm
Wall Thickness per table UG-45	tb3 = 11.3312 mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[11.3, \max(47.6, 5.5)]$$

$$= 11.3312 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(18.7, 11.3)$$

$$= 18.7073 \text{ mm}$$

Available Nozzle Neck Thickness = 34.9240 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 46492.7,	Allowable	: 147988.4 kPa	Passed
Expansion	: 0.0,	Allowable	: 323478.2 kPa	Passed
Occasional	: 34480.3,	Allowable	: 196824.6 kPa	Passed
Shear	: 7924.3,	Allowable	: 103591.9 kPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Gov. MDMT of the nozzle to shell joint welded assembly : -15 °C

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:

$$= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$$

$$= 185 / 0.5773$$

$$= 320.5612 \text{ mm}$$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40 Modified by Appendix 1-7, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1380.6000 mm
Parallel to Vessel Wall, opening length	d	690.3000 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	120.0000 mm



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FileName : 103-C2

Page 117 of 192

Nozzle Calcs.: N1- Tube inlet Nozl: 12 3:14pm Apr 10,2026

Reinforcement Areas for Large Nozzle (Per App 1-7(a)): (m²)

AREA AVAILABLE, A1 to A6		No Pad	With Pad
Area Required [Internal Pressure] AR		0.028	0.028
Area Available in Shell A1		0.001	0.001
Area Available in Nozzle Wall A2		0.004	0.004
Area Available in Inward Nozzle A3		0.000	0.000
Area Available in Welds A4		0.000	0.000
Area Available in Pad A5		0.000	0.000
Area Available in Hub A6		0.044	0.044
TOTAL AREA AVAILABLE Atot		0.049	0.049

The area available without a pad is Sufficient.

The Nozzle does not fall within the scope of the diameter requirements of App. 1-7(b). Therefore, the Membrane and Bending stress checks are not required.

Weld Size Calculations, Description: N1- Tube inlet

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

Required Thickness Actual Thickness
Nozzle Weld 6.0000 = Min per Code 10.6050 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$$

$$= \max(0, (0.042 - 0.0022 + 2 \cdot 217 \cdot 1 \cdot (1 \cdot 48 - 45.6)) \cdot 147988)$$

$$= 6031034.00 \text{ N}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2+A4-(W_i - \text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$$

$$= (0.048 + 0.00022 - 0 \cdot 1) \cdot 147988$$

$$= 7157562.50 \text{ N}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$$

$$= (0.048 + 0 + 0.00022 + (0.021)) \cdot 147988$$

$$= 10240373.00 \text{ N}$$

Weld Load [W3]:

$$= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$$

$$= (0.048 + 0 + 0.00022 + 0 + (0.021)) \cdot 147988$$

$$= 10240373.00 \text{ N}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.14/2.0) \cdot 1354 \cdot 15 \cdot 0.49 \cdot 147988$$

$$= 2314036. \text{ N}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$$

$$= (3.14 \cdot 569) \cdot (220 - 3) \cdot 0.7 \cdot 147988$$



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FileName : 103-C2

Page 118 of 192

Nozzle Calcs.: N1- Tube inlet Nozl: 12 3:14pm Apr 10,2026

= 40161204. N

Tension, Shell Groove Weld [Tngw]:

= $(\pi/2) \cdot D_{lo} \cdot (T - c_{as}) \cdot 0.74 \cdot S_{ng}$

= $(3.14/2.0) \cdot 1354 \cdot (50 - 2) \cdot 0.74 \cdot 147988$

= 11182934. N

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

= (SONW + SNW) = (2314036 + 40161204) = 42475240 N

Failure Path [Path 2-2]:

= (Sonw + Tpgw + Tngw + Sinw)

= (2314036 + 0 + 11182934 + 0) = 13496970 N

Failure Path [Path 3-3]:

= (Sonw + Tngw + Sinw)

= (2314036 + 11182934 + 0) = 13496970 N

Summary of Failure Path Calculations:

Path 1-1 = 42475240N , must exceed W = 6031034 N or W1 = 7157562 N

Path 2-2 = 13496970N , must exceed W = 6031034 N or W2 = 10240373 N

Path 3-3 = 13496970N , must exceed W = 6031034 N or W3 = 10240373 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 4.95 N/mm²

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 184.8322 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 666.8323 mm

This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 920.40 mm

Warning:

Review Appendix 1-7 (b)(3) for notes regarding externally applied loads to large nozzles and U-2(g).

Input Echo, WRC297 Item 1, Description: N1- Tube inlet:

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	2.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	2666.000	mm
Shell Thickness	50.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-387 11	
Vessel Cold S.I. Allowable	Smc 147988.41	kPa
Vessel Hot S.I. Allowable	Smh 147988.41	kPa

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	914.400	mm
Nozzle Thickness	220.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-336 F11	
Nozzle Cold S.I. Allowable	SNmc 147988.41	kPa
Nozzle Hot S.I. Allowable	SNmh 147988.41	kPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.



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FileName : 103-C2

Page 119 of 192

Nozzle Calcs.: N1- Tube inlet

Nozl: 12 3:14pm Apr 10,2026

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUSstained loads.

Design Internal Pressure	Dp	4.95	N/mm ²
Radial Load	P	-94000.02	N
Circumferential Shear	Vc	72000.22	N
Longitudinal Shear	Vl	72000.22	N
Circumferential Moment	Mc	202499.66	N-m
Longitudinal Moment	Ml	202499.66	N-m
Torsional Moment	Mt	243000.12	N-m

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, Dn/Tn (6.156) is < 10
check the Limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 1354.4 + 2 \cdot 1.65 \cdot \sqrt{1359.0(50.0 - 2.0)}$$

$$= 2197.238 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2718.000 mm
Nozzle Outside Diameter	(d) =	1354.400 mm
Vessel Thickness used	(T) =	48.000 mm
Nozzle Thickness used	(t) =	217.000 mm
T / t	=	0.221
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 3.750$$

Nr/P	=	0.051
Mr/P	=	0.087
M0/P	=	0.027
N0/P	=	0.057
MrD/Mc	=	0.257
NrDL/Mc	=	0.090
M0d/Mc	=	0.077
N0DL/Mc	=	0.085
MrD/Ml	=	0.071
NrDL/Ml	=	0.051
M0D/Ml	=	0.021
N0DL/Ml	=	0.062

Vessel Stresses (kPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au	Al	Bu	B1
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Membrane (P)	2323	2323	2323	2323
Outplane Bending (P)	6632	-6632	6632	-6632
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-4019	-4019	4019	4019
Outplane Bending (ML)	-8266	8266	8266	-8266
Normal Pressure Stress	135230	140181	135230	140181



PV Elite 28 Licensee: SL User

FileName : 103-C2

Page 120 of 192

Nozzle Calcs.: N1- Tube inlet Nozl: 12 3:14pm Apr 10,2026

Outplane Stress Summary 131900 140118 156472 131624

Vessel Stresses (kPa):

LONGITUDINAL PLANE
(Stresses parallel to longitudinal plane)

	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
--	----------------------	---------------------	-------------------------	------------------------

Inplane Membrane (P)	2089	2089	2089	2089
Inplane Bending (P)	21256	-21256	21256	-21256
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-3281	-3281	3281	3281
Inplane Bending (ML)	-27799	27799	27799	-27799
Inplane Pressure Stress	67615	67615	67615	67615

Inplane Stress Summary 59879 72965 122041 23931

Vessel Stresses (kPa):

LONGITUDINAL PLANE
(Shear stress normal to longitudinal plane)

	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
--	----------------------	---------------------	-------------------------	------------------------

Outplane Shear (Vc)	703	703	-703	-703
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	1751	1751	1751	1751

Shear Stress Summary 2454 2454 1047 1047

Vessel Stresses (kPa):

LONGITUDINAL PLANE
(Stress Intensities)

	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
--	----------------------	---------------------	-------------------------	------------------------

Two * Max Shear Stress 131983 140201 156500 131631

Vessel Stresses (kPa):

CIRCUMFERENTIAL PLANE
(Stresses Normal to circumferential plane)

	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Outplane Membrane (P)	2323	2323	2323	2323
Outplane Bending (P)	6632	-6632	6632	-6632
Outplane Membrane (Mc)	-5508	-5508	5508	5508
Outplane Bending (Mc)	-30005	30005	30005	-30005
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	67615	67615	67615	67615

Outplane Stress Summary 41057 87802 112085 38809

Vessel Stresses (kPa):

CIRCUMFERENTIAL PLANE
(Stresses parallel to circumferential plane)

	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Inplane Membrane (P)	2089	2089	2089	2089
Inplane Bending (P)	21256	-21256	21256	-21256
Inplane Membrane (Mc)	-5860	-5860	5860	5860
Inplane Bending (Mc)	-100171	100171	100171	-100171
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	135230	140181	135230	140181

Inplane Stress Summary 52543 215325 264607 26702



PV Elite 28 Licensee: SL User

FileName : 103-C2

Page 121 of 192

Nozzle Calcs.: N1- Tube inlet

Nozl: 12 3:14pm Apr 10,2026

Vessel Stresses (kPa):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)		Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)		0	0	0	0
Outplane Shear (Vl)		-703	-703	703	703
Torsional Shear (Mt)		1751	1751	1751	1751
Shear Stress Summary		1047	1047	2454	2454

Vessel Stresses (kPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)		Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress		52633	215332	264642	39285

Nozzle Stresses (kPa):

LONGITUDINAL PLANE (Stresses in the hoop direction)		Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)		2323	2323	2323	2323
Hoop Bending (P)		0	0	0	0
Hoop Membrane (Mc)		0	0	0	0
Hoop Bending (Mc)		0	0	0	0
Hoop Membrane (ML)		-4019	-4019	4019	4019
Hoop Bending (ML)		0	0	0	0
Hoop Pressure Stress		10190	16126	10190	16126
Hoop Stress Summary		8494	14430	16533	22469

Nozzle Stresses (kPa):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)		Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)		117	117	117	117
Axial Bending (P)		730	-730	730	-730
Axial Membrane (Mc)		0	0	0	0
Axial Bending (Mc)		0	0	0	0
Axial Membrane (ML)		-1047	-1047	1047	1047
Axial Bending (ML)		-875	875	875	-875
Axial Pressure Stress		5095	5095	5095	5095
Axial Stress Summary		4019	4309	7866	4653

Nozzle Stresses (kPa):

LONGITUDINAL PLANE (Shear stress)		Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)		151	151	-151	-151
Shear due to (Vl)		0	0	0	0
Shear due to Torsion		386	386	386	386
Shear Stress Summary		537	537	234	234

Nozzle Stresses (kPa):

LONGITUDINAL PLANE (Stress Intensities)		Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress		8556	14457	16533	22469

Nozzle Stresses (kPa):



PV Elite 28 Licensee: SL User

FileName : 103-C2

Page 122 of 192

Nozzle Calcs.: N1- Tube inlet Nozl: 12 3:14pm Apr 10,2026

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	2323	2323	2323	2323
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-5508	-5508	5508	5508
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	10190	16126	10190	16126
Hoop Stress Summary	7004	12941	18022	23958

Nozzle Stresses (kPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	117	117	117	117
Axial Bending (P)	730	-730	730	-730
Axial Membrane (Mc)	-1047	-1047	1047	1047
Axial Bending (Mc)	-4040	4040	4040	-4040
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	5095	5095	5095	5095
Axial Stress Summary	854	7473	11031	1489

Nozzle Stresses (kPa):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-151	-151	151	151
Shear due to Torsion	386	386	386	386
Shear Stress Summary	234	234	537	537

Nozzle Stresses (kPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	7011	12948	18056	23965

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (kPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	135230	140181	135230	140181	135230	140181	135230	140181
Circ. Pl (SUS)	-1696	-1696	6342	6342	-3771	-3771	7949	7949
Circ. Q (SUS)	-1634	1634	14898	-14898	-78915	78915	121427	-121427
Long. Pm (SUS)	67615	67615	67615	67615	67615	67615	67615	67615
Long. Pl (SUS)	-1192	-1192	5370	5370	-3185	-3185	7831	7831
Long. Q (SUS)	-6543	6543	49055	-49055	-23373	23373	36637	-36637
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	703	703	-703	-703	-703	-703	703	703
Shear Q (SUS)	1751	1751	1751	1751	1751	1751	1751	1751
Pm (SUS)	135230	140181	135230	140181	135230	140181	135230	140181



PV Elite 28 Licensee: SL User
 FileName : 103-C2 Page 123 of 192
 Nozzle Calcs.: N1- Tube inlet Nozl: 12 3:14pm Apr 10,2026

Pm+Pl (SUS)	133541	138491	141579	146529	131466	136416	143186	148136
Pm+Pl+Q (Total)	131983	140208	156501	131635	52638	215333	264645	39287

Vessel Stress Summation Comparison (kPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	140181	147988	Passed
Pm+Pl (SUS)	148136	221982	Passed
Pm+Pl+Q (TOTAL)	264645	443965	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (kPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	10190	16126	10190	16126	10190	16126	10190	16126
Circ. Pl (SUS)	-1696	-1696	6342	6342	-3185	-3185	7831	7831
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	5095	5095	5095	5095	5095	5095	5095	5095
Long. Pl (SUS)	-930	-930	1164	1164	-930	-930	1164	1164
Long. Q (SUS)	-145	145	1605	-1605	-3310	3310	4770	-4770
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	151	151	-151	-151	-151	-151	151	151
Shear Q (SUS)	386	386	386	386	386	386	386	386
Pm (SUS)	10190	16126	10190	16126	10190	16126	10190	16126
Pm+Pl (SUS)	8499	14432	16534	22469	7013	12943	18022	23958
Pm+Pl+Q (Total)	8557	14458	16538	22471	7013	12951	18062	23969

Nozzle Stress Summation Comparison (kPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	16126	147988	Passed
Pm+Pl (SUS)	23958	221982	Passed
Pm+Pl+Q (TOTAL)	23969	443965	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.



PV Elite 28 Licensee: SL User

FileName : 103-C2

Page 124 of 192

Nozzle Calcs.: N2- Tube outlet

Nozl: 13 3:14pm Apr 10,2026

Input, Nozzle Desc: N2- Tube outlet

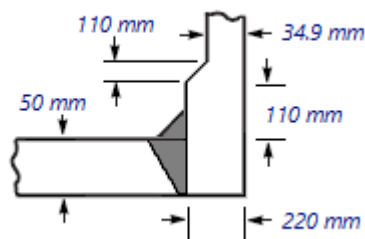
From: 20

Pressure for Reinforcement Calculations	P	4.9505	N/mm ²
Temperature for Internal Pressure	Temp	343	°C
Parent Material		SA-387 11	+IT
Parent Allowable Stress at Temperature	Sv	147988.41	kPa
Parent Allowable Stress At Ambient	Sva	147988.41	kPa
Inside Diameter of Cylindrical Shell	D	2666.00	mm
Shell Finished (Minimum) Thickness	t	50.0000	mm
Shell Internal Corrosion Allowance	c	2.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		1.10	m
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-336 F11	+IT
Material UNS Number		K11572	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	147988.41	kPa
Allowable Stress At Ambient	Sna	147988.41	kPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		36.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	34.9250	mm
Hub Height of Integral Nozzle	h	110.0000	mm
Height of Beveled Transition	L'	110.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		220.0000	mm
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	432.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	15.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	50.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: N2- Tube outlet



PV Elite 28 Licensee: SL User

FileName : 103-C2

Page 125 of 192

Nozzle Calcs.: N2- Tube outlet Nozl: 13 3:14pm Apr 10,2026

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 36.000 in.
Actual Thickness Used in Calculation 1.375 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $4.95 \cdot 1335 / (147988 \cdot 1 - 0.6 \cdot 4.95)$
= 45.5702 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $4.95 \cdot 460 / (147988 \cdot 1 - 0.6 \cdot 4.95)$
= 15.7089 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
= (Hub Thickness - Neck Thickness) / $\tan(30)$
= $185 / 0.5773$
= 320.5594 mm

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40 Modified by Appendix 1-7, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1840.8000	mm
Parallel to Vessel Wall, opening length	d	920.4000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	120.0000	mm

Weld Strength Reduction Factor [fr1]:
= $\min(1, S_n / S_v)$
= $\min(1, 147988 / 147988)$
= 1.000

Weld Strength Reduction Factor [fr2]:
= $\min(1, S_n / S_v)$
= $\min(1, 147988 / 147988)$
= 1.000

Weld Strength Reduction Factor [fr3]:
= $\min(fr2, fr4)$
= $\min(1, 1)$
= 1.000



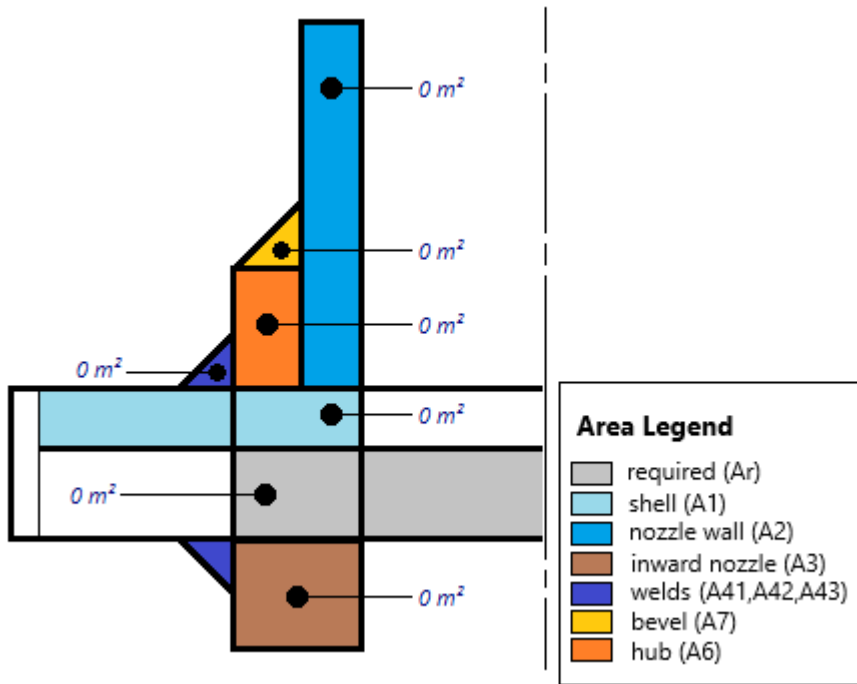
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FileName : 103-C2

Page 126 of 192

Nozzle Calcs.: N2- Tube outlet

Nozl: 13 3:14pm Apr 10, 2026



Results of Nozzle Reinforcement Area Calculations: (m²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	0.042	NA	NA
Area in Shell	A1	0.002	NA	NA
Area in Nozzle Wall	A2	0.004	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	0.000	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub + Bevel	A6	0.044	NA	NA
TOTAL AREA AVAILABLE	Atot	0.051	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (920 \cdot 45.6 \cdot 1 + 2 \cdot 217 \cdot 45.6 \cdot 1 \cdot (1 - 1))$$

$$= 0.042 \text{ m}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 920(1 \cdot 48 - 1 \cdot 45.6) - 2 \cdot 217$$

$$(1 \cdot 48 - 1 \cdot 45.6) \cdot (1 - 1)$$

$$= 0.002 \text{ m}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tnp)(tn - trn)fr2$$

$$= (2 \cdot 120)(31.9 - 15.7)1$$

$$= 0.004 \text{ m}^2$$



PV Elite 28 Licensee: SL User

FileName : 103-C2

Page 127 of 192

Nozzle Calcs.: N2- Tube outlet Nozl: 13 3:14pm Apr 10, 2026

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot fr_2 + (W_i - can / 0.707)^2 \cdot fr_2$$

$$= 15^2 \cdot 1 + (0)^2 \cdot 1$$

$$= 0.000 \text{ m}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lnp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot fr_2$$

$$= (2 \cdot \min(120, 432, 110)) \cdot (220 - 34.9) \cdot 1$$

$$= 0.044 \text{ m}^2 \text{ Includes Bevel Area of } 0.004 \text{ m}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 18.7089 mm
Wall Thickness per UG16(b),	tr16b = 5.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 47.5702 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 47.5702 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 5.5000 mm
Wall Thickness per table UG-45	tb3 = 11.3312 mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[11.3, \max(47.6, 5.5)]$$

$$= 11.3312 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(18.7, 11.3)$$

$$= 18.7089 \text{ mm}$$

Available Nozzle Neck Thickness = 34.9250 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 33511.4,	Allowable	: 147988.4 kPa	Passed
Expansion	: 0.0,	Allowable	: 336459.6 kPa	Passed
Occasional	: 34482.6,	Allowable	: 196824.6 kPa	Passed
Shear	: 7924.1,	Allowable	: 103591.9 kPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Gov. MDMT of the nozzle to shell joint welded assembly : -15 °C

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:

$$= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$$

$$= 185 / 0.5773$$

$$= 320.5594 \text{ mm}$$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40 Modified by Appendix 1-7, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1380.6000 mm
Parallel to Vessel Wall, opening length	d	690.3000 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	120.0000 mm



PV Elite 28 Licensee: SL User

FileName : 103-C2

Page 128 of 192

Nozzle Calcs.: N2- Tube outlet Nozl: 13 3:14pm Apr 10,2026

Reinforcement Areas for Large Nozzle (Per App 1-7(a)): (m²)

AREA AVAILABLE, A1 to A6		No Pad	With Pad
Area Required [Internal Pressure] AR		0.028	0.028
Area Available in Shell A1		0.001	0.001
Area Available in Nozzle Wall A2		0.004	0.004
Area Available in Inward Nozzle A3		0.000	0.000
Area Available in Welds A4		0.000	0.000
Area Available in Pad A5		0.000	0.000
Area Available in Hub A6		0.044	0.044
TOTAL AREA AVAILABLE Atot		0.049	0.049

The area available without a pad is Sufficient.

The Nozzle does not fall within the scope of the diameter requirements of App. 1-7(b). Therefore, the Membrane and Bending stress checks are not required.

Weld Size Calculations, Description: N2- Tube outlet

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

Required Thickness Actual Thickness
Nozzle Weld 6.0000 = Min per Code 10.6050 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$$

$$= \max(0, (0.042 - 0.0022 + 2 \cdot 217 \cdot 1 \cdot (1 \cdot 48 - 45.6)) \cdot 147988)$$

$$= 6031981.00 \text{ N}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2+A4-(W_i - \text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$$

$$= (0.048 + 0.00022 - 0 \cdot 1) \cdot 147988$$

$$= 7157506.00 \text{ N}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$$

$$= (0.048 + 0 + 0.00022 + (0.021)) \cdot 147988$$

$$= 10240316.00 \text{ N}$$

Weld Load [W3]:

$$= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$$

$$= (0.048 + 0 + 0.00022 + 0 + (0.021)) \cdot 147988$$

$$= 10240316.00 \text{ N}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.14/2.0) \cdot 1354 \cdot 15 \cdot 0.49 \cdot 147988$$

$$= 2314036. \text{ N}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$$

$$= (3.14 \cdot 569) \cdot (220 - 3) \cdot 0.7 \cdot 147988$$



PV Elite 28 Licensee: SL User

FileName : 103-C2

Page 129 of 192

Nozzle Calcs.: N2- Tube outlet Nozl: 13 3:14pm Apr 10, 2026

= 40161204. N

Tension, Shell Groove Weld [Tngw]:

= $(\pi/2) \cdot D_{lo} \cdot (T - c_{as}) \cdot 0.74 \cdot S_{ng}$

= $(3.14/2.0) \cdot 1354 \cdot (50 - 2) \cdot 0.74 \cdot 147988$

= 11182934. N

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

= (SONW + SNW) = (2314036 + 40161204) = 42475240 N

Failure Path [Path 2-2]:

= (Sonw + Tpgw + Tngw + Sinw)

= (2314036 + 0 + 11182934 + 0) = 13496970 N

Failure Path [Path 3-3]:

= (Sonw + Tngw + Sinw)

= (2314036 + 11182934 + 0) = 13496970 N

Summary of Failure Path Calculations:

Path 1-1 = 42475240N , must exceed W = 6031981 N or W1 = 7157506 N

Path 2-2 = 13496970N , must exceed W = 6031981 N or W2 = 10240316 N

Path 3-3 = 13496970N , must exceed W = 6031981 N or W3 = 10240316 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 4.95 N/mm²

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 184.8322 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 666.8323 mm

This nozzle or its diameter limit intersects weld seam # [2] on this element. The distance between the nozzle and seam is, 862.09 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 920.40 mm

Warning:

Review Appendix 1-7 (b)(3) for notes regarding externally applied loads to large nozzles and U-2(g).

Input Echo, WRC297 Item 1, Description: N2- Tube outle:

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	2.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	2666.000	mm
Shell Thickness	50.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-387 11	
Vessel Cold S.I. Allowable	Smc 147988.41	kPa
Vessel Hot S.I. Allowable	Smh 147988.41	kPa

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	914.400	mm
Nozzle Thickness	220.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-336 F11	
Nozzle Cold S.I. Allowable	SNmc 147988.41	kPa
Nozzle Hot S.I. Allowable	SNmh 147988.41	kPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.



PV Elite 28 Licensee: SL User

FileName : 103-C2

Page 130 of 192

Nozzle Calcs.: N2- Tube outlet Nozl: 13 3:14pm Apr 10,2026

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	4.95	N/mm ²
Radial Load	P	-94000.02	N
Circumferential Shear	Vc	72000.22	N
Longitudinal Shear	Vl	72000.22	N
Circumferential Moment	Mc	202.50	N-m
Longitudinal Moment	Ml	202.50	N-m
Torsional Moment	Mt	243000.12	N-m

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, Dn/Tn (6.156) is < 10
check the Limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 1354.4 + 2 \cdot 1.65 \cdot \sqrt{1359.0(50.0 - 2.0)}$$

$$= 2197.238 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2718.000 mm
Nozzle Outside Diameter	(d) =	1354.400 mm
Vessel Thickness used	(T) =	48.000 mm
Nozzle Thickness used	(t) =	217.000 mm
T / t	=	0.221
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 3.750$$

Nr/P	=	0.051
Mr/P	=	0.087
M0/P	=	0.027
N0/P	=	0.057
MrD/Mc	=	0.257
NrDL/Mc	=	0.090
M0d/Mc	=	0.077
N0DL/Mc	=	0.085
MrD/Ml	=	0.071
NrDL/Ml	=	0.051
M0D/Ml	=	0.021
N0DL/Ml	=	0.062

Vessel Stresses (kPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au	Al	Bu	B1
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Membrane (P)	2323	2323	2323	2323
Outplane Bending (P)	6632	-6632	6632	-6632
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (Ml)	0	0	0	0
Outplane Bending (Ml)	-6	6	6	-6
Normal Pressure Stress	135244	140194	135244	140194



PV Elite 28 Licensee: SL User

FileName : 103-C2

Page 131 of 192

Nozzle Calcs.: N2- Tube outlet Nozl: 13 3:14pm Apr 10,2026

Outplane Stress Summary 144193 135892 144207 135878

Vessel Stresses (kPa):

LONGITUDINAL PLANE
(Stresses parallel to longitudinal plane)

	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
--	----------------------	---------------------	-------------------------	------------------------

Inplane Membrane (P)	2089	2089	2089	2089
Inplane Bending (P)	21256	-21256	21256	-21256
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	-27	27	27	-27
Inplane Pressure Stress	67622	67622	67622	67622

Inplane Stress Summary 90939 48482 90994 48427

Vessel Stresses (kPa):

LONGITUDINAL PLANE
(Shear stress normal to longitudinal plane)

	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
--	----------------------	---------------------	-------------------------	------------------------

Outplane Shear (Vc)	703	703	-703	-703
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	1751	1751	1751	1751

Shear Stress Summary 2454 2454 1047 1047

Vessel Stresses (kPa):

LONGITUDINAL PLANE
(Stress Intensities)

	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
--	----------------------	---------------------	-------------------------	------------------------

Two * Max Shear Stress 144303 135954 144221 135885

Vessel Stresses (kPa):

CIRCUMFERENTIAL PLANE
(Stresses Normal to circumferential plane)

	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Outplane Membrane (P)	2323	2323	2323	2323
Outplane Bending (P)	6632	-6632	6632	-6632
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	-27	27	27	-27
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	67622	67622	67622	67622

Outplane Stress Summary 76550 63340 76605 63285

Vessel Stresses (kPa):

CIRCUMFERENTIAL PLANE
(Stresses parallel to circumferential plane)

	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Inplane Membrane (P)	2089	2089	2089	2089
Inplane Bending (P)	21256	-21256	21256	-21256
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	-96	96	96	-96
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	135244	140194	135244	140194

Inplane Stress Summary 158493 121124 158686 120931



PV Elite 28 Licensee: SL User

FileName : 103-C2

Page 132 of 192

Nozzle Calcs.: N2- Tube outlet Nozl: 13 3:14pm Apr 10,2026

Vessel Stresses (kPa):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)		Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)		0	0	0	0
Outplane Shear (Vl)		-703	-703	703	703
Torsional Shear (Mt)		1751	1751	1751	1751
Shear Stress Summary		1047	1047	2454	2454

Vessel Stresses (kPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)		Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress		158499	121138	158755	121034

Nozzle Stresses (kPa):

LONGITUDINAL PLANE (Stresses in the hoop direction)		Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)		2323	2323	2323	2323
Hoop Bending (P)		0	0	0	0
Hoop Membrane (Mc)		0	0	0	0
Hoop Bending (Mc)		0	0	0	0
Hoop Membrane (ML)		0	0	0	0
Hoop Bending (ML)		0	0	0	0
Hoop Pressure Stress		10190	16133	10190	16133
Hoop Stress Summary		12513	18456	12513	18456

Nozzle Stresses (kPa):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)		Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)		117	117	117	117
Axial Bending (P)		730	-730	730	-730
Axial Membrane (Mc)		0	0	0	0
Axial Bending (Mc)		0	0	0	0
Axial Membrane (ML)		0	0	0	0
Axial Bending (ML)		0	0	0	0
Axial Pressure Stress		5095	5095	5095	5095
Axial Stress Summary		5943	4481	5943	4481

Nozzle Stresses (kPa):

LONGITUDINAL PLANE (Shear stress)		Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)		151	151	-151	-151
Shear due to (Vl)		0	0	0	0
Shear due to Torsion		386	386	386	386
Shear Stress Summary		537	537	234	234

Nozzle Stresses (kPa):

LONGITUDINAL PLANE (Stress Intensities)		Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress		12555	18470	12520	18456

Nozzle Stresses (kPa):



PV Elite 28 Licensee: SL User

FileName : 103-C2

Page 133 of 192

Nozzle Calcs.: N2- Tube outlet Nozl: 13 3:14pm Apr 10,2026

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	2323	2323	2323	2323
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	10190	16133	10190	16133
Hoop Stress Summary	12513	18456	12513	18456

Nozzle Stresses (kPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	117	117	117	117
Axial Bending (P)	730	-730	730	-730
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	5095	5095	5095	5095
Axial Stress Summary	5943	4481	5943	4481

Nozzle Stresses (kPa):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-151	-151	151	151
Shear due to Torsion	386	386	386	386
Shear Stress Summary	234	234	537	537

Nozzle Stresses (kPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	12520	18456	12555	18470

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (kPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	135244	140194	135244	140194	135244	140194	135244	140194
Circ. Pl (SUS)	2323	2323	2323	2323	2089	2089	2089	2089
Circ. Q (SUS)	6626	-6626	6638	-6638	21160	-21160	21352	-21352
Long. Pm (SUS)	67622	67622	67622	67622	67622	67622	67622	67622
Long. Pl (SUS)	2089	2089	2089	2089	2323	2323	2323	2323
Long. Q (SUS)	21229	-21229	21283	-21283	6605	-6605	6659	-6659
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	703	703	-703	-703	-703	-703	703	703
Shear Q (SUS)	1751	1751	1751	1751	1751	1751	1751	1751
Pm (SUS)	135244	140194	135244	140194	135244	140194	135244	140194



PV Elite 28 Licensee: SL User
 FileName : 103-C2 Page 134 of 192
 Nozzle Calcs.: N2- Tube outlet Nozl: 13 3:14pm Apr 10,2026

Pm+Pl (SUS)	137574	142523	137574	142523	137340	142289	137340	142289
Pm+Pl+Q (Total)	144305	135959	144225	135891	158506	121142	158758	121035

Vessel Stress Summation Comparison (kPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	140194	147988	Passed
Pm+Pl (SUS)	142523	221982	Passed
Pm+Pl+Q (TOTAL)	158758	443965	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (kPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	10190	16133	10190	16133	10190	16133	10190	16133
Circ. Pl (SUS)	2323	2323	2323	2323	2323	2323	2323	2323
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	5095	5095	5095	5095	5095	5095	5095	5095
Long. Pl (SUS)	117	117	117	117	117	117	117	117
Long. Q (SUS)	730	-730	730	-730	730	-730	730	-730
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	151	151	-151	-151	-151	-151	151	151
Shear Q (SUS)	386	386	386	386	386	386	386	386
Pm (SUS)	10190	16133	10190	16133	10190	16133	10190	16133
Pm+Pl (SUS)	12516	18457	12516	18457	12516	18457	12516	18457
Pm+Pl+Q (Total)	12556	18476	12521	18459	12521	18459	12556	18476

Nozzle Stress Summation Comparison (kPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	16133	147988	Passed
Pm+Pl (SUS)	18457	221982	Passed
Pm+Pl+Q (TOTAL)	18476	443965	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.



PV Elite 28 Licensee: SL User

FileName : 103-C2

Page 135 of 192

Nozzle Calcs.: Shell outlet-1

Nozl: 14 3:14pm Apr 10,2026

Input, Nozzle Desc: Shell outlet-1

From: 40

Pressure for Reinforcement Calculations	P	14.2800	N/mm ²
Temperature for Internal Pressure	Temp	343	°C
Design External Pressure	Pext	0.10	N/mm ²
Temperature for External Pressure	Tempex	200	°C

Parent Material		SA-387 11	
Parent Allowable Stress at Temperature	Sv	147988.73	kPa
Parent Allowable Stress At Ambient	Sva	147988.73	kPa

Inside Diameter of Cylindrical Shell	D	2660.00	mm
Design Length of Section	L	6243.3643	mm
Shell Finished (Minimum) Thickness	t	142.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		4.20	m
-----------------------------------	--	------	---

User Entered Minimum Design Metal Temperature		-12.00	°C
---	--	--------	----

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	122834.11	kPa
Allowable Stress At Ambient	Sna	137989.50	kPa

Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		10.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	28.5750	mm

Hub Height of Integral Nozzle	h	120.0000	mm
Height of Beveled Transition	L'	80.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		160.0000	mm

Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	400.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	145.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
ASME Code Weld Type per UW-16			g

The Pressure Design option was MAWP + static head (to the nozzle).



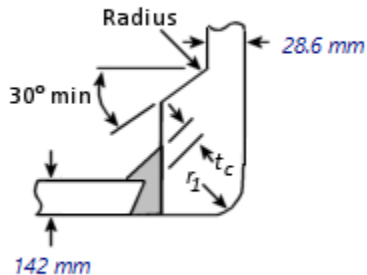
PV Elite 28 Licensee: SL User

FileName : 103-C2

Page 136 of 192

Nozzle Calcs.: Shell outlet-1

Nozl: 14 3:14pm Apr 10, 2026



UW-16.1 [g]

Reinforcement CALCULATION, Description: Shell outlet-1

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	10.000 in.
Actual Thickness Used in Calculation	1.125 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, t_r [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 14.3 \cdot 1333 / (147989 \cdot 1 - 0.6 \cdot 14.3)$
 $= 136.5228 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, t_{rn} [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 14.3 \cdot 127 / (122834 \cdot 1 + 0.4 \cdot 14.3)$
 $= 14.1075 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.9894 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 131 / 0.5773$
 $= 227.6349 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	532.0000 mm
Parallel to Vessel Wall	Rn+tn+t	266.0000 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	263.9375 mm

Weld Strength Reduction Factor [fr1]:

$= \min(1, S_n / S_v)$
 $= \min(1, 122834 / 147989)$
 $= 0.830$

Weld Strength Reduction Factor [fr2]:

$= \min(1, S_n / S_v)$
 $= \min(1, 122834 / 147989)$
 $= 0.830$



PV Elite 28 Licensee: SL User

FileName : 103-C2

Page 137 of 192

Nozzle Calcs.: Shell outlet-1

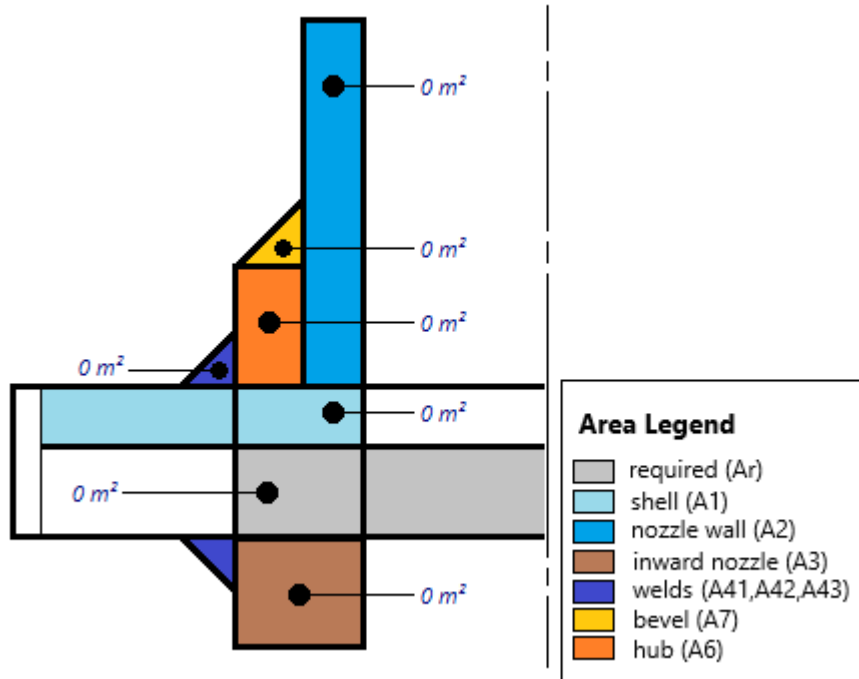
Nozl: 14 3:14pm Apr 10,2026

Weld Strength Reduction Factor [fr3]:

= min(fr2, fr4)

= min(0.83, 1)

= 0.830



Results of Nozzle Reinforcement Area Calculations: (m²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	0.035	0.002	NA
Area in Shell	A1	0.001	0.035	NA
Area in Nozzle Wall	A2	0.005	0.011	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	0.000	0.000	NA
Area in Element	A5	0.000	0.000	NA
Area in Hub	A6	0.035	0.035	NA
TOTAL AREA AVAILABLE	Atot	0.041	0.081	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

= (d • tr • F + 2 • tn • tr • F • (1-fr1)) UG-37(c)

= (203 • 137 • 1 + 2 • 157 • 137 • 1 • (1-0.83))

= 0.035 m²

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

= d(E1 • t - F • tr) - 2 • tn(E1 • t - F • tr) • (1 - fr1)

= 329 (1 • 139 - 1 • 137) - 2 • 157

(1 • 139 - 1 • 137) • (1 - 0.83)

= 0.001 m²



PV Elite 28 Licensee: SL User

FileName : 103-C2

Page 138 of 192

Nozzle Calcs.: Shell outlet-1 Nozl: 14 3:14pm Apr 10,2026

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot t_{np}) (t_n - t_{rn}) \cdot fr^2$$

$$= (2 \cdot 264) (25.6 - 14.1) \cdot 0.83$$

$$= 0.005 \text{ m}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= (W_o^2 - \text{Area Lost}) \cdot fr^2 + (W_i - \text{can}/0.707)^2 - \text{Area Lost}) \cdot fr^2$$

$$= (10^2 - 0.58792\text{E-}05) \cdot 0.83 + (0^2 - 0) \cdot 0.83$$

$$= 0.000 \text{ m}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{np}, h_o, \text{Hubht})) \cdot (\text{Hubtk} - t_n) \cdot fr^2$$

$$= (2 \cdot \min(264, 400, 120)) \cdot (160 - 28.6) \cdot 0.83$$

$$= 0.035 \text{ m}^2 \text{ Includes Bevel Area of } 0.009 \text{ m}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 17.1075 mm
Wall Thickness per UG16(b),	tr16b = 5.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 139.5228 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 139.5228 mm
Wall Thickness, shell/head, external pressure	trb2 = 3.9128 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 5.5000 mm
Wall Thickness per table UG-45	tb3 = 11.1026 mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[11.1, \max(140, 5.5)]$$

$$= 11.1026 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(17.1, 11.1)$$

$$= 17.1075 \text{ mm}$$

Available Nozzle Neck Thickness = 28.5750 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 51215.3,	Allowable	: 122834.1 kPa	Passed
Expansion	: 0.0,	Allowable	: 274814.2 kPa	Passed
Occasional	: 25144.0,	Allowable	: 163369.4 kPa	Passed
Shear	: 17566.8,	Allowable	: 85983.9 kPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: A

Warning:

The minimum metal temperature could not be computed because the governing thickness for this component was greater than 100.000 mm. This component must be impact tested.

Weld Size Calculations, Description: Shell outlet-1

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm



PV Elite 28 Licensee: SL User

FileName : 103-C2

Page 139 of 192

Nozzle Calcs.: Shell outlet-1

Nozl: 14 3:14pm Apr 10,2026

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 14.3 N/mm²

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0.1 N/mm²

The Drop for this Nozzle is : 6.0774 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 548.0775 mm

Input Echo, WRC297 Item 1, Description: Shell outlet-1:

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	2660.000	mm
Shell Thickness	142.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-387 11	
Vessel Cold S.I. Allowable	Smc 147988.73	kPa
Vessel Hot S.I. Allowable	Smh 147988.73	kPa
Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	516.850	mm
Nozzle Thickness	160.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-105	
Nozzle Cold S.I. Allowable	SNmc 137989.50	kPa
Nozzle Hot S.I. Allowable	SNmh 122834.11	kPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUSTAINED loads.

Design Internal Pressure	Dp	14.28	N/mm ²
Radial Load	P	-31250.05	N
Circumferential Shear	Vc	24999.98	N
Longitudinal Shear	Vl	24999.98	N
Circumferential Moment	Mc	18750.04	N-m
Longitudinal Moment	Ml	18750.04	N-m
Torsional Moment	Mt	21874.95	N-m

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, Dn/Tv, (3.640) is < 5
check the limitations of WRC bulletin 297.

Warning - The ratio, Dn/Tn (3.230) is < 10
check the limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{\text{Rmean}(t - ca)}$$

$$= 516.85 + 2 \cdot 1.65 \cdot \sqrt{1402.5(142.0 - 3.0)}$$

$$= 1973.894 \text{ mm}$$

WRC 297 Curve Access Parameters:



PV Elite 28 Licensee: SL User

FileName : 103-C2

Page 140 of 192

Nozzle Calcs.: Shell outlet-1 Nozl: 14 3:14pm Apr 10,2026

Vessel Mean Diameter (D) = 2805.000 mm
 Nozzle Outside Diameter (d) = 516.850 mm
 Vessel Thickness used (T) = 139.000 mm
 Nozzle Thickness used (t) = 157.000 mm
 T / t = 0.885
 d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.828$$

Nr/P = 0.135
 Mr/P = 0.221
 M0/P = 0.073
 N0/P = 0.134
 MrD/Mc = 0.291
 NrDL/Mc = 0.132
 M0d/Mc = 0.108
 N0DL/Mc = -0.043
 MrD/Ml = 0.218
 NrDL/Ml = 0.115
 M0D/Ml = 0.082
 N0DL/Ml = 0.093

Vessel Stresses (kPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	213	213	213	213
Outplane Bending (P)	703	-703	703	-703
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-172	-172	172	172
Outplane Bending (ML)	-916	916	916	-916
Normal Pressure Stress	130149	144428	130149	144428
Outplane Stress Summary	129977	144683	132155	143193

Vessel Stresses (kPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	213	213	213	213
Inplane Bending (P)	2144	-2144	2144	-2144
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-213	-213	213	213
Inplane Bending (ML)	-2447	2447	2447	-2447
Inplane Pressure Stress	65071	65071	65071	65071
Inplane Stress Summary	64767	65374	70090	60906

Vessel Stresses (kPa):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	220	220	-220	-220
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	372	372	372	372
Shear Stress Summary	592	592	151	151

Vessel Stresses (kPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
--------------------	----	----	----	----



PV Elite 28 Licensee: SL User

FileName : 103-C2

Page 141 of 192

Nozzle Calcs.: Shell outlet-1 Nozl: 14 3:14pm Apr 10,2026

(Stress Intensities	Top Outside	Top Inside	Bottom Outside	Bottom Inside
---------------------	----------------	---------------	-------------------	------------------

Two * Max Shear Stress	129977	144683	132155	143193
------------------------	--------	--------	--------	--------

Vessel Stresses (kPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Outplane Membrane (P)	213	213	213	213
Outplane Bending (P)	703	-703	703	-703
Outplane Membrane (Mc)	75	75	-75	-75
Outplane Bending (Mc)	-1213	1213	1213	-1213
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	65071	65071	65071	65071

Outplane Stress Summary	64850	65871	67125	63292
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Vessel Stresses (kPa):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Inplane Membrane (P)	213	213	213	213
Inplane Bending (P)	2144	-2144	2144	-2144
Inplane Membrane (Mc)	-248	-248	248	248
Inplane Bending (Mc)	-3268	3268	3268	-3268
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	130149	144428	130149	144428

Inplane Stress Summary	128991	145517	136023	139477
------------------------	--------	--------	--------	--------

Vessel Stresses (kPa):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-220	-220	220	220
Torsional Shear (Mt)	372	372	372	372

Shear Stress Summary	151	151	592	592
----------------------	-----	-----	-----	-----

Vessel Stresses (kPa):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Two * Max Shear Stress	128991	145517	136023	139477
------------------------	--------	--------	--------	--------

Nozzle Stresses (kPa):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
---	----------------------	---------------------	-------------------------	------------------------

Hoop Membrane (P)	213	213	213	213
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-172	-172	172	172
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	6239	23372	6239	23372



PV Elite 28 Licensee: SL User

FileName : 103-C2

Page 142 of 192

Nozzle Calcs.: Shell outlet-1

Nozl: 14 3:14pm Apr 10,2026

Hoop Stress Summary	6280	23414	6625	23758
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Nozzle Stresses (kPa):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
--	----------------------	---------------------	-------------------------	------------------------

Axial Membrane (P)	172	172	172	172
Axial Bending (P)	1165	-1165	1165	-1165
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-1413	-1413	1413	1413
Axial Bending (ML)	-1413	1413	1413	-1413
Axial Pressure Stress	3116	3116	3116	3116

Axial Stress Summary	1627	2123	7280	2123
----------------------	------	------	------	------

Nozzle Stresses (kPa):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
--------------------------------------	---------------	--------------	---------------	--------------

Shear due to (Vc)	193	193	-193	-193
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	330	330	330	330

Shear Stress Summary	523	523	137	137
----------------------	-----	-----	-----	-----

Nozzle Stresses (kPa):

LONGITUDINAL PLANE (Stress Intensities	Au Outside	Al Inside	Bu Outside	Bl Inside
---	---------------	--------------	---------------	--------------

Two * Max Shear Stress	6336	23420	7308	23758
------------------------	------	-------	------	-------

Nozzle Stresses (kPa):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Hoop Membrane (P)	213	213	213	213
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	75	75	-75	-75
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	6239	23372	6239	23372

Hoop Stress Summary	6529	23662	6377	23510
---------------------	------	-------	------	-------

Nozzle Stresses (kPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
---	-----------------------	----------------------	------------------------	-----------------------

Axial Membrane (P)	172	172	172	172
Axial Bending (P)	1165	-1165	1165	-1165
Axial Membrane (Mc)	-1413	-1413	1413	1413
Axial Bending (Mc)	-1978	1978	1978	-1978
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	3116	3116	3116	3116

Axial Stress Summary	1061	2688	7846	1558
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Nozzle Stresses (kPa):



PV Elite 28 Licensee: SL User

FileName : 103-C2

Page 143 of 192

Nozzle Calcs.: Shell outlet-1 Nozl: 14 3:14pm Apr 10,2026

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-193	-193	193	193
Shear due to Torsion	330	330	330	330
Shear Stress Summary	137	137	523	523

Nozzle Stresses (kPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	6529	23662	8011	23517

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (kPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	130149	144428	130149	144428	130149	144428	130149	144428
Circ. Pl (SUS)	41	41	385	385	-35	-35	461	461
Circ. Q (SUS)	-213	213	1619	-1619	-1124	1124	5412	-5412
Long. Pm (SUS)	65071	65071	65071	65071	65071	65071	65071	65071
Long. Pl (SUS)	0	0	426	426	288	288	138	138
Long. Q (SUS)	-303	303	4591	-4591	-510	510	1916	-1916
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	220	220	-220	-220	-220	-220	220	220
Shear Q (SUS)	372	372	372	372	372	372	372	372
Pm (SUS)	130149	144428	130149	144428	130149	144428	130149	144428
Pm+Pl (SUS)	130190	144469	130534	144813	130114	144393	130610	144889
Pm+Pl+Q (Total)	129982	144686	132153	143194	128990	145517	136027	139481

Vessel Stress Summation Comparison (kPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	144428	147988	Passed
Pm+Pl (SUS)	144889	221983	Passed
Pm+Pl+Q (TOTAL)	145517	443966	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (kPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	6239	23372	6239	23372	6239	23372	6239	23372
Circ. Pl (SUS)	41	41	385	385	288	288	138	138
Circ. Q (SUS)	0	0	0	0	0	0	0	0



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FileName : 103-C2

Page 144 of 192

Nozzle Calcs.: Shell outlet-1 Nozl: 14 3:14pm Apr 10,2026

Long. Pm (SUS)	3116	3116	3116	3116	3116	3116	3116	3116
Long. Pl (SUS)	-1241	-1241	1585	1585	-1241	-1241	1585	1585
Long. Q (SUS)	-248	248	2578	-2578	-813	813	3143	-3143
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	193	193	-193	-193	-193	-193	193	193
Shear Q (SUS)	330	330	330	330	330	330	330	330
Pm (SUS)	6239	23372	6239	23372	6239	23372	6239	23372
Pm+Pl (SUS)	6288	23414	6643	23758	6534	23661	6398	23511
Pm+Pl+Q (Total)	6338	23425	7306	23757	6530	23660	8011	23522

Nozzle Stress Summation Comparison (kPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	23372	122834	Passed
Pm+Pl (SUS)	23758	184251	Passed
Pm+Pl+Q (TOTAL)	23757	368502	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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FileName : 103-C2

Page 145 of 192

Nozzle Calcs.: Shell inlet-1

Nozl: 15 3:14pm Apr 10,2026

Input, Nozzle Desc: Shell inlet-1

From: 40

Pressure for Reinforcement Calculations	P	14.3061	N/mm ²
Temperature for Internal Pressure	Temp	343	°C
Design External Pressure	Pext	0.10	N/mm ²
Temperature for External Pressure	Tempex	200	°C

Parent Material		SA-387 11	
Parent Allowable Stress at Temperature	Sv	147988.73	kPa
Parent Allowable Stress At Ambient	Sva	147988.73	kPa

Inside Diameter of Cylindrical Shell	D	2660.00	mm
Design Length of Section	L	6243.3643	mm
Shell Finished (Minimum) Thickness	t	142.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		5.50	m
-----------------------------------	--	------	---

User Entered Minimum Design Metal Temperature		-12.00	°C
---	--	--------	----

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	122834.11	kPa
Allowable Stress At Ambient	Sna	137989.50	kPa

Diameter Basis (for tr calc only)		Outside	
Layout Angle		180.00	deg
Diameter		10.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	28.5750	mm

Hub Height of Integral Nozzle	h	120.0000	mm
Height of Beveled Transition	L'	80.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		160.0000	mm

Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	400.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	145.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
ASME Code Weld Type per UW-16			g

The Pressure Design option was MAWP + static head (to the nozzle).



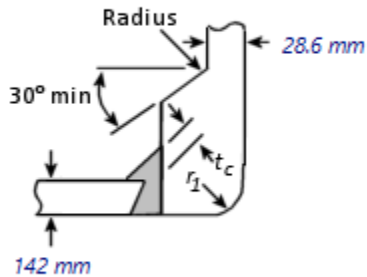
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FileName : 103-C2

Page 146 of 192

Nozzle Calcs.: Shell inlet-1

Nozl: 15 3:14pm Apr 10, 2026



UW-16.1 [g]

Reinforcement CALCULATION, Description: Shell inlet-1

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	10.000 in.
Actual Thickness Used in Calculation	1.125 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 14.3 \cdot 1333 / (147989 \cdot 1 - 0.6 \cdot 14.3)$
 $= 136.7874 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 14.3 \cdot 127 / (122834 \cdot 1 + 0.4 \cdot 14.3)$
 $= 14.1321 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.9894 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 131 / 0.5773$
 $= 227.6349 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	532.0000 mm
Parallel to Vessel Wall	Rn+tn+t	266.0000 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	263.9375 mm

Weld Strength Reduction Factor [fr1]:

$= \min(1, S_n / S_v)$
 $= \min(1, 122834 / 147989)$
 $= 0.830$

Weld Strength Reduction Factor [fr2]:

$= \min(1, S_n / S_v)$
 $= \min(1, 122834 / 147989)$
 $= 0.830$



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FileName : 103-C2

Page 147 of 192

Nozzle Calcs.: Shell inlet-1

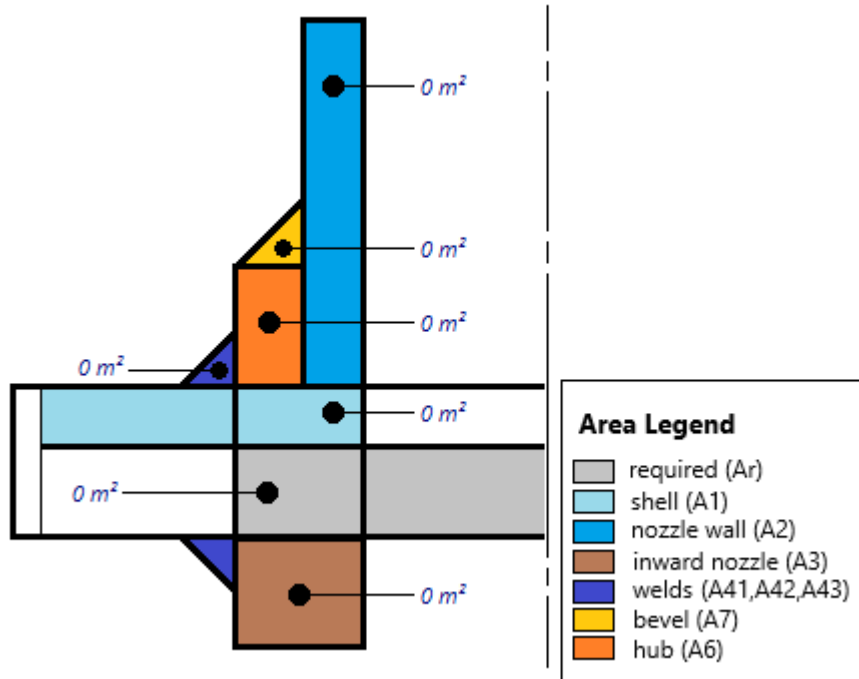
Nozl: 15 3:14pm Apr 10,2026

Weld Strength Reduction Factor [fr3]:

= min(fr2, fr4)

= min(0.83, 1)

= 0.830



Results of Nozzle Reinforcement Area Calculations: (m²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	0.035	0.002	NA
Area in Shell	A1	0.001	0.035	NA
Area in Nozzle Wall	A2	0.005	0.011	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	0.000	0.000	NA
Area in Element	A5	0.000	0.000	NA
Area in Hub	A6	0.035	0.035	NA
TOTAL AREA AVAILABLE	Atot	0.041	0.081	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

= (d • tr • F + 2 • tn • tr • F • (1-fr1)) UG-37(c)

= (203 • 137 • 1 + 2 • 157 • 137 • 1 • (1-0.83))

= 0.035 m²

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

= d (E1 • t - F • tr) - 2 • tn (E1 • t - F • tr) • (1 - fr1)

= 329 (1 • 139 - 1 • 137) - 2 • 157

(1 • 139 - 1 • 137) • (1 - 0.83)

= 0.001 m²



PV Elite 28 Licensee: SL User

FileName : 103-C2

Page 148 of 192

Nozzle Calcs.: Shell inlet-1 Nozl: 15 3:14pm Apr 10,2026

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot t_{np}) (t_n - t_{rn}) f_r^2$$

$$= (2 \cdot 264) (25.6 - 14.1) 0.83$$

$$= 0.005 \text{ m}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= (W_o^2 - \text{Area Lost}) \cdot f_r^2 + (W_i - \text{can}/0.707)^2 - \text{Area Lost} \cdot f_r^2$$

$$= (10^2 - 0.58792\text{E-}05) \cdot 0.83 + (0^2 - 0) \cdot 0.83$$

$$= 0.000 \text{ m}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(t_{np}, h_o, \text{Hubht})) \cdot (\text{Hubtk} - t_n) \cdot f_r^2$$

$$= (2 \cdot \min(264, 400, 120)) \cdot (160 - 28.6) \cdot 0.83$$

$$= 0.035 \text{ m}^2 \text{ Includes Bevel Area of } 0.009 \text{ m}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 17.1321 mm
Wall Thickness per UG16(b),	tr16b = 5.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 139.7874 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 139.7874 mm
Wall Thickness, shell/head, external pressure	trb2 = 3.9128 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 5.5000 mm
Wall Thickness per table UG-45	tb3 = 11.1026 mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[11.1, \max(140, 5.5)]$$

$$= 11.1026 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(17.1, 11.1)$$

$$= 17.1321 \text{ mm}$$

Available Nozzle Neck Thickness = 28.5750 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 51256.5,	Allowable	: 122834.1 kPa	Passed
Expansion	: 0.0,	Allowable	: 274773.0 kPa	Passed
Occasional	: 25189.9,	Allowable	: 163369.4 kPa	Passed
Shear	: 17566.5,	Allowable	: 85983.9 kPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: A

Warning:

The minimum metal temperature could not be computed because the governing thickness for this component was greater than 100.000 mm. This component must be impact tested.

Weld Size Calculations, Description: Shell inlet-1

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm



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FileName : 103-C2

Page 149 of 192

Nozzle Calcs.: Shell inlet-1

Nozl: 15 3:14pm Apr 10,2026

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 14.3 N/mm²

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0.1 N/mm²

The Drop for this Nozzle is : 6.0774 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 548.0775 mm

This nozzle or its diameter limit intersects weld seam # [2] on this element. The distance between the nozzle and seam is, 83.78 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 266.00 mm

Input Echo, WRC297 Item 1, Description: Shell inlet-1 :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	2660.000	mm
Shell Thickness	142.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-387 11	
Vessel Cold S.I. Allowable	Smc 147988.73	kPa
Vessel Hot S.I. Allowable	Smh 147988.73	kPa
Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	516.850	mm
Nozzle Thickness	160.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-105	
Nozzle Cold S.I. Allowable	SNmc 137989.50	kPa
Nozzle Hot S.I. Allowable	SNmh 122834.11	kPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:
These loads are assumed to be SUStained loads.

Design Internal Pressure	Dp	14.31	N/mm ²
Radial Load	P	-31241.91	N
Circumferential Shear	Vc	24995.49	N
Longitudinal Shear	Vl	24995.49	N
Circumferential Moment	Mc	18746.51	N-m
Longitudinal Moment	Ml	18746.51	N-m
Torsional Moment	Mt	21875.76	N-m
Include Axial Pressure Thrust		No	
Include Pressure Stress Indices per Div. 2		No	
Local Loads applied at end of Nozzle/Attachment		No	

Warning - The ratio, Dn/Tv, (3.640) is < 5
check the limitations of WRC bulletin 297.

Warning - The ratio, Dn/Tn (3.230) is < 10
check the limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 516.85 + 2 \cdot 1.65 \cdot \sqrt{1402.5(142.0 - 3.0)}$$



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FileName : 103-C2

Page 150 of 192

Nozzle Calcs.: Shell inlet-1

Nozl: 15 3:14pm Apr 10, 2026

= 1973.894 mm

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 2805.000 mm
 Nozzle Outside Diameter (d) = 516.850 mm
 Vessel Thickness used (T) = 139.000 mm
 Nozzle Thickness used (t) = 157.000 mm
 T / t = 0.885
 d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.828$$

Nr/P = 0.135
 Mr/P = 0.221
 M0/P = 0.073
 N0/P = 0.134
 MrD/Mc = 0.291
 NrDL/Mc = 0.132
 M0d/Mc = 0.108
 N0DL/Mc = -0.043
 MrD/Ml = 0.218
 NrDL/Ml = 0.115
 M0D/Ml = 0.082
 N0DL/Ml = 0.093

Vessel Stresses (kPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	213	213	213	213
Outplane Bending (P)	703	-703	703	-703
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-172	-172	172	172
Outplane Bending (ML)	-916	916	916	-916
Normal Pressure Stress	130383	144690	130383	144690
Outplane Stress Summary	130211	144945	132390	143455

Vessel Stresses (kPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	213	213	213	213
Inplane Bending (P)	2144	-2144	2144	-2144
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-213	-213	213	213
Inplane Bending (ML)	-2447	2447	2447	-2447
Inplane Pressure Stress	65188	65188	65188	65188
Inplane Stress Summary	64885	65491	70207	61024

Vessel Stresses (kPa):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	220	220	-220	-220
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	372	372	372	372



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FileName : 103-C2

Page 151 of 192

Nozzle Calcs.: Shell inlet-1 Nozl: 15 3:14pm Apr 10,2026

Shear Stress Summary	592	592	151	151
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Vessel Stresses (kPa):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Stress Intensities	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside

Two * Max Shear Stress	130211	144945	132390	143455
------------------------	--------	--------	--------	--------

Vessel Stresses (kPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Stresses Normal to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside

Outplane Membrane (P)	213	213	213	213
Outplane Bending (P)	703	-703	703	-703
Outplane Membrane (Mc)	75	75	-75	-75
Outplane Bending (Mc)	-1213	1213	1213	-1213
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	65188	65188	65188	65188

Outplane Stress Summary	64967	65988	67243	63409
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Vessel Stresses (kPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Stresses parallel to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside

Inplane Membrane (P)	213	213	213	213
Inplane Bending (P)	2144	-2144	2144	-2144
Inplane Membrane (Mc)	-248	-248	248	248
Inplane Bending (Mc)	-3268	3268	3268	-3268
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	130383	144690	130383	144690

Inplane Stress Summary	129225	145779	136257	139739
------------------------	--------	--------	--------	--------

Vessel Stresses (kPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Shear stress normal to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-220	-220	220	220
Torsional Shear (Mt)	372	372	372	372

Shear Stress Summary	151	151	592	592
----------------------	-----	-----	-----	-----

Vessel Stresses (kPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Stress Intensities	Left	Left	Right	Right
	Outside	Inside	Outside	Inside

Two * Max Shear Stress	129225	145779	136257	139739
------------------------	--------	--------	--------	--------

Nozzle Stresses (kPa):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Stresses in the	Top	Top	Bottom	Bottom
hoop direction)	Outside	Inside	Outside	Inside

Hoop Membrane (P)	213	213	213	213
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0



PV Elite 28 Licensee: SL User

FileName : 103-C2

Page 152 of 192

Nozzle Calcs.: Shell inlet-1 Nozl: 15 3:14pm Apr 10,2026

Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-172	-172	172	172
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	6246	23414	6246	23414

Hoop Stress Summary	6287	23455	6632	23800
---------------------	------	-------	------	-------

Nozzle Stresses (kPa):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	172	172	172	172
Axial Bending (P)	1165	-1165	1165	-1165
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-1413	-1413	1413	1413
Axial Bending (ML)	-1413	1413	1413	-1413
Axial Pressure Stress	3123	3123	3123	3123

Axial Stress Summary	1634	2130	7287	2130
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Nozzle Stresses (kPa):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	193	193	-193	-193
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	330	330	330	330

Shear Stress Summary	523	523	137	137
----------------------	-----	-----	-----	-----

Nozzle Stresses (kPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	6343	23462	7315	23800

Nozzle Stresses (kPa):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	213	213	213	213
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	75	75	-75	-75
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	6246	23414	6246	23414

Hoop Stress Summary	6536	23703	6384	23551
---------------------	------	-------	------	-------

Nozzle Stresses (kPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	172	172	172	172
Axial Bending (P)	1165	-1165	1165	-1165
Axial Membrane (Mc)	-1413	-1413	1413	1413
Axial Bending (Mc)	-1978	1978	1978	-1978
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	3123	3123	3123	3123



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FileName : 103-C2

Page 153 of 192

Nozzle Calcs.: Shell inlet-1 Nozl: 15 3:14pm Apr 10,2026

Axial Stress Summary	1068	2695	7852	1565
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Nozzle Stresses (kPa):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-193	-193	193	193
Shear due to Torsion	330	330	330	330

Shear Stress Summary	137	137	523	523
----------------------	-----	-----	-----	-----

Nozzle Stresses (kPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	6536	23703	8018	23558

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (kPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	130383	144690	130383	144690	130383	144690	130383	144690
Circ. Pl (SUS)	41	41	385	385	-35	-35	461	461
Circ. Q (SUS)	-213	213	1619	-1619	-1124	1124	5412	-5412
Long. Pm (SUS)	65188	65188	65188	65188	65188	65188	65188	65188
Long. Pl (SUS)	0	0	426	426	288	288	138	138
Long. Q (SUS)	-303	303	4591	-4591	-510	510	1916	-1916
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	220	220	-220	-220	-220	-220	220	220
Shear Q (SUS)	372	372	372	372	372	372	372	372
Pm (SUS)	130383	144690	130383	144690	130383	144690	130383	144690
Pm+Pl (SUS)	130424	144731	130768	145075	130348	144655	130844	145151
Pm+Pl+Q (Total)	130216	144948	132387	143456	129224	145779	136261	139743

Vessel Stress Summation Comparison (kPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	144690	147988	Passed
Pm+Pl (SUS)	145151	221983	Passed
Pm+Pl+Q (TOTAL)	145779	443966	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (kPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl



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FileName : 103-C2

Page 154 of 192

Nozzle Calcs.: Shell inlet-1 Nozl: 15 3:14pm Apr 10,2026

Circ. Pm (SUS)	6246	23414	6246	23414	6246	23414	6246	23414
Circ. Pl (SUS)	41	41	385	385	288	288	138	138
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	3123	3123	3123	3123	3123	3123	3123	3123
Long. Pl (SUS)	-1241	-1241	1585	1585	-1241	-1241	1585	1585
Long. Q (SUS)	-248	248	2578	-2578	-813	813	3143	-3143
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	193	193	-193	-193	-193	-193	193	193
Shear Q (SUS)	330	330	330	330	330	330	330	330
Pm (SUS)	6246	23414	6246	23414	6246	23414	6246	23414
Pm+Pl (SUS)	6295	23456	6650	23800	6541	23703	6405	23553
Pm+Pl+Q (Total)	6345	23467	7313	23799	6537	23702	8018	23564

Nozzle Stress Summation Comparison (kPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	23414	122834	Passed
Pm+Pl (SUS)	23800	184251	Passed
Pm+Pl+Q (TOTAL)	23799	368502	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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FileName : 103-C2

Page 155 of 192

ASME TS Calc: DESIGN CASE

Case: 1 3:14p Apr 10,2026

Input Echo, Tubesheet Number 1, Description: Tubesheet

Shell Data:

Main Shell Description: Shell

Shell Maximum Design Pressure	Psd,max	14.28 N/mm ²
Shell Maximum Operating Pressure	Psox,max	14.28 N/mm ²
Shell Minimum Operating Pressure	Psox,min	12.55 N/mm ²
Shell Thickness	ts	142.0000 mm
Shell Internal Corrosion Allowance	cas	3.0000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	2660.000 mm
Shell Temperature for Internal Pressure	Ts	343.00 °C
Shell Material		SA-387 11

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K11789
Shell Allowable Stress at Temperature	Ss	147988.73 kPa
Shell Allowable Stress at Ambient		147988.73 kPa

Channel Data:

Channel Description: Channel shell

Channel Type:		Cylinder
Channel Maximum Design Pressure	Ptd,max	4.95 N/mm ²
Channel Maximum Operating Pressure	Ptox,max	4.95 N/mm ²
Channel Minimum Operating Pressure	Ptox,min	4.44 N/mm ²
Channel Thickness	tc	50.0000 mm
Channel Internal Corrosion Allowance	cac	2.0000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	2666.000 mm
Channel Design Temperature	TEMPC	343.00 °C
Channel Flange Matl		SA-387 11

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K11789
Channel Allowable Stress at Temperature	Sc	147988.73 kPa
Channel Allowable Stress at Ambient		147988.73 kPa

Tube Data:

Number of Tube Holes	Nt	5546
Tube Wall Thickness	et	3.0480 mm
Tube Outside Diameter	D	25.4000 mm
Total Straight Tube Length	Lt	5500.00 mm
Straight Tube Length (bet. inner tubsht faces)	L	5000.00 mm
Design Temperature of the Tubes		343.00 °C
Tube Material		SA-213 TP316L
Tube Material UNS Number		S31603
Is this a Welded Tube		No
Tube Material Specification used		Smls. tube
Tube Allowable Stress at Temperature		94542.65 kPa
Tube Allowable Stress At Ambient		114991.25 kPa
Tube Yield Stress At design Temperature	Syt	105554.28 kPa
Tube Pitch (Center to Center Spacing)	P	31.7500 mm
Tube Layout Pattern		Triangular
Fillet Weld Leg	af	3.0000 mm
Groove Weld Leg	ag	3.0000 mm
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3
Is the Tube Inset		No
Radius to Outermost Tube Hole Center	ro	1272.760 mm
Largest Center-to-Center Tube Distance	U1	101.6000 mm



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FileName : 103-C2

Page 156 of 192

ASME TS Calc: DESIGN CASE

Case: 1 3:14p Apr 10,2026

Length of Expanded Portion of Tube	ltx	50.0000 mm
Tube-side pass partition groove depth	hg	5.0000 mm
Area of Tube Layout	Ap	5.0 m ²
Perimeter of Tube Layout	Cp	8219.2 mm

Tubesheet Data:

Tubesheet TYPE: U-tube, Integral both sides, Conf. a

Tubesheet Design Metal Temperature	T	343.00 °C
Tubesheet Material		SA-336 F11

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

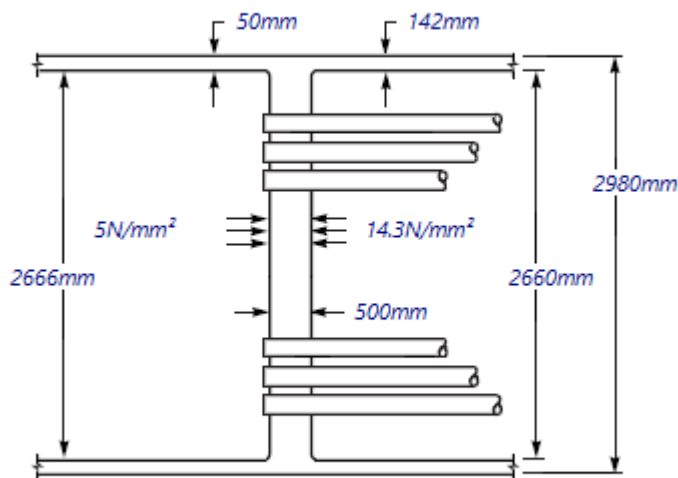
Tubesheet Material UNS Number		K11572
Tubesheet Allowable Stress at Temperature	S	147988.73 kPa
Tubesheet Allowable Stress at Ambient	Tt	147988.73 kPa
Thickness of Tubesheet	h	500.0000 mm
Tubesheet Corr. Allowance (Shell side)	Cats	3.0000 mm
Tubesheet Corr. Allowance (Channel side)	Catc	13.0000 mm
Tubesheet Outside Diameter	A	2980.000 mm

Area of the Untubed Lanes	AL	0.203 m ²
---------------------------	----	----------------------

Junction Stress Reduction option	Perform Plastic Calculation
Perform Differential Pressure Design	No
Run Multiple Load Cases	Yes

Tubesheet Integral with	Both
Tubesheet Extended as Flange	No

Tubesheet Integral w/Shell and Channel



Configuration a:

ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties :

Shell - TM-1 Material Group C



PV Elite 28 Licensee: SL User

FileName : 103-C2

Page 157 of 192

ASME TS Calc: DESIGN CASE

Case: 1 3:14p Apr 10, 2026

Elastic Mod. at Design Temperature	343.0 °C	0.18309E+06 N/mm ²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20409E+06 N/mm ²

Channel - TM-1 Material Group C

Elastic Mod. at Design Temperature	343.0 °C	0.18309E+06 N/mm ²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20409E+06 N/mm ²

Tubes - TM-1 Material Group G

Elastic Mod. at Tubsht. Design Temp.	343.0 °C	0.17274E+06 N/mm ²
Elastic Mod. at Ambient Temperature	21.1 °C	0.19513E+06 N/mm ²

TubeSheet - TM-1 Material Group C

Elastic Mod. at Design Temperature	343.0 °C	0.18309E+06 N/mm ²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20409E+06 N/mm ²

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:

$$= (P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P) \text{ per Appendix 1-1 (a)(1)}$$

$$= (4.95 \cdot (25.4/2 - 0)) / (94543 \cdot 1 + 0.4 \cdot 4.95)$$

$$= 0.6513 + 0.0000 = 0.6513 \text{ mm}$$

Tube Required Thickness under External Pressure (Shellside pressure) :

Determine Parameter [Mx]:

$$= L / \sqrt{(Ro \cdot t)}$$

$$= 6273 / \sqrt{(12.7 \cdot 3.05)}$$

$$= 1008.206$$

Compute the Factor [Ch]:

$$\text{Since } Mx > 2 \cdot (Do/t)0.94, Ch = 0.55 / (Do/t)$$

$$= 0.55 / (25.4/3.05)$$

$$= 0.0660$$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$$= 1.6 \cdot Ch \cdot Ey (t / Do)$$

$$= 1.6 \cdot 0.066 \cdot 172742 (3.05/25.4)$$

$$= 2188862.000 \text{ kPa}$$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$$= Fhe / Fy$$

$$= 2188862.0/105554.3$$

$$= 20.737$$

Predicted Buckling Stress [Fic]:

$$\text{Since } Fr \geq 2.439, Fic = Fy$$

$$= 105554.281 \text{ kPa}$$

For non-Carbon and Low Alloy Steel Materials re-compute [Fic]:

Factor A [A]:

$$= F/Emod, \text{ see 2286-6 Step 1}$$

$$= 2188862/172742$$

$$= 0.01267$$

From the External Pressure Chart, HA-4, determine the Stress [B]:

$$= 53229.980 \text{ kPa}$$

The Tangent Modulus from 3-D.5.2, [Et]:

$$= 2B/A$$

$$= 2 \cdot 53230/0.013$$

$$= 8401.7 \text{ N/mm}^2$$



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FileName : 103-C2

Page 158 of 192

ASME TS Calc: DESIGN CASE

Case: 1 3:14p Apr 10, 2026

Predicted Buckling Stress [Fic]:

$$= (F \cdot Et) / (FS \cdot Emod)$$

$$= (2188862 \cdot 8402) / (1 \cdot 172742)$$

$$= 106459.961 \text{ kPa}$$

Factor of Safety [FS]:

$$\text{Since } Fic [106460] \geq Fy [105554], FS = 1.667$$

Allowable Circumferential Compressive Stress [Fha]:

$$\text{Since } Fr > 2.439, Fha = Fy / FS$$

$$= 105554.281 / 1.667$$

$$= 63319.898 \text{ kPa}$$

Allowable External Pressure at the given Thickness [Pa]:

$$= 2 \cdot Fha (t / Do)$$

$$= 2 \cdot 63.3 (3.05 / 25.4)$$

$$= 15.198 \text{ N/mm}^2$$

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	2.8693 mm
Allowable Internal Pressure at Corroded thickness	25.10 N/mm ²
Required Internal Design Pressure	4.95 N/mm ²
Allowable External Pressure at Corroded thickness	15.20 N/mm ²
Required External Design Pressure	14.28 N/mm ²
Required Thickness due to Shell Side pressure	2.8693 mm

Detailed Results for load Case D1 un-corr. (Psd,min + Ptd,max)

Results for ASME U-tube Tubesheet Calculations for Configuration a:

Per Edition 2025, Original Thickness :

ASME Section VIII-2 Part 4.18.7.4 Step 1:

Equivalent outer tube limit circle diameter [Do]:

$$= 2 \cdot ro + dt$$

$$= 2 \cdot 1273 + 25.4$$

$$= 2570.920 \text{ mm}$$

Basic ligament efficiency for shear [μ]:

$$= (p - dt) / p$$

$$= (31.8 - 25.4) / 31.8$$

$$= 0.200$$

ASME Section VIII-2 Part 4.18.7.4 Step 2:

Ratio [ps]:

$$= Ds / Do \text{ (Configurations a, b, c)}$$

$$= 2660 / 2571$$

$$= 1.0346$$

Ratio [pc]:

$$= Dc / Do \text{ (Configurations a, e, f)}$$

$$= 2666 / 2571$$

$$= 1.0370$$

ASME Section VIII-2 Part 4.18.7.4 Step 3:

Ratio [ρ]:

$$= ltx / h$$

$$= 50 / 500$$



PV Elite 28 Licensee: SL User

FileName : 103-C2

Page 159 of 192

ASME TS Calc: DESIGN CASE

Case: 1 3:14p Apr 10, 2026

$$= 0.100 \text{ (must be } 0 \leq \rho \leq 1 \text{)}$$

Effective tube hole Diameter [d*]:

$$\begin{aligned} &= \max(dt - 2tt(EtT/E) \cdot (StT/S) \cdot (\rho); dt - 2tt) \\ &= \max(25.4 - 2 \cdot 3.05 (172742/183094) \cdot \\ &\quad (94543/147989) \cdot (0.1); 25.4 - 2 \cdot 3.05) \\ &= 25.0326 \text{ mm} \end{aligned}$$

Effective tube pitch [p*]:

$$\begin{aligned} &= p / \sqrt{ (1 - 4 \cdot \min(A1; 4 \cdot Do \cdot p) / (\pi \cdot Do^2)) } \\ &= 31.8 / \sqrt{ (1 - 4 \cdot \min(0.2; 4 \cdot 2571 \cdot 31.8) / (\pi \cdot 2571^2)) } \\ &= 32.3886 \text{ mm} \end{aligned}$$

Effective ligament efficiency for bending [μ^*]:

$$\begin{aligned} &= (p^* - d^*) / p^* \\ &= (32.4 - 25) / 32.4 \\ &= 0.2271 \end{aligned}$$

$$h/p = 15.748032 ; \mu^* = 0.227118 ; E^* = 34428 \text{ N/mm}^2$$

$$E^*/E = 0.188040 ; \nu^* = 0.422203$$

Note: As h/p (15.748) is > 2, data values for h/p = 2 were used.

ASME Section VIII-2 Part 4.18.7.4 Step 4:

Shell coefficient [β_s]:

$$\begin{aligned} &= ((12(1 - \nu_s^2))^{1/4}) / \sqrt{ ((D_s + t_s) t_s) } \\ &= ((12(1 - 0.3^2))^{1/4}) / \sqrt{ ((2660 + 142) 142) } \\ &= 0.0029 \text{ 1/mm} \end{aligned}$$

Shell coefficient [k_s]:

$$\begin{aligned} &= \beta_s \cdot E_s \cdot t_s^3 / (6(1 - \nu_s^2)) \\ &= 0.0029 \cdot 183094 \cdot 142^3 / (6(1 - 0.3^2)) \\ &= 276708960.0000 \text{ N/mm}^2 \cdot \text{mm}^2 \end{aligned}$$

Shell coefficient [λ_s]:

$$\begin{aligned} &= 6(D_s + t_s) \cdot k_s / h^3 \cdot (1 + h \cdot \beta_s + (h \cdot \beta_s)^2 / 2) \\ &= 6(2660 + 142) \cdot 276708960 / 500^3 \cdot (1 + 500 \cdot 0.0029 + (500 \cdot 0.0029)^2 / 2) \\ &= 129479.3 \text{ N/mm}^2 \end{aligned}$$

Shell coefficient [δ_s]:

$$\begin{aligned} &= D_s^2 / (4 \cdot E_s \cdot t_s) \cdot (1 - D_s \cdot \nu_s / (2(D_s + t_s))) \\ &= 2660^2 / (4 \cdot 183094 \cdot 142) \cdot (1 - 2660 \cdot 0.3 / (2(2660 + 142))) \\ &= 0.58352E-04 \text{ mm/kPa} \end{aligned}$$

Parameter [w_s]:

$$\begin{aligned} &= p_s \cdot k_s \cdot \beta_s \cdot \delta_s (1 + h \cdot \beta_s) \\ &= 1.03 \cdot 276708960 \cdot 0.0029 \cdot 0.58352E-04 (1 + 500 \cdot 0.0029) \\ &= 117511.0781 \text{ mm}^2 \end{aligned}$$

Channel coefficient [β_c]:

$$\begin{aligned} &= ((12(1 - \nu_c^2))^{1/4}) / \sqrt{ ((D_c + t_c) t_c) } \\ &= ((12(1 - 0.3^2))^{1/4}) / \sqrt{ ((2666 + 50) 50) } \\ &= 0.0049 \text{ 1/mm} \end{aligned}$$

Channel coefficient [k_c]:

$$\begin{aligned} &= \beta_c \cdot E_c \cdot t_c^3 / (6(1 - \nu_c^2)) \\ &= 0.0049 \cdot 183094 \cdot 50^3 / (6(1 - 0.3^2)) \\ &= 20677430.0000 \text{ N/mm}^2 \cdot \text{mm}^2 \end{aligned}$$

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:

$$CqDelta = CqTheta = CmTheta = CmDelta = 1$$

Channel coefficient [λ_c]:

$$= 6(D_c + t_c / \cos(\theta)) \cdot k_c / h^3 \cdot (CmTheta +$$



PV Elite 28 Licensee: SL User

FileName : 103-C2

Page 160 of 192

ASME TS Calc: DESIGN CASE

Case: 1 3:14p Apr 10, 2026

$$\begin{aligned} & h \cdot \beta c \cdot (Cm\Delta + CqTheta)/2 + (h \cdot \beta c)^2/2 \cdot Cq\Delta \\ &= 6(2666 + 50/\cos(0)) \cdot 20677430/500^3 \cdot (1 + \\ & \quad 500 \cdot 0.0049 \cdot (1 + 1)/2 + (500 \cdot 0.0049)^2/2 \cdot 1) \\ &= 17544.0 \text{ N/mm}^2 \end{aligned}$$

Channel coefficient [δc]:

$$\begin{aligned} &= Dc^2/(4 \cdot Ec \cdot tc) \cdot (1 - Dc \cdot vc/(2(Dc + tc))) \\ &= 2666^2/(4 \cdot 183094 \cdot 50) \cdot (1 - 2666 \cdot 0.3/(2(2666 + 50))) \\ &= 0.16553E-03 \text{ mm/kPa} \end{aligned}$$

Channel coefficient [zeta αc]:

$$zeta\alpha c = 0$$

Parameter [ωc]:

$$\begin{aligned} &= \rho c \cdot kc[\delta c(CmTheta \cdot \beta c + Cq\Delta \cdot h \cdot \beta c^2) - \\ & \quad \zeta c(CmTheta + h \cdot CqTheta \cdot h \cdot \beta c/2)] \\ &= 1.04 \cdot 20677430 [0.000165527(1 + 0.0049 \cdot 1 \cdot 500) - \\ & \quad 0.000(1/500 + 1 \cdot 0.0049/2)] \\ &= 60688.4023 \text{ mm}^2 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 5:

Diameter ratio [K]:

$$\begin{aligned} &= A / Do \\ &= 2980/2571 \\ &= 1.1591 \end{aligned}$$

Coefficient [F]:

$$\begin{aligned} &= (1 - \nu^*)/E^* \cdot (\lambda s + \lambda c + E \cdot \ln(K)) \\ &= (1 - 0.42)/34429 \cdot (129479 + 17544 + \\ & \quad 183094 \cdot \ln(1.16)) \\ &= 2.9211 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 6:

Moment on tubesheet due to pressures (Ps, Pt) [Mts]:

$$\begin{aligned} &= Do^2/16 \cdot [(ps-1) \cdot (ps^2+1) \cdot Ps - (pc-1) \cdot (pc^2+1) \cdot Pt] \\ &= 2571^2/16 \cdot [(1.03-1) \cdot (1.03^2+1) \cdot 0 - \\ & \quad (1.04-1) \cdot (1.04^2+1) \cdot 4.95] \\ &= -156955.4219 \text{ N/mm}^2 \cdot \text{mm}^2 \end{aligned}$$

Moment acting on unperforated tubesheet rim [M*]:

$$\begin{aligned} &= Mts + \omega c \cdot Pt - \omega s \cdot Ps \\ &= -156955 + 60688 \cdot 4.95 - 117511 \cdot 0 \\ &= 143469.6094 \text{ N/mm}^2 \cdot \text{mm}^2 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 7:

Maximum bending moment acting on periphery of tubesheet [Mp]:

$$\begin{aligned} &= ((M^*) - Do^2/32 \cdot F(Ps - Pt)) / (1 + F) \\ &= ((143470) - 2571^2/32 \cdot 2.92(0 - 4.95)) / (1 + 2.92) \\ &= 798310.6875 \text{ N/mm}^2 \cdot \text{mm}^2 \end{aligned}$$

Maximum bending moment acting on center of tubesheet [Mo]:

$$\begin{aligned} &= Mp + Do^2/64 \cdot (3 + \nu^*) \cdot (Ps - Pt) \\ &= 798311 + 2571^2/64 \cdot (3 + 0.42) \cdot (0 - 4.95) \\ &= -951267.0000 \text{ N/mm}^2 \cdot \text{mm}^2 \end{aligned}$$

Maximum bending moment acting on tubesheet [M]:

$$\begin{aligned} &= \max(|Mp|; |Mo|) \\ &= \max(|798311|; |-951267|) \\ &= 951267.0000 \text{ N/mm}^2 \cdot \text{mm}^2 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 8:



PV Elite 28 Licensee: SL User

FileName : 103-C2

Page 161 of 192

ASME TS Calc: DESIGN CASE

Case: 1 3:14p Apr 10, 2026

Tubesheet Bending Stress at original thickness:

$$\begin{aligned}
 &= 6 \cdot M / (\mu \cdot (h - h_g')^2) \\
 &= 6 \cdot 951267 / (0.23 \cdot (500 - 5)^2) \\
 &= 102557.3047 \text{ kPa}
 \end{aligned}$$

Allowable Tubesheet Bending Stress [σ_{All}]:

$$\begin{aligned}
 &= 2 \cdot S \\
 &= 2 \cdot 147989 \\
 &= 295977.47 \text{ kPa}
 \end{aligned}$$

Tubesheet bending stress at final thickness [σ]:

$$\begin{aligned}
 &= 6 \cdot M / (\mu \cdot (h - h_g')^2) \\
 &= 6 \cdot 916922 / (0.24 \cdot (286 - 5)^2) \\
 &= 295872.0625 \text{ kPa}
 \end{aligned}$$

Required tubesheet thickness for Bending Stress [H_{reqB}]:

$$\begin{aligned}
 &= h + Cats + Catc \\
 &= 286 + 0 + 0 \\
 &= 285.9173 \text{ mm}
 \end{aligned}$$

Required tubesheet thickness for given loadings (includes CA) [H_{req}]:

$$\begin{aligned}
 &= \max(H_{reqB}; H_{reqS}) \\
 &= \max(286; 126) \\
 &= 285.9173 \text{ mm}
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 9:

Absolute pressure differential [$|P_s - P_t|$]:

$$\begin{aligned}
 &= |P_s - P_t| \\
 &= |0 - 4.95| \\
 &= 4.950 \text{ N/mm}^2
 \end{aligned}$$

Shear Stress check [τ_{limit}]:

$$\begin{aligned}
 &= 4 \cdot \mu \cdot h / D_o \cdot \min(0.8 \cdot S; 0.533 \cdot S_y) \\
 &= 4 \cdot 0.2 \cdot 500 / 2571 \cdot \min(0.8 \cdot 147989; 0.533 \cdot 239106) \\
 &= 18420.020 \text{ kPa}
 \end{aligned}$$

$|P_s - P_t|$ is $\leq$ the Shear Stress limit (τ_{limit}). Therefore, the required thickness for shear is for information only.

Minimum required thickness for Shear [H_{reqS}]:

$$\begin{aligned}
 &= 1 / (4 \cdot \mu \cdot \min(0.8 \cdot S; 0.533 \cdot S_y)) \cdot (4 \cdot A_p / C_p) \cdot |P_s - P_t| + Cats + Catc \\
 &= 1 / (4 \cdot 0.2 \cdot 118391) \cdot (4 \cdot 4.95 / 8219) \cdot |0 - 4.95| + 0 \\
 &= 126.0141 \text{ mm}
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 10:

Note: For a given Shell thickness of: 142.000mm

Min. Shell len. adjacent to the tubesheet is: 1106.261 mm

Axial Shell Membrane Stress [$\sigma_{s,m}$]:

$$\begin{aligned}
 &= D_s^2 / (4 \cdot t_s (D_s + t_s)) \cdot P_s \\
 &= 2660^2 / (4 \cdot 142 \cdot (2660 + 142)) \cdot 0 \\
 &= 0.0000 \text{ kPa}
 \end{aligned}$$

Axial Shell Bending Stress [$\sigma_{s,b}$]:

$$\begin{aligned}
 &= 6 \cdot k_s / t_s^2 \cdot [\beta S \cdot P_s + 6(1 - \nu) / E \cdot \\
 &\quad D_o / h^3 \cdot (1 + h \cdot \beta S / 2) \cdot (M_p + D_o^2 / 32 \cdot (P_s - P_t))] \\
 &= 6 \cdot 62204464 / 142^2 \cdot [0.0029 \cdot 0 + 6(1 - 0.42) / 34429 \cdot \\
 &\quad 2571 / 500^3 \cdot (1 + 500 \cdot 0.0029 / 2) \cdot (798311 + 2571^2 / 32 \cdot \\
 &\quad (0 - 4.95))] \\
 &= -65764.273 \text{ kPa}
 \end{aligned}$$

Shell Membrane + Bending Stress [σ_s]:

$$= |\sigma_{s,m}| + |\sigma_{s,b}| \text{ (Should be } \leq 1.5 \cdot S_s)$$



PV Elite 28 Licensee: SL User

FileName : 103-C2

Page 162 of 192

ASME TS Calc: DESIGN CASE

Case: 1 3:14p Apr 10, 2026

= 65764.273 kPa (Should be $\leq$ 221983.11)

Note: For a given Channel thickness of: 50.000 mm
Min. Channel len. adjacent to the tubesheet is: 657.185mm

Axial Channel Membrane Stress [σ_c, m]:

$$= D_c^2 / (4 \cdot t_c (D_c + t_c) \cdot \cos(\theta)) \cdot P_t$$

$$= 2666^2 / (4 \cdot 50 (2666 + 50) \cdot \cos(0)) \cdot 4.95$$

$$= 64768.781 \text{ kPa}$$

Axial Channel Bending Stress [σ_c, b]:

$$= 6 \cdot k_c / t_c^2 \cdot [(\delta_c \cdot C_m \Delta \cdot \beta_c - \zeta_c \cdot C_m \theta) \cdot P_t - 6((1 - \nu^*) / E^*) \cdot D_o / h^3 \cdot (C_m \theta + h \cdot \beta_c / 2 \cdot C_m \Delta) \cdot (M_p + D_o^2 / 32 \cdot (P_s - P_t))]$$

$$= 6 \cdot 4648307 / 50^2 \cdot [(0.000165527 \cdot 1 \cdot 0.0049 - 0.000 \cdot 1) \cdot 4.95 - 6((1 - 0.42) / 34429) \cdot 2571 / 500^3 \cdot (1 + 0.0049 \cdot 500 / 2 \cdot 1) \cdot (798311 + 2571^2 / 32 \cdot (0 - 4.95))]$$

$$= 252018.844 \text{ kPa}$$

Channel Membrane + Bending Stress [σ_c]:

$$= | \sigma_{c,m} | + | \sigma_{c,b} | \text{ (Should be } \leq 1.5 \cdot S_c)$$

$$= 316787.625 \text{ kPa (Should be } \leq 221983.11)$$

Step 11, The Cylinder-to-Tubesheet Junction is Overstressed.

Performing a simplified Elastic-Plastic calculation

(option 3) to reduce the overstress condition.

Modify E_s and/or E_c and recompute from Step 4 onwards.

Modified Elasticity [E_c]:

$$= E_c \cdot \sqrt{1.5 \cdot S_c / \sigma_c}$$

$$= 183094 \cdot \sqrt{1.5 \cdot 147989 / 316788}$$

$$= 153267.2 \text{ N/mm}^2$$

Tubesheet Bending Stress at Original Thk., after the Elastic-Plastic iteration:

$$= 6 \cdot M / (\mu \cdot (h - h'g)^2)$$

$$= 6 \cdot 954043 / (0.23 (500 - 5)^2)$$

$$= 102856.6 \text{ kPa (Should be } \leq 295977.5)$$



Note: The tubesheet is Not overstressed after the Elastic-Plastic iteration.

The design is acceptable. Recomputing tubesheet required thickness.

Required Tubesheet Thk., for Bending Stress after the Elastic-Plastic iteration [H_{reqB}]:

$$= h + C_{ats} + C_{atc} = 286 + 0 + 0 = 286 \text{ mm}$$

Required Tubesheet Thk. after the Elastic-Plastic iteration (includes CA) [H_{req}]:

$$= \max(H_{reqB}; H_{reqS}) = \max(286; 126) = 286 \text{ mm}$$

Tube Metal Area [A_t]:

$$= \pi (d_o - t) t$$

$$= \pi (25.4 - 3.05) 3.05$$

$$= 0.000 \text{ m}^2$$

Tube Weld Size Results per UW-20.3:

Tube Strength [F_t]:

$$= \pi \cdot t (d_o - t) S$$

$$= \pi \cdot 3.05 (25.4 - 3.05) 94543$$

$$= 20234.723 \text{ N}$$

Groove Weld Strength [F_g]:

$$= 0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w \text{ (but not } > F_t)$$



PV Elite 28 Licensee: SL User

FileName : 103-C2

Page 163 of 192

ASME TS Calc: DESIGN CASE

Case: 1 3:14p Apr 10, 2026

$$= 0.85 \cdot \pi \cdot 3 (25.4 + 0.67 \cdot 3) 94543$$

$$= 20234.7227 \text{ N}$$

Weld Strength Factor [fw]:

$$= S / S_w \text{ where } S_w = \min(S \text{ and } S_t)$$

$$= 94543/94543$$

$$= 1.000$$

Design Strength Ratio [fd]:

= 1 for full strength welds

Fillet Weld Strength [Ff]:

$$= 0.55 \cdot \pi \cdot a_f (d_o + 0.67 \cdot a_f) S_w \text{ (but not } > F_t)$$

$$= 0.55 \cdot \pi \cdot 3 (25.4 + 0.67 \cdot 3) 94543$$

$$= 13432.5635 \text{ N}$$

Minimum Length of Weld [ar]:

$$= 2 \left(\sqrt{(0.75 \cdot d_o)^2 + 1.07 \cdot t \cdot (d_o - t) f_w \cdot f_d} - 0.75 \cdot d_o \right)$$

$$= 2 \left(\sqrt{(0.75 \cdot 25.4)^2 + 1.07 \cdot 3.05 \cdot (25.4 - 3.05) 1 \cdot 1} - 0.75 \cdot 25.4 \right)$$

$$= 3.652 \text{ mm}$$

Minimum Required Fillet Weld Leg [afr]: 1.8288 mm

Minimum Required Groove Weld Leg [agr]: 1.8288 mm

Tube-Tubesheet Jt allowable, 20235 is >= tube strength 20235 N

Note: This tube-tubesheet joint is a Full Strength joint

Stress/Force summary for loadcase D1 un-corr. (Psd,min + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet bend. stress	102856.6 <=	295977.5 kPa	Ok
Tubesheet shear stress	29837.9 <=	118391.0 kPa	Ok
Stress in Shell at Tubesheet	65764.3 <=	221983.1 kPa	Ok
Stress in Channel at Tubesheet	316787.6 <=	478235.3 kPa	Ok

Thickness results for loadcase D1 un-corr. (Psd,min + Ptd,max):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	285.917	500.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.829	3.000	Ok
Tube-Tubesheet Groove Weld Leg :	1.829	3.000	Ok

Detailed Results for load Case D2 un-corr. (Psd,max + Ptd,min)

Results for ASME U-tube Tubesheet Calculations for Configuration a:

Per Edition 2025, Original Thickness :

ASME Section VIII-2 Part 4.18.7.4 Step 1:

Equivalent outer tube limit circle diameter [Do]:

$$= 2 \cdot r_o + d_t$$

$$= 2 \cdot 1273 + 25.4$$

$$= 2570.920 \text{ mm}$$

Basic ligament efficiency for shear [μ]:

$$= (p - d_t) / p$$



PV Elite 28 Licensee: SL User

FileName : 103-C2

Page 164 of 192

ASME TS Calc: DESIGN CASE

Case: 1 3:14p Apr 10, 2026

$$= (31.8 - 25.4) / 31.8$$

$$= 0.200$$

ASME Section VIII-2 Part 4.18.7.4 Step 2:

Ratio [ps]:

$$= D_s / D_o \text{ (Configurations a, b, c)}$$

$$= 2660 / 2571$$

$$= 1.0346$$

Ratio [pc]:

$$= D_c / D_o \text{ (Configurations a, e, f)}$$

$$= 2666 / 2571$$

$$= 1.0370$$

ASME Section VIII-2 Part 4.18.7.4 Step 3:

Ratio [p]:

$$= l_{tx} / h$$

$$= 50 / 500$$

$$= 0.100 \text{ (must be } 0 \leq p \leq 1 \text{)}$$

Effective tube hole Diameter [d*]:

$$= \max(dt - 2tt(E_t T / E) \cdot (StT / S) \cdot (p); dt - 2tt)$$

$$= \max(25.4 - 2 \cdot 3.05 (172742 / 183094) \cdot (94543 / 147989) \cdot (0.1); 25.4 - 2 \cdot 3.05)$$

$$= 25.0326 \text{ mm}$$

Effective tube pitch [p*]:

$$= p / \sqrt{ 1 - 4 \cdot \min(A_l; 4 \cdot D_o \cdot p) / (\pi \cdot D_o^2) }$$

$$= 31.8 / \sqrt{ 1 - 4 \cdot \min(0.2; 4 \cdot 2571 \cdot 31.8) / (\pi \cdot 2571^2) }$$

$$= 32.3886 \text{ mm}$$

Effective ligament efficiency for bending [μ^*]:

$$= (p^* - d^*) / p^*$$

$$= (32.4 - 25) / 32.4$$

$$= 0.2271$$

$$h/p = 15.748032 ; \mu^* = 0.227118 ; E^* = 34428 \text{ N/mm}^2$$

$$E^*/E = 0.188040 ; \nu^* = 0.422203$$

Note: As h/p (15.748) is > 2, data values for h/p = 2 were used.

ASME Section VIII-2 Part 4.18.7.4 Step 4:

Shell coefficient [β_s]:

$$= ((12(1 - \nu_s^2))^{1/4}) / \sqrt{ (D_s + t_s) t_s }$$

$$= ((12(1 - 0.3^2))^{1/4}) / \sqrt{ (2660 + 142) 142 }$$

$$= 0.0029 \text{ 1/mm}$$

Shell coefficient [k_s]:

$$= \beta_s \cdot E_s \cdot t_s^3 / (6(1 - \nu_s^2))$$

$$= 0.0029 \cdot 183094 \cdot 142^3 / (6(1 - 0.3^2))$$

$$= 276708960.0000 \text{ N/mm}^2 \cdot \text{mm}^2$$

Shell coefficient [λ_s]:

$$= 6(D_s + t_s) \cdot k_s / h^3 \cdot (1 + h \cdot \beta_s + (h \cdot \beta_s)^2 / 2)$$

$$= 6(2660 + 142) \cdot 276708960 / 500^3 \cdot (1 + 500 \cdot 0.0029 + (500 \cdot 0.0029)^2 / 2)$$

$$= 129479.3 \text{ N/mm}^2$$

Shell coefficient [δ_s]:

$$= D_s^2 / (4 \cdot E_s \cdot t_s) \cdot (1 - D_s \cdot \nu_s / (2(D_s + t_s)))$$

$$= 2660^2 / (4 \cdot 183094 \cdot 142) \cdot (1 - 2660 \cdot 0.3 / (2(2660 + 142)))$$

$$= 0.58352\text{E-}04 \text{ mm/kPa}$$



PV Elite 28 Licensee: SL User

FileName : 103-C2

Page 165 of 192

ASME TS Calc: DESIGN CASE

Case: 1 3:14p Apr 10, 2026

Parameter [ws]:

$$= ps \cdot ks \cdot \beta_s \cdot \delta_s (1 + h \cdot \beta_s)$$

$$= 1.03 \cdot 276708960 \cdot 0.0029 \cdot 0.58352E-04 (1 + 500 \cdot 0.0029)$$

$$= 117511.0781 \text{ mm}^2$$

Channel coefficient [β_c]:

$$= \left(\left(12(1 - v_c^2) \right)^{1/4} \right) / \sqrt{\left((Dc + tc)tc \right)}$$

$$= \left(\left(12(1 - 0.3^2) \right)^{1/4} \right) / \sqrt{(2666 + 50)50}$$

$$= 0.0049 \text{ 1/mm}$$

Channel coefficient [kc]:

$$= \beta_c \cdot Ec \cdot tc^3 / (6(1 - v_s^2))$$

$$= 0.0049 \cdot 183094 \cdot 50^3 / (6(1 - 0.3^2))$$

$$= 20677430.0000 \text{ N/mm}^2 \cdot \text{mm}^2$$

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:

$$CqDelta = CqTheta = CmTheta = CmDelta = 1$$

Channel coefficient [λ_c]:

$$= 6(Dc + tc/\cos(\theta)) \cdot kc/h^3 \cdot (CmTheta + h \cdot \beta_c \cdot (Cm\Delta + CqTheta)/2 + (h \cdot \beta_c)^2/2 \cdot Cq\Delta)$$

$$= 6(2666 + 50/\cos(0)) \cdot 20677430/500^3 \cdot (1 + 500 \cdot 0.0049 \cdot (1 + 1)/2 + (500 \cdot 0.0049)^2/2 \cdot 1)$$

$$= 17544.0 \text{ N/mm}^2$$

Channel coefficient [δ_c]:

$$= Dc^2 / (4 \cdot Ec \cdot tc) \cdot (1 - Dc \cdot v_c / (2(Dc + tc)))$$

$$= 2666^2 / (4 \cdot 183094 \cdot 50) \cdot (1 - 2666 \cdot 0.3 / (2(2666 + 50)))$$

$$= 0.16553E-03 \text{ mm/kPa}$$

Channel coefficient [zetac]:

$$zetac = 0$$

Parameter [wc]:

$$= pc \cdot kc [\delta_c (CmTheta \cdot \beta_c + Cq\Delta \cdot h \cdot \beta_c^2) - \zeta_c (CmTheta + h \cdot CqTheta \cdot h \cdot \beta_c / 2)]$$

$$= 1.04 \cdot 20677430 [0.000165527 (1 + 0.0049 \cdot 1 \cdot 500) - 0.000 (1/500 + 1 \cdot 0.0049/2)]$$

$$= 60688.4023 \text{ mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 5:

Diameter ratio [K]:

$$= A / Do$$

$$= 2980/2571$$

$$= 1.1591$$

Coefficient [F]:

$$= (1 - v^*) / E^* \cdot (\lambda_s + \lambda_c + E \cdot \ln(K))$$

$$= (1 - 0.42) / 34429 \cdot (129479 + 17544 + 183094 \cdot \ln(1.16))$$

$$= 2.9211$$

ASME Section VIII-2 Part 4.18.7.4 Step 6:

Moment on tubesheet due to pressures (Ps, Pt) [Mts]:

$$= Do^2/16 \cdot [(ps-1) \cdot (ps^2+1) \cdot Ps - (pc-1) \cdot (pc^2+1) \cdot Pt]$$

$$= 2571^2/16 \cdot [(1.03-1) \cdot (1.03^2+1) \cdot 14.3 - (1.04-1) \cdot (1.04^2+1) \cdot 0]$$

$$= 423231.4375 \text{ N/mm}^2 \cdot \text{mm}^2$$

Moment acting on unperforated tubesheet rim [M*]:

$$= Mts + wc \cdot Pt - ws \cdot Ps$$

$$= 423231 + 60688 \cdot 0 - 117511 \cdot 14.3$$

$$= -1254924.2500 \text{ N/mm}^2 \cdot \text{mm}^2$$



PV Elite 28 Licensee: SL User

FileName : 103-C2

Page 166 of 192

ASME TS Calc: DESIGN CASE

Case: 1 3:14p Apr 10, 2026

ASME Section VIII-2 Part 4.18.7.4 Step 7:

Maximum bending moment acting on periphery of tubesheet [Mp]:

$$= ((M \bullet) - Do^2/32 \bullet F (Ps - Pt)) / (1 + F)$$

$$= ((-1254924) - 2571^2/32 \bullet 2.92 (14.3 - 0)) / (1 + 2.92)$$

$$= -2517495.0000 \text{ N/mm}^2 \bullet \text{mm}^2$$

Maximum bending moment acting on center of tubesheet [Mo]:

$$= Mp + Do^2/64 \bullet (3 + \nu \bullet) \bullet (Ps - Pt)$$

$$= -2517495 + 2571^2/64 \bullet (3 + 0.42) \bullet (14.3 - 0)$$

$$= 2529772.0000 \text{ N/mm}^2 \bullet \text{mm}^2$$

Maximum bending moment acting on tubesheet [M]:

$$= \max(| Mp | ; | Mo |)$$

$$= \max(| -2517495 | ; | 2529772 |)$$

$$= 2529772.0000 \text{ N/mm}^2 \bullet \text{mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 8:

Tubesheet Bending Stress at original thickness:

$$= 6 \bullet M / (\mu \bullet (h - hg')^2)$$

$$= 6 \bullet 2529772 / (0.23 (500 - 5)^2)$$

$$= 272737.9688 \text{ kPa}$$

Allowable Tubesheet Bending Stress [σ_{All}]:

$$= 2 \bullet S$$

$$= 2 \bullet 147989$$

$$= 295977.47 \text{ kPa}$$

Tubesheet bending stress at final thickness [σ]:

$$= 6 \bullet M / (\mu \bullet (h - hg')^2)$$

$$= 6 \bullet 2528826 / (0.23 (480 - 5)^2)$$

$$= 295500.3438 \text{ kPa}$$

Required tubesheet thickness for Bending Stress [HreqB]:

$$= h + Cats + Catc$$

$$= 480 + 0 + 0$$

$$= 479.9693 \text{ mm}$$

Required tubesheet thickness for given loadings (includes CA) [Hreq]:

$$= \max(HreqB ; HreqS)$$

$$= \max(480 ; 364)$$

$$= 479.9693 \text{ mm}$$

ASME Section VIII-2 Part 4.18.7.4 Step 9:

Absolute pressure differential [| Ps - Pt |]:

$$= | Ps - Pt |$$

$$= | 14.3 - 0 |$$

$$= 14.281 \text{ N/mm}^2$$

Shear Stress check [τ_{limit}]:

$$= 4 \bullet \mu \bullet h/Do \bullet \min(0.8 \bullet S ; 0.533 \bullet Sy)$$

$$= 4 \bullet 0.2 \bullet 500/2571 \bullet \min(0.8 \bullet 147989 ; 0.533 \bullet 239106)$$

$$= 18420.020 \text{ kPa}$$

| Ps - Pt | is $\leq$ the Shear Stress limit (τ_{limit}). Therefore, the required thickness for shear is for information only.

Minimum required thickness for Shear [HreqS]:

$$= 1 / (4 \bullet \mu \bullet \min(0.8 \bullet S ; 0.533 \bullet Sy)) \bullet (4 \bullet Ap/Cp) \bullet | Ps - Pt | + Cats + Catc$$

$$= 1 / (4 \bullet 0.2 \bullet 118391) \bullet (4 \bullet 4.95/8219) \bullet | 14.3 - 0 | + 0$$

$$= 363.5316 \text{ mm}$$



PV Elite 28 Licensee: SL User

FileName : 103-C2

Page 167 of 192

ASME TS Calc: DESIGN CASE

Case: 1 3:14p Apr 10, 2026

ASME Section VIII-2 Part 4.18.7.4 Step 10:

Note: For a given Shell thickness of: 142.000mm
Min. Shell len. adjacent to the tubesheet is: 1106.261 mm

Axial Shell Membrane Stress [σ_s, m]:

$$= D_s^2 / (4 \cdot t_s (D_s + t_s)) \cdot P_s$$

$$= 2660^2 / (4 \cdot 142 (2660 + 142)) \cdot 14.3$$

$$= 63485.5625 \text{ kPa}$$

Axial Shell Bending Stress [σ_s, b]:

$$= 6 \cdot k_s / t_s^2 \cdot [\beta S \cdot P_s + 6(1 - \nu^*) / E^* \cdot$$

$$Do / h^3 \cdot (1 + h \cdot \beta S / 2) \cdot (M_p + Do^2 / 32 \cdot (P_s - P_t))]$$

$$= 6 \cdot 62204464 / 142^2 \cdot [0.0029 \cdot 0.83 + 6(1 - 0.42) / 34429 \cdot$$

$$2571 / 500^3 \cdot (1 + 500 \cdot 0.0029 / 2) \cdot (-2517495 + 2571^2 / 32 \cdot$$

$$(14.3 - 0))]$$

$$= 324508.062 \text{ kPa}$$

Shell Membrane + Bending Stress [σ_s]:

$$= | \sigma_{s, m} | + | \sigma_{s, b} | \text{ (Should be } \leq 1.5 \cdot S_s)$$

$$= 387993.625 \text{ kPa (Should be } \leq 221983.11)$$

Note: For a given Channel thickness of: 50.000 mm
Min. Channel len. adjacent to the tubesheet is: 657.185mm

Axial Channel Membrane Stress [σ_c, m]:

$$= D_c^2 / (4 \cdot t_c (D_c + t_c) \cdot \cos(\theta)) \cdot P_t$$

$$= 2666^2 / (4 \cdot 50 (2666 + 50) \cdot \cos(0)) \cdot 0$$

$$= 0.000 \text{ kPa}$$

Axial Channel Bending Stress [σ_c, b]:

$$= 6 \cdot k_c / t_c^2 \cdot [(\delta_c \cdot C_m \Delta + \beta_c - \zeta_c \cdot C_m \theta) \cdot P_t -$$

$$6((1 - \nu^*) / E^*) \cdot Do / h^3 \cdot (C_m \theta +$$

$$h \cdot \beta_c / 2 \cdot C_m \Delta) \cdot (M_p + Do^2 / 32 \cdot (P_s - P_t))]$$

$$= 6 \cdot 4648307 / 50^2 \cdot [(0.000165527 \cdot 1 \cdot 0.0049 - 0.000 \cdot 1) \cdot 0 -$$

$$6((1 - 0.42) / 34429) \cdot 2571 / 500^3 \cdot (1 +$$

$$0.0049 \cdot 500 / 2 \cdot 1) \cdot (-2517495 + 2571^2 / 32 \cdot (14.3 - 0))]$$

$$= -99198.773 \text{ kPa}$$

Channel Membrane + Bending Stress [σ_c]:

$$= | \sigma_{c, m} | + | \sigma_{c, b} | \text{ (Should be } \leq 1.5 \cdot S_c)$$

$$= 99198.773 \text{ kPa (Should be } \leq 221983.11)$$

Step 11, The Cylinder-to-Tubesheet Junction is Overstressed.
Performing a simplified Elastic-Plastic calculation
(option 3) to reduce the overstress condition.
Modify E_s and/or E_c and recompute from Step 4 onwards.

Modified Elasticity [E_s]:

$$= E_s \cdot \sqrt{(1.5 \cdot S_s / \sigma_s)}$$

$$= 183094 \cdot \sqrt{(1.5 \cdot 147989 / 387994)}$$

$$= 138490.9 \text{ N/mm}^2$$

Tubesheet Bending Stress at Original Thk., after the Elastic-Plastic iteration:

$$= 6 \cdot M / (\mu^* (h - h'g)^2)$$

$$= 6 \cdot 2597228 / (0.23 (500 - 5)^2)$$

$$= 280010.5 \text{ kPa (Should be } \leq 295977.5)$$



Note: The tubesheet is Not overstressed after the Elastic-Plastic iteration.
The design is acceptable. Recomputing tubesheet required thickness.

Required Tubesheet Thk., for Bending Stress after the Elastic-Plastic iteration [H_{reqB}]:



PV Elite 28 Licensee: SL User

FileName : 103-C2

Page 168 of 192

ASME TS Calc: DESIGN CASE

Case: 1 3:14p Apr 10, 2026

$$= h + \text{Cats} + \text{Catc} = 486 + 0 + 0 = 486 \text{ mm}$$

Required Tubesheet Thk. after the Elastic-Plastic iteration (includes CA) [Hreq]:

$$= \max(\text{HreqB}; \text{HreqS}) = \max(486; 364) = 486 \text{ mm}$$

Tube Metal Area [At]:

$$= \pi (d_o - t) t$$

$$= \pi (25.4 - 3.05) 3.05$$

$$= 0.000 \text{ m}^2$$

Tube Weld Size Results per UW-20.3:

Tube Strength [Ft]:

$$= \pi \cdot t (d_o - t) S$$

$$= \pi \cdot 3.05 (25.4 - 3.05) 94543$$

$$= 20234.723 \text{ N}$$

Groove Weld Strength [Fg]:

$$= 0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w \text{ (but not } > \text{ Ft)}$$

$$= 0.85 \cdot \pi \cdot 3 (25.4 + 0.67 \cdot 3) 94543$$

$$= 20234.7227 \text{ N}$$

Weld Strength Factor [fw]:

$$= S / S_w \text{ where } S_w = \min(S \text{ and } S_t)$$

$$= 94543/94543$$

$$= 1.000$$

Design Strength Ratio [fd]:

$$= 1 \text{ for full strength welds}$$

Fillet Weld Strength [Ff]:

$$= 0.55 \cdot \pi \cdot a_f (d_o + 0.67 \cdot a_f) S_w \text{ (but not } > \text{ Ft)}$$

$$= 0.55 \cdot \pi \cdot 3 (25.4 + 0.67 \cdot 3) 94543$$

$$= 13432.5635 \text{ N}$$

Minimum Length of Weld [ar]:

$$= 2 \left(\sqrt{((0.75 \cdot d_o)^2 + 1.07 \cdot t \cdot (d_o - t) f_w \cdot f_d) - 0.75 \cdot d_o} \right)$$

$$= 2 \left(\sqrt{((0.75 \cdot 25.4)^2 + 1.07 \cdot 3.05 \cdot (25.4 - 3.05) 1 \cdot 1) - 0.75 \cdot 25.4} \right)$$

$$= 3.652 \text{ mm}$$

Minimum Required Fillet Weld Leg [afr]: 1.8288 mm

Minimum Required Groove Weld Leg [agr]: 1.8288 mm

Tube-Tubesheet Jt allowable, 20235 is $\geq$ tube strength 20235 N

Note: This tube-tubesheet joint is a Full Strength joint

Stress/Force summary for loadcase D2 un-corr. (Psd,max + Ptd,min):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet bend. stress	280010.5	$\leq$ 295977.5 kPa	Ok
Tubesheet shear stress	86077.7	$\leq$ 118391.0 kPa	Ok
Stress in Shell at Tubesheet	387993.6	$\leq$ 478235.3 kPa	Ok
Stress in Channel at Tubesheet	99198.8	$\leq$ 221983.1 kPa	Ok

Thickness results for loadcase D2 un-corr. (Psd,max + Ptd,min):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	486.326	500.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.829	3.000	Ok



PV Elite 28 Licensee: SL User

FileName : 103-C2

Page 169 of 192

ASME TS Calc: DESIGN CASE

Case: 1 3:14p Apr 10, 2026

Tube-Tubesheet Groove Weld Leg :	1.829	3.000	Ok
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Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)

Results for ASME U-tube Tubesheet Calculations for Configuration a:

Per Edition 2025, Original Thickness :

ASME Section VIII-2 Part 4.18.7.4 Step 1:

Equivalent outer tube limit circle diameter [Do]:

$$= 2 \cdot r_o + d_t$$

$$= 2 \cdot 1273 + 25.4$$

$$= 2570.920 \text{ mm}$$

Basic ligament efficiency for shear [μ]:

$$= (p - d_t) / p$$

$$= (31.8 - 25.4) / 31.8$$

$$= 0.200$$

ASME Section VIII-2 Part 4.18.7.4 Step 2:

Ratio [ps]:

$$= D_s / D_o \text{ (Configurations a, b, c)}$$

$$= 2660 / 2571$$

$$= 1.0346$$

Ratio [pc]:

$$= D_c / D_o \text{ (Configurations a, e, f)}$$

$$= 2666 / 2571$$

$$= 1.0370$$

ASME Section VIII-2 Part 4.18.7.4 Step 3:

Ratio [p]:

$$= l_{tx} / h$$

$$= 50 / 500$$

$$= 0.100 \text{ (must be } 0 \leq p \leq 1 \text{)}$$

Effective tube hole Diameter [d*]:

$$= \max(d_t - 2t_t(E_t/E) \cdot (S_t/S) \cdot (p); d_t - 2t_t)$$

$$= \max(25.4 - 2 \cdot 3.05 (172742/183094) \cdot (94543/147989) \cdot (0.1); 25.4 - 2 \cdot 3.05)$$

$$= 25.0326 \text{ mm}$$

Effective tube pitch [p*]:

$$= p / \sqrt{ 1 - 4 \cdot \min(A_l; 4 \cdot D_o \cdot p) / (\pi \cdot D_o^2) }$$

$$= 31.8 / \sqrt{ 1 - 4 \cdot \min(0.2; 4 \cdot 2571 \cdot 31.8) / (\pi \cdot 2571^2) }$$

$$= 32.3886 \text{ mm}$$

Effective ligament efficiency for bending [μ^*]:

$$= (p^* - d^*) / p^*$$

$$= (32.4 - 25) / 32.4$$

$$= 0.2271$$

$h/p = 15.748032$; $\mu^* = 0.227118$; $E^* = 34428 \text{ N/mm}^2$

$E^*/E = 0.188040$; $v^* = 0.422203$

Note: As h/p (15.748) is > 2 , data values for $h/p = 2$ were used.

ASME Section VIII-2 Part 4.18.7.4 Step 4:



PV Elite 28 Licensee: SL User

FileName : 103-C2

Page 170 of 192

ASME TS Calc: DESIGN CASE

Case: 1 3:14p Apr 10, 2026

Shell coefficient [β_s]:

$$= \left(\frac{12(1 - \nu_s^2)}{E_s} \right)^{1/4} / \sqrt{\frac{(D_s + t_s)t_s}{6}}$$

$$= \left(\frac{12(1 - 0.3^2)}{183094} \right)^{1/4} / \sqrt{\frac{(2660 + 142)142}{6}}$$

$$= 0.0029 \text{ 1/mm}$$

Shell coefficient [k_s]:

$$= \beta_s \cdot E_s \cdot t_s^3 / (6(1 - \nu_s^2))$$

$$= 0.0029 \cdot 183094 \cdot 142^3 / (6(1 - 0.3^2))$$

$$= 276708960.0000 \text{ N/mm}^2 \cdot \text{mm}^2$$

Shell coefficient [λ_s]:

$$= 6(D_s + t_s) \cdot k_s / h^3 \cdot (1 + h \cdot \beta_s + (h \cdot \beta_s)^2 / 2)$$

$$= 6(2660 + 142) \cdot 276708960 / 500^3 \cdot (1 + 500 \cdot 0.0029 + (500 \cdot 0.0029)^2 / 2)$$

$$= 129479.3 \text{ N/mm}^2$$

Shell coefficient [δ_s]:

$$= D_s^2 / (4 \cdot E_s \cdot t_s) \cdot (1 - D_s \cdot \nu_s / (2(D_s + t_s)))$$

$$= 2660^2 / (4 \cdot 183094 \cdot 142) \cdot (1 - 2660 \cdot 0.3 / (2(2660 + 142)))$$

$$= 0.58352\text{E-}04 \text{ mm/kPa}$$

Parameter [ω_s]:

$$= \rho_s \cdot k_s \cdot \beta_s \cdot \delta_s (1 + h \cdot \beta_s)$$

$$= 1.03 \cdot 276708960 \cdot 0.0029 \cdot 0.58352\text{E-}04 (1 + 500 \cdot 0.0029)$$

$$= 117511.0781 \text{ mm}^2$$

Channel coefficient [β_c]:

$$= \left(\frac{12(1 - \nu_c^2)}{E_c} \right)^{1/4} / \sqrt{\frac{(D_c + t_c)t_c}{6}}$$

$$= \left(\frac{12(1 - 0.3^2)}{183094} \right)^{1/4} / \sqrt{\frac{(2666 + 50)50}{6}}$$

$$= 0.0049 \text{ 1/mm}$$

Channel coefficient [k_c]:

$$= \beta_c \cdot E_c \cdot t_c^3 / (6(1 - \nu_c^2))$$

$$= 0.0049 \cdot 183094 \cdot 50^3 / (6(1 - 0.3^2))$$

$$= 20677430.0000 \text{ N/mm}^2 \cdot \text{mm}^2$$

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:

$$CqDelta = CqTheta = CmTheta = CmDelta = 1$$

Channel coefficient [λ_c]:

$$= 6(D_c + t_c / \cos(\theta)) \cdot k_c / h^3 \cdot (CmTheta + h \cdot \beta_c \cdot (CmDelta + CqTheta) / 2 + (h \cdot \beta_c)^2 / 2 \cdot CqDelta)$$

$$= 6(2666 + 50 / \cos(0)) \cdot 20677430 / 500^3 \cdot (1 + 500 \cdot 0.0049 \cdot (1 + 1) / 2 + (500 \cdot 0.0049)^2 / 2 \cdot 1)$$

$$= 17544.0 \text{ N/mm}^2$$

Channel coefficient [δ_c]:

$$= D_c^2 / (4 \cdot E_c \cdot t_c) \cdot (1 - D_c \cdot \nu_c / (2(D_c + t_c)))$$

$$= 2666^2 / (4 \cdot 183094 \cdot 50) \cdot (1 - 2666 \cdot 0.3 / (2(2666 + 50)))$$

$$= 0.16553\text{E-}03 \text{ mm/kPa}$$

Channel coefficient [zetac]:

$$\text{zetac} = 0$$

Parameter [ω_c]:

$$= \rho_c \cdot k_c [\delta_c (CmTheta \cdot \beta_c + CqDelta \cdot h \cdot \beta_c^2) - \zeta_c (CmTheta + h \cdot CqTheta \cdot h \cdot \beta_c / 2)]$$

$$= 1.04 \cdot 20677430 [0.000165527 (1 + 0.0049 \cdot 1 \cdot 500) - 0.000 (1/500 + 1 \cdot 0.0049/2)]$$

$$= 60688.4023 \text{ mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 5:

Diameter ratio [K]:

$$= A / D_o$$

$$= 2980 / 2571$$



PV Elite 28 Licensee: SL User

FileName : 103-C2

Page 171 of 192

ASME TS Calc: DESIGN CASE

Case: 1 3:14p Apr 10, 2026

$$= 1.1591$$

Coefficient [F]:

$$\begin{aligned} &= (1 - \nu^*) / E^* \cdot (\lambda_s + \lambda_c + E \cdot \ln(K)) \\ &= (1 - 0.42) / 34429 \cdot (129479 + 17544 + 183094 \cdot \ln(1.16)) \\ &= 2.9211 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 6:

Moment on tubesheet due to pressures (Ps, Pt) [Mts]:

$$\begin{aligned} &= Do^2/16 \cdot [(ps-1) \cdot (ps^2+1) \cdot Ps - (pc-1) \cdot (pc^2+1) \cdot Pt] \\ &= 2571^2/16 \cdot [(1.03 - 1) \cdot (1.03^2 + 1) \cdot 14.3 - (1.04 - 1) \cdot (1.04^2 + 1) \cdot 4.95] \\ &= 266276.0312 \text{ N/mm}^2 \cdot \text{mm}^2 \end{aligned}$$

Moment acting on unperforated tubesheet rim [M*]:

$$\begin{aligned} &= Mts + \omega_c \cdot Pt - \omega_s \cdot Ps \\ &= 266276 + 60688 \cdot 4.95 - 117511 \cdot 14.3 \\ &= -1111454.7500 \text{ N/mm}^2 \cdot \text{mm}^2 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 7:

Maximum bending moment acting on periphery of tubesheet [Mp]:

$$\begin{aligned} &= (M^* - Do^2/32 \cdot F(Ps - Pt)) / (1 + F) \\ &= (-1111454) - 2571^2/32 \cdot 2.92 (14.3 - 4.95) / (1 + 2.92) \\ &= -1719184.2500 \text{ N/mm}^2 \cdot \text{mm}^2 \end{aligned}$$

Maximum bending moment acting on center of tubesheet [Mo]:

$$\begin{aligned} &= Mp + Do^2/64 \cdot (3 + \nu^*) \cdot (Ps - Pt) \\ &= -1719184 + 2571^2/64 \cdot (3 + 0.42) \cdot (14.3 - 4.95) \\ &= 1578505.1250 \text{ N/mm}^2 \cdot \text{mm}^2 \end{aligned}$$

Maximum bending moment acting on tubesheet [M]:

$$\begin{aligned} &= \max(|Mp|; |Mo|) \\ &= \max(|-1719184|; |1578505|) \\ &= 1719184.2500 \text{ N/mm}^2 \cdot \text{mm}^2 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 8:

Tubesheet Bending Stress at original thickness:

$$\begin{aligned} &= 6 \cdot M / (\mu^* (h - hg')^2) \\ &= 6 \cdot 1719184 / (0.23 (500 - 5)^2) \\ &= 185347.4531 \text{ kPa} \end{aligned}$$

Allowable Tubesheet Bending Stress [σ_{All}]:

$$\begin{aligned} &= 2 \cdot S \\ &= 2 \cdot 147989 \\ &= 295977.47 \text{ kPa} \end{aligned}$$

Tubesheet bending stress at final thickness [σ]:

$$\begin{aligned} &= 6 \cdot M / (\mu^* (h - hg')^2) \\ &= 6 \cdot 1744123 / (0.23 (397 - 5)^2) \\ &= 295924.3125 \text{ kPa} \end{aligned}$$

Required tubesheet thickness for Bending Stress [HreqB]:

$$\begin{aligned} &= h + Cats + Catc \\ &= 397 + 0 + 0 \\ &= 397.0499 \text{ mm} \end{aligned}$$

Required tubesheet thickness for given loadings (includes CA) [Hreq]:

$$\begin{aligned} &= \max(HreqB; HreqS) \\ &= \max(397; 238) \\ &= 397.0499 \text{ mm} \end{aligned}$$



PV Elite 28 Licensee: SL User

FileName : 103-C2

Page 172 of 192

ASME TS Calc: DESIGN CASE

Case: 1 3:14p Apr 10, 2026

ASME Section VIII-2 Part 4.18.7.4 Step 9:

Absolute pressure differential [$|P_s - P_t|$]:

$$= |P_s - P_t|$$

$$= |14.3 - 4.95|$$

$$= 9.331 \text{ N/mm}^2$$

Shear Stress check [τ_{limit}]:

$$= 4 \cdot \mu \cdot h / D_o \cdot \min(0.8 \cdot S; 0.533 \cdot S_y)$$

$$= 4 \cdot 0.2 \cdot 500 / 2571 \cdot \min(0.8 \cdot 147989; 0.533 \cdot 239106)$$

$$= 18420.020 \text{ kPa}$$

$|P_s - P_t|$ is $\leq$ the Shear Stress limit (τ_{limit}). Therefore, the required thickness for shear is for information only.

Minimum required thickness for Shear [H_{reqS}]:

$$= 1 / (4 \cdot \mu \cdot \min(0.8 \cdot S; 0.533 \cdot S_y)) \cdot (4 \cdot A_p / C_p) \cdot |P_s - P_t| + C_{ats} + C_{atc}$$

$$= 1 / (4 \cdot 0.2 \cdot 118391) \cdot (4 \cdot 4.95 / 8219) \cdot |14.3 - 4.95| + 0$$

$$= 237.5175 \text{ mm}$$

ASME Section VIII-2 Part 4.18.7.4 Step 10:

Note: For a given Shell thickness of: 142.000mm

Min. Shell len. adjacent to the tubesheet is: 1106.261 mm

Axial Shell Membrane Stress [$\sigma_{s,m}$]:

$$= D_s^2 / (4 \cdot t_s (D_s + t_s)) \cdot P_s$$

$$= 2660^2 / (4 \cdot 142 (2660 + 142)) \cdot 14.3$$

$$= 63485.5625 \text{ kPa}$$

Axial Shell Bending Stress [$\sigma_{s,b}$]:

$$= 6 \cdot k_s / t_s^2 \cdot [\beta S \cdot P_s + 6(1 - \nu^*) / E^* \cdot D_o / h^3 \cdot (1 + h \cdot \beta S / 2) \cdot (M_p + D_o^2 / 32 \cdot (P_s - P_t))]$$

$$= 6 \cdot 62204464 / 142^2 \cdot [0.0029 \cdot 0.83 + 6(1 - 0.42) / 34429 \cdot 2571 / 500^3 \cdot (1 + 500 \cdot 0.0029 / 2) \cdot (-1719184 + 2571^2 / 32 \cdot (14.3 - 4.95))]$$

$$= 258743.859 \text{ kPa}$$

Shell Membrane + Bending Stress [σ_s]:

$$= |\sigma_{s,m}| + |\sigma_{s,b}| \text{ (Should be } \leq 1.5 \cdot S_s)$$

$$= 322229.406 \text{ kPa (Should be } \leq 221983.11)$$

Note: For a given Channel thickness of: 50.000 mm

Min. Channel len. adjacent to the tubesheet is: 657.185mm

Axial Channel Membrane Stress [$\sigma_{c,m}$]:

$$= D_c^2 / (4 \cdot t_c (D_c + t_c)) \cdot \cos(\theta) \cdot P_t$$

$$= 2666^2 / (4 \cdot 50 (2666 + 50)) \cdot \cos(0) \cdot 4.95$$

$$= 64768.781 \text{ kPa}$$

Axial Channel Bending Stress [$\sigma_{c,b}$]:

$$= 6 \cdot k_c / t_c^2 \cdot [(\delta_c \cdot C_m \Delta \cdot \beta_c - \zeta_c \cdot C_m \theta) \cdot P_t - 6((1 - \nu^*) / E^*) \cdot D_o / h^3 \cdot (C_m \theta + h \cdot \beta_c / 2 \cdot C_m \Delta) \cdot (M_p + D_o^2 / 32 \cdot (P_s - P_t))]$$

$$= 6 \cdot 4648307 / 50^2 \cdot [(0.000165527 \cdot 1 \cdot 0.0049 - 0.000 \cdot 1) \cdot 4.95 - 6((1 - 0.42) / 34429) \cdot 2571 / 500^3 \cdot (1 + 0.0049 \cdot 500 / 2 \cdot 1) \cdot (-1719184 + 2571^2 / 32 \cdot (14.3 - 4.95))]$$

$$= 152819.984 \text{ kPa}$$

Channel Membrane + Bending Stress [σ_c]:

$$= |\sigma_{c,m}| + |\sigma_{c,b}| \text{ (Should be } \leq 1.5 \cdot S_c)$$

$$= 217588.781 \text{ kPa (Should be } \leq 221983.11)$$

Step 11, The Cylinder-to-Tubesheet Junction is Overstressed.
Performing a simplified Elastic-Plastic calculation



PV Elite 28 Licensee: SL User

FileName : 103-C2

Page 173 of 192

ASME TS Calc: DESIGN CASE

Case: 1 3:14p Apr 10, 2026

(option 3) to reduce the overstress condition.
Modify Es and/or Ec and recompute from Step 4 onwards.

Modified Elasticity [Es]:

$$= E_s \cdot \sqrt{(1.5 \cdot S_s / \sigma_s)}$$

$$= 183094 \cdot \sqrt{(1.5 \cdot 147989 / 322229)}$$

$$= 151967.5 \text{ N/mm}^2$$

Tubesheet Bending Stress at Original Thk., after the Elastic-Plastic iteration:

$$= 6 \cdot M / (\mu \cdot (h - h'g)^2)$$

$$= 6 \cdot 1697545 / (0.23 \cdot (500 - 5)^2)$$

$$= 183014.6 \text{ kPa (Should be } \leq 295977.5 \text{)}$$



Note: The tubesheet is Not overstressed after the Elastic-Plastic iteration.
The design is acceptable. Recomputing tubesheet required thickness.

Required Tubesheet Thk., for Bending Stress after the Elastic-Plastic iteration [HreqB]:

$$= h + C_{ats} + C_{atc} = 397 + 0 + 0 = 397 \text{ mm}$$

Required Tubesheet Thk. after the Elastic-Plastic iteration (includes CA) [Hreq]:

$$= \max(H_{reqB}; H_{reqS}) = \max(397; 238) = 397 \text{ mm}$$

Tube Metal Area [At]:

$$= \pi (d_o - t) t$$

$$= \pi (25.4 - 3.05) 3.05$$

$$= 0.000 \text{ m}^2$$

Tube Weld Size Results per UW-20.3:

Tube Strength [Ft]:

$$= \pi \cdot t (d_o - t) S$$

$$= \pi \cdot 3.05 (25.4 - 3.05) 94543$$

$$= 20234.723 \text{ N}$$

Groove Weld Strength [Fg]:

$$= 0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w \text{ (but not } > F_t)$$

$$= 0.85 \cdot \pi \cdot 3 (25.4 + 0.67 \cdot 3) 94543$$

$$= 20234.7227 \text{ N}$$

Weld Strength Factor [fw]:

$$= S / S_w \text{ where } S_w = \min(S \text{ and } S_t)$$

$$= 94543 / 94543$$

$$= 1.000$$

Design Strength Ratio [fd]:

$$= 1 \text{ for full strength welds}$$

Fillet Weld Strength [Ff]:

$$= 0.55 \cdot \pi \cdot a_f (d_o + 0.67 \cdot a_f) S_w \text{ (but not } > F_t)$$

$$= 0.55 \cdot \pi \cdot 3 (25.4 + 0.67 \cdot 3) 94543$$

$$= 13432.5635 \text{ N}$$

Minimum Length of Weld [ar]:

$$= 2 \left(\sqrt{((0.75 \cdot d_o)^2 + 1.07 \cdot t \cdot (d_o - t) f_w \cdot f_d)} - 0.75 \cdot d_o \right)$$

$$= 2 \left(\sqrt{((0.75 \cdot 25.4)^2 + 1.07 \cdot 3.05 \cdot (25.4 - 3.05) 1 \cdot 1)} - 0.75 \cdot 25.4 \right)$$

$$= 3.652 \text{ mm}$$

Minimum Required Fillet Weld Leg [afr]: 1.8288 mm

Minimum Required Groove Weld Leg [agr]: 1.8288 mm



PV Elite 28 Licensee: SL User

FileName : 103-C2

Page 174 of 192

ASME TS Calc: DESIGN CASE

Case: 1 3:14p Apr 10,2026

Tube-Tubesheet Jt allowable, 20235 is >= tube strength 20235 N

Note: This tube-tubesheet joint is a Full Strength joint

Stress/Force summary for loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet bend. stress	183014.6	<= 295977.5 kPa	Ok
Tubesheet shear stress	56239.9	<= 118391.0 kPa	Ok
Stress in Shell at Tubesheet	322229.4	<= 478235.3 kPa	Ok
Stress in Channel at Tubesheet	217588.8	<= 221983.1 kPa	Ok

Thickness results for loadcase D3 un-corr. (Psd,max + Ptd,max):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	397.050	500.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.829	3.000	Ok
Tube-Tubesheet Groove Weld Leg :	1.829	3.000	Ok

Detailed Results for load Case D1 corr. (Psd,min + Ptd,max)

Results for ASME U-tube Tubesheet Calculations for Configuration a:

Per Edition 2025, Original Thickness :

ASME Section VIII-2 Part 4.18.7.4 Step 1:

Equivalent outer tube limit circle diameter [Do]:

$$= 2 \cdot r_o + dt$$

$$= 2 \cdot 1273 + 25.4$$

$$= 2570.920 \text{ mm}$$

Basic ligament efficiency for shear [μ]:

$$= (p - dt) / p$$

$$= (31.8 - 25.4) / 31.8$$

$$= 0.200$$

ASME Section VIII-2 Part 4.18.7.4 Step 2:

Ratio [ps]:

$$= D_s / D_o \text{ (Configurations a, b, c)}$$

$$= 2666 / 2571$$

$$= 1.0370$$

Ratio [pc]:

$$= D_c / D_o \text{ (Configurations a, e, f)}$$

$$= 2670 / 2571$$

$$= 1.0385$$

ASME Section VIII-2 Part 4.18.7.4 Step 3:

Ratio [p]:

$$= l_{tx} / h$$

$$= 50 / 484$$

$$= 0.103 \text{ (must be } 0 \leq p \leq 1 \text{)}$$

Effective tube hole Diameter [d*]:

$$= \max(dt - 2tt(E_t/E) \cdot (StT/S) \cdot (p); dt - 2tt)$$

$$= \max(25.4 - 2 \cdot 3.05 (172742 / 183094) \cdot$$



PV Elite 28 Licensee: SL User

FileName : 103-C2

Page 175 of 192

ASME TS Calc: DESIGN CASE

Case: 1 3:14p Apr 10, 2026

$$\begin{aligned} & (94543/147989) \cdot (0.1); 25.4 - 2 \cdot 3.05 \\ & = 25.0204 \text{ mm} \end{aligned}$$

Effective tube pitch [p*]:

$$\begin{aligned} & = p / \sqrt{(1 - 4 \cdot \min(AI; 4 \cdot Do \cdot p) / (\pi \cdot Do^2))} \\ & = 31.8 / \sqrt{(1 - 4 \cdot \min(0.2; 4 \cdot 2571 \cdot 31.8) / (\pi \cdot 2571^2))} \\ & = 32.3886 \text{ mm} \end{aligned}$$

Effective ligament efficiency for bending [μ*]:

$$\begin{aligned} & = (p^* - d^*) / p^* \\ & = (32.4 - 25) / 32.4 \\ & = 0.2275 \end{aligned}$$

$$h/p = 15.244095; \mu^* = 0.227493; E^* = 34513 \text{ N/mm}^2$$

$$E^*/E = 0.188502; \nu^* = 0.421746$$

Note: As h/p (15.244) is > 2, data values for h/p = 2 were used.

ASME Section VIII-2 Part 4.18.7.4 Step 4:

Shell coefficient [βs]:

$$\begin{aligned} & = ((12(1 - \nu_s^2))^{1/4} / \sqrt{(Ds + ts)ts}) \\ & = ((12(1 - 0.3^2))^{1/4} / \sqrt{(2666 + 139)139}) \\ & = 0.0029 \text{ 1/mm} \end{aligned}$$

Shell coefficient [ks]:

$$\begin{aligned} & = \beta_s \cdot E_s \cdot ts^3 / (6(1 - \nu_s^2)) \\ & = 0.0029 \cdot 183094 \cdot 139^3 / (6(1 - 0.3^2)) \\ & = 262184528.0000 \text{ N/mm}^2 \cdot \text{mm}^2 \end{aligned}$$

Shell coefficient [λs]:

$$\begin{aligned} & = 6(Ds + ts) \cdot ks / h^3 \cdot (1 + h \cdot \beta_s + (h \cdot \beta_s)^2/2) \\ & = 6(2666 + 139) \cdot 262184528 / 484^3 \cdot (1 + 484 \cdot 0.0029 + (484 \cdot 0.0029)^2/2) \\ & = 132391.4 \text{ N/mm}^2 \end{aligned}$$

Shell coefficient [δs]:

$$\begin{aligned} & = Ds^2 / (4 \cdot E_s \cdot ts) \cdot (1 - Ds \cdot \nu_s / (2(Ds + ts))) \\ & = 2666^2 / (4 \cdot 183094 \cdot 139) \cdot (1 - 2666 \cdot 0.3 / (2(2666 + 139))) \\ & = 0.59868\text{E-}04 \text{ mm/kPa} \end{aligned}$$

Parameter [ωs]:

$$\begin{aligned} & = \rho_s \cdot ks \cdot \beta_s \cdot \delta_s (1 + h \cdot \beta_s) \\ & = 1.04 \cdot 262184528 \cdot 0.0029 \cdot 0.59868\text{E-}04 (1 + 484 \cdot 0.0029) \\ & = 114150.8047 \text{ mm}^2 \end{aligned}$$

Channel coefficient [βc]:

$$\begin{aligned} & = ((12(1 - \nu_c^2))^{1/4} / \sqrt{(Dc + tc)tc}) \\ & = ((12(1 - 0.3^2))^{1/4} / \sqrt{(2670 + 48)48}) \\ & = 0.0050 \text{ 1/mm} \end{aligned}$$

Channel coefficient [kc]:

$$\begin{aligned} & = \beta_c \cdot E_c \cdot tc^3 / (6(1 - \nu_c^2)) \\ & = 0.005 \cdot 183094 \cdot 48^3 / (6(1 - 0.3^2)) \\ & = 18664434.0000 \text{ N/mm}^2 \cdot \text{mm}^2 \end{aligned}$$

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:

$$CqDelta = CqTheta = CmTheta = CmDelta = 1$$

Channel coefficient [λc]:

$$\begin{aligned} & = 6(Dc + tc / \cos(\theta)) \cdot kc / h^3 \cdot (CmTheta + \\ & \quad h \cdot \beta_c \cdot (Cm\Delta + CqTheta) / 2 + (h \cdot \beta_c)^2/2 \cdot Cq\Delta) \\ & = 6(2670 + 48 / \cos(0)) \cdot 18664434 / 484^3 \cdot (1 + \\ & \quad 484 \cdot 0.005 \cdot (1 + 1) / 2 + (484 \cdot 0.005)^2/2 \cdot 1) \\ & = 17188.5 \text{ N/mm}^2 \end{aligned}$$



PV Elite 28 Licensee: SL User

FileName : 103-C2

Page 176 of 192

ASME TS Calc: DESIGN CASE

Case: 1 3:14p Apr 10, 2026

Channel coefficient [δ_c]:

$$= D_c^2 / (4 \cdot E_c \cdot t_c) \cdot (1 - D_c \cdot v_c / (2(D_c + t_c)))$$

$$= 2670^2 / (4 \cdot 183094 \cdot 48) \cdot (1 - 2670 \cdot 0.3 / (2(2670 + 48)))$$

$$= 0.17292E-03 \text{ mm/kPa}$$

Channel coefficient [zetac]:

$$\text{zetac} = 0$$

Parameter [ω_c]:

$$= \rho_c \cdot k_c [\delta_c (C_m \Theta + \beta_c + C_q \Delta \cdot h \cdot \beta_c^2) - \zeta_c (C_m \Theta + h \cdot C_q \Theta \cdot h \cdot \beta_c / 2)]$$

$$= 1.04 \cdot 18664434 [0.000172919 (1 + 0.005 \cdot 1 \cdot 484) - 0.000 (1/484 + 1 \cdot 0.005/2)]$$

$$= 57956.7070 \text{ mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 5:

Diameter ratio [K]:

$$= A / D_o$$

$$= 2980/2571$$

$$= 1.1591$$

Coefficient [F]:

$$= (1 - v^*) / E^* \cdot (\lambda_s + \lambda_c + E \cdot \ln(K))$$

$$= (1 - 0.42) / 34514 \cdot (132391 + 17188 + 183094 \cdot \ln(1.16))$$

$$= 2.9591$$

ASME Section VIII-2 Part 4.18.7.4 Step 6:

Moment on tubesheet due to pressures (Ps, Pt) [Mts]:

$$= D_o^2 / 16 \cdot [(p_s - 1) \cdot (p_s^2 + 1) \cdot P_s - (p_c - 1) \cdot (p_c^2 + 1) \cdot P_t]$$

$$= 2571^2 / 16 \cdot [(1.04 - 1) \cdot (1.04^2 + 1) \cdot 0 - (1.04 - 1) \cdot (1.04^2 + 1) \cdot 4.95]$$

$$= -163812.7344 \text{ N/mm}^2 \cdot \text{mm}^2$$

Moment acting on unperforated tubesheet rim [M*]:

$$= M_{ts} + \omega_c \cdot P_t - \omega_s \cdot P_s$$

$$= -163813 + 57957 \cdot 4.95 - 114151 \cdot 0$$

$$= 123089.6094 \text{ N/mm}^2 \cdot \text{mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 7:

Maximum bending moment acting on periphery of tubesheet [Mp]:

$$= (M^*) - D_o^2 / 32 \cdot F (P_s - P_t) / (1 + F)$$

$$= (123090) - 2571^2 / 32 \cdot 2.96 (0 - 4.95) / (1 + 2.96)$$

$$= 795313.8125 \text{ N/mm}^2 \cdot \text{mm}^2$$

Maximum bending moment acting on center of tubesheet [Mo]:

$$= M_p + D_o^2 / 64 \cdot (3 + v^*) \cdot (P_s - P_t)$$

$$= 795314 + 2571^2 / 64 \cdot (3 + 0.42) \cdot (0 - 4.95)$$

$$= -954030.1875 \text{ N/mm}^2 \cdot \text{mm}^2$$

Maximum bending moment acting on tubesheet [M]:

$$= \max(|M_p|; |M_o|)$$

$$= \max(|795314|; |-954030|)$$

$$= 954030.1875 \text{ N/mm}^2 \cdot \text{mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 8:

Tubesheet Bending Stress at original thickness:

$$= 6 \cdot M / (\mu^* (h - h_g')^2)$$

$$= 6 \cdot 954030 / (0.23 (484 - 0)^2)$$

$$= 107406.2266 \text{ kPa}$$



PV Elite 28 Licensee: SL User

FileName : 103-C2

Page 177 of 192

ASME TS Calc: DESIGN CASE

Case: 1 3:14p Apr 10, 2026

Allowable Tubesheet Bending Stress [σ_{All}]:

$$= 2 \cdot S$$

$$= 2 \cdot 147989$$

$$= 295977.47 \text{ kPa}$$

Tubesheet bending stress at final thickness [σ]:

$$= 6 \cdot M / (\mu \cdot (h - h_g')^2)$$

$$= 6 \cdot 914770 / (0.24 \cdot (280 - 0)^2)$$

$$= 295888.2500 \text{ kPa}$$

Required tubesheet thickness for Bending Stress [H_{reqB}]:

$$= h + Cats + Catc$$

$$= 280 + 3 + 13$$

$$= 296.3454 \text{ mm}$$

Required tubesheet thickness for given loadings (includes CA) [H_{req}]:

$$= \max(H_{reqB}; H_{reqS})$$

$$= \max(296 ; 142)$$

$$= 296.3454 \text{ mm}$$

ASME Section VIII-2 Part 4.18.7.4 Step 9:

Absolute pressure differential [$|P_s - P_t|$]:

$$= |P_s - P_t|$$

$$= |0 - 4.95|$$

$$= 4.950 \text{ N/mm}^2$$

Shear Stress check [τ_{limit}]:

$$= 4 \cdot \mu \cdot h / Do \cdot \min(0.8 \cdot S; 0.533 \cdot S_y)$$

$$= 4 \cdot 0.2 \cdot 484 / 2571 \cdot \min(0.8 \cdot 147989 ; 0.533 \cdot 239106)$$

$$= 17830.578 \text{ kPa}$$

$|P_s - P_t|$ is $\leq$ the Shear Stress limit (τ_{limit}). Therefore, the required thickness for shear is for information only.

Minimum required thickness for Shear [H_{reqS}]:

$$= 1 / (4 \cdot \mu \cdot \min(0.8 \cdot S; 0.533 \cdot S_y)) \cdot (4 \cdot A_p / C_p) \cdot |P_s - P_t| + Cats + Catc$$

$$= 1 / (4 \cdot 0.2 \cdot 118391) \cdot (4 \cdot 4.95 / 8219) \cdot |0 - 4.95| + 16$$

$$= 142.0141 \text{ mm}$$

ASME Section VIII-2 Part 4.18.7.4 Step 10:

Note: For a given Shell thickness of: 139.000mm

Min. Shell len. adjacent to the tubesheet is: 1095.746 mm

Axial Shell Membrane Stress [$\sigma_{s,m}$]:

$$= D_s^2 / (4 \cdot t_s (D_s + t_s)) \cdot P_s$$

$$= 2666^2 / (4 \cdot 139 (2666 + 139)) \cdot 0$$

$$= 0.0000 \text{ kPa}$$

Axial Shell Bending Stress [$\sigma_{s,b}$]:

$$= 6 \cdot k_s / t_s^2 \cdot [\beta S \cdot P_s + 6(1 - \nu^*) / E^* \cdot Do / h^3 \cdot (1 + h \cdot \beta S / 2) \cdot (M_p + Do^2 / 32 \cdot (P_s - P_t))]$$

$$= 6 \cdot 58939352 / 139^2 \cdot [0.0029 \cdot 0 + 6(1 - 0.42) / 34514 \cdot 2571 / 484^3 \cdot (1 + 484 \cdot 0.0029 / 2) \cdot (795314 + 2571^2 / 32 \cdot (0 - 4.95))]$$

$$= -71861.547 \text{ kPa}$$

Shell Membrane + Bending Stress [σ_s]:

$$= | \sigma_{s,m} | + | \sigma_{s,b} | \text{ (Should be } \leq 1.5 \cdot S_s)$$

$$= 71861.547 \text{ kPa (Should be } \leq 221983.11)$$

Note: For a given Channel thickness of: 48.000 mm

Min. Channel len. adjacent to the tubesheet is: 644.390mm



PV Elite 28 Licensee: SL User

FileName : 103-C2

Page 178 of 192

ASME TS Calc: DESIGN CASE

Case: 1 3:14p Apr 10, 2026

Axial Channel Membrane Stress [σ_c, m]:

$$= D_c^2 / (4 \cdot t_c (D_c + t_c) \cdot \cos(\theta)) \cdot P_t$$

$$= 2670^2 / (4 \cdot 48 (2670 + 48) \cdot \cos(0)) \cdot 4.95$$

$$= 67620.289 \text{ kPa}$$

Axial Channel Bending Stress [σ_c, b]:

$$= 6 \cdot k_c / t_c^2 \cdot [(\delta_c \cdot C_m \Delta \cdot \beta_c - \zeta_c \cdot C_m \theta) \cdot P_t - 6((1 - \nu^*) / E^*) \cdot D_o / h^3 \cdot (C_m \theta + h \cdot \beta_c / 2 \cdot C_m \Delta) \cdot (M_p + D_o^2 / 32 \cdot (P_s - P_t))]$$

$$= 6 \cdot 4195784 / 48^2 \cdot [(0.000172919 \cdot 1 \cdot 0.005 - 0.000 \cdot 1) \cdot 4.95 - 6((1 - 0.42) / 34514) \cdot 2571 / 484^3 \cdot (1 + 0.005 \cdot 484 / 2 \cdot 1) \cdot (795314 + 2571^2 / 32 \cdot (0 - 4.95))]$$

$$= 265192.312 \text{ kPa}$$

Channel Membrane + Bending Stress [σ_c]:

$$= | \sigma_{c,m} | + | \sigma_{c,b} | \text{ (Should be } \leq 1.5 \cdot S_c)$$

$$= 332812.594 \text{ kPa (Should be } \leq 221983.11)$$

Step 11, The Cylinder-to-Tubesheet Junction is Overstressed.

Performing a simplified Elastic-Plastic calculation (option 3) to reduce the overstress condition.

Modify E_s and/or E_c and recompute from Step 4 onwards.

Modified Elasticity [E_c]:

$$= E_c \cdot \sqrt{1.5 \cdot S_c / \sigma_c}$$

$$= 183094 \cdot \sqrt{1.5 \cdot 147989 / 332813}$$

$$= 149531.8 \text{ N/mm}^2$$

Tubesheet Bending Stress at Original Thk., after the Elastic-Plastic iteration:

$$= 6 \cdot M / (\mu \cdot (h - h'g)^2)$$

$$= 6 \cdot 957100 / (0.23 (484 - 0)^2)$$

$$= 107751.8 \text{ kPa (Should be } \leq 295977.5)$$



Note: The tubesheet is Not overstressed after the Elastic-Plastic iteration.
The design is acceptable. Recomputing tubesheet required thickness.

Required Tubesheet Thk., for Bending Stress after the Elastic-Plastic iteration [H_{reqB}]:

$$= h + C_{ats} + C_{atc} = 280 + 3 + 13 = 296 \text{ mm}$$

Required Tubesheet Thk. after the Elastic-Plastic iteration (includes CA) [H_{req}]:

$$= \max(H_{reqB}, H_{reqS}) = \max(296; 142) = 296 \text{ mm}$$

Tube Metal Area [A_t]:

$$= \pi (d_o - t) t$$

$$= \pi (25.4 - 3.05) 3.05$$

$$= 0.000 \text{ m}^2$$

Tube Weld Size Results per UW-20.3:

Tube Strength [F_t]:

$$= \pi \cdot t (d_o - t) S$$

$$= \pi \cdot 3.05 (25.4 - 3.05) 94543$$

$$= 20234.723 \text{ N}$$

Groove Weld Strength [F_g]:

$$= 0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w \text{ (but not } > F_t)$$

$$= 0.85 \cdot \pi \cdot 3 (25.4 + 0.67 \cdot 3) 94543$$

$$= 20234.7227 \text{ N}$$

Weld Strength Factor [f_w]:

$$= S / S_w \text{ where } S_w = \min(S \text{ and } S_t)$$



PV Elite 28 Licensee: SL User

FileName : 103-C2

Page 179 of 192

ASME TS Calc: DESIGN CASE

Case: 1 3:14p Apr 10,2026

$$= 94543/94543$$

$$= 1.000$$

Design Strength Ratio [fd]:
= 1 for full strength welds

Fillet Weld Strength [Ff]:

$$= 0.55 \cdot \pi \cdot af (do + 0.67 \cdot af) Sw \text{ (but not } > Ft)$$

$$= 0.55 \cdot \pi \cdot 3 (25.4 + 0.67 \cdot 3) 94543$$

$$= 13432.5635 \text{ N}$$

Minimum Length of Weld [ar]:

$$= 2 \left(\sqrt{(0.75 \cdot do)^2 + 1.07 \cdot t \cdot (do - t) \cdot fw \cdot fd} - 0.75 \cdot do \right)$$

$$= 2 \left(\sqrt{(0.75 \cdot 25.4)^2 + 1.07 \cdot 3.05 \cdot (25.4 - 3.05) \cdot 1 \cdot 1} - 0.75 \cdot 25.4 \right)$$

$$= 3.652 \text{ mm}$$

Minimum Required Fillet Weld Leg [afr]: 1.8288 mm
Minimum Required Groove Weld Leg [agr]: 1.8288 mm

Tube-Tubesheet Jt allowable, 20235 is >= tube strength 20235 N

Note: This tube-tubesheet joint is a Full Strength joint

Stress/Force summary for loadcase D1 corr. (Psd,min + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet bend. stress	107751.8 <=	295977.5 kPa	Ok
Tubesheet shear stress	30824.2 <=	118391.0 kPa	Ok
Stress in Shell at Tubesheet	71861.5 <=	221983.1 kPa	Ok
Stress in Channel at Tubesheet	332812.6 <=	478235.3 kPa	Ok

Thickness results for loadcase D1 corr. (Psd,min + Ptd,max):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	296.345	500.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.829	3.000	Ok
Tube-Tubesheet Groove Weld Leg :	1.829	3.000	Ok

Detailed Results for load Case D2 corr. (Psd,max + Ptd,min)

Results for ASME U-tube Tubesheet Calculations for Configuration a:

Per Edition 2025, Original Thickness :

ASME Section VIII-2 Part 4.18.7.4 Step 1:

Equivalent outer tube limit circle diameter [Do]:

$$= 2 \cdot ro + dt$$

$$= 2 \cdot 1273 + 25.4$$

$$= 2570.920 \text{ mm}$$

Basic ligament efficiency for shear [μ]:

$$= (p - dt) / p$$

$$= (31.8 - 25.4) / 31.8$$

$$= 0.200$$

ASME Section VIII-2 Part 4.18.7.4 Step 2:



PV Elite 28 Licensee: SL User

FileName : 103-C2

Page 180 of 192

ASME TS Calc: DESIGN CASE

Case: 1 3:14p Apr 10, 2026

Ratio [ps]:

$$= D_s / D_o \text{ (Configurations a, b, c)}$$

$$= 2666/2571$$

$$= 1.0370$$

Ratio [pc]:

$$= D_c / D_o \text{ (Configurations a, e, f)}$$

$$= 2670/2571$$

$$= 1.0385$$

ASME Section VIII-2 Part 4.18.7.4 Step 3:

Ratio [p]:

$$= l_{tx} / h$$

$$= 50/484$$

$$= 0.103 \text{ (must be } 0 \leq p \leq 1 \text{)}$$

Effective tube hole Diameter [d*]:

$$= \max(dt - 2tt(E_t T/E) \cdot (StT/S) \cdot (p); dt - 2tt)$$

$$= \max(25.4 - 2 \cdot 3.05 (172742/183094) \cdot (94543/147989) \cdot (0.1); 25.4 - 2 \cdot 3.05)$$

$$= 25.0204 \text{ mm}$$

Effective tube pitch [p*]:

$$= p / \sqrt{ (1 - 4 \cdot \min(A_l; 4 \cdot D_o \cdot p) / (\pi \cdot D_o^2)) }$$

$$= 31.8 / \sqrt{ (1 - 4 \cdot \min(0.2; 4 \cdot 2571 \cdot 31.8) / (\pi \cdot 2571^2)) }$$

$$= 32.3886 \text{ mm}$$

Effective ligament efficiency for bending [μ^*]:

$$= (p^* - d^*) / p^*$$

$$= (32.4 - 25) / 32.4$$

$$= 0.2275$$

$h/p = 15.244095$; $\mu^* = 0.227493$; $E^* = 34513 \text{ N/mm}^2$

$E^*/E = 0.188502$; $\nu^* = 0.421746$

Note: As h/p (15.244) is > 2, data values for $h/p = 2$ were used.

ASME Section VIII-2 Part 4.18.7.4 Step 4:

Shell coefficient [β_s]:

$$= ((12(1 - \nu_s^2))^{1/4}) / \sqrt{ (D_s + t_s) t_s }$$

$$= ((12(1 - 0.3^2))^{1/4}) / \sqrt{ (2666 + 139) 139 }$$

$$= 0.0029 \text{ 1/mm}$$

Shell coefficient [k_s]:

$$= \beta_s \cdot E_s \cdot t_s^3 / (6(1 - \nu_s^2))$$

$$= 0.0029 \cdot 183094 \cdot 139^3 / (6(1 - 0.3^2))$$

$$= 262184528.0000 \text{ N/mm}^2 \cdot \text{mm}^2$$

Shell coefficient [λ_s]:

$$= 6(D_s + t_s) \cdot k_s / h^3 \cdot (1 + h \cdot \beta_s + (h \cdot \beta_s)^2 / 2)$$

$$= 6(2666 + 139) \cdot 262184528 / 484^3 \cdot (1 + 484 \cdot 0.0029 + (484 \cdot 0.0029)^2 / 2)$$

$$= 132391.4 \text{ N/mm}^2$$

Shell coefficient [δ_s]:

$$= D_s^2 / (4 \cdot E_s \cdot t_s) \cdot (1 - D_s \cdot \nu_s / (2(D_s + t_s)))$$

$$= 2666^2 / (4 \cdot 183094 \cdot 139) \cdot (1 - 2666 \cdot 0.3 / (2(2666 + 139)))$$

$$= 0.59868\text{E-}04 \text{ mm/kPa}$$

Parameter [ω_s]:

$$= p_s \cdot k_s \cdot \beta_s \cdot \delta_s (1 + h \cdot \beta_s)$$

$$= 1.04 \cdot 262184528 \cdot 0.0029 \cdot 0.59868\text{E-}04 (1 + 484 \cdot 0.0029)$$

$$= 114150.8047 \text{ mm}^2$$



PV Elite 28 Licensee: SL User

FileName : 103-C2

Page 181 of 192

ASME TS Calc: DESIGN CASE

Case: 1 3:14p Apr 10, 2026

Channel coefficient [β_c]:

$$= \left(\left(12(1 - \nu_c^2) \right)^{1/4} / \sqrt{(D_c + t_c)t_c} \right)$$

$$= \left(\left(12(1 - 0.3^2) \right)^{1/4} / \sqrt{(2670 + 48)48} \right)$$

$$= 0.0050 \text{ 1/mm}$$

Channel coefficient [k_c]:

$$= \beta_c \cdot E_c \cdot t_c^3 / (6(1 - \nu_s^2))$$

$$= 0.005 \cdot 183094 \cdot 48^3 / (6(1 - 0.3^2))$$

$$= 18664434.0000 \text{ N/mm}^2 \cdot \text{mm}^2$$

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:

$$CqDelta = CqTheta = CmTheta = CmDelta = 1$$

Channel coefficient [λ_c]:

$$= 6(D_c + t_c / \cos(\theta)) \cdot k_c / h^3 \cdot (CmTheta + h \cdot \beta_c \cdot (CmDelta + CqTheta) / 2 + (h \cdot \beta_c)^2 / 2 \cdot CqDelta)$$

$$= 6(2670 + 48 / \cos(0)) \cdot 18664434 / 48^3 \cdot (1 + 484 \cdot 0.005 \cdot (1 + 1) / 2 + (484 \cdot 0.005)^2 / 2 \cdot 1)$$

$$= 17188.5 \text{ N/mm}^2$$

Channel coefficient [δ_c]:

$$= D_c^2 / (4 \cdot E_c \cdot t_c) \cdot (1 - D_c \cdot \nu_c / (2(D_c + t_c)))$$

$$= 2670^2 / (4 \cdot 183094 \cdot 48) \cdot (1 - 2670 \cdot 0.3 / (2(2670 + 48)))$$

$$= 0.17292\text{E-}03 \text{ mm/kPa}$$

Channel coefficient [zetac]:

$$\text{zetac} = 0$$

Parameter [ω_c]:

$$= \rho_c \cdot k_c [\delta_c (CmTheta \cdot \beta_c + CqDelta \cdot h \cdot \beta_c^2) - \zeta_c (CmTheta + h \cdot CqTheta \cdot h \cdot \beta_c / 2)]$$

$$= 1.04 \cdot 18664434 [0.000172919 (1 + 0.005 \cdot 1 \cdot 484) - 0.000 (1/484 + 1 \cdot 0.005/2)]$$

$$= 57956.7070 \text{ mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 5:

Diameter ratio [K]:

$$= A / D_o$$

$$= 2980 / 2571$$

$$= 1.1591$$

Coefficient [F]:

$$= (1 - \nu_s) / E_s \cdot (\lambda_s + \lambda_c + E \cdot \ln(K))$$

$$= (1 - 0.42) / 34514 \cdot (132391 + 17188 + 183094 \cdot \ln(1.16))$$

$$= 2.9591$$

ASME Section VIII-2 Part 4.18.7.4 Step 6:

Moment on tubesheet due to pressures (Ps, Pt) [Mts]:

$$= D_o^2 / 16 \cdot [(\rho_s - 1) \cdot (\rho_s^2 + 1) \cdot P_s - (\rho_c - 1) \cdot (\rho_c^2 + 1) \cdot P_t]$$

$$= 2571^2 / 16 \cdot [(1.04 - 1) \cdot (1.04^2 + 1) \cdot 14.3 - (1.04 - 1) \cdot (1.04^2 + 1) \cdot 0]$$

$$= 452792.6562 \text{ N/mm}^2 \cdot \text{mm}^2$$

Moment acting on unperforated tubesheet rim [M*]:

$$= Mts + \omega_c \cdot P_t - \omega_s \cdot P_s$$

$$= 452793 + 57957 \cdot 0 - 114151 \cdot 14.3$$

$$= -1177375.6250 \text{ N/mm}^2 \cdot \text{mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 7:

Maximum bending moment acting on periphery of tubesheet [Mp]:

$$= (M^*) - D_o^2 / 32 \cdot F (P_s - P_t) / (1 + F)$$



PV Elite 28 Licensee: SL User

FileName : 103-C2

Page 182 of 192

ASME TS Calc: DESIGN CASE

Case: 1 3:14p Apr 10, 2026

$$= ((-1177375) - 2571^2/32 \cdot 2.96 (14.3 - 0)) / (1 + 2.96)$$

$$= -2502054.2500 \text{ N/mm}^2 \cdot \text{mm}^2$$

Maximum bending moment acting on center of tubesheet [Mo]:

$$= M_p + D_o^2/64 \cdot (3 + \nu) \cdot (P_s - P_t)$$

$$= -2502054 + 2571^2/64 \cdot (3 + 0.42) \cdot (14.3 - 0)$$

$$= 2544538.5000 \text{ N/mm}^2 \cdot \text{mm}^2$$

Maximum bending moment acting on tubesheet [M]:

$$= \max(| M_p | ; | M_o |)$$

$$= \max(| -2502054 | ; | 2544538 |)$$

$$= 2544538.5000 \text{ N/mm}^2 \cdot \text{mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 8:

Tubesheet Bending Stress at original thickness:

$$= 6 \cdot M / (\mu \cdot (h - h_g')^2)$$

$$= 6 \cdot 2544538 / (0.23 (484 - 0)^2)$$

$$= 286468.1875 \text{ kPa}$$

Allowable Tubesheet Bending Stress [σ_{All}]:

$$= 2 \cdot S$$

$$= 2 \cdot 147989$$

$$= 295977.47 \text{ kPa}$$

Tubesheet bending stress at final thickness [σ]:

$$= 6 \cdot M / (\mu \cdot (h - h_g')^2)$$

$$= 6 \cdot 2539216 / (0.23 (476 - 0)^2)$$

$$= 295108.1250 \text{ kPa}$$

Required tubesheet thickness for Bending Stress [HreqB]:

$$= h + C_{ats} + C_{atc}$$

$$= 476 + 3 + 13$$

$$= 492.1614 \text{ mm}$$

Required tubesheet thickness for given loadings (includes CA) [Hreq]:

$$= \max(H_{reqB} ; H_{reqS})$$

$$= \max(492 ; 380)$$

$$= 492.1614 \text{ mm}$$

ASME Section VIII-2 Part 4.18.7.4 Step 9:

Absolute pressure differential [$| P_s - P_t |$]:

$$= | P_s - P_t |$$

$$= | 14.3 - 0 |$$

$$= 14.281 \text{ N/mm}^2$$

Shear Stress check [τ_{limit}]:

$$= 4 \cdot \mu \cdot h / D_o \cdot \min(0.8 \cdot S ; 0.533 \cdot S_y)$$

$$= 4 \cdot 0.2 \cdot 484 / 2571 \cdot \min(0.8 \cdot 147989 ; 0.533 \cdot 239106)$$

$$= 17830.578 \text{ kPa}$$

$| P_s - P_t |$ is $\leq$ the Shear Stress limit (τ_{limit}). Therefore, the required thickness for shear is for information only.

Minimum required thickness for Shear [HreqS]:

$$= 1 / (4 \cdot \mu \cdot \min(0.8 \cdot S ; 0.533 \cdot S_y)) \cdot (4 \cdot A_p / C_p) \cdot | P_s - P_t | + C_{ats} + C_{atc}$$

$$= 1 / (4 \cdot 0.2 \cdot 118391) \cdot (4 \cdot 4.95 / 8219) \cdot | 14.3 - 0 | + 16$$

$$= 379.5316 \text{ mm}$$

ASME Section VIII-2 Part 4.18.7.4 Step 10:

Note: For a given Shell thickness of: 139.000mm

Min. Shell len. adjacent to the tubesheet is: 1095.746 mm



PV Elite 28 Licensee: SL User

FileName : 103-C2

Page 183 of 192

ASME TS Calc: DESIGN CASE

Case: 1 3:14p Apr 10, 2026

Axial Shell Membrane Stress [σ_s, m]:

$$= D_s^2 / (4 \cdot t_s (D_s + t_s)) \cdot P_s$$

$$= 2666^2 / (4 \cdot 139 (2666 + 139)) \cdot 14.3$$

$$= 65078.9961 \text{ kPa}$$

Axial Shell Bending Stress [σ_s, b]:

$$= 6 \cdot k_s / t_s^2 \cdot [\beta S \cdot P_s + 6(1 - \nu^*) / E^* \cdot$$

$$Do / h^3 \cdot (1 + h \cdot \beta S / 2) \cdot (M_p + Do^2 / 32 \cdot (P_s - P_t))]$$

$$= 6 \cdot 58939352 / 139^2 \cdot [0.0029 \cdot 0.85 + 6(1 - 0.42) / 34514 \cdot$$

$$2571 / 484^3 \cdot (1 + 484 \cdot 0.0029 / 2) \cdot (-2502054 + 2571^2 / 32 \cdot$$

$$(14.3 - 0))]$$

$$= 344243.125 \text{ kPa}$$

Shell Membrane + Bending Stress [σ_s]:

$$= | \sigma_{s,m} | + | \sigma_{s,b} | \text{ (Should be } \leq 1.5 \cdot S_s)$$

$$= 409322.094 \text{ kPa (Should be } \leq 221983.11)$$

Note: For a given Channel thickness of: 48.000 mm

Min. Channel len. adjacent to the tubesheet is: 644.390mm

Axial Channel Membrane Stress [σ_c, m]:

$$= D_c^2 / (4 \cdot t_c (D_c + t_c) \cdot \cos(\theta)) \cdot P_t$$

$$= 2670^2 / (4 \cdot 48 (2670 + 48) \cdot \cos(0)) \cdot 0$$

$$= 0.000 \text{ kPa}$$

Axial Channel Bending Stress [σ_c, b]:

$$= 6 \cdot k_c / t_c^2 \cdot [(\delta_c \cdot C_m \Delta \cdot \beta_c - \zeta_c \cdot C_m \theta) \cdot P_t -$$

$$6((1 - \nu^*) / E^*) \cdot Do / h^3 \cdot (C_m \theta +$$

$$h \cdot \beta_c / 2 \cdot C_m \Delta) \cdot (M_p + Do^2 / 32 \cdot (P_s - P_t))]$$

$$= 6 \cdot 4195784 / 48^2 \cdot [(0.000172919 \cdot 1 \cdot 0.005 - 0.000 \cdot 1) \cdot 0 -$$

$$6((1 - 0.42) / 34514) \cdot 2571 / 484^3 \cdot (1 +$$

$$0.005 \cdot 484 / 2 \cdot 1) \cdot (-2502054 + 2571^2 / 32 \cdot (14.3 - 0))]$$

$$= -110000.328 \text{ kPa}$$

Channel Membrane + Bending Stress [σ_c]:

$$= | \sigma_{c,m} | + | \sigma_{c,b} | \text{ (Should be } \leq 1.5 \cdot S_c)$$

$$= 110000.328 \text{ kPa (Should be } \leq 221983.11)$$

Step 11. The Cylinder-to-Tubesheet Juncture is Overstressed.

Performing a simplified Elastic-Plastic calculation

(option 3) to reduce the overstress condition.

Modify E_s and/or E_c and recompute from Step 4 onwards.

Modified Elasticity [E_s]:

$$= E_s \cdot \sqrt{(1.5 \cdot S_s / \sigma_s)}$$

$$= 183094 \cdot \sqrt{(1.5 \cdot 147989 / 409322)}$$

$$= 134834.4 \text{ N/mm}^2$$

Tubesheet Bending Stress at Original Thk., after the Elastic-Plastic iteration:

$$= 6 \cdot M / (\mu^* (h - h'g)^2)$$

$$= 6 \cdot 2622100 / (0.23 (484 - 0)^2)$$

$$= 295200.2 \text{ kPa (Should be } \leq 295977.5)$$



Note: The tubesheet is Not overstressed after the Elastic-Plastic iteration.

The design is acceptable. Recomputing tubesheet required thickness.

Required Tubesheet Thk., for Bending Stress after the Elastic-Plastic iteration [H_{reqB}]:

$$= h + C_{ats} + C_{atc} = 484 + 3 + 13 = 500 \text{ mm}$$

Required Tubesheet Thk. after the Elastic-Plastic iteration (includes CA) [H_{req}]:

$$= \max(H_{reqB}; H_{reqS}) = \max(500; 380) = 500 \text{ mm}$$



PV Elite 28 Licensee: SL User

FileName : 103-C2

Page 184 of 192

ASME TS Calc: DESIGN CASE

Case: 1 3:14p Apr 10, 2026

Tube Metal Area [At]:

$$= \pi (d_o - t) t$$

$$= \pi (25.4 - 3.05) 3.05$$

$$= 0.000 \text{ m}^2$$

Tube Weld Size Results per UW-20.3:

Tube Strength [Ft]:

$$= \pi \cdot t (d_o - t) S$$

$$= \pi \cdot 3.05 (25.4 - 3.05) 94543$$

$$= 20234.723 \text{ N}$$

Groove Weld Strength [Fg]:

$$= 0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w \text{ (but not } > F_t)$$

$$= 0.85 \cdot \pi \cdot 3 (25.4 + 0.67 \cdot 3) 94543$$

$$= 20234.7227 \text{ N}$$

Weld Strength Factor [fw]:

$$= S / S_w \text{ where } S_w = \min(S \text{ and } S_t)$$

$$= 94543 / 94543$$

$$= 1.000$$

Design Strength Ratio [fd]:

$$= 1 \text{ for full strength welds}$$

Fillet Weld Strength [Ff]:

$$= 0.55 \cdot \pi \cdot a_f (d_o + 0.67 \cdot a_f) S_w \text{ (but not } > F_t)$$

$$= 0.55 \cdot \pi \cdot 3 (25.4 + 0.67 \cdot 3) 94543$$

$$= 13432.5635 \text{ N}$$

Minimum Length of Weld [ar]:

$$= 2 \left(\sqrt{(0.75 \cdot d_o)^2 + 1.07 \cdot t \cdot (d_o - t) f_w \cdot f_d} - 0.75 \cdot d_o \right)$$

$$= 2 \left(\sqrt{(0.75 \cdot 25.4)^2 + 1.07 \cdot 3.05 \cdot (25.4 - 3.05) 1 \cdot 1} - 0.75 \cdot 25.4 \right)$$

$$= 3.652 \text{ mm}$$

Minimum Required Fillet Weld Leg [afr]: 1.8288 mm

Minimum Required Groove Weld Leg [agr]: 1.8288 mm

Tube-Tubesheet Jt allowable, 20235 is $\geq$ tube strength 20235 N

Note: This tube-tubesheet joint is a Full Strength joint

Stress/Force summary for loadcase D2 corr. (Psd,max + Ptd,min):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet bend. stress	295200.2	$\leq$ 295977.5 kPa	Ok
Tubesheet shear stress	88923.3	$\leq$ 118391.0 kPa	Ok
Stress in Shell at Tubesheet	409322.1	$\leq$ 478235.3 kPa	Ok
Stress in Channel at Tubesheet	110000.3	$\leq$ 221983.1 kPa	Ok

Thickness results for loadcase D2 corr. (Psd,max + Ptd,min):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	500.000	500.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.829	3.000	Ok
Tube-Tubesheet Groove Weld Leg :	1.829	3.000	Ok

Detailed Results for load Case D3 corr. (Psd,max + Ptd,max)



PV Elite 28 Licensee: SL User

FileName : 103-C2

Page 185 of 192

ASME TS Calc: DESIGN CASE

Case: 1 3:14p Apr 10, 2026

Results for ASME U-tube Tubesheet Calculations for Configuration a:

Per Edition 2025, Original Thickness :

ASME Section VIII-2 Part 4.18.7.4 Step 1:

Equivalent outer tube limit circle diameter [Do]:

$$\begin{aligned} &= 2 \cdot r_o + dt \\ &= 2 \cdot 1273 + 25.4 \\ &= 2570.920 \text{ mm} \end{aligned}$$

Basic ligament efficiency for shear [μ]:

$$\begin{aligned} &= (p - dt) / p \\ &= (31.8 - 25.4) / 31.8 \\ &= 0.200 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 2:

Ratio [ρ_s]:

$$\begin{aligned} &= D_s / D_o \text{ (Configurations a, b, c)} \\ &= 2666 / 2571 \\ &= 1.0370 \end{aligned}$$

Ratio [ρ_c]:

$$\begin{aligned} &= D_c / D_o \text{ (Configurations a, e, f)} \\ &= 2670 / 2571 \\ &= 1.0385 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 3:

Ratio [ρ]:

$$\begin{aligned} &= l_{tx} / h \\ &= 50 / 484 \\ &= 0.103 \text{ (must be } 0 \leq \rho \leq 1 \text{)} \end{aligned}$$

Effective tube hole Diameter [d^*]:

$$\begin{aligned} &= \max(dt - 2tt(E_t/E) \cdot (StT/S) \cdot (\rho); dt - 2tt) \\ &= \max(25.4 - 2 \cdot 3.05 (172742 / 183094) \cdot (0.1); 25.4 - 2 \cdot 3.05) \\ &= 25.0204 \text{ mm} \end{aligned}$$

Effective tube pitch [p^*]:

$$\begin{aligned} &= p / \sqrt{ (1 - 4 \cdot \min(A_l; 4 \cdot D_o \cdot p) / (\pi \cdot D_o^2)) } \\ &= 31.8 / \sqrt{ (1 - 4 \cdot \min(0.2; 4 \cdot 2571 \cdot 31.8) / (\pi \cdot 2571^2)) } \\ &= 32.3886 \text{ mm} \end{aligned}$$

Effective ligament efficiency for bending [μ^*]:

$$\begin{aligned} &= (p^* - d^*) / p^* \\ &= (32.4 - 25) / 32.4 \\ &= 0.2275 \end{aligned}$$

$h/p = 15.244095$; $\mu^* = 0.227493$; $E^* = 34513 \text{ N/mm}^2$

$E^*/E = 0.188502$; $v^* = 0.421746$

Note: As h/p (15.244) is > 2 , data values for $h/p = 2$ were used.

ASME Section VIII-2 Part 4.18.7.4 Step 4:

Shell coefficient [β_s]:

$$\begin{aligned} &= ((12(1 - v_s^2))^{1/4}) / \sqrt{ (D_s + t_s) t_s } \\ &= ((12(1 - 0.3^2))^{1/4}) / \sqrt{ (2666 + 139) 139 } \\ &= 0.0029 \text{ 1/mm} \end{aligned}$$



PV Elite 28 Licensee: SL User

FileName : 103-C2

Page 186 of 192

ASME TS Calc: DESIGN CASE

Case: 1 3:14p Apr 10, 2026

Shell coefficient [ks]:

$$= \beta_s \cdot E_s \cdot t_s^3 / (6(1 - \nu_s^2))$$

$$= 0.0029 \cdot 183094 \cdot 139^3 / (6(1 - 0.3^2))$$

$$= 262184528.0000 \text{ N/mm}^2 \cdot \text{mm}^2$$

Shell coefficient [λs]:

$$= 6(D_s + t_s) \cdot k_s / h^3 \cdot (1 + h \cdot \beta_s + (h \cdot \beta_s)^2 / 2)$$

$$= 6(2666 + 139) \cdot 262184528 / 484^3 \cdot (1 + 484 \cdot 0.0029 + (484 \cdot 0.0029)^2 / 2)$$

$$= 132391.4 \text{ N/mm}^2$$

Shell coefficient [δs]:

$$= D_s^2 / (4 \cdot E_s \cdot t_s) \cdot (1 - D_s \cdot \nu_s / (2(D_s + t_s)))$$

$$= 2666^2 / (4 \cdot 183094 \cdot 139) \cdot (1 - 2666 \cdot 0.3 / (2(2666 + 139)))$$

$$= 0.59868\text{E-}04 \text{ mm/kPa}$$

Parameter [ws]:

$$= p_s \cdot k_s \cdot \beta_s \cdot \delta_s (1 + h \cdot \beta_s)$$

$$= 1.04 \cdot 262184528 \cdot 0.0029 \cdot 0.59868\text{E-}04 (1 + 484 \cdot 0.0029)$$

$$= 114150.8047 \text{ mm}^2$$

Channel coefficient [βc]:

$$= ((12(1 - \nu_c^2))^{1/4} / \sqrt{(D_c + t_c)t_c})$$

$$= ((12(1 - 0.3^2))^{1/4} / \sqrt{(2670 + 48)48})$$

$$= 0.0050 \text{ 1/mm}$$

Channel coefficient [kc]:

$$= \beta_c \cdot E_c \cdot t_c^3 / (6(1 - \nu_s^2))$$

$$= 0.005 \cdot 183094 \cdot 48^3 / (6(1 - 0.3^2))$$

$$= 18664434.0000 \text{ N/mm}^2 \cdot \text{mm}^2$$

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:

$$CqDelta = CqTheta = CmTheta = CmDelta = 1$$

Channel coefficient [λc]:

$$= 6(D_c + t_c / \cos(\theta)) \cdot k_c / h^3 \cdot (CmTheta + h \cdot \beta_c \cdot (CmDelta + CqTheta) / 2 + (h \cdot \beta_c)^2 / 2 \cdot CqDelta)$$

$$= 6(2670 + 48 / \cos(0)) \cdot 18664434 / 484^3 \cdot (1 + 484 \cdot 0.005 \cdot (1 + 1) / 2 + (484 \cdot 0.005)^2 / 2 \cdot 1)$$

$$= 17188.5 \text{ N/mm}^2$$

Channel coefficient [δc]:

$$= D_c^2 / (4 \cdot E_c \cdot t_c) \cdot (1 - D_c \cdot \nu_c / (2(D_c + t_c)))$$

$$= 2670^2 / (4 \cdot 183094 \cdot 48) \cdot (1 - 2670 \cdot 0.3 / (2(2670 + 48)))$$

$$= 0.17292\text{E-}03 \text{ mm/kPa}$$

Channel coefficient [zetac]:

$$\text{zetac} = 0$$

Parameter [wc]:

$$= p_c \cdot k_c [\delta_c (CmTheta \cdot \beta_c + CqDelta \cdot h \cdot \beta_c^2) - \zeta_c (CmTheta + h \cdot CqTheta \cdot h \cdot \beta_c / 2)]$$

$$= 1.04 \cdot 18664434 [0.000172919 (1 + 0.005 \cdot 1 \cdot 484) - 0.000 (1/484 + 1 \cdot 0.005/2)]$$

$$= 57956.7070 \text{ mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 5:

Diameter ratio [K]:

$$= A / D_o$$

$$= 2980 / 2571$$

$$= 1.1591$$

Coefficient [F]:

$$= (1 - \nu^*) / E^* \cdot (\lambda_s + \lambda_c + E \cdot \ln(K))$$

$$= (1 - 0.42) / 34514 \cdot (132391 + 17188 +$$



PV Elite 28 Licensee: SL User

FileName : 103-C2

Page 187 of 192

ASME TS Calc: DESIGN CASE

Case: 1 3:14p Apr 10, 2026

$$183094 \cdot \ln(1.16) \\ = 2.9591$$

ASME Section VIII-2 Part 4.18.7.4 Step 6:

Moment on tubesheet due to pressures (Ps, Pt) [Mts]:

$$= Do^2/16 \cdot [(ps-1) \cdot (ps^2+1) \cdot Ps - (pc-1) \cdot (pc^2+1) \cdot Pt] \\ = 2571^2/16 \cdot [(1.04 - 1) \cdot (1.04^2 + 1) \cdot 14.3 - \\ (1.04 - 1) \cdot (1.04^2 + 1) \cdot 4.95] \\ = 288979.9062 \text{ N/mm}^2 \cdot \text{mm}^2$$

Moment acting on unperforated tubesheet rim [M*]:

$$= Mts + wc \cdot Pt - ws \cdot Ps \\ = 288980 + 57957 \cdot 4.95 - 114151 \cdot 14.3 \\ = -1054285.8750 \text{ N/mm}^2 \cdot \text{mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 7:

Maximum bending moment acting on periphery of tubesheet [Mp]:

$$= ((M*) - Do^2/32 \cdot F (Ps - Pt)) / (1 + F) \\ = ((-1054285) - 2571^2/32 \cdot 2.96 (14.3 - 4.95)) / (1 + 2.96) \\ = -1706740.5000 \text{ N/mm}^2 \cdot \text{mm}^2$$

Maximum bending moment acting on center of tubesheet [Mo]:

$$= Mp + Do^2/64 \cdot (3 + \nu) \cdot (Ps - Pt) \\ = -1706740 + 2571^2/64 \cdot (3 + 0.42) \cdot (14.3 - 4.95) \\ = 1590508.2500 \text{ N/mm}^2 \cdot \text{mm}^2$$

Maximum bending moment acting on tubesheet [M]:

$$= \max(| Mp |, | Mo |) \\ = \max(| -1706740 |, | 1590508 |) \\ = 1706740.5000 \text{ N/mm}^2 \cdot \text{mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 8:

Tubesheet Bending Stress at original thickness:

$$= 6 \cdot M / (\mu \cdot (h - hg')^2) \\ = 6 \cdot 1706740 / (0.23 (484 - 0)^2) \\ = 192147.5625 \text{ kPa}$$

Allowable Tubesheet Bending Stress [σ_{All}]:

$$= 2 \cdot S \\ = 2 \cdot 147989 \\ = 295977.47 \text{ kPa}$$

Tubesheet bending stress at final thickness [σ]:

$$= 6 \cdot M / (\mu \cdot (h - hg')^2) \\ = 6 \cdot 1734162 / (0.23 (391 - 0)^2) \\ = 295958.4375 \text{ kPa}$$

Required tubesheet thickness for Bending Stress [HreqB]:

$$= h + Cats + Catc \\ = 391 + 3 + 13 \\ = 406.7094 \text{ mm}$$

Required tubesheet thickness for given loadings (includes CA) [Hreq]:

$$= \max(HreqB; HreqS) \\ = \max(407; 254) \\ = 406.7094 \text{ mm}$$

ASME Section VIII-2 Part 4.18.7.4 Step 9:

Absolute pressure differential [| Ps - Pt |]:

$$= | Ps - Pt | \\ = | 14.3 - 4.95 |$$



PV Elite 28 Licensee: SL User

FileName : 103-C2

Page 188 of 192

ASME TS Calc: DESIGN CASE

Case: 1 3:14p Apr 10, 2026

$$= 9.331 \text{ N/mm}^2$$

Shear Stress check [τ limit]:

$$= 4 \cdot \mu \cdot h / D_o \cdot \min(0.8 \cdot S; 0.533 \cdot S_y)$$

$$= 4 \cdot 0.2 \cdot 484 / 2571 \cdot \min(0.8 \cdot 147989; 0.533 \cdot 239106)$$

$$= 17830.578 \text{ kPa}$$

| $P_s - P_t$ | is $\leq$ the Shear Stress limit (τ limit). Therefore, the required thickness for shear is for information only.

Minimum required thickness for Shear [H_{reqS}]:

$$= 1 / (4 \cdot \mu \cdot \min(0.8 \cdot S; 0.533 \cdot S_y)) \cdot (4 \cdot A_p / C_p) \cdot |P_s - P_t| + C_{ats} + C_{atc}$$

$$= 1 / (4 \cdot 0.2 \cdot 118391) \cdot (4 \cdot 4.95 / 8219) \cdot |14.3 - 4.95| + 16$$

$$= 253.5175 \text{ mm}$$

ASME Section VIII-2 Part 4.18.7.4 Step 10:

Note: For a given Shell thickness of: 139.000mm

Min. Shell len. adjacent to the tubesheet is: 1095.746 mm

Axial Shell Membrane Stress [σ_s, m]:

$$= D_s^2 / (4 \cdot t_s (D_s + t_s)) \cdot P_s$$

$$= 2666^2 / (4 \cdot 139 (2666 + 139)) \cdot 14.3$$

$$= 65078.9961 \text{ kPa}$$

Axial Shell Bending Stress [σ_s, b]:

$$= 6 \cdot k_s / t_s^2 \cdot [\beta S \cdot P_s + 6(1 - \nu^*) / E^* \cdot$$

$$D_o / h^3 \cdot (1 + h \cdot \beta S / 2) \cdot (M_p + D_o^2 / 32 \cdot (P_s - P_t))]$$

$$= 6 \cdot 58939352 / 139^2 \cdot [0.0029 \cdot 0.85 + 6(1 - 0.42) / 34514 \cdot$$

$$2571 / 484^3 \cdot (1 + 484 \cdot 0.0029 / 2) \cdot (-1706740 + 2571^2 / 32 \cdot$$

$$(14.3 - 4.95))]$$

$$= 272381.594 \text{ kPa}$$

Shell Membrane + Bending Stress [σ_s]:

$$= |\sigma_s, m| + |\sigma_s, b| \text{ (Should be } \leq 1.5 \cdot S_s)$$

$$= 337460.594 \text{ kPa (Should be } \leq 221983.11)$$

Note: For a given Channel thickness of: 48.000 mm

Min. Channel len. adjacent to the tubesheet is: 644.390mm

Axial Channel Membrane Stress [σ_c, m]:

$$= D_c^2 / (4 \cdot t_c (D_c + t_c)) \cdot \cos(\theta) \cdot P_t$$

$$= 2670^2 / (4 \cdot 48 (2670 + 48)) \cdot \cos(0) \cdot 4.95$$

$$= 67620.289 \text{ kPa}$$

Axial Channel Bending Stress [σ_c, b]:

$$= 6 \cdot k_c / t_c^2 \cdot [(\delta_c \cdot C_m \Delta \cdot \beta_c - \zeta_c \cdot C_m \theta) \cdot P_t -$$

$$6((1 - \nu^*) / E^*) \cdot D_o / h^3 \cdot (C_m \theta +$$

$$h \cdot \beta_c / 2 \cdot C_m \Delta) \cdot (M_p + D_o^2 / 32 \cdot (P_s - P_t))]$$

$$= 6 \cdot 4195784 / 48^2 \cdot [(0.000172919 \cdot 1 \cdot 0.005 - 0.000 \cdot 1) \cdot 4.95 -$$

$$6((1 - 0.42) / 34514) \cdot 2571 / 484^3 \cdot (1 +$$

$$0.005 \cdot 484 / 2 \cdot 1) \cdot (-1706740 + 2571^2 / 32 \cdot (14.3 - 4.95))]$$

$$= 155191.953 \text{ kPa}$$

Channel Membrane + Bending Stress [σ_c]:

$$= |\sigma_c, m| + |\sigma_c, b| \text{ (Should be } \leq 1.5 \cdot S_c)$$

$$= 222812.250 \text{ kPa (Should be } \leq 221983.11)$$

Step 11, The Cylinder-to-Tubesheet Junction is Overstressed.

Performing a simplified Elastic-Plastic calculation

(option 3) to reduce the overstress condition.

Modify E_s and/or E_c and recompute from Step 4 onwards.

Modified Elasticity [E_s]:



PV Elite 28 Licensee: SL User

FileName : 103-C2

Page 189 of 192

ASME TS Calc: DESIGN CASE

Case: 1 3:14p Apr 10, 2026

$$= E_s \cdot \sqrt{(1.5 \cdot S_s / \sigma_s)}$$

$$= 183094 \cdot \sqrt{(1.5 \cdot 147989 / 337461)}$$

$$= 148498.4 \text{ N/mm}^2$$

Modified Elasticity [Ec]:

$$= E_c \cdot \sqrt{(1.5 \cdot S_c / \sigma_c)}$$

$$= 183094 \cdot \sqrt{(1.5 \cdot 147989 / 222812)}$$

$$= 182752.7 \text{ N/mm}^2$$

Tubesheet Bending Stress at Original Thk., after the Elastic-Plastic iteration:

$$= 6 \cdot M / (\mu \cdot (h - h'g)^2)$$

$$= 6 \cdot 1680598 / (0.23 \cdot (484 - 0)^2)$$

$$= 189204.4 \text{ kPa (Should be } \leq 295977.5)$$



Note: The tubesheet is Not overstressed after the Elastic-Plastic iteration.
The design is acceptable. Recomputing tubesheet required thickness.

Required Tubesheet Thk., for Bending Stress after the Elastic-Plastic iteration [HreqB]:

$$= h + C_{ats} + C_{atc} = 391 + 3 + 13 = 407 \text{ mm}$$

Required Tubesheet Thk. after the Elastic-Plastic iteration (includes CA) [Hreq]:

$$= \max(H_{reqB}; H_{reqS}) = \max(407; 254) = 407 \text{ mm}$$

Tube Metal Area [At]:

$$= \pi (d_o - t) t$$

$$= \pi (25.4 - 3.05) 3.05$$

$$= 0.000 \text{ m}^2$$

Tube Weld Size Results per UW-20.3:

Tube Strength [Ft]:

$$= \pi \cdot t (d_o - t) S$$

$$= \pi \cdot 3.05 (25.4 - 3.05) 94543$$

$$= 20234.723 \text{ N}$$

Groove Weld Strength [Fg]:

$$= 0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w \text{ (but not } > F_t)$$

$$= 0.85 \cdot \pi \cdot 3 (25.4 + 0.67 \cdot 3) 94543$$

$$= 20234.7227 \text{ N}$$

Weld Strength Factor [fw]:

$$= S / S_w \text{ where } S_w = \min(S \text{ and } S_t)$$

$$= 94543 / 94543$$

$$= 1.000$$

Design Strength Ratio [fd]:

$$= 1 \text{ for full strength welds}$$

Fillet Weld Strength [Ff]:

$$= 0.55 \cdot \pi \cdot a_f (d_o + 0.67 \cdot a_f) S_w \text{ (but not } > F_t)$$

$$= 0.55 \cdot \pi \cdot 3 (25.4 + 0.67 \cdot 3) 94543$$

$$= 13432.5635 \text{ N}$$

Minimum Length of Weld [ar]:

$$= 2 \left(\sqrt{(0.75 \cdot d_o)^2 + 1.07 \cdot t \cdot (d_o - t) f_w \cdot f_d} - 0.75 \cdot d_o \right)$$

$$= 2 \left(\sqrt{(0.75 \cdot 25.4)^2 + 1.07 \cdot 3.05 \cdot (25.4 - 3.05) 1 \cdot 1} - 0.75 \cdot 25.4 \right)$$

$$= 3.652 \text{ mm}$$

Minimum Required Fillet Weld Leg [afr]: 1.8288 mm

Minimum Required Groove Weld Leg [agr]: 1.8288 mm



PV Elite 28 Licensee: SL User

FileName : 103-C2

Page 190 of 192

ASME TS Calc: DESIGN CASE

Case: 1 3:14p Apr 10,2026

Tube-Tubesheet Jt allowable, 20235 is >= tube strength 20235 N

Note: This tube-tubesheet joint is a Full Strength joint

Stress/Force summary for loadcase D3 corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet bend. stress	189204.4	<= 295977.5 kPa	Ok
Tubesheet shear stress	58099.0	<= 118391.0 kPa	Ok
Stress in Shell at Tubesheet	337460.6	<= 478235.3 kPa	Ok
Stress in Channel at Tubesheet	222812.2	<= 478235.3 kPa	Ok

Thickness results for loadcase D3 corr. (Psd,max + Ptd,max):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	406.709	500.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.829	3.000	Ok
Tube-Tubesheet Groove Weld Leg :	1.829	3.000	Ok

U-Tube Tubesheet results per ASME Div. 2 Part 4.18.7 2025

Results for 6 Load Cases:

Case#	--Reqd. Thk. + CA Tbsht Extnsn	----	Tubesheet Bend Allwd	Stresses Shear Allwd	Case Type	Pass/ Fail
D1uc	285.917	...	102857	295977	29838 118391	Ps+Pt D1 Ok
D2uc	486.326	...	280011	295977	86078 118391	Ps+Pt D2 Ok
D3uc	397.050	...	183015	295977	56240 118391	Ps+Pt D3 Ok
D1c	296.345	...	107752	295977	30824 118391	Ps+Pt-c D1 Ok
D2c	500.000	...	295200	295977	88923 118391	Ps+Pt-c D2 Ok
D3c	406.709	...	189204	295977	58099 118391	Ps+Pt-c D3 Ok
Max:	500.0000	... mm	0.997	0.751	(Str. Ratio)	

Load Case Definitions:

[Ps & Pt]:

Shell-side and Tube-side Design or Operating Pressures
derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the
Load Case Tables

[c]:

With or Without Corrosion Allowance

[D1, D2, D3]:

Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:

Design Load Case using the Minimum (Vacuum) Pressures (if specified)

Shell and Channel Stress Summary:

Case#	-- Integral Cyl. Shell	Reqd. Thk. Channel	Shell Stress	Stress Allwd	Channel Stress	Stress Allwd	Pass Fail
D1uc	...	...	65764	221983	316788	478235	Ok
D2uc	...	...	387994	478235	99199	221983	Ok
D3uc	...	...	322229	478235	217589	221983	Ok
D1c	...	...	71862	221983	332813	478235	Ok
D2c	...	...	409322	478235	110000	221983	Ok
D3c	...	...	337461	478235	222812	478235	Ok



PV Elite 28 Licensee: SL User

FileName : 103-C2

Page 191 of 192

ASME TS Calc: DESIGN CASE

Case: 1 3:14p Apr 10,2026

Max ... 0.856 0.980

Note:

Because there was no net pressure differential across the tubesheet due to the specified pressures psd,min and ptd,min; load case D4 was not processed as the resulting tubesheet bending stress would be zero.

Summary of Thickness Comparisons for 6 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	500.000	500.000	Ok
Tube thickness :	2.869	3.048	Ok
Tube-tubesheet fillet weld leg :	1.829	3.000	Ok
Tube-tubesheet groove weld leg :	1.829	3.000	Ok

Given Shell Thickness: 142.0000 mm

Given Channel Thickness: 50.0000 mm

Min Shell length of thk, (142.000) adj. to tubesheet: 1106.261 mm

Min Channel length of thk, (50.000) adj. to tubesheet: 657.185 mm

Note: This is a full strength Tube to Tubesheet Joint.

Tubesheet MAWP used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAWP	Ø shellside Stress Rat.	Shellside MAWP	Ø tubeside Stress Rat.
Tubesheet Bending Stress	13.641	1.000	14.274	0.997
Tubesheet Shear Stress	19.013	1.000	19.013	1.000
Shell Stress (Axial, Junction)	19.013	0.577	14.274	0.997
Tubesheet-Channel Junction	7.113	1.000	14.274	0.997
Tube Pressure Stress	25.101	1.000	15.198	1.000
Minimum MAWP	7.113		14.274	

Tubesheet MAPnc used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAPnc	Ø shellside Stress Rat.	Shellside MAPnc	Ø tubeside Stress Rat.
Tubesheet Bending Stress	14.405	0.997	15.227	1.000
Tubesheet Shear Stress	19.642	1.000	19.642	1.000
Shell Stress (Axial, Junction)	19.642	0.420	19.642	1.305
Tubesheet-Channel Junction	9.692	1.000	19.642	1.305
Tube Pressure Stress	30.530	1.000	21.069	1.000
Minimum MAPnc	9.692		15.227	

(*) All load cases were analyzed to compute the MAWP for determining the test pressure.

Tubesheet MDMT Calculations:

Shell Side MDMT calculation:

Governing thickness on the shell side per figure UCS-66.3 (e):

= max(tubesheet thk/4, min(tubesheet thk, shell thickness))
 = max(500/4, min(500, 142))
 = 142.000 mm



PV Elite 28 Licensee: SL User

FileName : 103-C2

Page 192 of 192

ASME TS Calc: DESIGN CASE

Case: 1 3:14p Apr 10, 2026

MDMT reduction ratio per UCS 66 (b)(1)(b):

= min(ps/Shellside MAPnc, 1)

= min(14.3/15.2, 1)

= 0.938

Warning:

The minimum metal temperature could not be computed because the governing thickness for this component was greater than 100.000 mm.

This component must be impact tested.

Channel Side MDMT calculation:

Governing Channel side thickness per figure UCS-66.3 (e):

= max(tubesheet thk/4, min(tubesheet thk, channel thickness))

= max(500/4, min(500, 50))

= 125.000 mm

MDMT reduction ratio per UCS 66 (b)(1)(b):

= min(pt/Tubeside MAPnc, 1)

= min(4.95/9.69, 1)

= 0.511

Warning:

The minimum metal temperature could not be computed because the governing thickness for this component was greater than 100.000 mm.

This component must be impact tested.

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